BR234

SAP S/4HANA Service: Subscription Order Management

PARTICIPANT HANDBOOK
INSTRUCTOR-LED TRAINING

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Typographic Conventions

American English is the standard used in this handbook.

The following typographic conventions are also used.

This information is displayed in the instructor's presentation	
Demonstration	
Procedure	
Warning or Caution	
Hint	
Related or Additional Information	
Facilitated Discussion	
User interface control	*Example text*
Window title	*Example text*

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Course Overview

TARGET AUDIENCE
This course is intended for the following audiences:

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UNIT 1 Introduction into Subscription Order Management (SOM)

Lesson 1

Introduction in Subscription Business 3

Lesson 2

Architecture of SAP S/4HANA Service SOM 7

UNIT OBJECTIVES

- Understand Business Examples for Subscription Business
- Understand Subscriptions Change Economy
- Explain our Industry examples
- Provide and Overview of SAP Billing and Revenue Innovation Management
- Understand an Overview of SOM Processes
- Understand the business example Cloud Selection Service

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Unit 1
Lesson 1

Introduction in Subscription Business

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Understand Business Examples for Subscription Business
- Understand Subscriptions Change Economy
- Explain our Industry examples

Introduction to Subscription Business

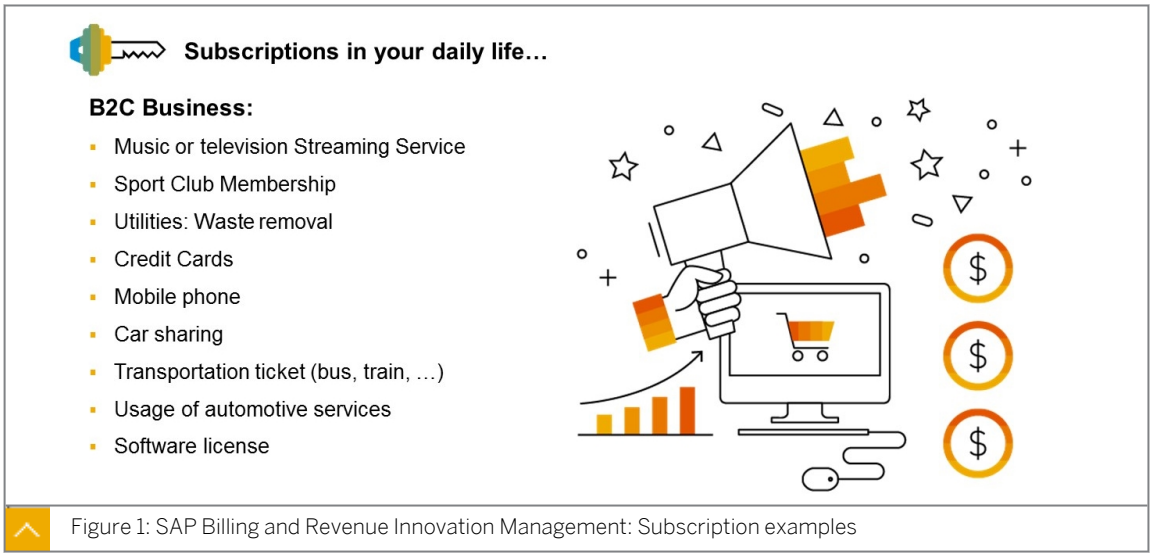

Figure 1: SAP Billing and Revenue Innovation Management: Subscription examples

Subscription business is found as well in B2B Business. Here are some examples:

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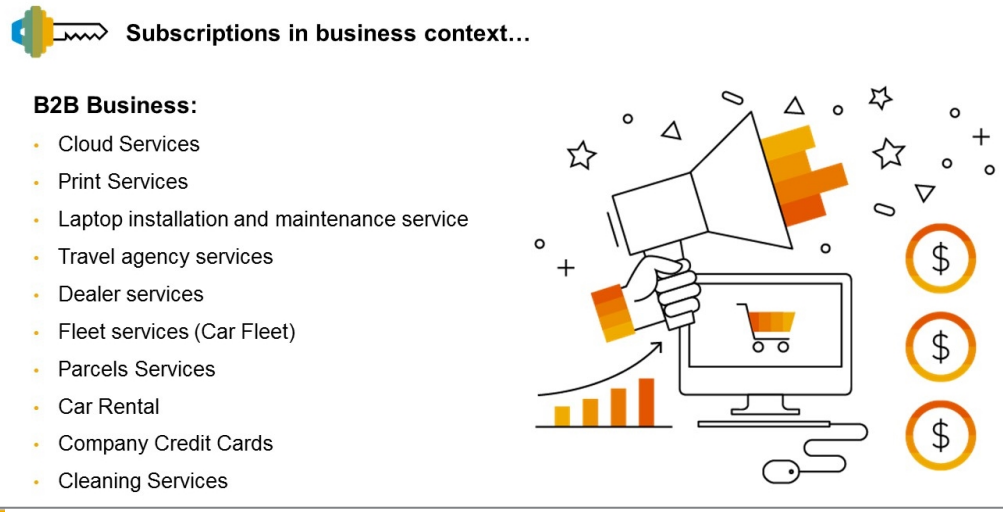

Figure 2: Business subscription examples

Subscriptions Change Economy

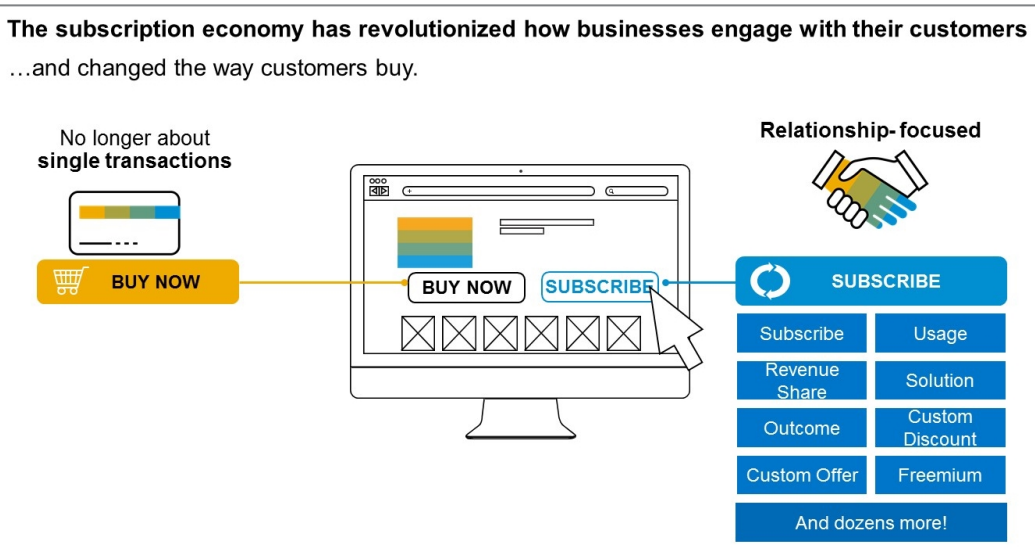

Figure 3: Subscription Economy Market Trends

These innovations in business models are with the singular aim of retaining customers by building long term relationships with them and not just focusing on single one-off transactions.

In the following we analyze some industry examples that use SAP Billing and Revenue Innovation Management (BRIM):

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Industry Examples

Parcel X is postal company in North America. Parcel X customer are B2B Business Customer with customer hierachies:
Online Shop New York, Fresh Food New York, Postal Shop New York, etc. Master Agreement for customer groups are realised in BRIM Solution.

- Postal industry needs a configurable price model with characteristics:
  - Surcharge (CO2, Insurance, destination)
  - Fees depending on delivery zones
  - Dimensions of goods (weight, length, …)
  - Channels (Road, Airfreight,…)
  - Servicelevel (Tracking level, …)

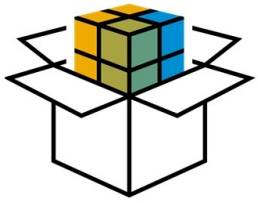

Figure 4: Business Example Postal

The SAP Billing and Revenue Innovation Management (BRIM) solution allows you to:

- Introduce industry and/or customer-specific attributes
- Use industry-specific pricing models
- Scale the Solution architecture to high complexity

Railway X is operating a large railway net in Europe. The company offers different tariff types for different customer groups: Prepaid Tariffs, subscription based discounting (e.g. Flat Rate or 50% price reduction)

Revenue sharing with other operating companies like local Bus Companies, local taxis, Railway operating companies in other countries is daily work.

- Usage is represented by a journey from point A to point B.

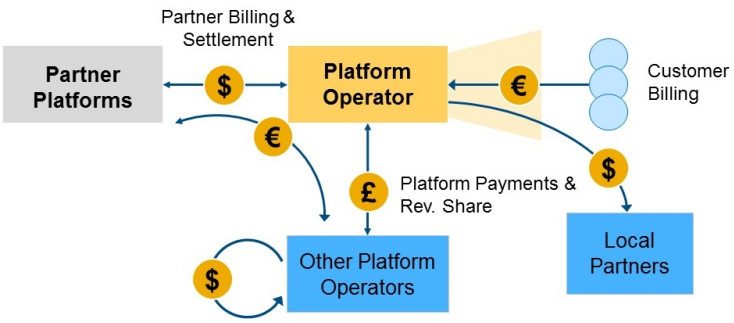

Figure 5: Business Example Railway B2B Postpaid

The SAP BRIM solution allows you to:

- Handle partner contract terms with just as much flexibility as customer pricing and using the same toolset and skills
- Interlink customer pricing with partner revenue share and commissioning agreements to correctly model your business
- Tailor your offers to partners by allowing special incentives for your most valuable partners while still maintaining the overall manageability of your pricing catalog

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- Rationalize your infrastructure by calculating and settling amounts due to partners through the same rating and billing solution as for customer transactions
- Gain partner loyalty and trust with full transparency for partners through customizable detailed reporting and prompt payments

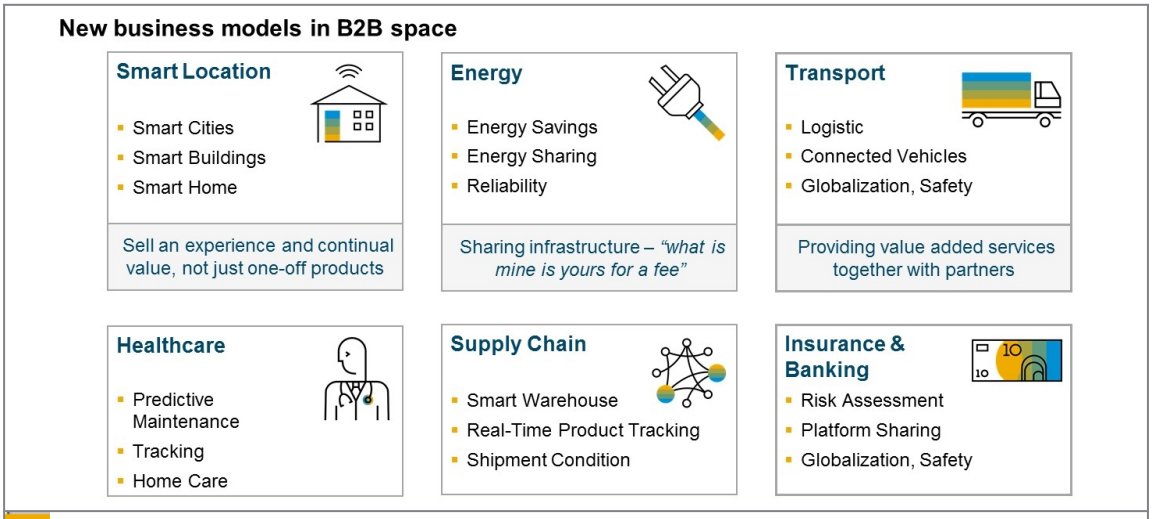

Figure 6: Subscription Economy Market Trends

Various new business models have sprung from this new engagement with customers and partners. One example is sharing infrastructure and enriching partners to serve customers better.

LESSON SUMMARY

You should now be able to:

- Understand Business Examples for Subscription Business
- Understand Subscriptions Change Economy
- Explain our Industry examples

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Architecture of SAP S/4HANA Service SOM

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Provide and Overview of SAP Billing and Revenue Innovation Management
- Understand an Overview of SOM Processes
- Understand the business example Cloud Selection Service

SAP Billing Revenue innovation Management

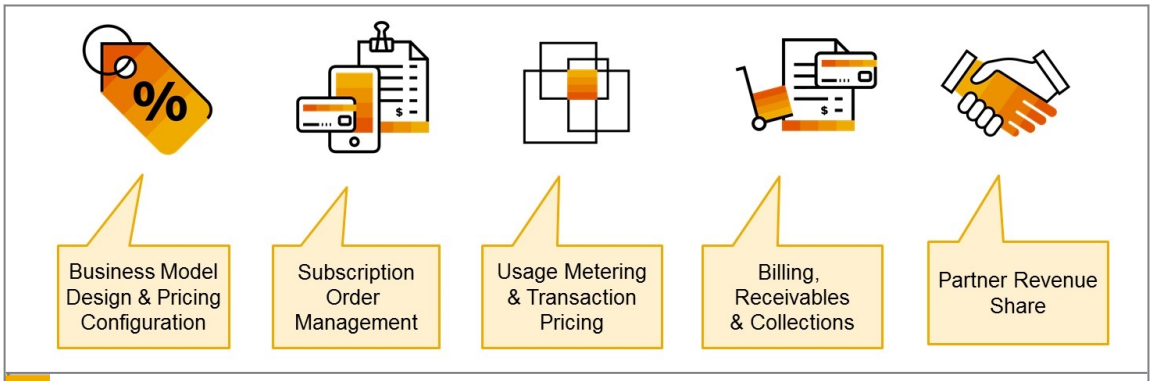

Figure 7: SAP BRIM Overview

Key Capabilities

Monetize new business models: Move from product sales to selling services and outcomes based on subscriptions and real-time usagebased services

Support your business model: Support B2B/B2C/B2B2C models in either a prepaid or postpaid strategy leveraging business rules–based automation for invoicing and accounting processes

Support multisided revenue models: Unlock the revenue potential of your ecosystem, including with transparent revenue-sharing models

Scale to the volumes of the future: Support digital business, the IoT, and connected devices with high-volume automated and transparent processing

Figure 8: SAP Billing and Revenue Innovation Management (BRIM) product description

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With SAP Billing and Revenue Innovation Management, service providers can bring innovative pricing offers to market faster than the competition with an intuitive user interface that requires zero coding.

The solution supports prepaid, postpaid and hybrid models to monitor customer accounts in real-time and improve customer sentiment by giving subscribers control over their spending.

Pricing for complex B2C scenarios, such as family plans, are supported out-of-the-box, as are B2B scenarios like credit pooling and sophisticated tiered models. Calculation of partner revenue share can be configured right along with customer pricing.

Marketing and sales are empowered to craft pricing models that reward loyalty, offer partners flexible terms, and ensure accurate settlement.

Finally, operations can consolidate pricing for multiple services and multiple lines of business on a single, high-performance platform that dynamically scales to extreme transaction volumes with 99.999% high availability.

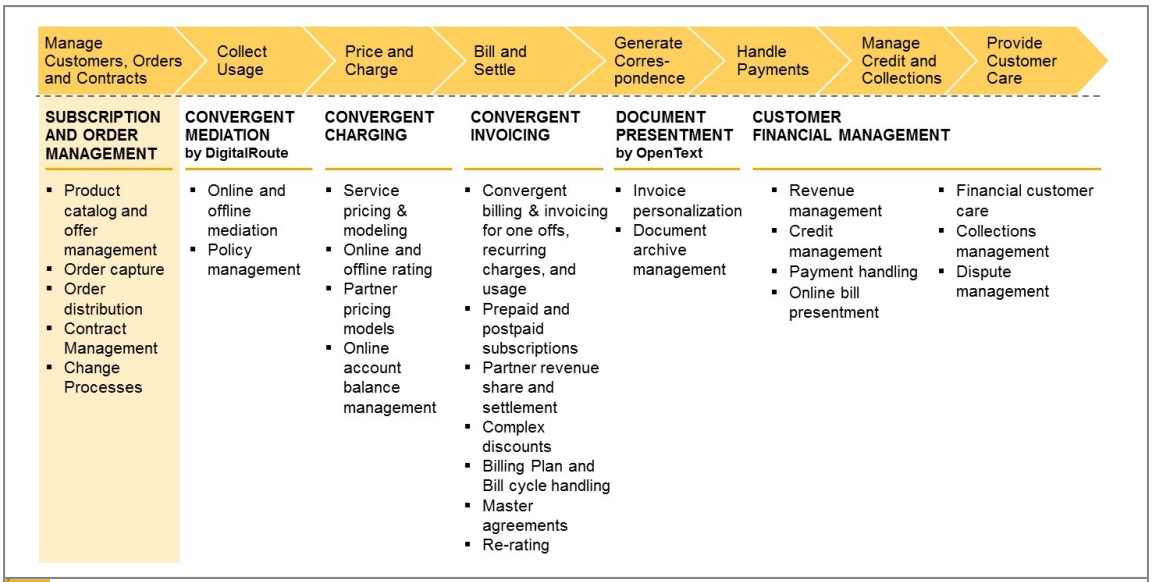

Figure 9: SAP BRIM Solution Landscape

Overview Subscription Order Management (SOM) Processes

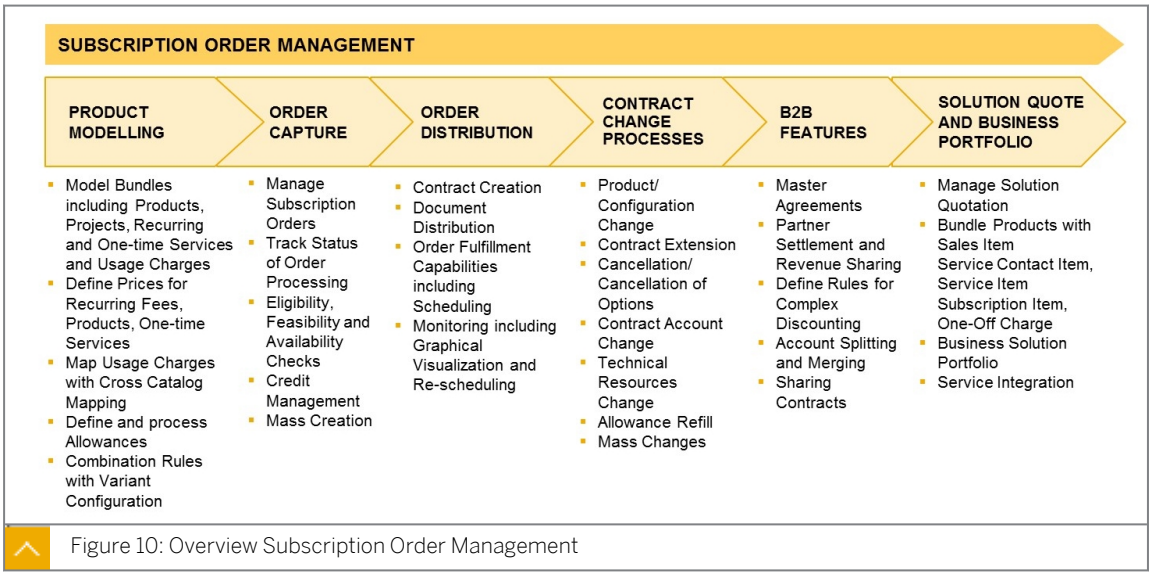

Figure 10: Overview Subscription Order Management

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The subscription order management component of SAP Billing and Revenue Innovation Management (BRIM) helps you explore new B2C or B2B markets or extend market opportunities, by addressing new customer segments in new verticals and geographies.

Additional benefits are:

- Design new subscription offerings that are instantly available for quoting and ordering across all channels.
- Handle the complexity of dynamically bundled service packages, to ensure that what is ordered can be delivered and billed with subscription order capture and automated management of contract changes, renewals, and extensions.
- Manage B2B master agreements and hierarchies with large enterprise customers, including specific terms for pricing, shared allowances or credit pooling, who can order what, volume discounts, and invoicing.
- Give your service agents a 360° view of customer interactions across the entire order-to-cash process, allowing them to provide prompt and expert service with a personal touch.

Looking closer into the functional areas of Subscription and Order Management, the sales process is located in the SOM system which means, campaign, offer, order and contract management are managed in the same system. There are different kinds of contracts which will be covered throughout this training. There are subscription contracts, partner agreements, and master agreements.

The SOM also provides master data, for example, business partners, contract accounts, and products. Change processes to existing contracts are managed in the SOM system and, last but not least, the fulfillment is triggered from the SOM system.

All of these aspects will be covered in later units in detail.

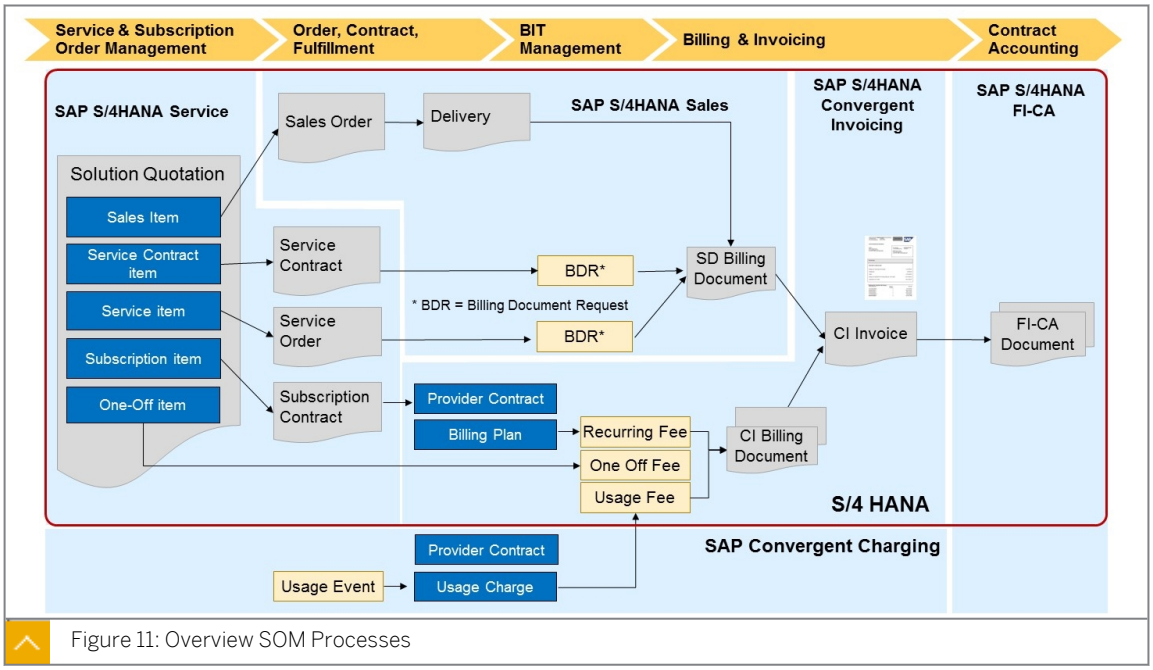

Figure 11: Overview SOM Processes

Subscription order management is part of SAP S/4 HANA Service. In combination with a solution quotation, the link between sales and distribution is triggered.

As a business example for our training we use a Cloud Selection Service:

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Cloud Selection Service

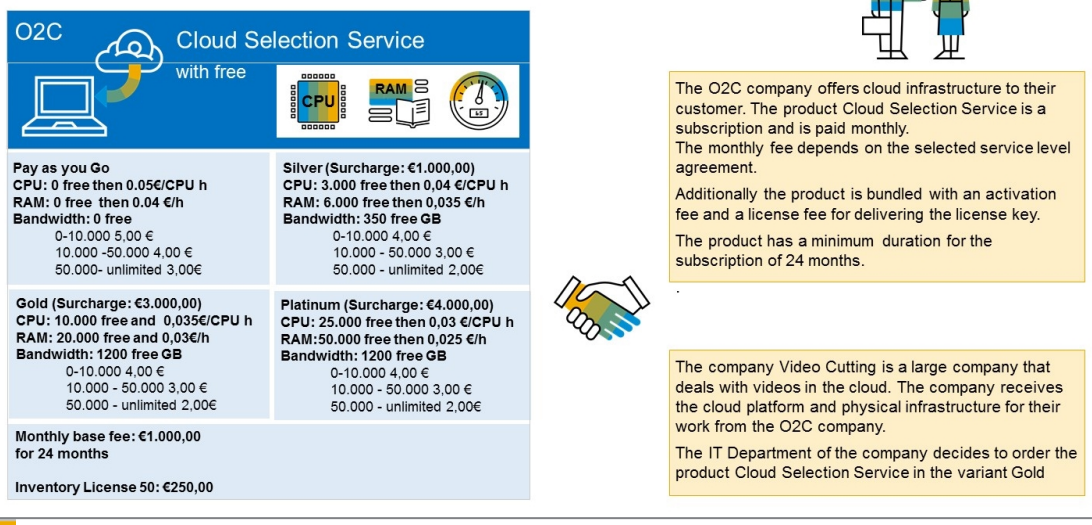

Figure 12: Business Example Cloud Service Provider

The O2C company offers cloud infrastructure to their customers. The product Cloud Selection Service is a subscription and is paid monthly.

The monthly fee depends on the selected service level agreement. Four options are possible: Pay As You Go, Silver, Gold, and Platinum. Depending on the selected service level, the customer can consume a number of CPU hours, RAM hours, and Gigabytes of bandwidth for free.

Additionally the product is bundled with an activation fee and a license fee for delivering the license key. The customer can choose between License50 and License100. The product has a minimum duration for the subscription of 24 months.

We will follow this Business Example with Cloud Service Provider in the exercises of the training.

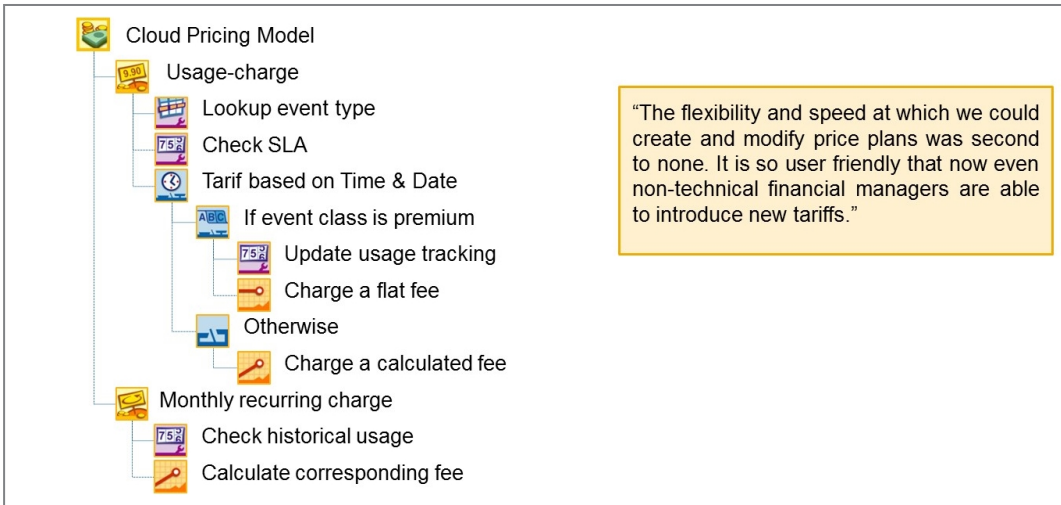

Figure 13: Express Your Business Model And Evolve It Rapidly

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Pricing and charging model design in SAP Billing and Revenue Innovation Management (BRIM) is done with decision trees (simplified example graphic on slide). Decision trees allow for truly flexible expression of configurable pricing and charging logic. They are easy to learn and easy to use.

This is not a "fill in the blanks" model, where the pricing logic is predefined and all you can do is set price levels or tiers. Some competing solutions may tout their simple UI, but it often greatly constrains the type of pricing models that you can implement. With such solutions when you need to make significant changes, that were not covered in the initial vendor-hardcoded design, then you either have to get your hands dirty by custom-coding with lengthy development cycles or worse, you have to go back to the vendor for a change request.

In the SAP BRIM billing solution, pricing and charging logic is built up by assembling decision trees containing simple components.

These components are like simple building blocks that you can put together in many different ways. There are no constraints to the depth of the tree or the way in which you choose to sequence the different decision building blocks. In this way you can create exactly what you need, without requiring custom coding or change requests from SAP.

Multisided Revenue Models: This solution allows for both customer pricing and partner settlement to be managed and calculated in the same engine, using the exact same approach to pricing configuration. Competitive products either do not handle partner-centric pricing at all or handle it with a disjointed, separate solution. This solution requires different configuration expertise and comes with the risk that customer pricing and partner settlement cannot be properly maintained in sync.

Charging Flexibility: The charging model design offers true charging convergence and can manage prepaid, postpaid, and hybrid charging models. The determination of the account/wallet or allowance to be charged is made using flexible decision tree logic and can change charging event-by-event, based on transaction characteristics. Competitive solutions tend to have a much more static and inflexible method of assigning charges to customer or partner balances.

Service Control with Allowances: This solution includes powerful new capabilities for stacking and sharing allowances/entitlements/quote. These are vital to allow service providers to give subscribers, families, and businesses full control over their spending, while permitting the rapid introduction of promotional packages (data, minutes, and service credits).

Cloud Selection Service Industry Examples

Telcom ONE Business Customer is a large Asian telecommunication provider with the largest 4G LTE data network.

- Telcom ONE has a high volume of end customer. Core business are prepaid (80%) telecommunication services.
- Telcom ONE needs realtime rating for prepaid and postpaid (credit check handling) services and a multiple-scalable architecture for availability, volume and country-size.
- Family and other group tariffs are in place. Corporate tariffs also offered.
- Products: Data at high speed, Voice, SMS, free Smart Phones with specific plans (only small security deposit necessary).

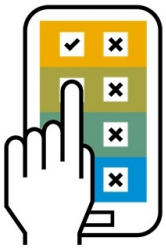

Figure 14: Business Example: B2C Telecommunication Prepaid

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As you bring partner solutions and reseller partners along in multisided business models, revenue capture and revenue share processes must be interlinked to ensure transparent win-win relationships.

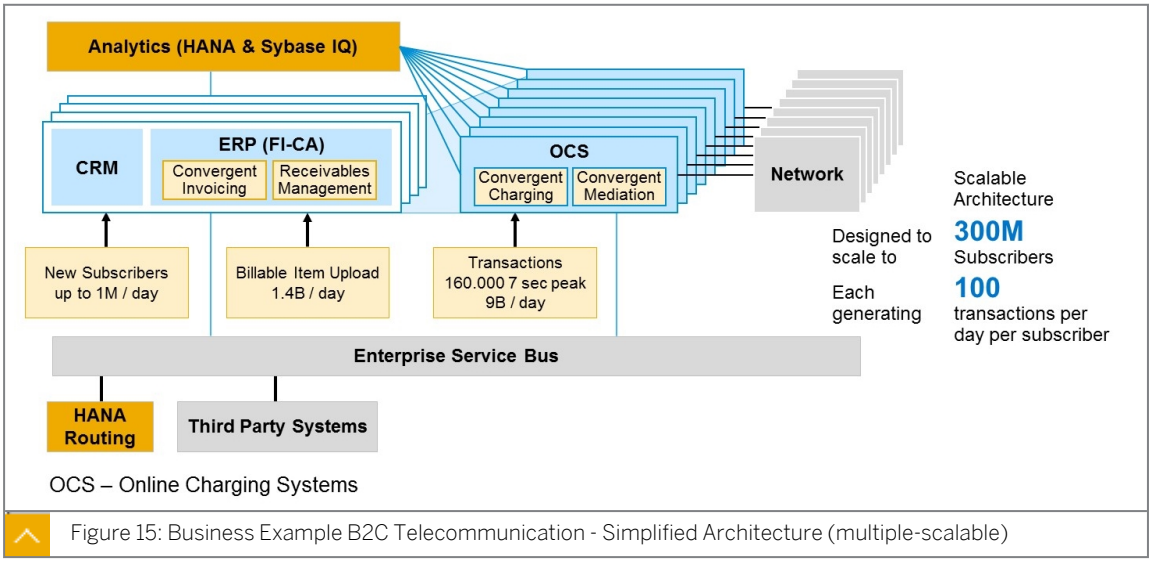

Figure 15: Business Example B2C Telecommunication - Simplified Architecture (multiple-scalable)

LESSON SUMMARY

You should now be able to:

- Provide and Overview of SAP Billing and Revenue Innovation Management
- Understand an Overview of SOM Processes
- Understand the business example Cloud Selection Service

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Learning Assessment

1. In this training context, SOM is the abbreviation for ...?

   Choose the correct answer.

   - ☐ A Sales and Order Management
   - ☐ B Search and Order Management
   - ☐ C Service and Order Management
   - ☐ D Subscription Order Management

2. Which of the following is SOM part of?

   Choose the correct answer.

   - ☐ A SAP S/4HANA FI-CA
   - ☐ B SAP S/4HANA Service
   - ☐ C SAP S/4HANA Sales and Distribution
   - ☐ D SAP Convergent Charging

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Learning Assessment - Answers

1. In this training context, SOM is the abbreviation for ...?

   Choose the correct answer.

   - [ ] A Sales and Order Management
   - [ ] B Search and Order Management
   - [ ] C Service and Order Management
   - [x] D Subscription Order Management

   That is correct! In this training context, SOM is the abbreviation for Subscription Order Management.

2. Which of the following is SOM part of?

   Choose the correct answer.

   - [ ] A SAP S/4HANA FI-CA
   - [x] B SAP S/4HANA Service
   - [ ] C SAP S/4HANA Sales and Distribution
   - [ ] D SAP Convergent Charging

   That is correct! SOM is part of SAP S/4HANA Service.

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UNIT 2

Master Data: Business Partner and Contract Account

Lesson 1

Business Partner and Organizational Structure 17

Lesson 2

Contract Account 23

UNIT OBJECTIVES

- Understand the Business Partner Concept
- Explain the Business Partner Role Concept
- Organizational Structure
- Describe Contract Account
- Describe Master Data in BRIM

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Unit 2
Lesson 1

Business Partner and Organizational Structure

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Understand the Business Partner Concept
- Explain the Business Partner Role Concept
- Organizational Structure

Business Example

You are responsible for maintaining business partner, master data in your company. You will need to change an existing contract account for an existing business partner.

Business Partner Concept

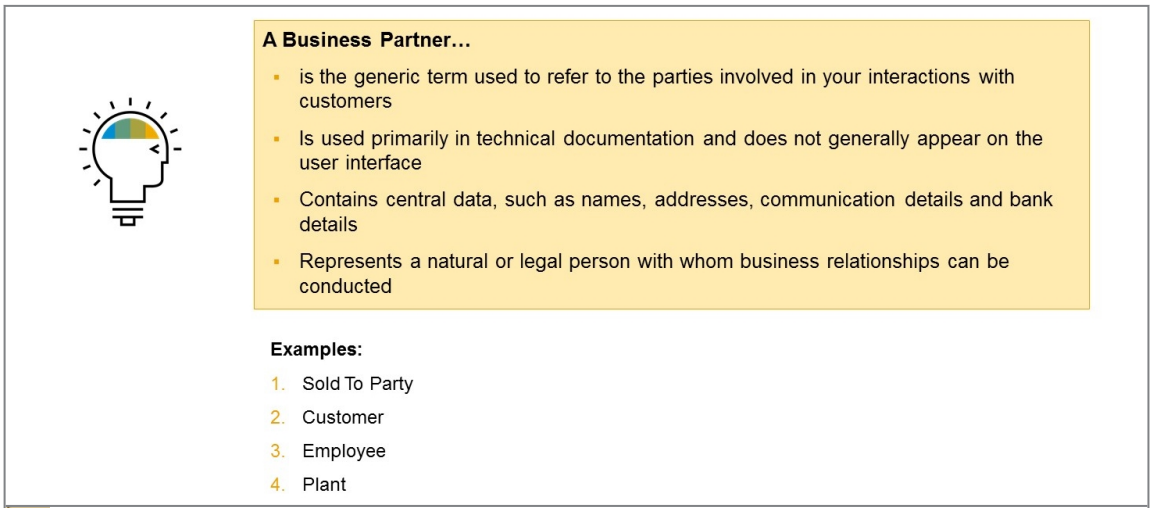

Figure 16: Definition Business Partner

Business partner is the generic term used to refer to the parties involved in your interactions with customers. Business partners are created in the app, Manage Business Partner Data. In general terms a business partner is called an account.

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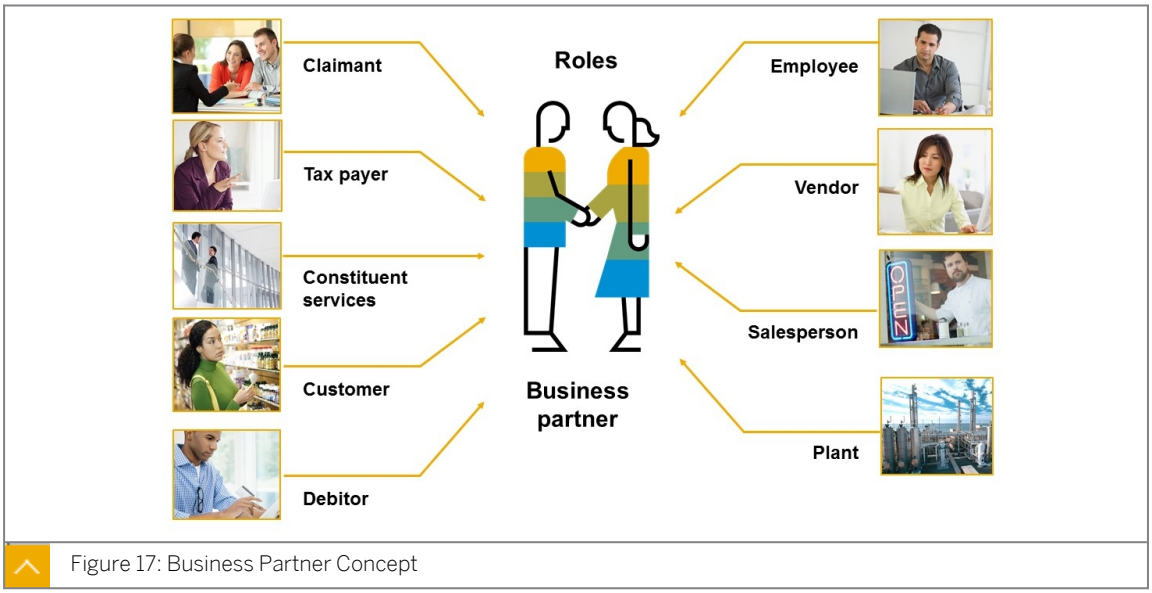

Figure 17: Business Partner Concept

The term business partner is used primarily in technical documentation and does not generally appear on the UI.

- A business partner contains central data, such as names, addresses, communication details, and bank details.
- A business partner represents a natural or a legal person with whom business relationships can be conducted.
- A business partner is used in the roles Sold-to-Party / Contract Partner.

A business partner role, for example, contact, sold-to party or supplier, is used to classify a business partner in business function. The roles that are assigned to a business partner reflect the functions it has and the business transactions in which it is likely to be involved.

The classification of a business partner as a natural person is called the business partner category. Examples of this are, a private person, a group or an organization, such as a legal person or part of a legal person, for example, a department.

A group specifies a shared living arrangement, a married couple, or an executive board. An organization represents units such as a company, a department of a company, a club, or an association. In addition to a legal person, parts of a legal person can be mapped as a business partner.

Business partner type refers to the classification of B2C customers and B2B customers. In our example, the company Headquarter is defined as Business Customer.

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Business Partner Role Concept

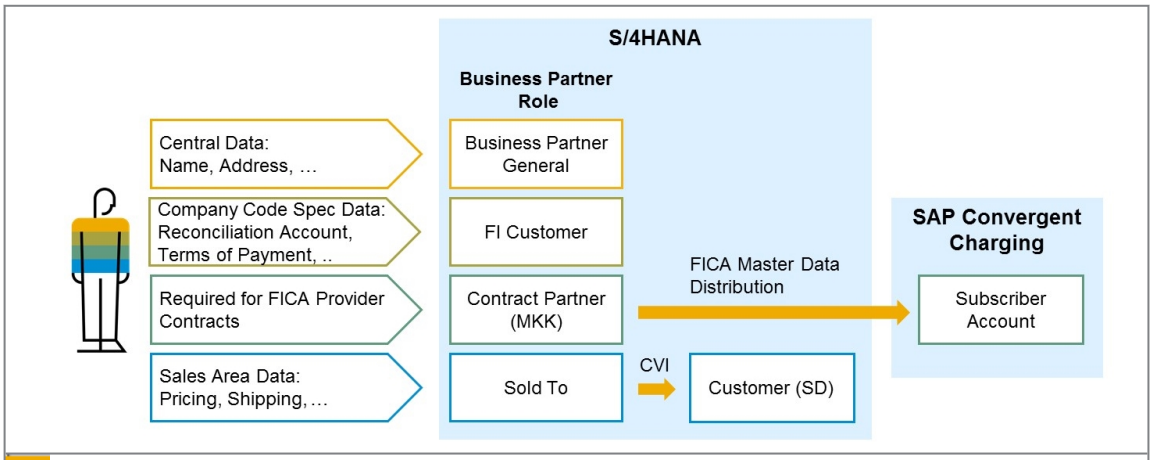

Figure 18: Business Partner in SOM

The business partner is created as a central object in SAP S/4HANA. Depending on the business requirements, different roles can be assigned to the business partner.

For Subscription Order Management (SOM), as a part of SAP Billing and Revenue Innovation Management (BRIM), using provider contracts in SAP Contract Accounts Receivables and Payable (FI-CA), the role MKK is required for the business partner. Business partners having this role will also be replicated to SAP Convergent Charging as subscriber accounts.

If processes in SD are involved, the Sold To Role is required. The SD customer is automatically created in the background by customer vendor integration in S/4HANA. Business partners represent all parties involved in business transactions, such as persons, groups of persons, and organizations (customer, debtor, creditor, or supplier).

Activities

Create business partners using the Manage Business Partner app on the SAP Fiori Launchpad.

Using this business role, you can search for and create business partners on the master data page, using a corporate account, an individual account (for creation), and accounts (for searching). You must assign the appropriate role(s) to the account as below:

- For business partners who should be used as contract partners in subscription order processing, add the role Customer (FLCU01) or any other role belonging to this category.
- For business partners associated with sales processes executed in SAP S/4HANA, add the role FI Customer (FLCU00).
- For business partners working with provider contracts in SAP Contract Accounts Receivables and Payable (FI-CA), add the role Contract Partner (MKK).
- When working in a partner settlement scenario, add the role Vendor (FLVN00).

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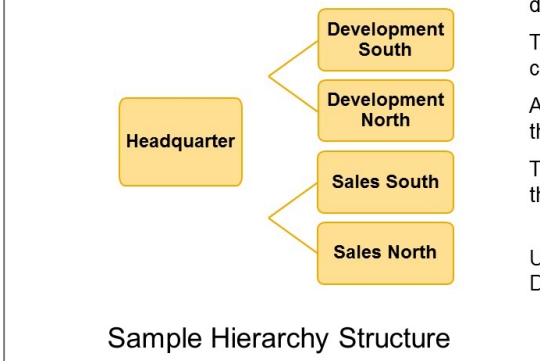

Business Customer “Headquarter” consists of 4 departments.

The development departments shall get special conditions for a cloud storage.

A Master agreement for this hierarchy is agreed between the Headquarter and the service provider.

The Headquarter is the owner of the master agreement, the allowed contract partner is the designers department.

Use a customer hierarchy in S/4 HANA Sales and Distribution for this use case.

Figure 19: Customer Hierarchy: Sample Use Case

Customer Hierarchy Maintenance: TA VDH1N

Define the Organizational Structure

With an Organizational Structure

- You can define your Marketing, Sales and Service Structure
- Characterize your Organizational units with organizational and general attributes
- Set up determination rules
- You can Assign Employees

In SOM Context

- Is needed to create all relevant SOM Transactions

Examples:

1. Sales Structures: Online and onsite business
2. Service Regions: MEE, North America,…
3. Financial Structure: Company Codes

Figure 20: Definition Organizational Structure

The organizational structure represents your company structure. Billing relevance data and organizational determination setup is linked to this structure.

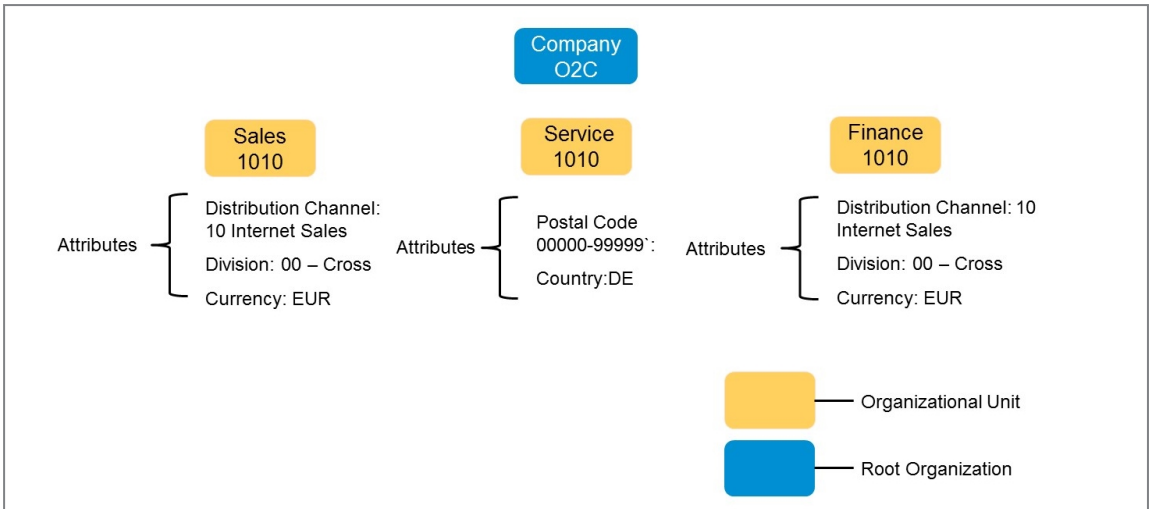

Figure 21: Example Organizational Structure B2B

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Organizational Structure for Subscription Order Management (SOM) - Steps to proceed:

- Define sales organization, if no SD organizational structure is in place
- Define service organization, for example,. for solution quotation
- Define organizational attributes as currency, distribution channel, division, and postal codes
- Transfer sales and distribution organizational structure to SOM (Customizing)

This organizational structure represents the current training environment.

An organizational unit needs to be set up with different attributes, as shown in the slides. Mandatory attributes are; the distribution channel, the division, and the document currency. These enable the Sales Organizational view and the **Organizational Determination** for the overall SOM process.

The Service Organization for SOM purpose is necessary to set up the solution quotation process, as service orders are as a result of it.

LESSON SUMMARY
You should now be able to:

- Understand the Business Partner Concept
- Explain the Business Partner Role Concept
- Organizational Structure

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Contract Account

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Describe Contract Account
- Describe Master Data in BRIM

Contract Account

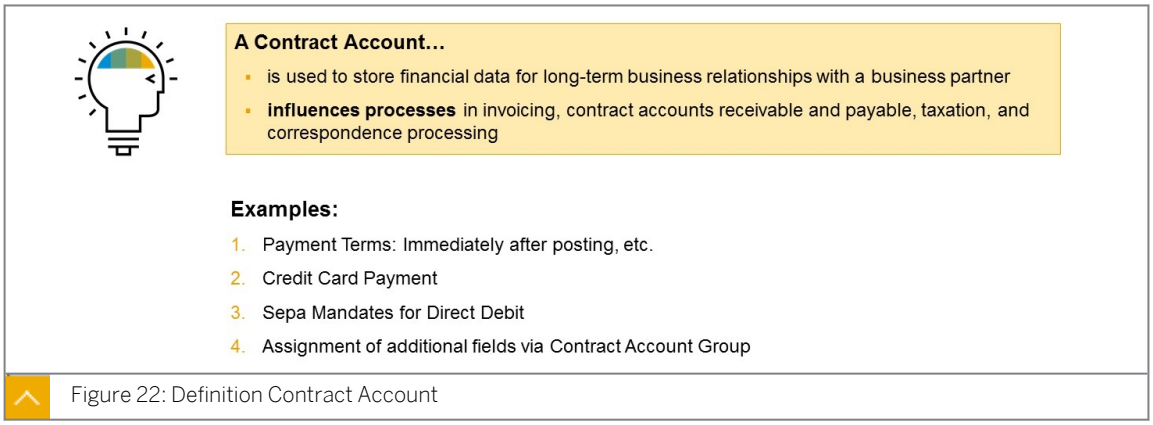

Figure 22: Definition Contract Account

The Contract account is used to store controlling data for long term business relationships with a business partner. The data contained in a contract account influences the processes in invoicing, contract account and receivable and payable, taxation, and corresponding processes. In the case of business transactions to a contract account, invoicing documents for the subscription contract, as an example, are submitted to the active Contract Accounts Receivable and Payable (FI-CA). When processing a business transaction, such as subscription order or subscription contract, you can make sure the system determines a suitable contract account.

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An account in which posting data for contracts or contract items are processed in Contract Accounts Receivable and Payable (FI-CA). In addition you define and maintain master data for Business Partner-specific procedures, such as payment methods, that can be defined and applied during further processing of the postings.

Distribution Contract Account

Figure 23: Contract Account

In SAP S/4HANA, only the contract account is used both in FI-CA and in Subscription Order Management (SOM). Business agreements are no longer required and therefore no longer available.

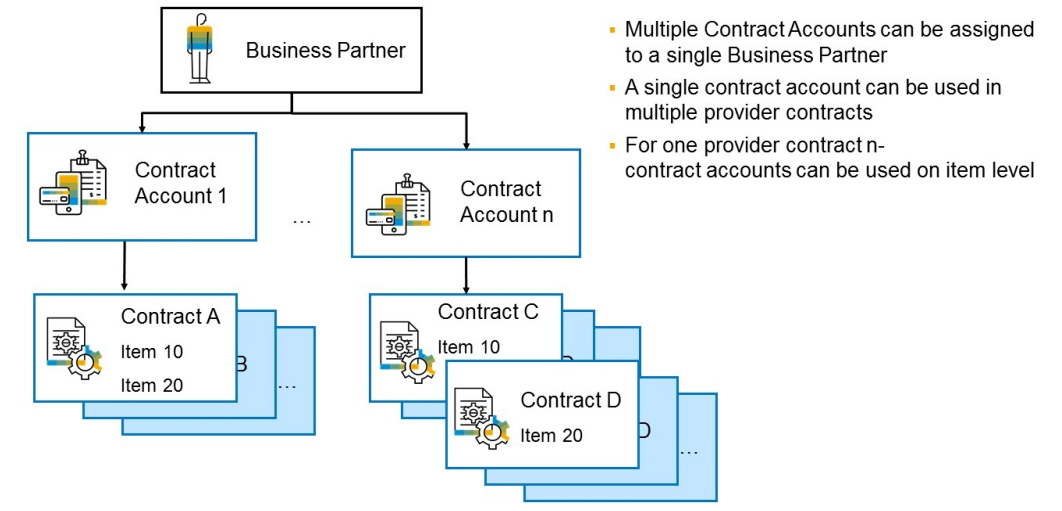

Figure 24: Relation Business Partner - Contract Account - Contract

A business partner can be assigned to more than one contract account to store different payment information for different use cases. For each item in a subscription order / contract a different contract account related to the business partner can be assigned.

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Master Data in SAP Billing and Revenue Innovation Management

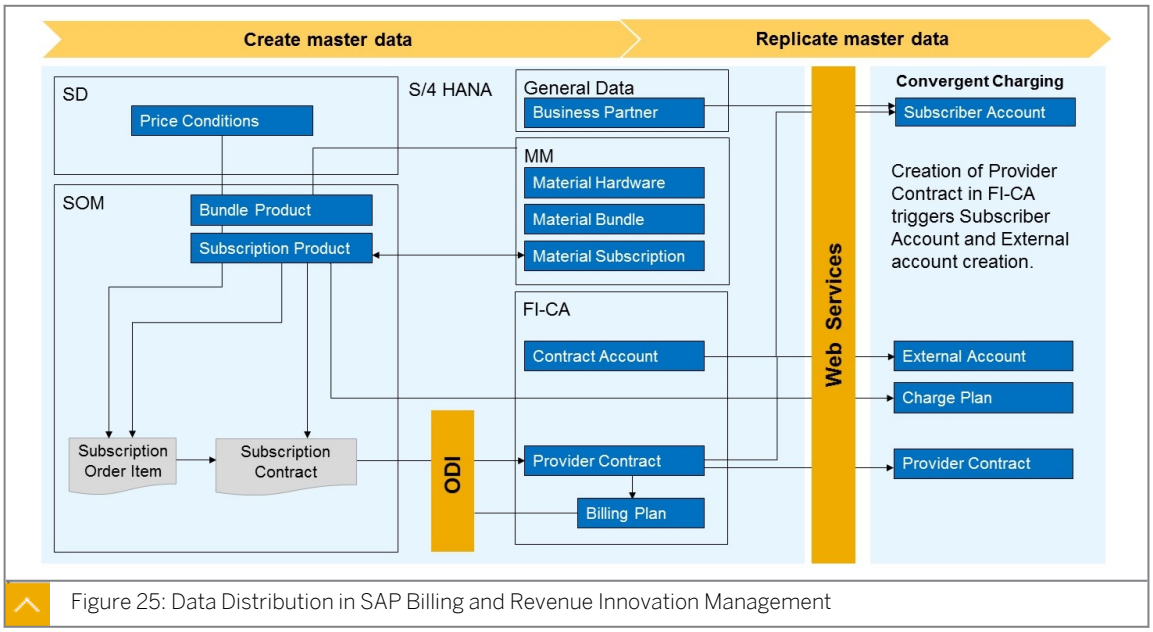

Figure 25: Data Distribution in SAP Billing and Revenue Innovation Management

Several master data objects are replicated between SAP S/4HANA and SAP Convergent Charging. Similarly, invoice products for rate plans or fees, for example, Cloud Selection Service can be exchanged between these systems.

In addition to that, the following Subscription Order Management (SOM) objects can be replicated to Contract Accounts Payable and Receivable (FI-CA) of SAP S/4HANA and SAP CC:

- Business Partner, called subscriber accounts in SAP CC
- Contract Account in FI-CA called external accounts in SAP CC

The subscription contract created for the IT department is replicated uni-directional from SAP SOM to FI-CA of SAP S/4HANA and to SAP CC. The contract is sent to the common object layer in FI-CA, from where the contract generation in SAP FI-CA itself, as well as in SAP CC, is triggered. Only in the case where the contract can be stored in both systems, will the contract be created.

On the SAP CC site, each provider contract is linked to the corresponding charge plans defined in the Cross Catalog Mapping.

LESSON SUMMARY

You should now be able to:

- Describe Contract Account
- Describe Master Data in BRIM

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Learning Assessment

1. Which of the following roles does a Business Partner need when FI-CA is used as a sub-ledger for SAP S/4HANA SOM?

   Choose the correct answer.

   - [ ] **A** MKK
   - [ ] **B** FLUC00
   - [ ] **C** General

2. The FI-CA Master Object Contract Account includes a parameter, Micro Services, for payment and dunning control.

   Determine whether this statement is true or false.

   - [ ] True
   - [ ] False

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Learning Assessment - Answers

1. Which of the following roles does a Business Partner need when FI-CA is used as a sub-ledger for SAP S/4HANA SOM?

   Choose the correct answer.

   - [X] A MKK
   - [ ] B FLUC00
   - [ ] C General

   That is correct! A Business Partner needs the MKK role when FI-CA is used as a sub-leger for SAP S/4HANA SOM.

2. The FI-CA Master Object Contract Account includes a parameter, Micro Services, for payment and dunning control.

   Determine whether this statement is true or false.

   - [X] True
   - [ ] False

   That is correct!

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UNIT 3 Subscription Order Capture

Lesson 1

Subscription Order Creation 31

Lesson 2

Customizing of Subscription Order 39

UNIT OBJECTIVES

- Understand Subscription Order Management Process Overview
- Understand Subscription Product Overview
- Proceed with Subscription Order Creation
- Explain the BRIM Data Model
- Understand the Customizing of Subscription Order

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Subscription Order Creation

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Understand Subscription Order Management Process Overview
- Understand Subscription Product Overview
- Proceed with Subscription Order Creation

Business Example

A subscription order for Cloud Selection Service is captured for an existing customer in SAP Billing and Revenue Innovation Management (BRIM) Subscription Order Management (SOM). The customer already has another subscription contract in the system.

The customer also has a contract account assigned as default for further subscription contracts. The department of the company orders a cloud selection service starting at 01.01.2020. They decide to select the service level gold out of pay as you go, silver, gold, and platinum. Additionally there is an activation fee of 399 Euro assigned for the installation of the service. The order includes a product offering combining a service, activation fee, and hardware.

Use the IP address as a technical identifier for your cloud service.

Understand Subscription Order Management Process Overview

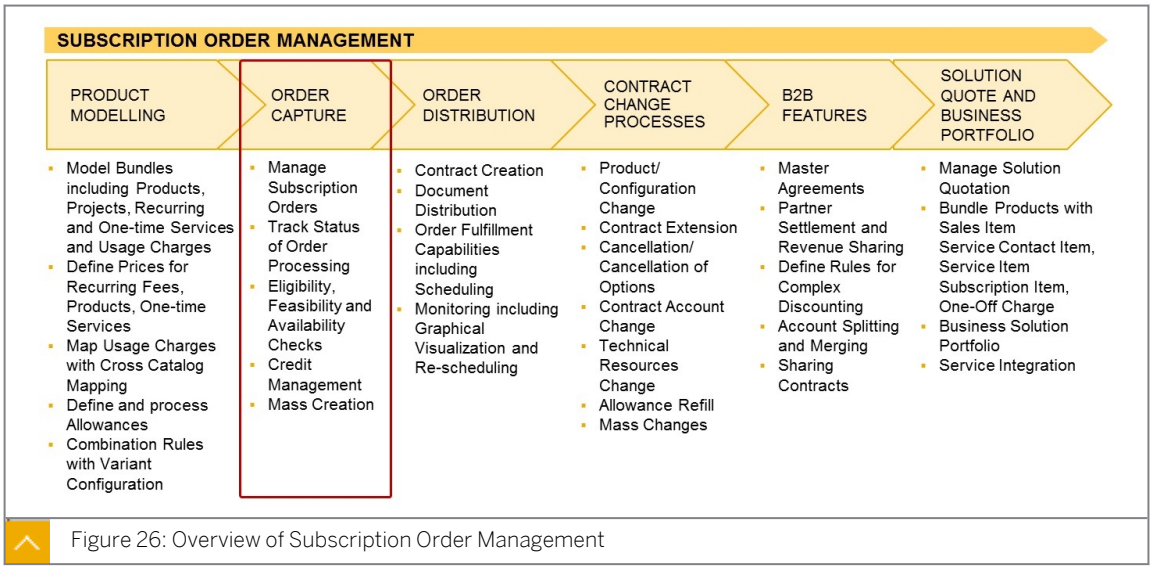

Figure 26: Overview of Subscription Order Management

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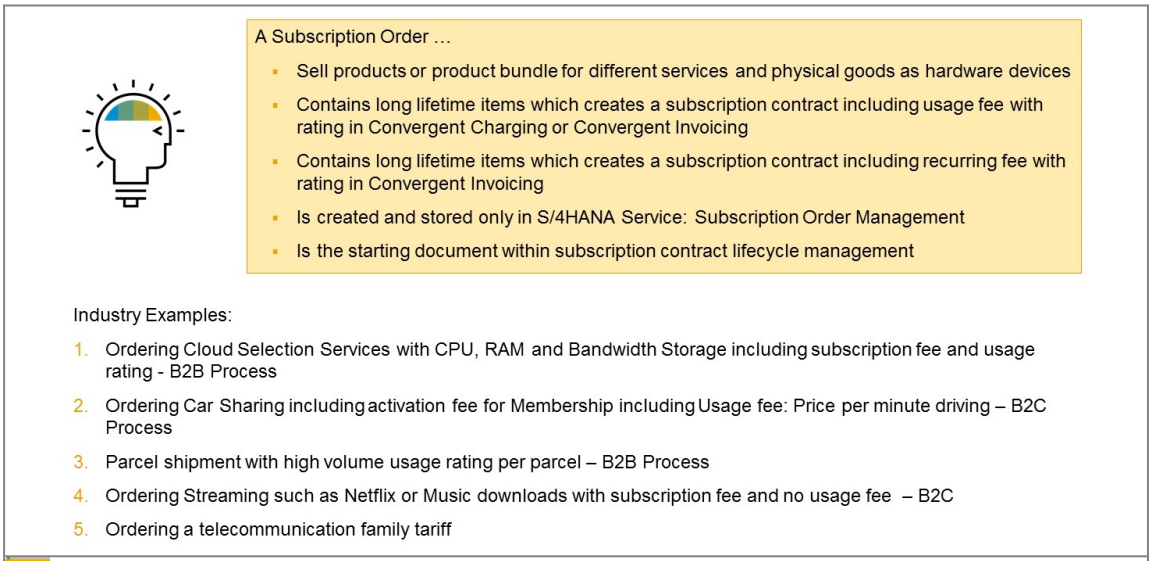

Figure 27: Definition of a Subscription Order

The subscription order can be used as follows:

- A subscription order for selling products and services for the creation of a subscription contract, such as usage-based charges, recurring, and one-time charges.
- Change order for subscription contract changes.

When you confirm the subscription order, the application automatically creates or updates the relevant subscription contract, according to the status.

You can:

- Search for subscription orders
- Create and maintain subscription orders
- Release a subscription order for subscription contract creation
- Maintain change orders

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Understand Subscription Product Overview

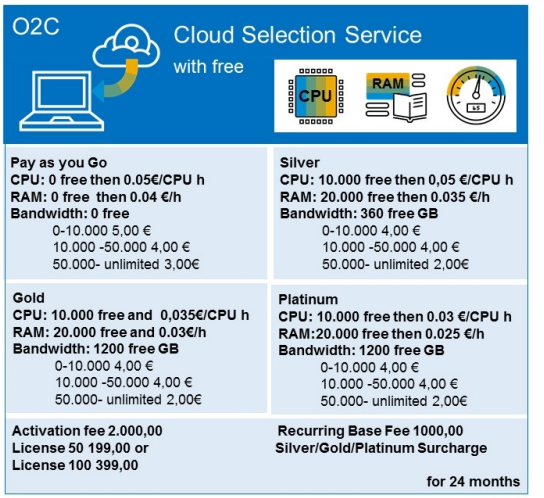

Figure 28: A Business Example of Cloud Service Provider

We will follow this business example of cloud service provider in the exercises of the training.

Business example cloud service product for Subscription Order Management (SOM). We are selling a product bundle with an activation and usage fee.

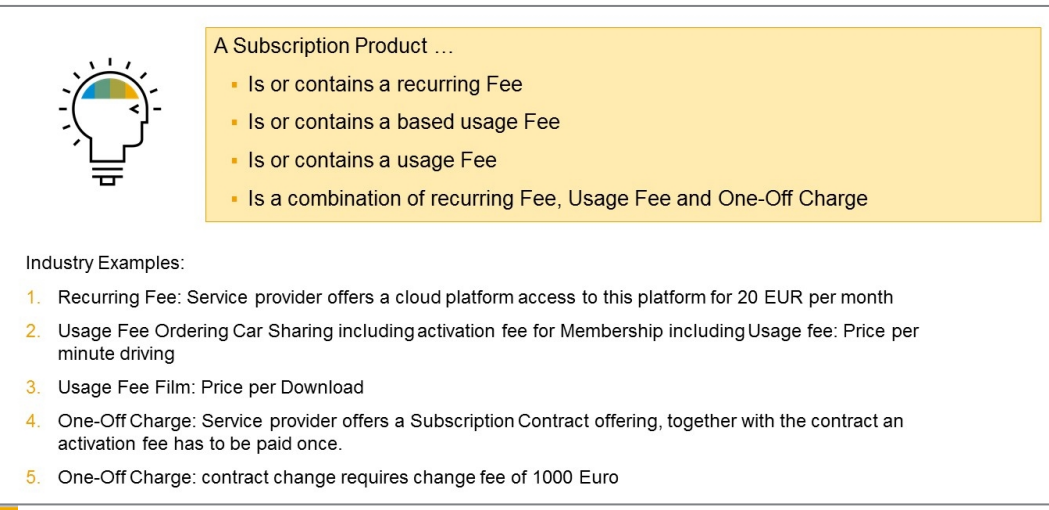

Figure 29: Definition Subscription Product

Subscription products that are used in subscription orders, may need to be invoiced periodically like a monthly fee for a cloud subscription.

When a subscription order with a recurring fee is submitted, a subscription contract is created. During contract distribution to Contract Accounts Receivable and Payable (FI-CA), contract items, for which recurring fees exist in the pricing elements, are identified. A dependent billing plan is created in SAP Convergent Invoicing for these items, in addition to the provider contract item.

This billing plan holds the information about the billing cycle, reference to the provider contract item, payment amount, and payment currency. You can now use this billing plan to request the recurring fees, using the corresponding features in SAP Convergent Invoicing.

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Subscription products that are used in subscription orders, may have an assignment to a charge plan in SAP Convergent Charging, via cross catalog mapping, to support complex consumption based pricing models.

When a subscription order, with a usage based fee, is submitted, a subscription contract is created, During contract distribution to FI-CA, contract items for which a cross catalog mapping exists are identified and a provider contract with the corresponding assigned charge plan is created. Incoming usage will then identify the provider contract in SAP Convergent Charging during charging, derive the price, and send the resulting billable items to SAP Convergent Invoicing.

A one-off charge is a charge which has only to be paid once, for example, an activation fee or suspension fee. The one-off charges are delivered to SAP Convergent Invoicing in the form of billable items and later provided, either on the customer's invoice together with usage charges, or on a separate invoice.

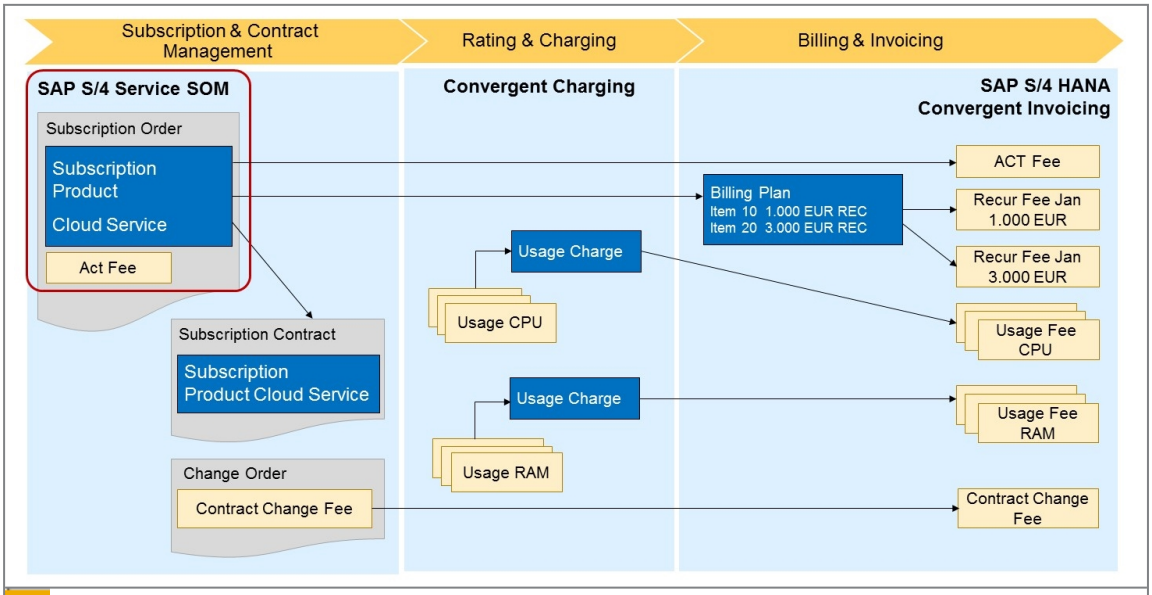

Figure 30: Subscription Process Overview

Explain Subscription Order Creation

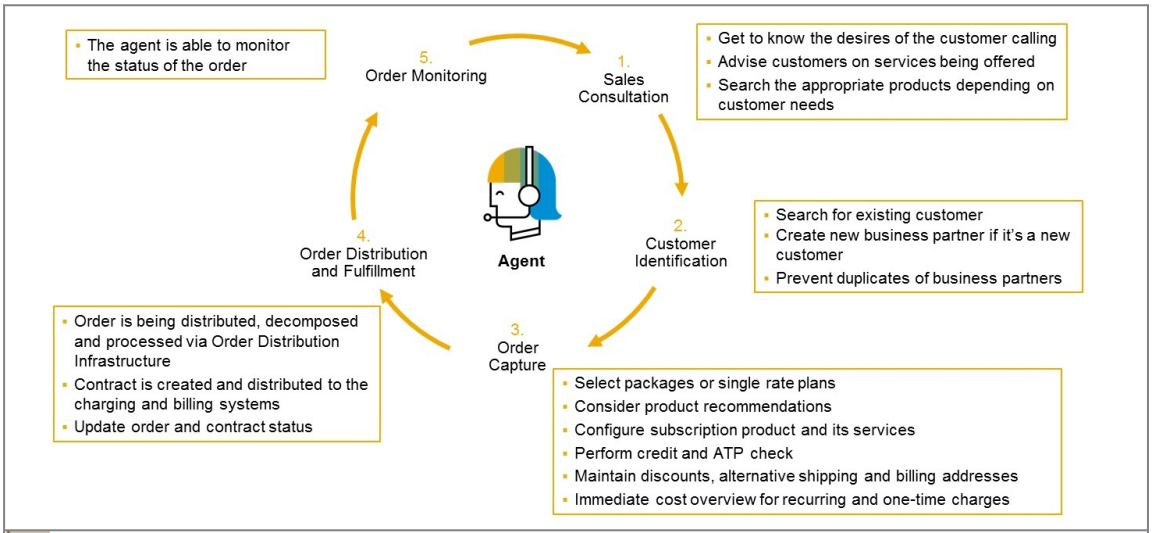

Figure 31: Complete Sales Process

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SAP Subscription and Order Management is the application that handles the sales process in the SAP Billing and Revenue Innovation Management (BRIM) solution. The complete sales process is explained, from sales consultation, to identifying the customer, and to order capturing.

This process example shows the different steps an agent performs to capture and complete an order (step 1-3). After the order is captured and submitted by the agent, the order is automatically processed based on events in Order Distribution Infrastructure (ODI) (step 4).

The agent can track the progress of order processing at any time (step 5).

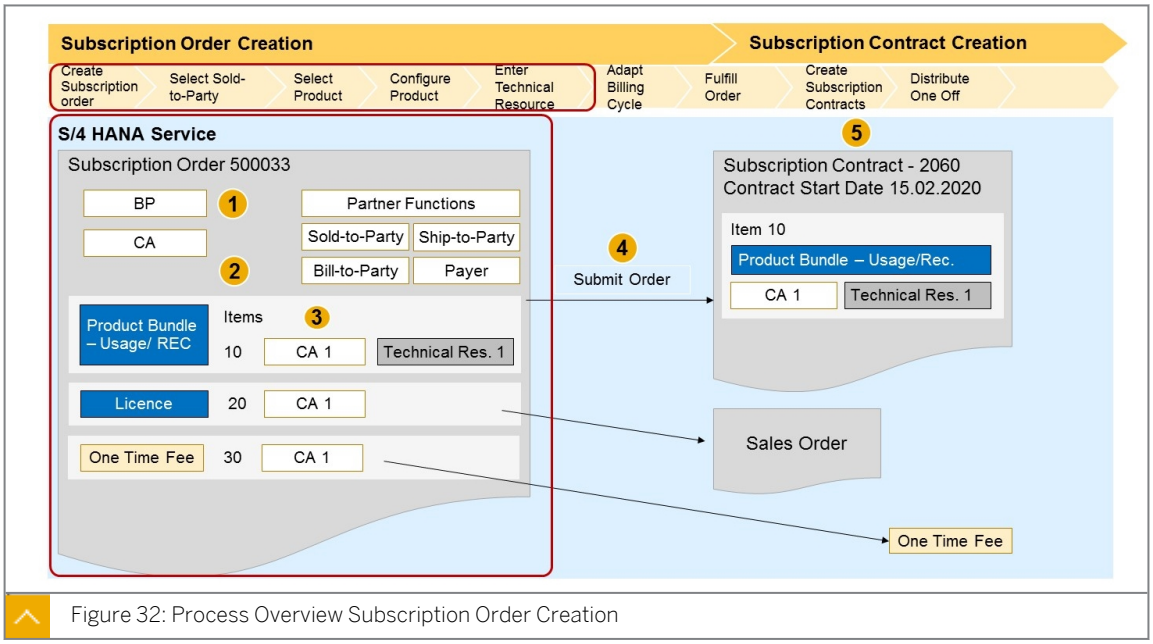

Figure 32: Process Overview Subscription Order Creation

This business scenario shows the creation of a subscription order.

The subscription order is a special sales order that can be used to sell product bundles that consist of physical products and services. Subscription order has additional data structure, like technical data, billing cycle, contract start date, and contract duration. In connection with the subscription contract, it supports the sale of special products and services, in the way that they are assembled and offered.

When the subscription order has been released, a new subscription contract is automatically created, whereby items with a short lifetime such as, sales items or one-time fees stay in the order and items with a long lifetime become part of the contract. In the first step you log on to the Fiori Launchpad and go to the section Subscription Order Management. Here you select the app Manage Subscription Order.

Subscription Order Creation

Selection of the transaction type PRVO subscription order.

Step 1. At header level you need to select the following:

Customer Data:

Select an existing Sold-to Party. The Sold-to Party entered is also used for the required partner functions: Ship-to Party, Bill-to Party, and Payer and determined via the partner determination PROV0001 within customizing.

Organizational Data:

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The Organizational Data is automatically determined according to the entered business partner. This is done via customizing of the organizational profile for the subscription order with PROVHEADER.

Contract Account:

This contains the information about the payment and financial data for long term business. The data contained in the contract account influences processes within invoicing, taxation, and correspondence processing. If Contract Accounts Payable and Receivable (FI-CA) is activated a contract account needs to be maintained.

At item level you need to select the following:

Step 2. Product Data

The products entered in the subscription order are represented as items. Products can have complex models that produce additional sub items - this is called a product bundle. Enter the product or item related data, contract start date, product configuration, or technical resources and choose the contract accounts.

Step 3. Technical Resources

Enter additional information that is necessary for the sale of the service. For example, in the telecommunication industry technical connection data such as mobile number or SIM card number is entered. The technical resource is unique for each contract.

SAP Convergent Charging Tables

These tables are mapping or range tables in SAP Convergent Charging (CC) system. They are used in the SAP CC for pricing and in the charge plan for price determination. These tables are referenced in the Cross Catalog Mapping (CCM) within the product. In the provider order creation they can be redefined at item level.

Step 4. Submit Order

After the subscription order is captured you need to submit it to trigger the order fulfillment. The result of the order fulfillment is the subscription contract creation. This leads to Step 5

Only products / services which are contract relevant are copied into the contract, because they occur with a recurring frequency. An activation fee and a change fee are not contract relevant, they are considered as one time fees, hence they are not copied into the contract.

With respect to the business example described above, the IT department of the Cloud Business company calls the O2C company and orders the product Cloud Selection Service. The agent of the O2C company captures the order in the system.

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The billing cycle…
- the derivation of the billing period for periodic bills
- the scheduling in the billing plan used for creation of recurring charges
- Contract account can have a billing cycle assigned
- Subscription products have a billing cycle assigned
- Billing cycles are defined by a cycle start date, a frequency, a period length and optionally, special rules for period ends
- is Master date in BRIM process - defined in FI-CA
- Yearly, quarterly, (bi-)monthly, (bi-)weekly frequencies possible

Industry Examples:

1. Product BSU_CLOUD_SEL_SERV has assigned Billing cycle-monthly at the end of the month.
2. Billing Cycle for Sport Subscription is Yearly in advanced or monthly. The price depends on the billing cycle.
3. Waste Fees: Billing Plan and Invoice Cycle can be different, e.g. monthly creation of recurring fee, invoice yearly.

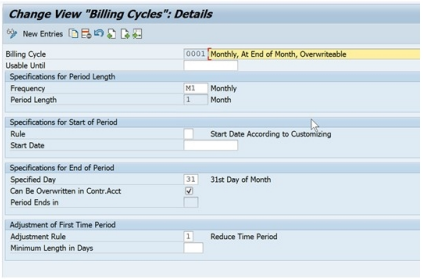

Figure 33: Definition Billing Cycle

Flexible scheduling of recurring billing plan Items with the help of the billing cycle.

The billing cycle of the billing plan can be monthly, while the customer gets a monthly invoice.

The other way round is also possible. A customer can get yearly billing plan items but monthly invoices.

Convergent Charging uses Usage Charges for the creation of subscription related fees. An access object will be created if the charge contains a usage rate.
- The Technical Ressource = CustomerID is a unique identifier of the subscriber account/provider contract.
- The ServiceID is a unique technical identifier of the service (charge). The access is the combination of UserID and Service ID. The usage Events can be assigned to the usage charge with help of the access.

Industry Examples:

1. Product BSU_CLOUD_SEL_SERV has as a Customer user identifier the IP address assigned. The Service Identifier is CPU, RAM or Data Packet. Per Contract item 3 access pairs (Customer ID/Service ID) are assigned to the chargeplan.
2. Car plate can be used as a Technical Ressource for a car rental service, in order to identify the car used.
3. Technical Resource can be a list of identifier e.g. a telecommunication equipment can have a list of phone numbers assigned. Create a assignments scheme for the technical resource.

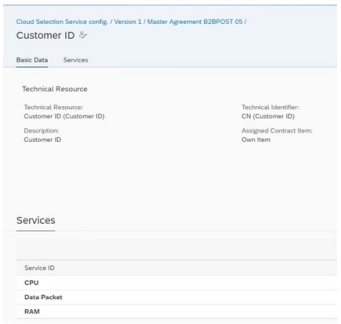

Figure 34: Definition Access

The technical data allows you to define the identifier under which a customer consumes a service, such as a phone number, or a car plate. This identifier is used by the system to rate the usages of this service.

Note:
The same identifier can be used for rating the usages of different services like Voice, SMS, or Data.

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LESSON SUMMARY

You should now be able to:

- Understand Subscription Order Management Process Overview
- Understand Subscription Product Overview
- Proceed with Subscription Order Creation

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Customizing of Subscription Order

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explain the BRIM Data Model
- Understand the Customizing of Subscription Order

BRIM Data Model

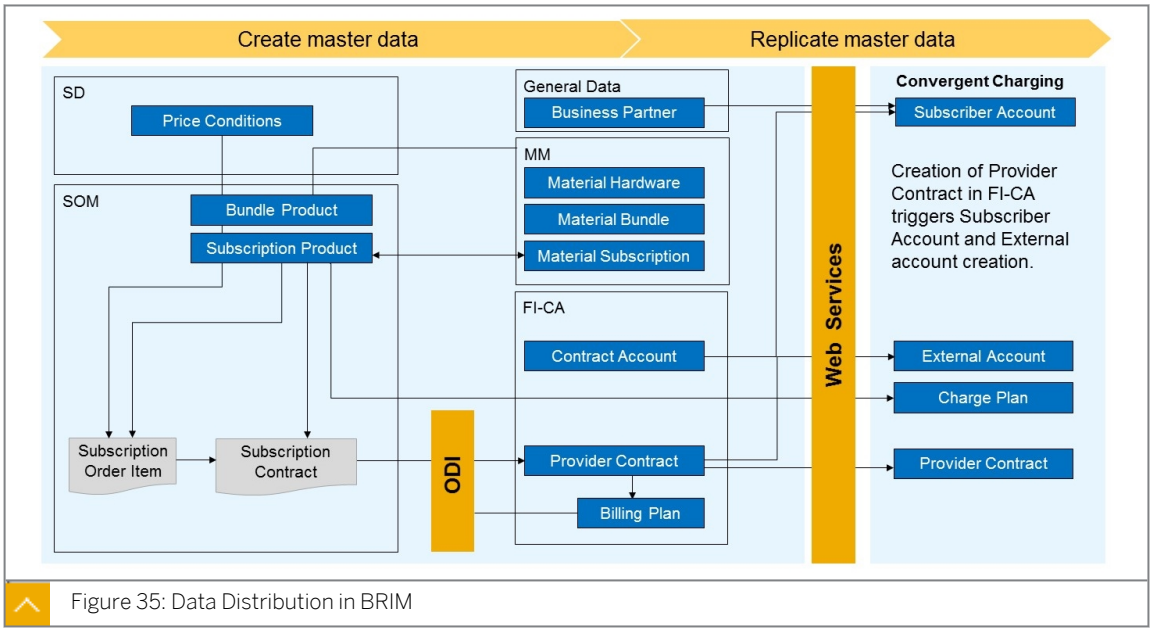

Figure 35: Data Distribution in BRIM

Several master data objects are replicated between SAP S/4HANA and SAP Convergent Charging (CC).

In addition to that, the following Subscription Order Management (SOM) objects can be replicated to the Contract Accounts Receivable and Payable (FI-CA) of SAP S/4HANA and SAP CC:

- Business Partner, called subscriber accounts in SAP CC
- Contract accounts in FI-CA and external accounts in SAP CC
- The subscription contract created for the Business Partner is replicated uni-directionally from SAP SOM to FI-CA of SAP S/4HANA and to SAP CC. The contract is sent to the common object layer in SAP FI-CA, from where the contract generation is triggered in SAP FI-CA itself as well as in SAP CC. The contract will only be created if it can be stored in both systems. Otherwise, there will be a rollback in both systems.

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On SAP CC site, each provider contract is linked to the corresponding charge plans defined in the cross catalog mapping.

Understand the Customizing of Subscription Order

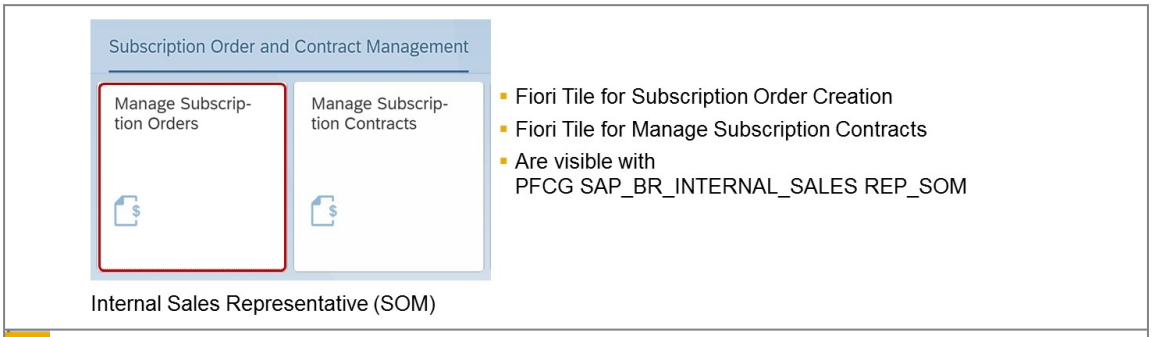

Figure 36: Fiori Business Roles in Fiori Launchpad

Fiori Business Role is available: *Internal Sales Representative (SOM)*

Fiori Business Role: *SAP_BR_INTERNAL_SALES REP_SOM*

The internal sales rep can access either role via Fiori Launchpad or via Web Client UI with role Internal Sales Rep. (Subscr.)

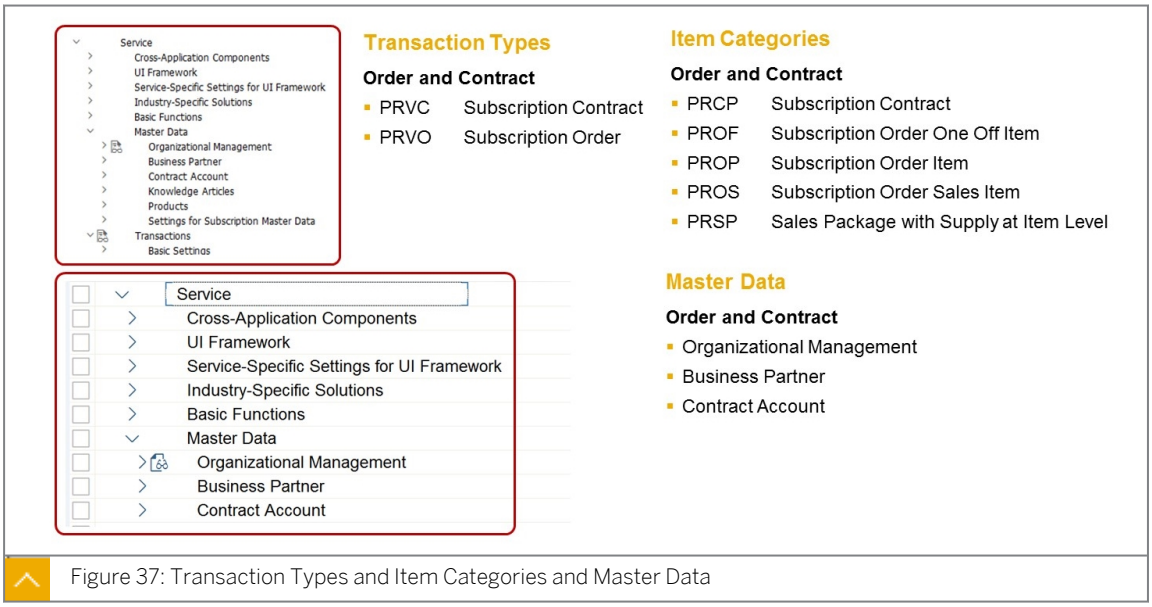

Figure 37: Transaction Types and Item Categories and Master Data

In this customizing path *SPRO → Service → Transactions → Basic Settings*, you set up your subscription order and subscription contract as well as the corresponding item category determination.

The main settings for Subscription Order Management (SOM) can be found under *SPRO → Service → Master Data → Settings for Subscription Master Data* and *SPRO → Service → Transactions → Settings for Subscription Transactions*

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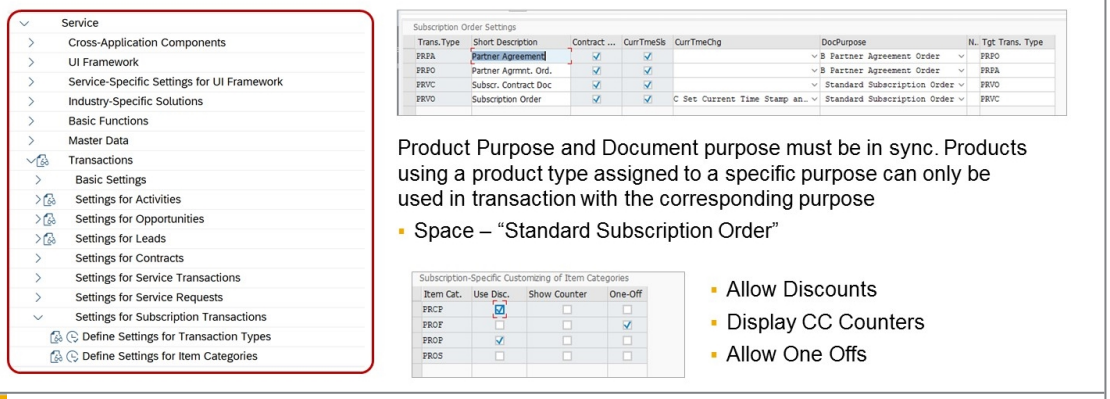

Figure 38: Settings for Subscription Transactions and Item Categories

You can maintain the setting in Implementation Guide (IMG) *SPRO → Service → Transactions → Settings for Subscription Transactions*. In this customizing path you match your transaction types for the document purpose, for example, if the subscription order is a standard one. Furthermore for the item categories you define subscription specific customizing, for example, to allow discounts on item level.

Table						
Transaction Type	**Description**	**Text Procedure**	**Partner Determination**	**Org. Data Profile**	**Date Profile**	**Action Profile**
PRVO	**Subscription Order**	ORDER001	PROV0001	IST000000001	PROVHEADER	IST_ACTION_ORDER_HEAD
PRVC	**Subscription Contract**	ORDER001	PROV0001	IST000000001	PROVHEADER	

Tables for Subscription Order and Subscription Contract are located in SE16:
- CRMS4D_SERV_H (Header)
- CRMS4D_SERV_I

Figure 39: Subscription Order and Contract Customizing overview

This overview shows the general transaction customizing of subscription order and subscription contract.

LESSON SUMMARY

You should now be able to:

- Explain the BRIM Data Model
- Understand the Customizing of Subscription Order

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Learning Assessment

1. Which of the following data do you need to assign to the subscription order during the subscription ordering process?

   Choose the correct answers.

   - [ ] **A** Organizational Data
   - [ ] **B** Usage Records
   - [ ] **C** Invoices
   - [ ] **D** Sold-to-Party
   - [ ] **E** Bundle Product
   - [ ] **F** Contract Account

2. A subscription order is distributed via the IDO framework.

   Determine whether this statement is true or false.

   - [ ] True
   - [ ] False

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Learning Assessment - Answers

1. Which of the following data do you need to assign to the subscription order during the subscription ordering process?

   Choose the correct answers.

   - [X] A Organizational Data
   - [ ] B Usage Records
   - [ ] C Invoices
   - [X] D Sold-to-Party
   - [X] E Bundle Product
   - [X] F Contract Account

   That is correct! Usage records and invoices are not part of the ordering process.

2. A subscription order is distributed via the IDO framework.

   Determine whether this statement is true or false.

   - [ ] True
   - [X] False

   That is correct! A subscription order is distributed, not via the IDO framework, but via the ODI framework.

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UNIT 4 Subscription Contract

Lesson 1

Subscription Contract Creation in the BRIM Landscape 47

Lesson 2

Phased Contracts 53

Lesson 3

Order Distribution Infrastructure 61

Lesson 4

Device as a Service 67

UNIT OBJECTIVES

- Understand Subscription Contract Creation in the BRIM Landscape
- Understand Phased Contracts
- Understand the Order Distribution Infrastructure
- Explain Fulfillment Monitor
- Describe Customizing the Order Infrastructure
- Understand Device as a Service (DaaS)

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Unit 4
Lesson 1

Subscription Contract Creation in the BRIM Landscape

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Understand Subscription Contract Creation in the BRIM Landscape

Business Example

For the Subscription Order, which you created in the lesson with the product cloud selection service, one Subscription Contract with two billable items was created, after submitting your Order fulfillment.

Understand Subscription Contract Creation

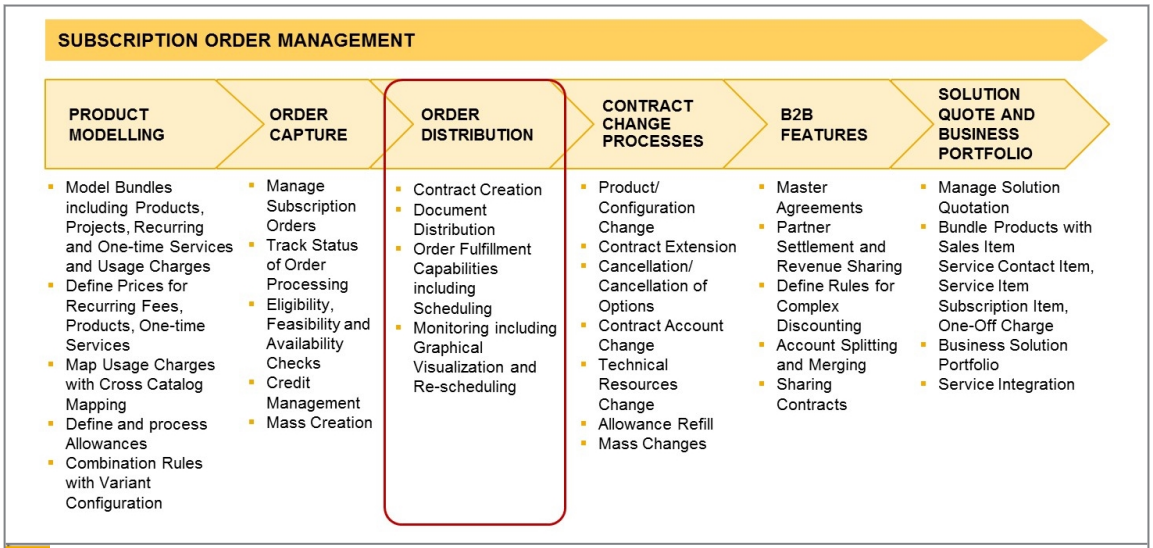

Figure 40: Overview Subscription Order Management

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A Subscription Contract …

- Results of the subscription order or a solution quotation within the BRIM landscape
- Includes: Contract Account, Billing Plan with billing lifecycle, prices, assigned with a technical resource for unique identification, contract duration, allowed services, discounts on usage
- Includes time slices within the contract lifecycle → Changes on contract
- Has a billing plan assigned in Convergent Invoicing
- Is the basis for rating in Convergent Charging and Convergent invoicing
- Is distributed to SAP S/4 HANA Convergent Invoicing as a provider contract
- Is distributed to SAP Convergent Charging as a provider contract

Industry Examples:

1. Contract for Cloud Selection Service with CPU, RAM and Bandwidth Storage including subscription fee and usage – B2B Process
2. Contract for Car Sharing with usage rating – B2C Process
3. Parcel shipment with high volume usage rating per parcel – B2B Process
4. Streaming membership contract including monthly subscription e.g. music and TV - optional usage fee for additional services
5. Telecommunication contract including 4 listed mobile numbers and a counter pool for free usage

Figure 41: Definition Subscription Contract

A Subscription Contract manages the **long term agreement** for a technical service that the providing company offers to a customer. As a result of a subscription order, one or more Subscription Contract documents are created. A Subscription Contract document is automatically created when a subscription order is successfully released. Contract-relevant data is copied from the order to the contract document.

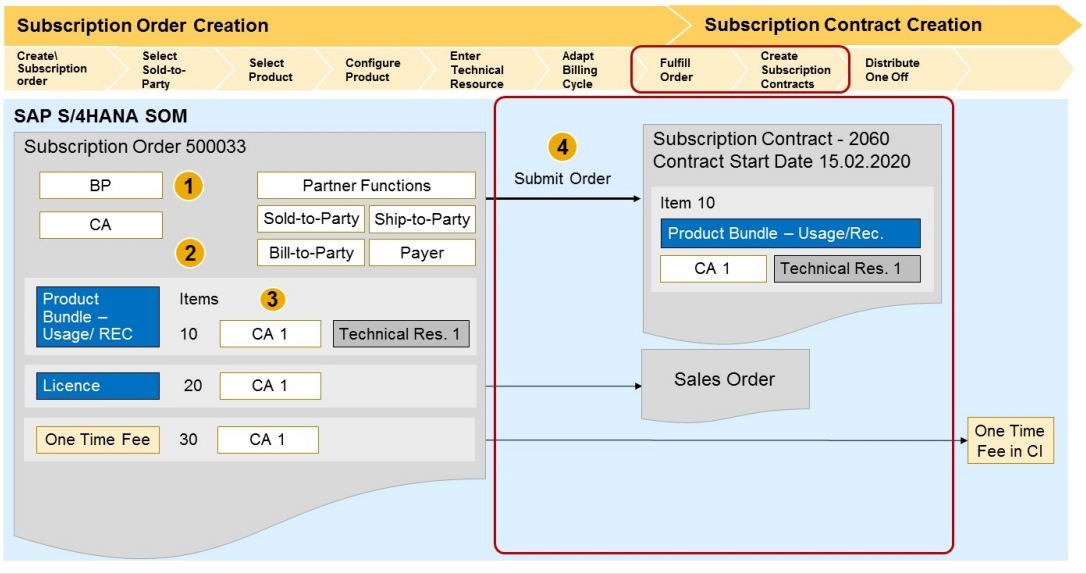

Figure 42: Process Overview Subscription Contract Creation (1/3)

This business scenario shows the creation of one Subscription Contract with one contract item.

This Subscription Contract example shows the creation of a Subscription Order with one Subscription Contract. The product bundle used in the Subscription Order is our training example *BSU_CLOUD_SEL_SERV*, which contains a usage and recurring fee, a license, and a one-time fee. The usage and the recurring fee is transmitted into the Subscription Contract after submitting the Subscription Order (Step 4), because it is contract-relevant. The technical resource and contract account are also transmitted to the Subscription Contract (Step 5).

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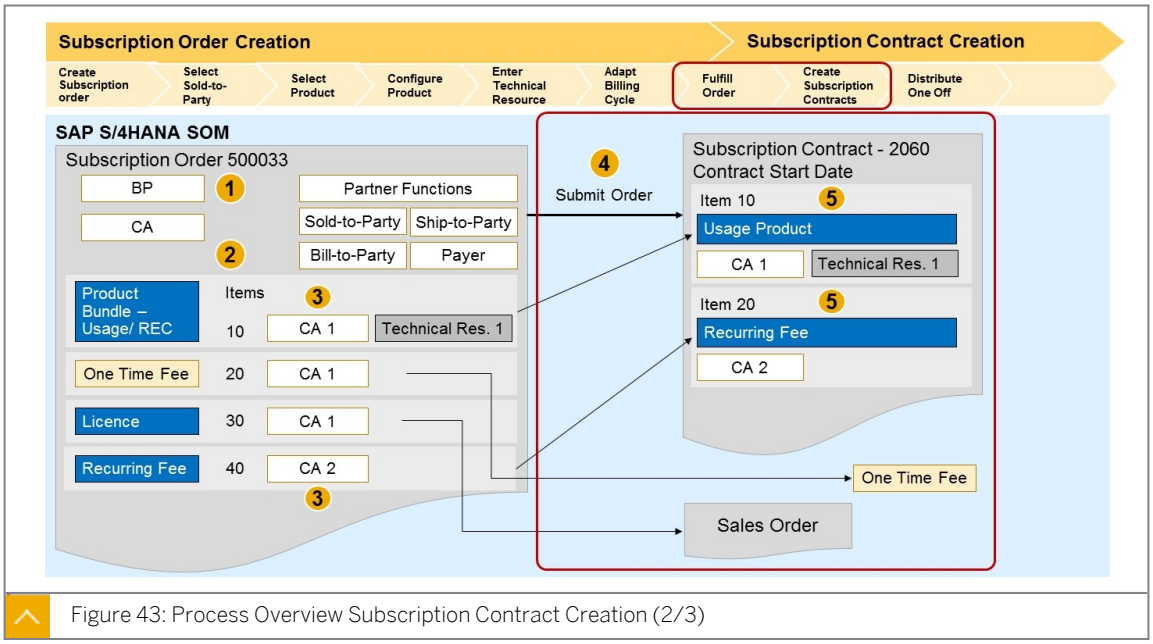

Figure 43: Process Overview Subscription Contract Creation (2/3)

This business scenario shows the creation of one Subscription Contract with two items.

This Subscription Contract example shows the creation of a Subscription Order with one Subscription Contract that includes two contract-relevant items. The product bundle used in the Subscription Order contains a usage fee and a recurring fee. The usage fee and recurring fee are transmitted into one Subscription Contract after submitting the Subscription Order (Step 4), because they are contract-relevant. The technical resource and contract accounts are also transmitted to the Subscription Contract (Step 5).

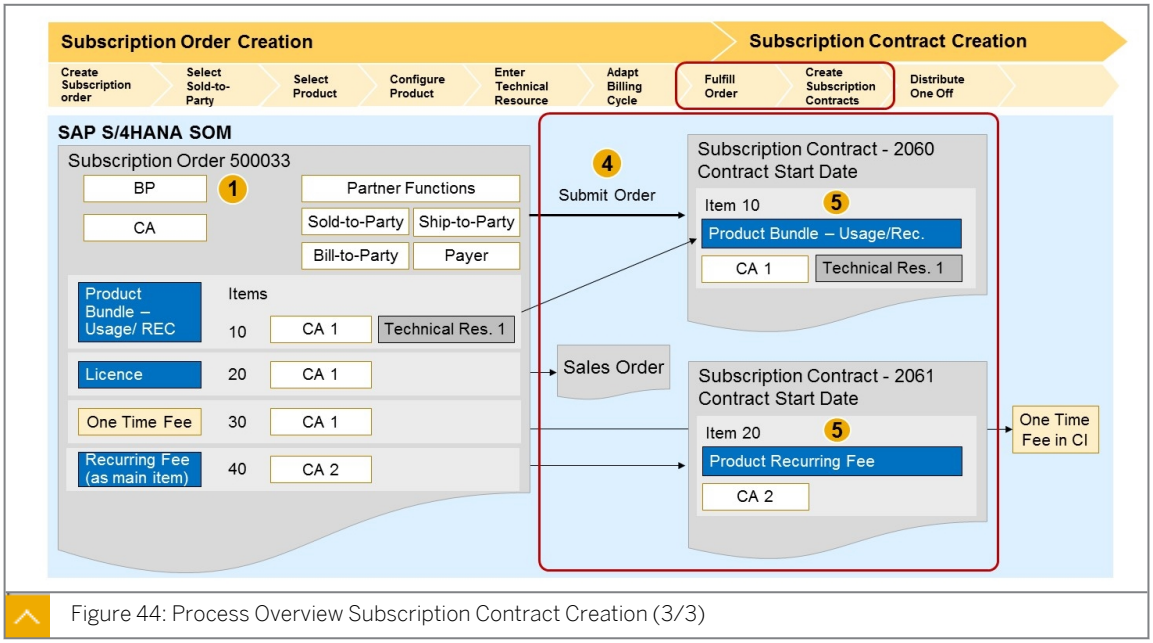

Figure 44: Process Overview Subscription Contract Creation (3/3)

This business scenario shows the creation of two Subscription contracts.

This Subscription Contract example shows the creation of a Subscription Order with two Subscription Contracts creation. One product bundle used within the Subscription Order contains a usage fee, and the other product bundle contains the recurring fee. The usage fee and the recurring fee are transmitted into a Subscription Contract, after submitting the

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Subscription Order (Step 4), as they are contract-relevant. The technical resource and contract accounts are also transmitted to the Subscription Contracts (Step 5).

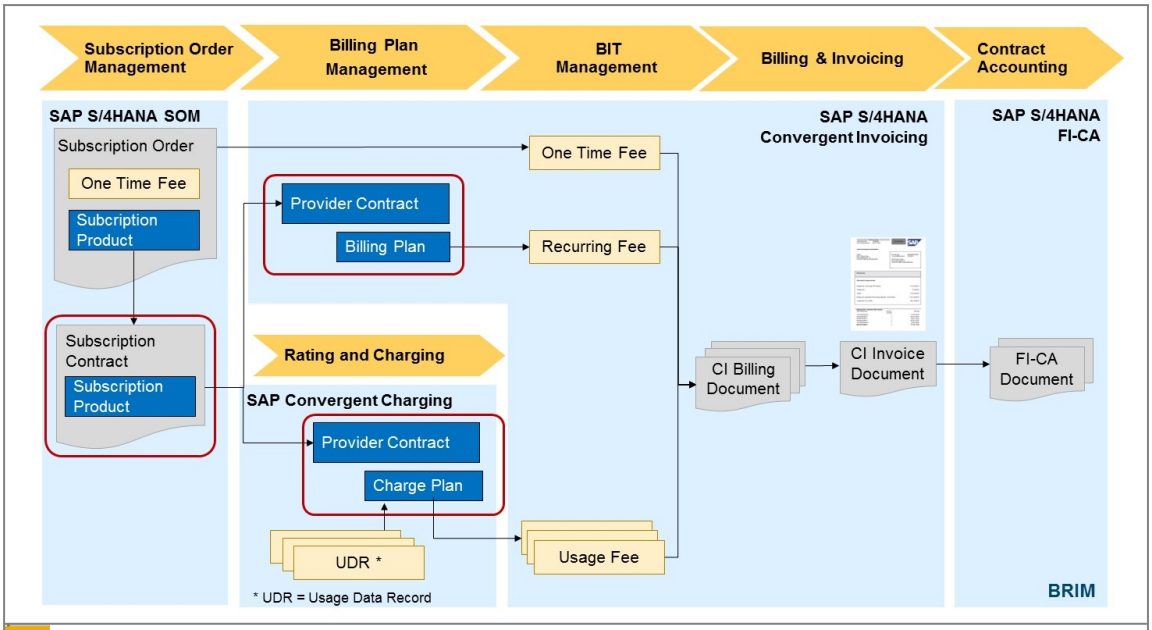

Figure 45: Understanding Subscription Contract Creation

This overview shows the document flow of the Subscription Contract through Subscription Order Management (SOM), Convergent Invoicing and Contract Accounts Receivable and Payable (FI-CA). The Subscription Contract is transmitted to Convergent Invoicing, named Provider Contract and for the items, a CI Billing Document and Invoice is generated. The Invoice is stored in FI-CA.

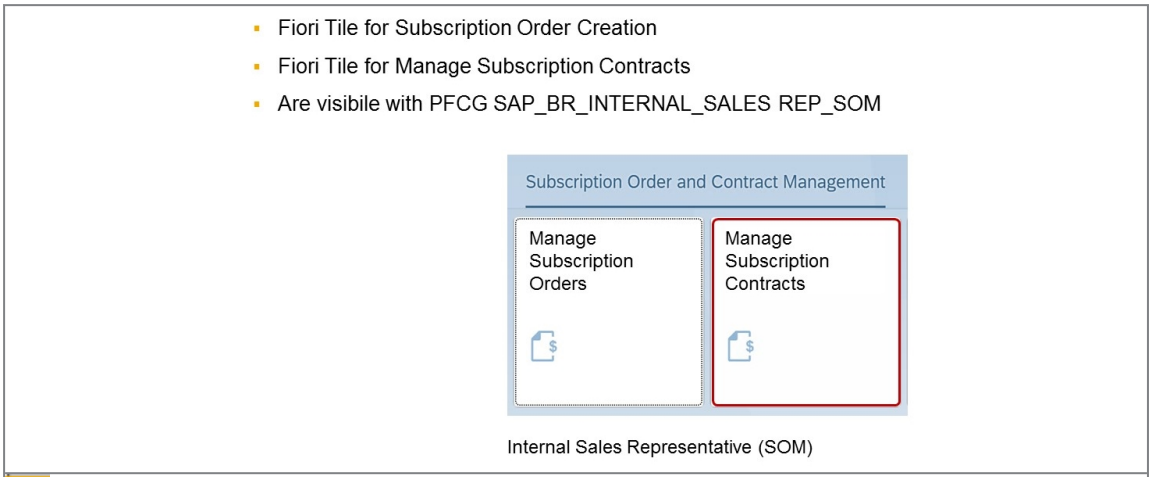

Figure 46: Related Fiori Business Roles in Fiori Launchpad

To perform these tasks, the required Fiori Business Role is *Internal Sales Representative (SOM)*

Fiori Business Role: *SAP_BR_INTERNAL_SALES REP_SOM*.

The internal Sales Rep can access either from the Fiori Launchpad or from the Web Client UI with role Internal Sales Rep. (Subscr.)

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LESSON SUMMARY

You should now be able to:

- Understand Subscription Contract Creation in the BRIM Landscape

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Phased Contracts

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Understand Phased Contracts

Explain Phased Subscription Contracts

Overview of Phased Subscription Contracts

The phased contracts feature enables you to split a contract item into multiple phases, with attributes such as item quantity, price, and configuration varying across each phase.

Phases can have different phase lengths, but neither overlaps nor gaps between individual phases of an item are allowed.

It is only possible to create subscription contract items with phases using the solution quotation API, but not using any UI.

Features:

- Receive data for a phased contract from the Solution Quotation OData API
- Create subscription contracts with phased contract items as a follow up
- Display phase data in a subscription item UI
- Display phases in a subscription contract hierarchy UI
- Distribute active phase to FI-CA Provider Contract & Billing Plan / SAP Convergent Charging
- Distribute future phases to FI-CA Provider Contract & Billing Plan

Figure 47: Overview of Phased Subscription Contracts

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Phased Contract Phases

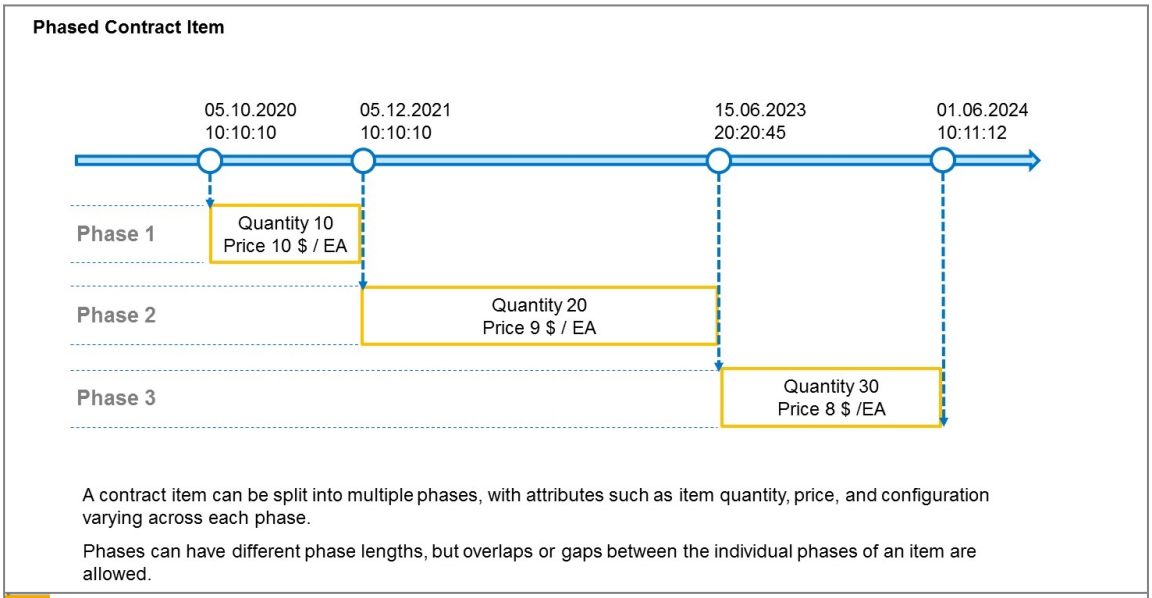

Figure 48: Phased Contract Phases

Phased contracts are subscription contracts where customers prefer to split the contract into various phases, with varied requirements across each phase. Each phase is defined by a start and end date, such that one phase begins right after the end of the previous phase. All phases need not be of the same length, but they must be continuous, that is, without any gaps or overlaps.

Attributes such as quantity, configuration, and contract account can vary across each phase.

Contract Phases in a Solution Quotation

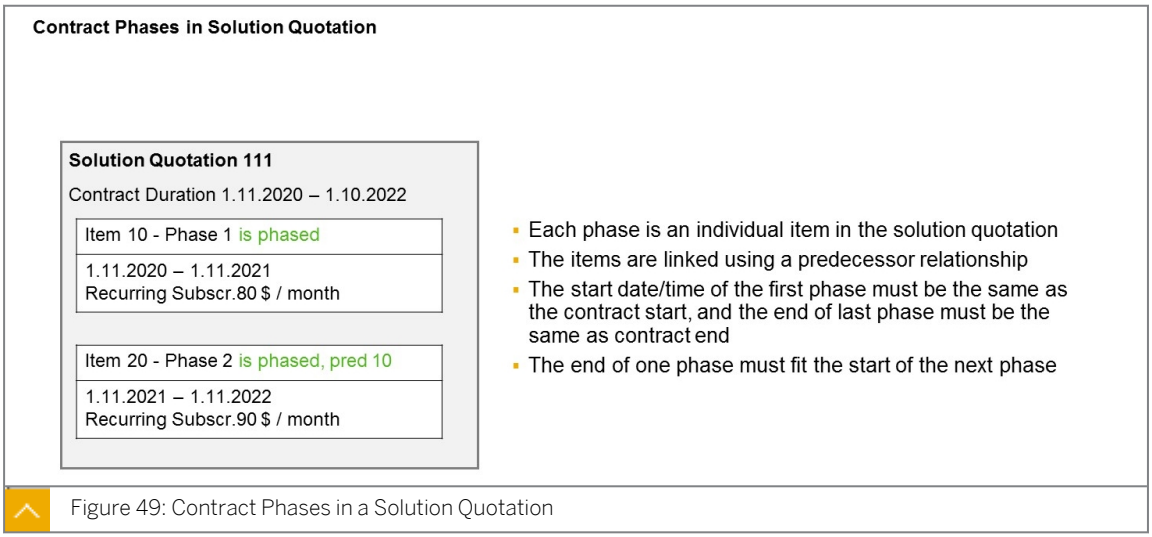

Figure 49: Contract Phases in a Solution Quotation

The following are the prerequisites for phased contracts in Subscription Order Management:

- The Payer and Sold-to Party must be the same across all phases of a single subscription contract.

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- Phases can have different lengths, but there must be no gaps or overlaps between the phases.
- The start of the time slice for the first phase and the end of the time slice for the last phase must be the same as the start and end dates of the contract, respectively.
- The phase indicator and predecessor item number must be provided in the API structure of the subscription item.
- Information for all phases, such as the FICA contract number and item number in subscription contract, must be supplied during a contract change together with the change process information.

Using the API to Create a Phased Contract

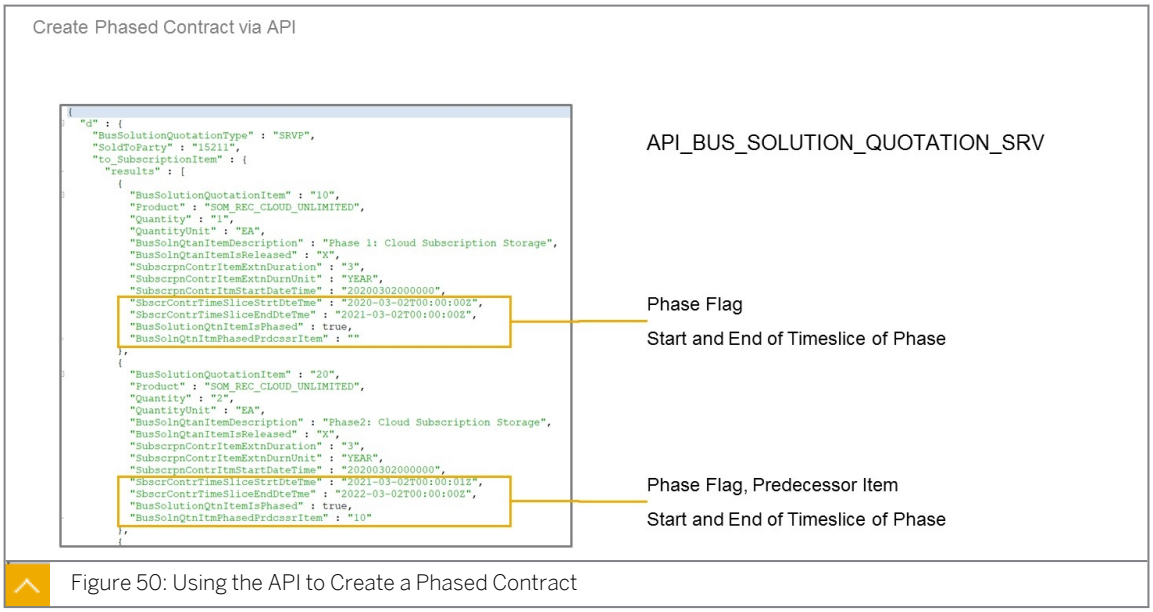

Figure 50: Using the API to Create a Phased Contract

Phased contracts have the following features:

- You can create and change subscription contracts using the Solution Quotation API only.
  - Phased subscription contracts with individual products are created as a followup of the solution quotation.
  - To change existing active and future phases of a phased contract, you must pass data for all the phases along with the required changes.
- The following sets of dates are important to phased contracts and are visible under the *Dates* assignment block.
  - Contract start and end dates indicate the original contract duration.
  - Term start and end dates represent the planned contract duration.
  - Time slice start and end dates help to define the length of a phase.
- You can view the phases on the subscription contract UI and the phase data on the subscription item UI.

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E2E Data Flow Phased Contracts

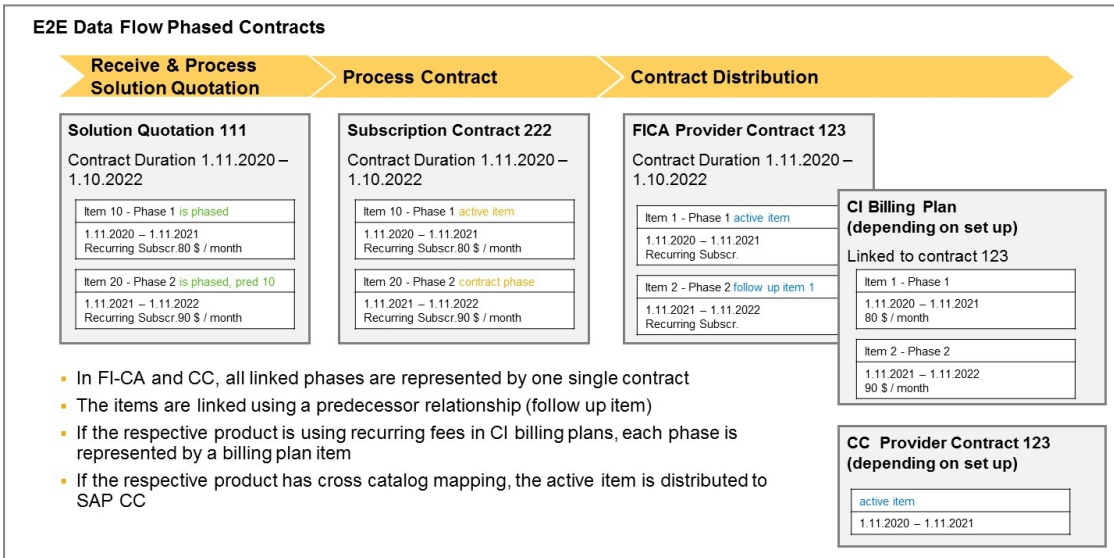

Figure 51: E2E Data Flow Phased Contracts

The active phases of a phased contract are distributed to the FI-CA contract, CI billing plan, and SAP Convergent Charging. Future phases are distributed to the SAP FI-CA provider contract and the CI billing plan only.

Phased Contract Items in a Solution Quotation

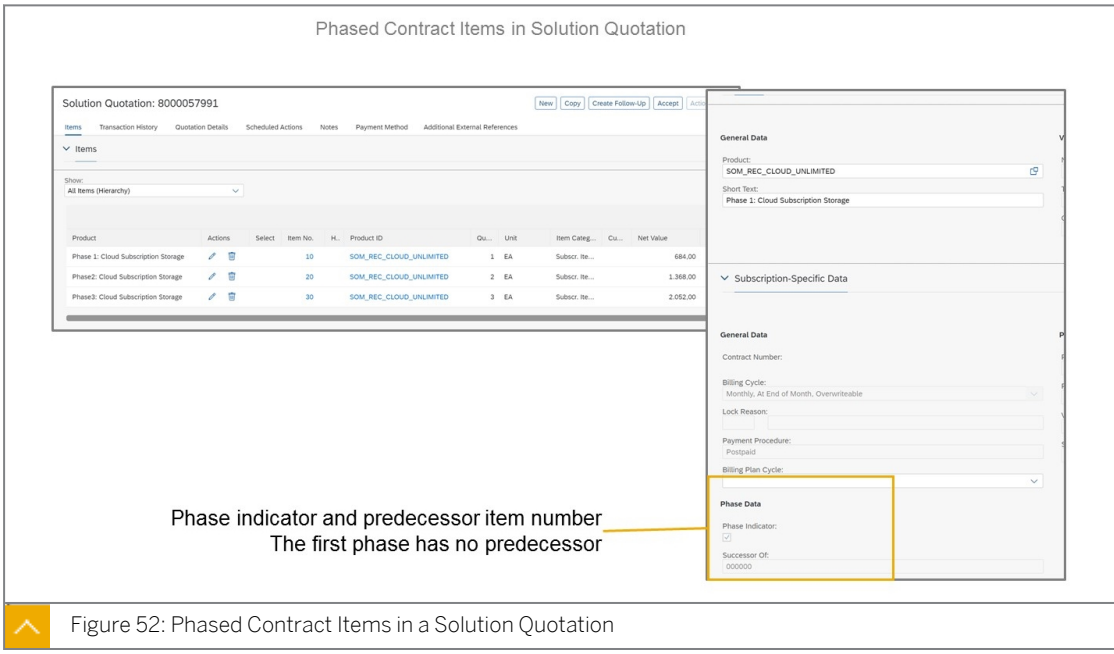

Figure 52: Phased Contract Items in a Solution Quotation

Observe the following:

- You can create phased contracts using the Solution Quotation OData API **only**.
- Once defined the **phase structure is fixed**.
- Phases can have different phase lengths, but overlaps and gaps between the individual phases of an item are not allowed.

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- The start date and time of first phase must be the same as the contract start date and time, and the end of last phase must be same as the contract end date and time.
- A contract phase always represents the full state of an item.
  - For example, in year 1 you plan 100 PC, in year 2 you plan 500 PC
  - Then, your items must look like year 1 – 100 PC, year 2 – 500 PC
- Only **main items** are supported as phased items. This means that sub-items cannot be combined together with phases.
- When you **change** a contract, you must always send all active and future phases over the API.
- If you are working with the **FI-CA distribution scenario**, you must plan the ODI distribution report CRM_ISX_MSG_START_PLANNED to trigger the distribution of future phases.

Contract Phases in a Subscription Contract

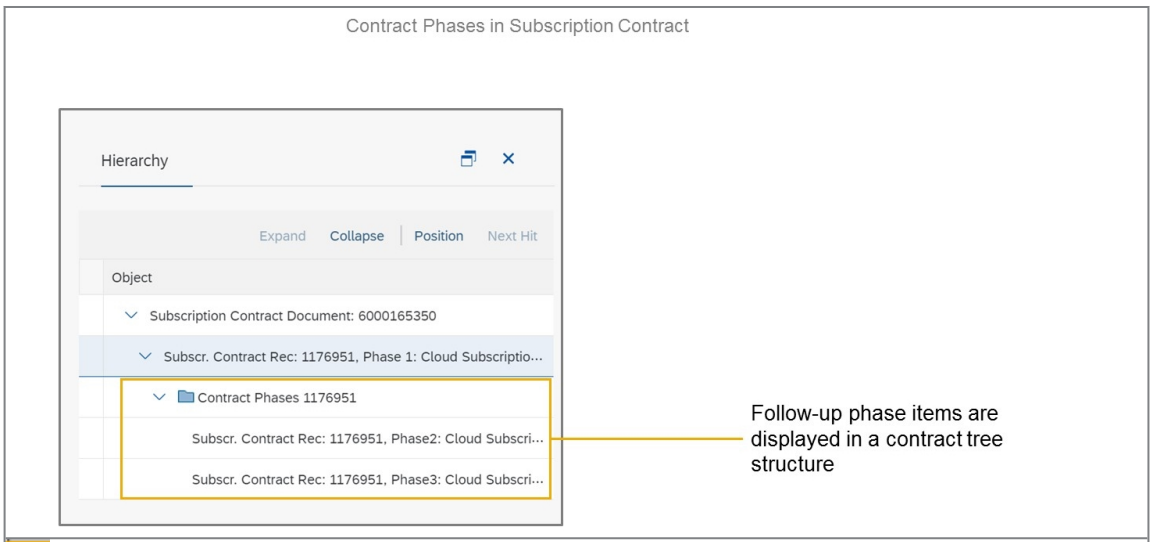

Figure 53: Contract Phases in a Subscription Contract

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Phases in an FI-CA Contract and Billing Plan

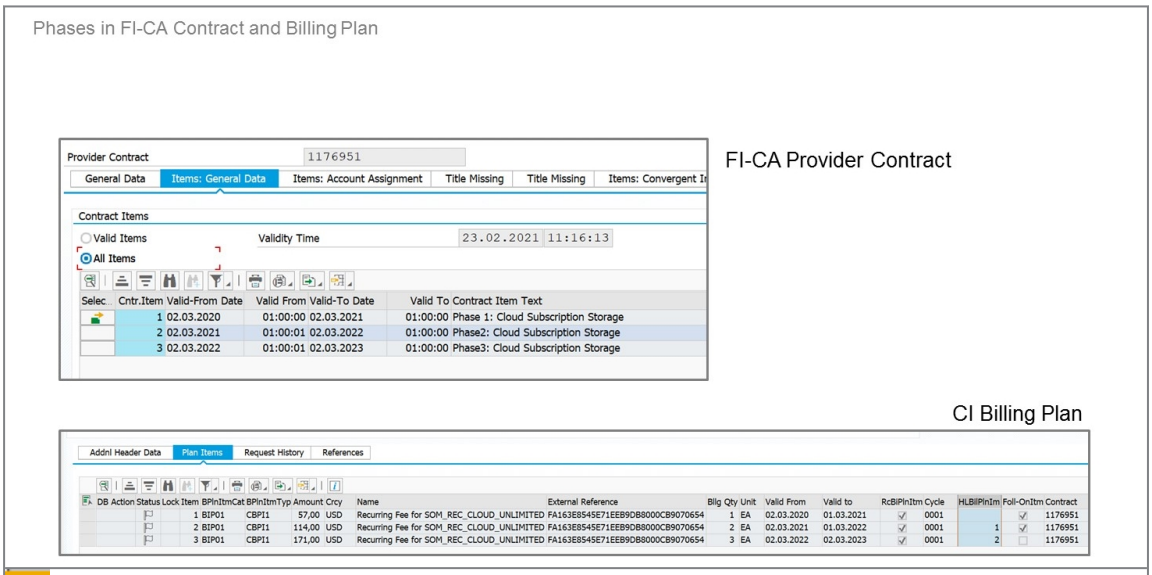

Figure 54: Phases in an FI-CA Contract and Billing Plan

Combining Phased Items and Non-Phased Items in a Solution Quotation

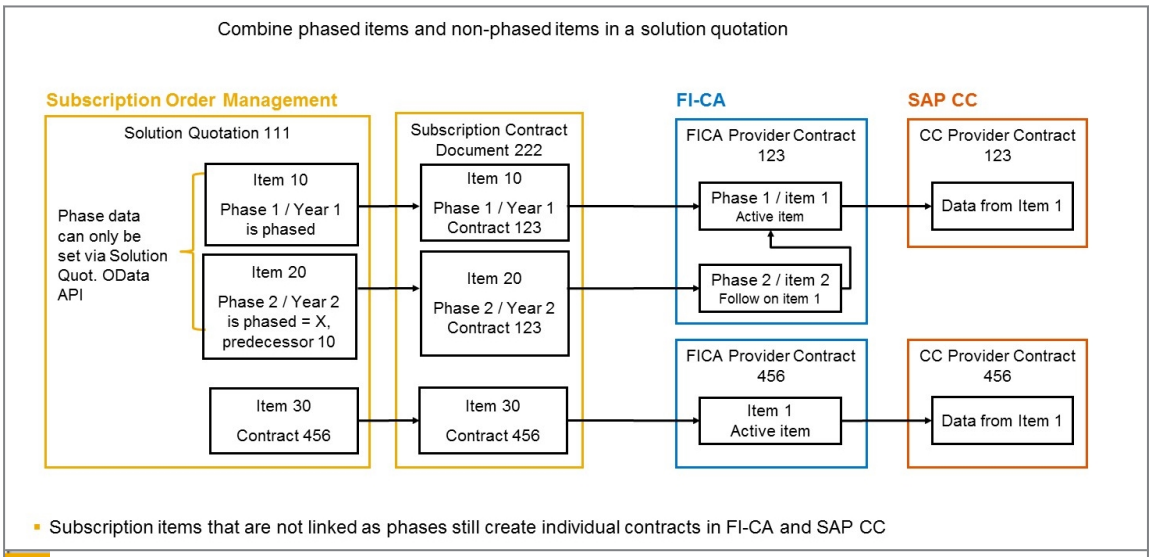

Figure 55: Combining Phased Items and Non-Phased Items in a Solution Quotation

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Distribution of Initial Phase and Next Phase

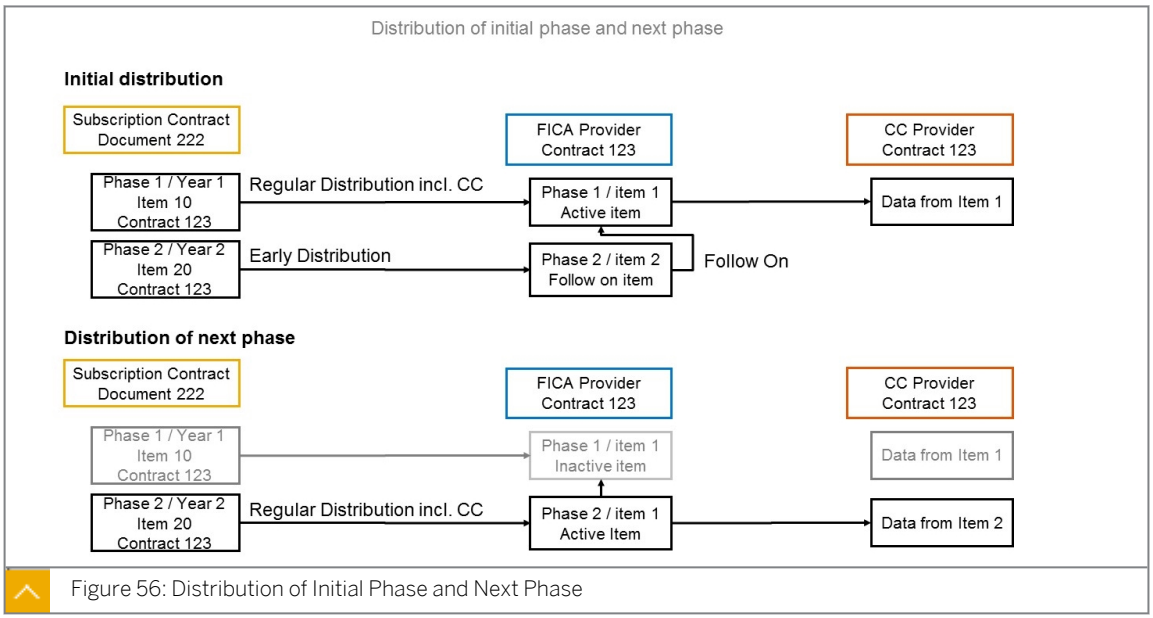

Figure 56: Distribution of Initial Phase and Next Phase

Phases in the FI-CA contract and Billing Plan are created with early distribution.

The CC Contract is only updated with the active time slice using a normal distribution.

LESSON SUMMARY

You should now be able to:

- Understand Phased Contracts

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Unit 4
Lesson 3

Order Distribution Infrastructure

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Understand the Order Distribution Infrastructure
- Explain Fulfillment Monitor
- Describe Customizing the Order Infrastructure

Business Example

After submitting your Subscription Contract, you need to check the Document Distribution – Order Fulfillment.

Understand Order Distribution

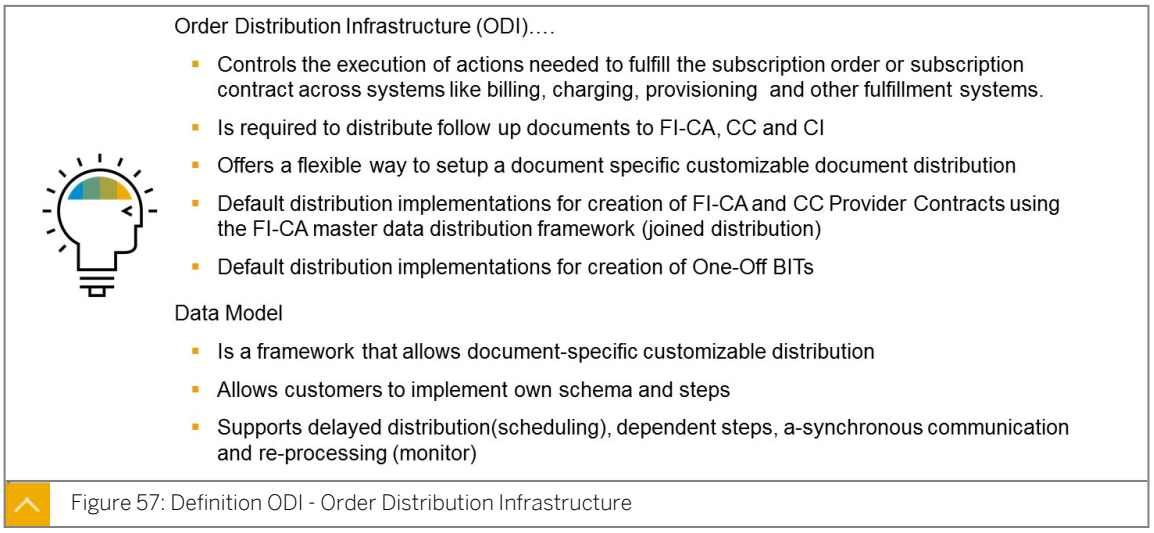

Figure 57: Definition ODI - Order Distribution Infrastructure

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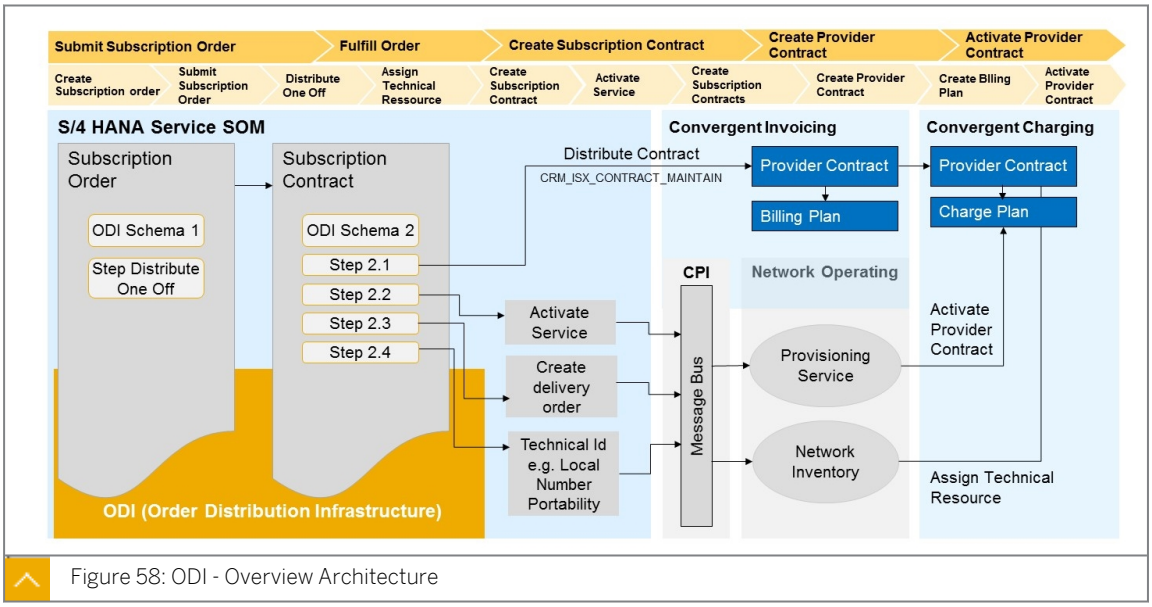

Figure 58: ODI - Overview Architecture

In this Order Definition Overview (ODI) overview you see the different ODI steps with Schemas assigned to the Subscription Order and Subscription Contract. The results of the steps in the outcome is marked below. While submitting a Subscription Order, a one off BIT is send to the Convergent Invoicing (CI). The Subscription Contract ODI triggers a Contract in Contract Accounts Receivable and Payable (FI-CA) and Convergent Charging (SAP CC).

Schema		Step Type		Category	
PO01	Subscription Order	O001	Always true activation	O1	Order Distribution
POOF	Subscription Order One-off Distribution	ONEF	Distribute One-off Items from Order	O1	Order Distribution
PERP	Subscription Contract Distribution	PCWA	Wait for activation	P0	Subscription Contract waits for activate (Batch)
		PCEA	Call FI-CA Message contract activation	P1	Subscription Contract Activation
		PCOF	Distribute One-off Items	P1	Subscription Contract Activation
		PCWD	Wait for deactivation	P5	Subscription Contract waits for deactivation (Batch)
		PCED	Call FI-CA Message contract deactivation	P6	Subscription Contract Deactivation
0001	Subscription Contract Direct activation	PCWA	Wait for activation	P0	Subscription Contract waits for activate (Batch)
		0011	Always true activation	P1	Subscription Contract Activation
		PCWD	Wait for deactivation	P5	Subscription Contract waits for deactivation (Batch)
		0012	Always true deactivation	P6	Subscription Contract Deactivation
PCER	Subscription Contract Early Distribution	PCER PCWA PCEA PCWD PCED	Call Message for Early Distribution Wait for activation Call FICA message contract activation Wait for deactivation Call FICA Message contract deactivation	P0 P0 P1 P5 P6	Subscription Contract waits for activate Subscription Contract waits for activate Subscription Contract Activation Subscription Contract waits for deactive Subscription Contract Deactivation

Figure 59: ODI Schemas and Steps

This table shows the ODI Schemas, Step Types and Categories. All information is located within the Customizing of the Order Distribution.

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Schema		Step Type		Category	
PCAL	Subscription Contract with Allowances	ALLW	Call Message for Allowance Creation	P1	Subscription Contract Activation
		CCMU	Contract update	C 1	Subscription Contract Update
		PCEA	Call FICA message contract activation	P1	Subscription Contract Activation
		PCED	Call FICA Message contract deactivation	P6	Subscription Contract Deactivation
		PCOF	Distribute One-off Items	P1	Subscription Contract Activation
		PCWA	Wait for activation	P0	Subscription Contract waits for activate (Batch
		PCWD	Wait for deactivation	P5	Subscription Contract waits for deactive

Figure 60: ODI Schemas and Steps

This table shows the ODI Schemas, Step Types and Categories. All information are located within the Customizing of the Order Distribution.

Explain Fulfillment Monitor

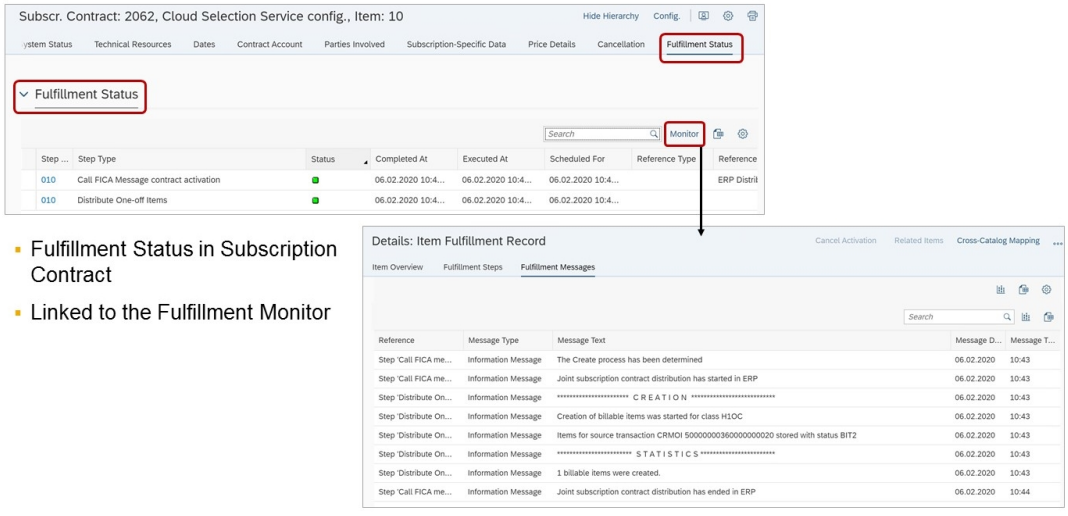

Figure 61: Fulfillment Monitor

In the Subscription Contract, the Fulfillment Status of the Contract is shown. It shows up the ODI (Order Distribution) steps that must proceed for the contract creation. Over the Fulfillment Monitor, detailed step information is located. The navigation takes place over the fulfillment status.

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Describe Customizing Order Distribution

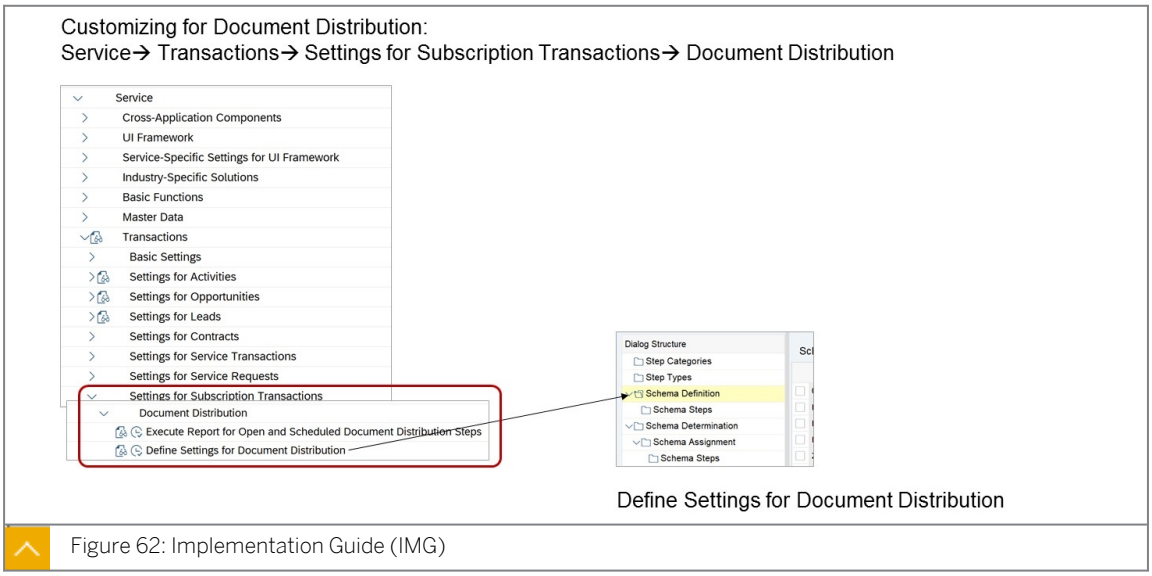

Figure 62: Implementation Guide (IMG)

Settings for document distribution can be done in *IMG Service → Transactions → Settings for Subscription Transactions → Document Distribution*.

Detailed explanations for the single elements of document distribution can be found in the IMG documentation and the F1 helps of the single fields.

The IMG entry *"Execute Report for Open and Scheduled Document Distribution Steps"*, points to the report which can be used to trigger asynchronous Order Distribution Infrastructure (ODI) steps.

Early Distribution:

- For a subscription contract with a future activation date, the distribution of the contract is typically executed only on the activation date. In certain cases, it may be required that the contract be distributed at the time of creation itself, such as when revenue accounting and reporting is involved. This can be reached by an early distribution scenario.

Directly after creation:

- Step **PCER** (Call Message for Early Distribution) of category P0 is executed.

  Create only a Contract Accounts Receivable and Payable (FI-CA) provider contract, no Convergent Charging (CC) provider contract

- Step **PCWA** (Wait for Activation) of category P0 is **scheduled**.

  Default setting is scheduled date = contract activation date (ISTCONTSRTTS)

On contract activation date:

- Step **PCWA** (Wait for activation) of category P0 is processed by the batch report *CRM_ISX_MSG_START_PLANNED*, which results in the start of processing of category P1 steps

- Step **PCEA** (Call FICA message contract activation) of category P1 is executed. Update FI-CA provider contract and create CC provider contract

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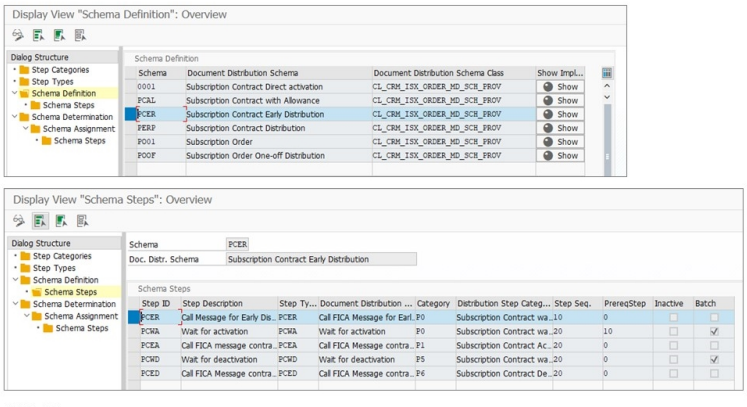

Figure 63: Customizing Early Distribution

Customizing to set up for the early distribution.

LESSON SUMMARY

You should now be able to:

- Understand the Order Distribution Infrastructure
- Explain Fulfillment Monitor
- Describe Customizing the Order Infrastructure

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Unit 4
Lesson 4

Device as a Service

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Understand Device as a Service (DaaS)

Device as a Service (DaaS)

Equipment Integration Overview

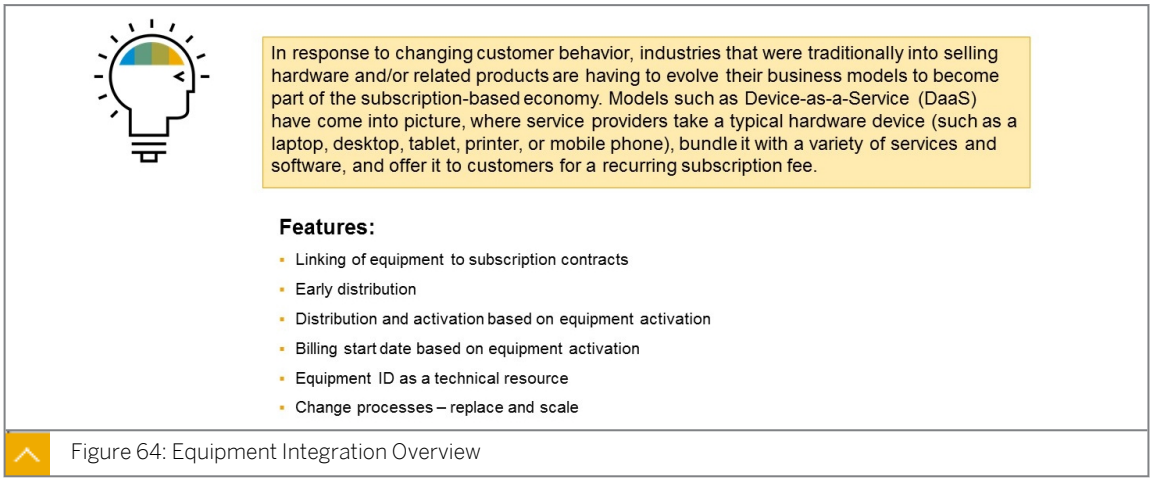

Figure 64: Equipment Integration Overview

Service Providers take a typical hardware device (such as a laptop, desktop, tablet, printer, or mobile phone), bundle it with a variety of services and software, and offer it to a customer for a monthly subscription fee. It gives customers one contract and just one provider to hold accountable.

Examples:

- A company that offers laptop devices, IT support, and software (including 3rd party software) bundled into a subscription on a fixed recurring price per device.
- A company that sells exercise equipment outright, but offers related training classes on a subscription basis.

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Equipment ID Maintenance on Contract and Trigger Contract Start

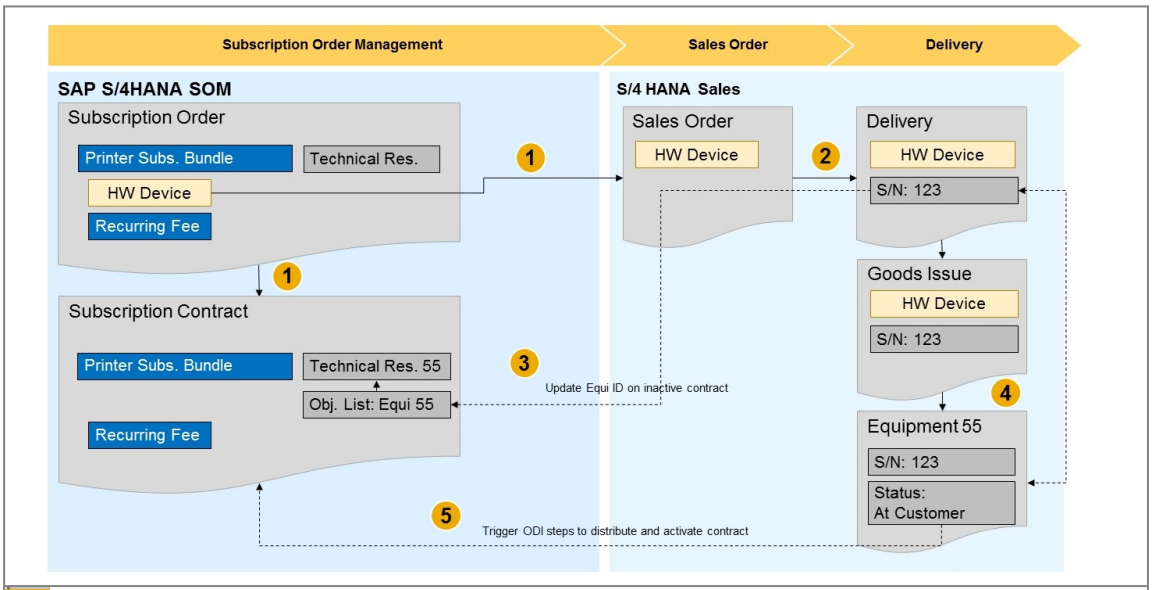

Figure 65: Equipment ID Maintenance on Contract and Trigger Contract Start

The flow of information is as follows:

1. A subscription order is created with a subscription bundle, which contains a sales-relevant item. When the order is submitted, a subscription contract is created with all subscription items. In addition, a sales order is created with the sales-relevant item.
2. A delivery is created as a follow-up of the sales order. The delivery has the picking quantity and a serial number assigned to it. The serial number is for a piece of equipment.
3. When the delivery is saved, the equipment is assigned to the object list of the subscription contract main item. The equipment number is also copied to a technical resource of type equipment.
4. When the delivery is posted goods issue, the status of the equipment is changed to “At Customer”.
5. The new status of the equipment triggers the ODI steps of the subscription contract to distribute and activate the contract. The billing start date is set to the date when the equipment is at the customer site.

When serial numbers are assigned in a delivery and the document is saved, the CREATED/CHANGED event for the LIKP object is raised.

If equipment is created and activated when serial numbers for a product are created, then during delivery, it is possible to identify the relevant equipment IDs.

By tracking back from the delivery, it should be possible to identify the relevant sales item in the subscription order or solution quotation.

If the products are structured in such a way that the sales item is a sub-item of a subscription item or vice versa, then that subscription item is the one on which maintenance of equipment is required.

By registering an event receiver, and leveraging the still open for change subscription order or the new Maintain Equipment IDs change process, equipment IDs can be maintained on the relevant subscription contract.

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When equipment status is changed, the CHANGED event for the EQUI object is raised.

If this equipment, which is changed, is maintained in the object list of a subscription contract, then potentially depending on the status, different processes can be triggered on the subscription contract.

In this example, the At Customer status is being checked for and if it is set, an attempt is made to trigger the ODI step of the relevant subscription contract to start the distribution and activation process.

Sample Product

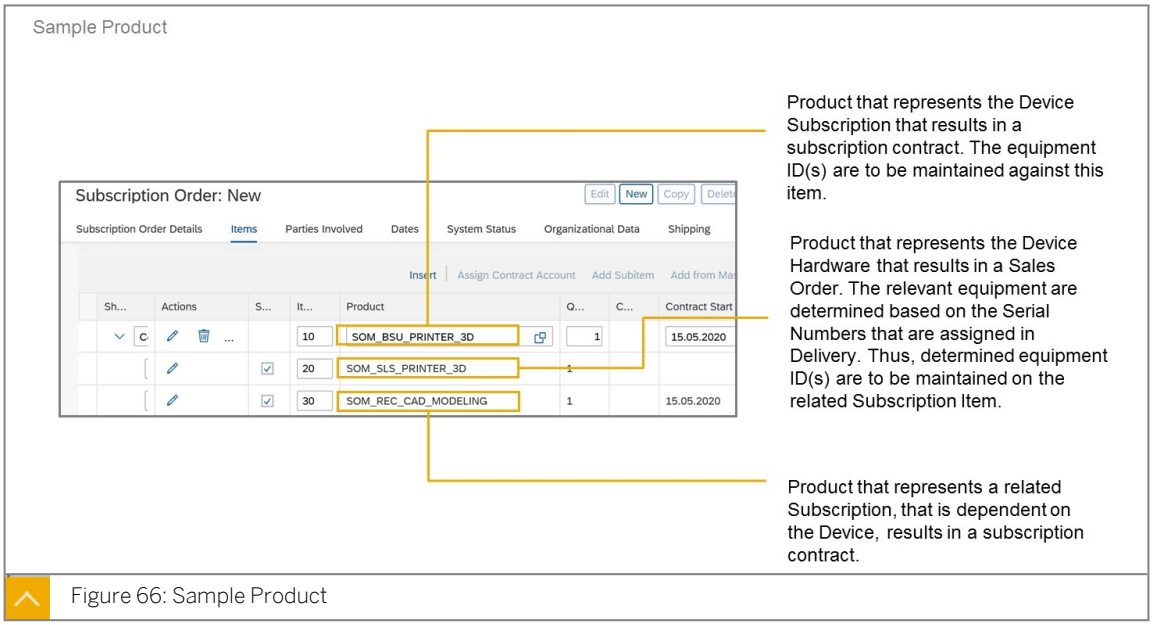

Figure 66: Sample Product

To support models where the subscription is for the hardware or the subscription(s) are dependent on the hardware, the feature of integration between the equipment master and the subscription contract is provided. The ability to bundle different kinds of products (hardware, service, and subscription) already existed and hardware products resulted in the creation of SD Sales Orders from which downstream processes such as Delivery, Goods Issue, and so on, could be performed. This has been taken further now, where if the hardware resulted in the creation of an Equipment item, then such an Equipment item can be linked back to the related Subscription Contract.

With this integration, dependent processes such as the follow can be realized:

- Activation of a subscription and start of billing only after the hardware is delivered to the customer and installed
- Cancellation of a subscription if the related hardware is deactivated

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Equipment ID on a Subscription Item

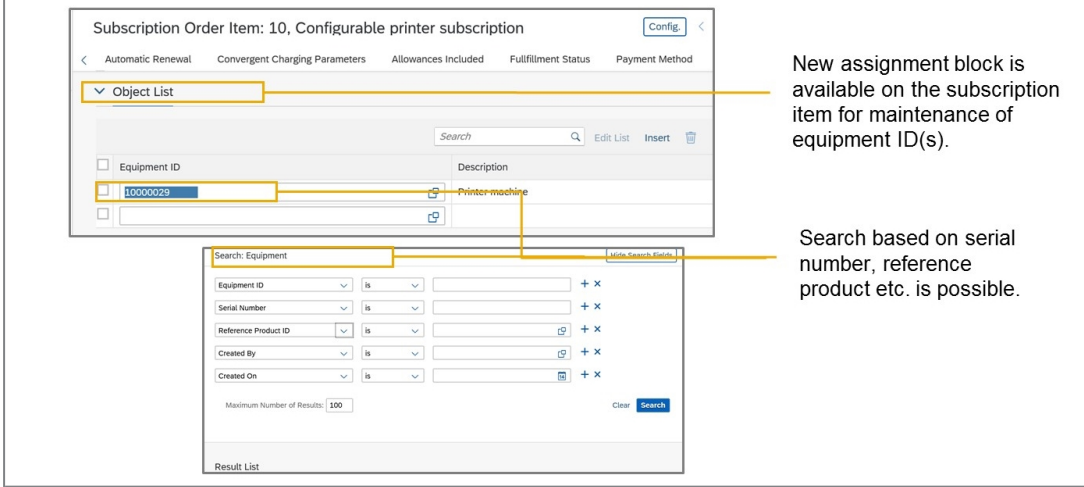

Figure 67: Equipment ID on a Subscription Item

Equipment IDs for a subscription item in a subscription order or a solution quotation can be maintained under the Object List. Additionally, you can search for equipment based on the Equipment ID, Serial Number, Reference Product ID, and so on. The maintained Equipment IDs are copied to the Subscription Contract. If the equipment IDs are unknown during the creation of the subscription order or the solution quotation, the subscription contract is created without the equipment IDs, while the contract remains inactive. Equipment IDs that are determined after downstream processes, such as delivery of the hardware, must be maintained on the related Subscription Contract. The Maintain Equipment IDs change process is available to support the maintenance of equipment IDs later in the process.

Note:
This process can only be used if the contract is inactive, or the preceding document is not open (for example, Solution Quotation) and distribution is not in progress.

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Equipment ID as a Technical Resource

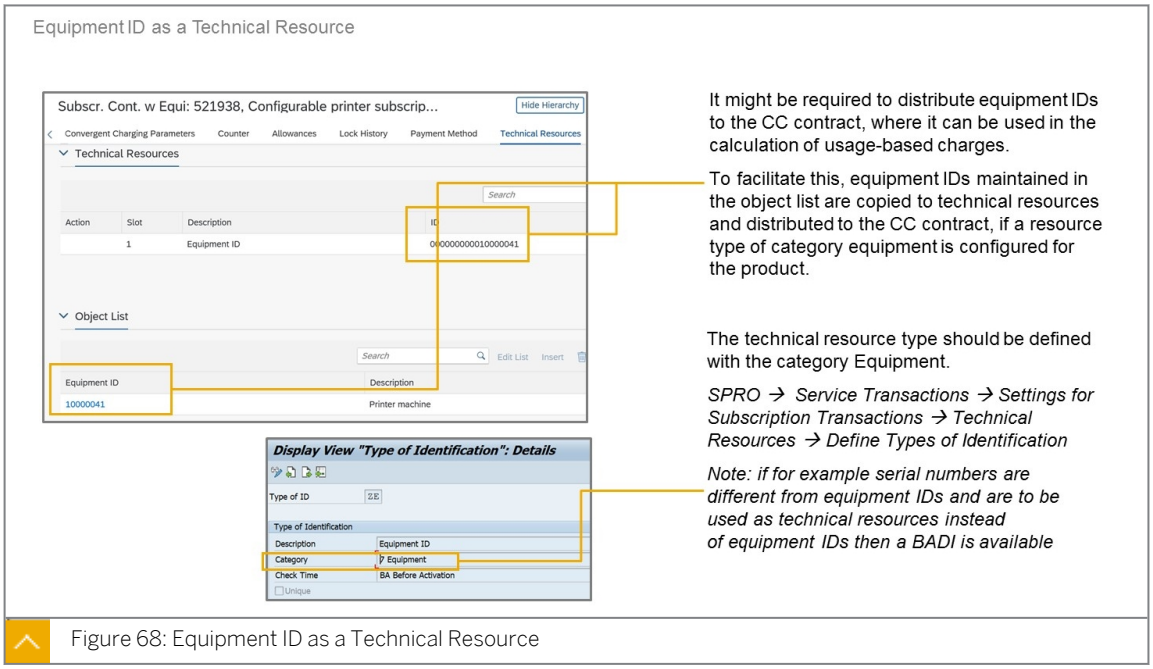

Figure 68: Equipment ID as a Technical Resource

The ID associated with the hardware (serial number or equipment ID) may be relevant in the calculation of usage-based charges in CC. To facilitate this, equipment IDs maintained in the object list are copied to technical resources and distributed to the CC contract. The prerequisite for such a copy is that the Technical Resource Type should be defined with the category Equipment. The standard implementation copies equipment IDs, as they are, to technical resources.

In scenarios where the serial number is set up to be different from equipment IDs and the serial number is to be used as a technical resource, or if the technical resource is to be derived based on the equipment ID, then this can be done by implementing the CRMS4_SOM_BADI_EQUI_INTEG BADI. In scenarios where the equipment IDs are not available at the time of subscription order or solution quotation creation, then the technical resource should be set up in such a way that the time of allocation check should be set to BA-Before Activation to suppress the check during order capture.

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Equipment ID on an Inactive Subscription Contract

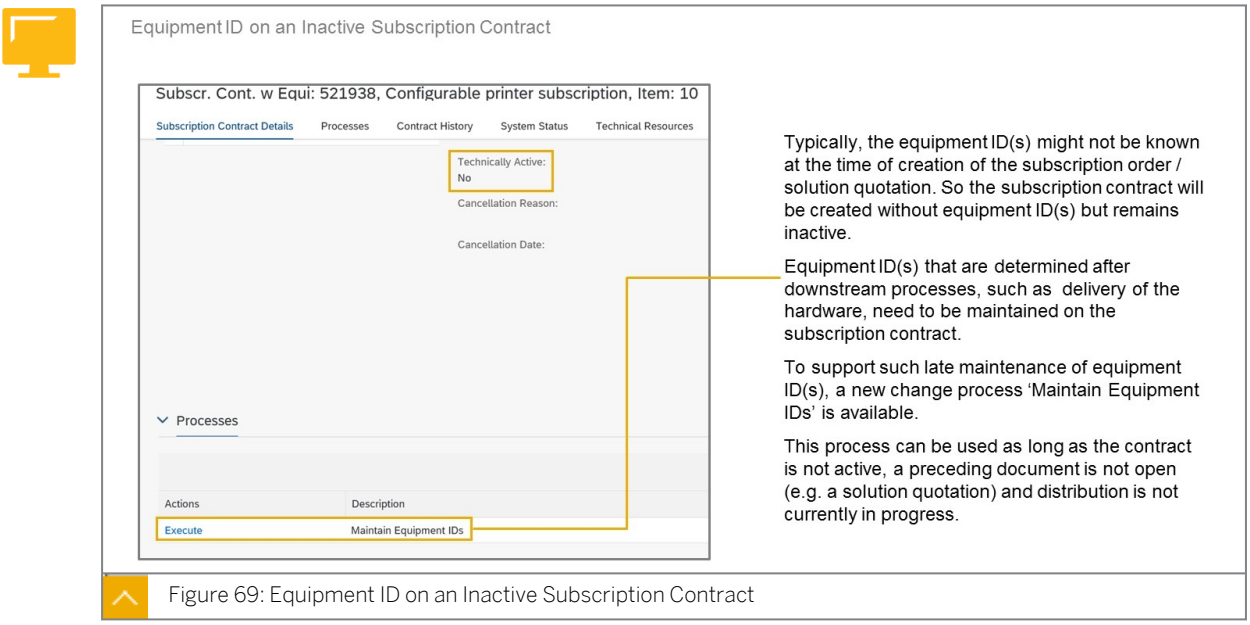

Figure 69: Equipment ID on an Inactive Subscription Contract

Early Distribution

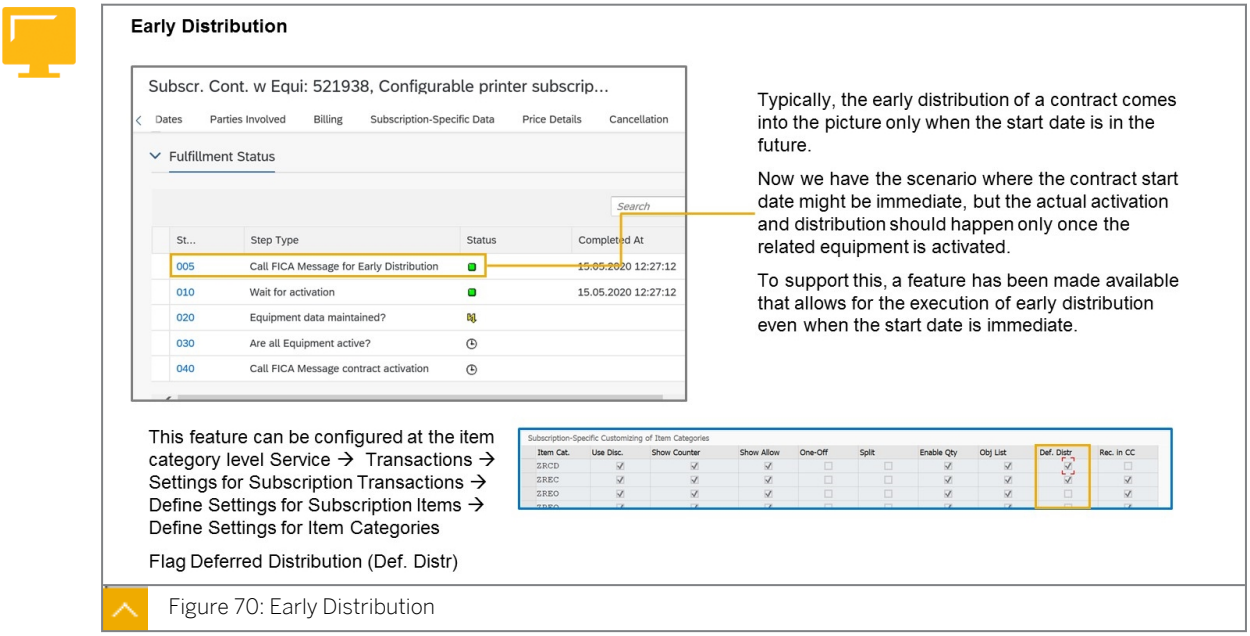

Figure 70: Early Distribution

The previous figure shows early distribution when equipment activation is delayed.

Typically, early distribution of subscription contract is applicable only when the start date is in the future. With equipment, we have the scenario where the contract start date may be immediate, but the actual activation and distribution should happen only when the related equipment is activated, which can be in the future. With Revenue Accounting and Reporting in the picture, an early view into such a contract may be required even though it is not yet active. To support this, a feature is provided that allows for execution of early distribution even when the start date is immediate, but the actual distribution is to be deferred. This can be configured at the item category level.

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Distribution and Activation Based on Equipment Activation

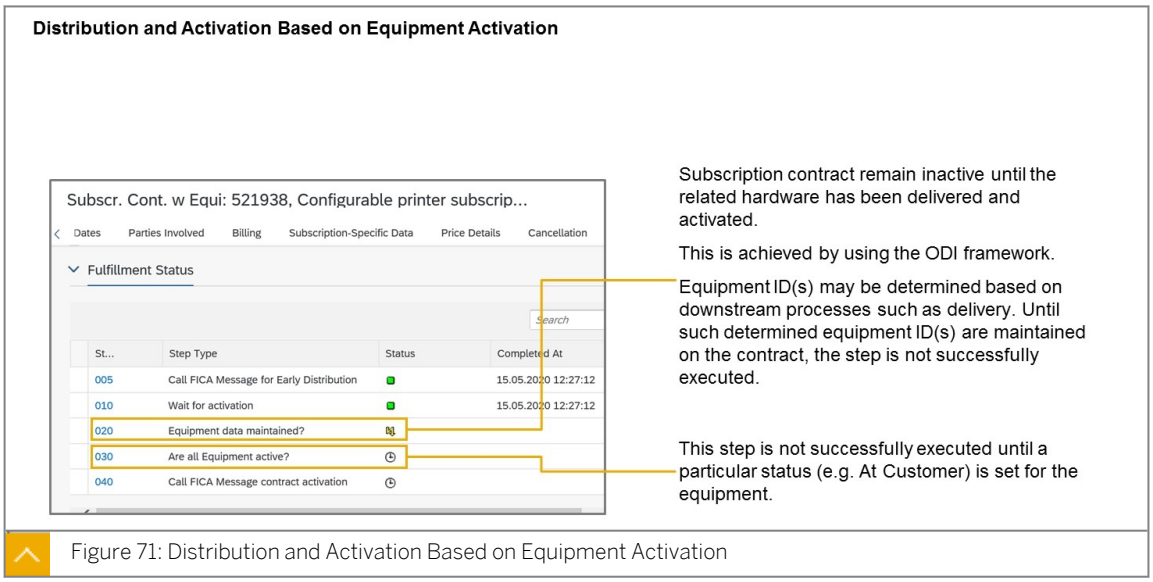

Figure 71: Distribution and Activation Based on Equipment Activation

An example implementation of how equipment IDs can be automatically maintained on a subscription contract based on serial number assignment in the corresponding delivery is provided. If the equipment master is created and activated when serial numbers for a product are created, then during the delivery, it is possible to identify the relevant equipment IDs. By tracking back from the delivery, it should be possible to identify the relevant sales item in the subscription order or solution quotation.

If the products are structured in such a way that the sales item is a sub-item of a subscription item or vice versa, then that subscription item is the one on which maintenance of equipment IDs is required. By leveraging the "still open for change" subscription order or the Maintain Equipment IDs change process, equipment IDs can be maintained on the relevant subscription contract. This system-based auto update is achieved using event type linkages (see transaction `SWE2`). When serial numbers are assigned in a delivery and the document is saved, the CREATED/CHANGED event for the LIKP object is raised. A SOM_EQUI_DELV event receiver is registered against this event, which facilitates the maintenance of the equipment IDs on the subscription contract.

This is strictly an example only as each customer may implement these processes in the manner they see fit: Some examples:

- Customer A has implemented the creation of equipment master data when serial numbers are created for a product. In this case, at the time of delivery creation, based on the serial numbers, it is possible to identify the relevant equipment IDs so as to trigger the maintenance of the same on the inactive subscription contract. The example implementation mentioned previously caters for this.
- Customer B has implemented the creation of equipment master data only when the delivery of the hardware to the end user is complete. In this case, at the time of delivery creation, it is not possible to identify the relevant equipment. A custom implementation can be built to support this.

Distribution and activation of subscription contract: The subscription contract is not distributed and activated until the equipment IDs are maintained and the relevant status (for example, At Customer, Ready for Use, and so on) is set on the equipment. This is achieved

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using the ODI framework. A standard example EQUI schema is provided that includes the following steps:

- EQUM (Equipment data maintained?) – The status of this step is set to executed only if the equipment IDs are maintained on the subscription contract.
- EQUA (Are all Equipment active?) – The status of this step is set to executed only if the linked equipment has a certain status. Standard implementation checks for ECUS(At Customer).
- PCEA (Call FICA Message contract activation) – The standard distribution step that triggers the creation of the FICA and CC contracts, which is executed only when the two previous steps are successful.

Billing Start Date Based on Equipment Activation

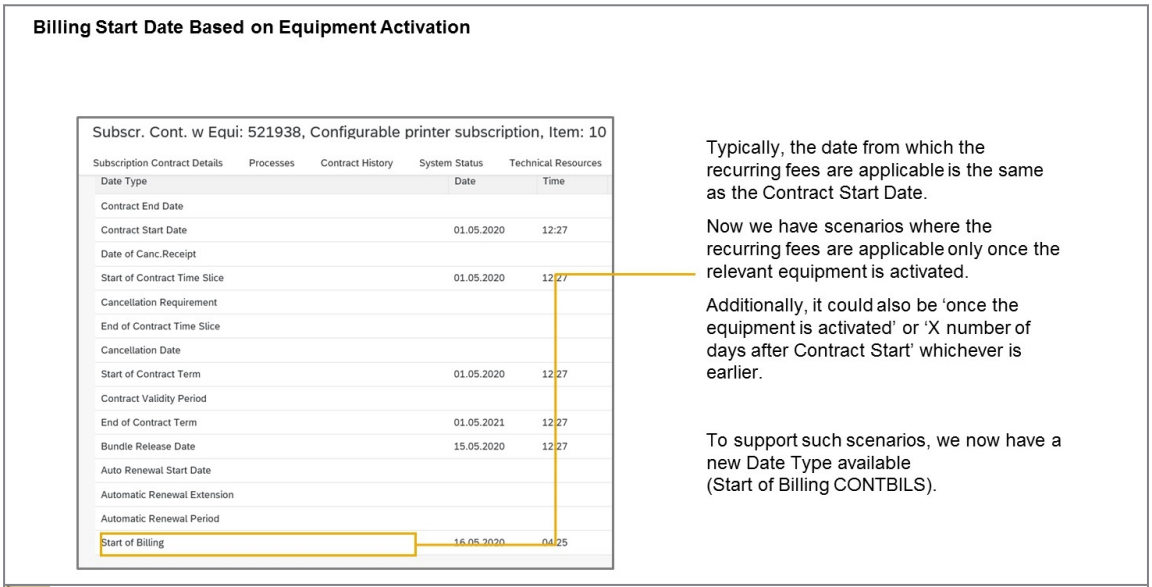

Figure 72: Billing Start Date Based on Equipment Activation

Trigger subscription contract activation on equipment activation: Similar to the previous scenario, an example implementation of how subscription contract distribution and activation can be triggered based on a status on the equipment is provided. When the status of equipment changes, the CHANGED event for the EQUI object is raised. If the ID of the equipment that is changed is maintained on a subscription contract, then potentially, depending on the status, different processes can be triggered.

In this example shown, when the status of equipment is set to ECUS(At Customer), the distribution and activation process of the subscription contract is triggered. A SOM_EQUI_ACTV event receiver is registered against this event, which facilitates the trigger.

> Note:
> The standard example implementation is based on status ECUS, but this can be changed by implementing the CRMS4_SOM_BADI_EQUI_INTEG BADI.

Billing Start Date based on equipment activation: Typically with a subscription contract, the date from which the recurring charges are applicable is the same as the contract start date. When equipment is involved, we have scenarios where the recurring charges are applicable only when the relevant equipment is activated. Additionally, it can also be "once the

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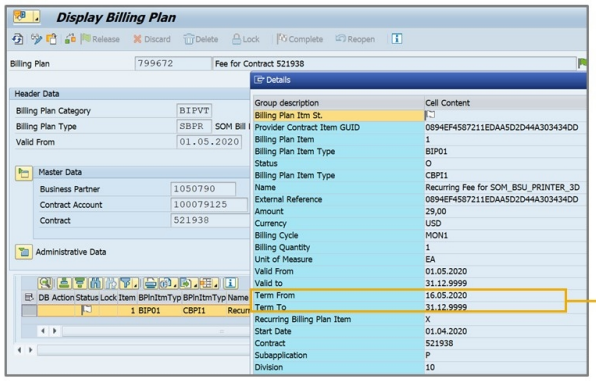
equipment is activated" or "X number of days after contract start", whichever is earlier. To support such scenarios, a CONTBILS(Start of Billing) date type is provided. As standard, this date type is set to the date on which the step EQUA is executed successfully, but this can be changed by implementing the CRMS4_SOM_BADI_EQUI_INTEG BADI.

Billing Start Date Based on Equipment Activation

Billing Start Date Based on Equipment Activation



The 'Start of Billing' date of the subscription contract is mapped to the 'Term From' date of the billing plan item.

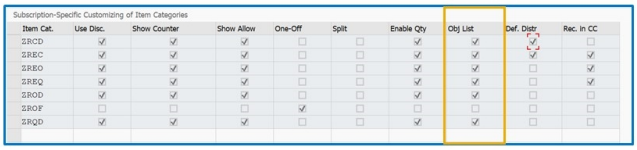
The amount in the BIT that is generated will be prorated based on the 'Term From'.

Note: In the 2020 release, there will be a gap with respect to RAR.

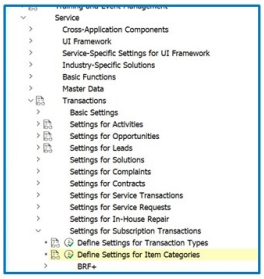
Figure 73: Billing Start Date Based on Equipment Activation

Customizing for Object List in Subscription Items

Customizing for Object List in Subscription Items



Equipment can be assigned to subscription items in the assignment block object list.



Service → Transactions → Settings for Subscription Transactions → Define Settings for Subscription Items → Define Settings for Item Categories

Figure 74: Customizing for Object List in Subscription Items

Prerequisite: Item category is marked as relevant for the equipment link.

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Customizing ODI for Equipment

Customizing ODI for Equipment

Schema Steps

Step ID	Step Description	Step Type	Document Distribution Step Ty...	Category	Distribution Step Category	Step Seq.	PrereqStep	Inactive	Batch
EQUA	Are all Equipment active?	EQUA	Are all Equipment active?	P1	Subscription Contract Activation	30	20	☐	☐
EQUM	Equipment data maintained?	EQUM	Equipment data maintained?	P1	Subscription Contract Activation	20		☐	☐
EQUW	Wait for Contract Start date	EQUW	Wait for Contract Start date - no.	P0	Subscription Contract waits for a.	20	10	☐	☐
PCEA	Call FICA Message contract activ.	PCEA	Call FICA Message contract activ.	P1	Subscription Contract Activation	40	30	☐	☐
PCER	Call FICA Message for Early Distr.	PCER	Call FICA Message for Early Distr.	P0	Subscription Contract waits for a.	10		☐	☐

There may also be a business scenario where the equipment ID is not known upfront. The hardware device would be delivered to the customer and the activation of a subscription contract would depend on the equipment delivery status.

An example implementation is provided for this scenario where the physical device is a sales sub-item of the contract main item. The sales item is processed for delivery where the equipment ID, linked to the serial number of the sales product, is assigned in the delivery document. On POST GOODS ISSUE of the delivery document, the equipment status is updated to ECUS.

The ODI schema EQUI has been delivered with the following steps which are updated based on certain events when a delivery document is created and processed.

Figure 75: Customizing ODI for Equipment

LESSON SUMMARY

You should now be able to:

- Understand Device as a Service (DaaS)

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Learning Assessment

1. A subscription contract is distributed to SAP FI-CA as a provider contract.
   Determine whether this statement is true or false.

   ☐ True

   ☐ False

2. Which of the following objects are included in a subscription contract?
   Choose the correct answers.

   ☐ **A** Technical Identifier

   ☐ **B** Contract Account

   ☐ **C** Business Partner

   ☐ **D** Cross Selling Products

3. It is possible to have gaps between contract phases?
   Determine whether this statement is true or false.

   ☐ True

   ☐ False

4. Which of the following are the key features of equipment integration in SOM?
   Choose the correct answers.

   ☐ **A** Billing start is dependant on equipment integration

   ☐ **B** Equipment items are created from subscription orders

   ☐ **C** Equipment items are technical resources

   ☐ **D** Equipment items are shown in a contract's object list

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Learning Assessment - Answers

1. A subscription contract is distributed to SAP FI-CA as a provider contract.

   Determine whether this statement is true or false.

   [X] True

   [ ] False

   That is correct! A subscription contract is distributed to SAP FI-CA as a provider contract.

2. Which of the following objects are included in a subscription contract?

   Choose the correct answers.

   [X] A Technical Identifier

   [X] B Contract Account

   [X] C Business Partner

   [ ] D Cross Selling Products

   That is correct! All of these objects are included in a subscription contract with the exception of Cross Selling Products, which are not included.

3. It is possible to have gaps between contract phases?

   Determine whether this statement is true or false.

   [ ] True

   [X] False

   That is correct! Gaps between contract phases are not permitted.

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4. Which of the following are the key features of equipment integration in SOM?

   Choose the correct answers.

   [X] A Billing start is dependant on equipment integration

   [ ] B Equipment items are created from subscription orders

   [X] C Equipment items are technical resources

   [X] D Equipment items are shown in a contract's object list

   That is correct! Billing start is dependant on equipment integration, equipment items are technical resources, and equipment items are shown in a contract's object list. These are all key features of equipment integration in SOM.

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UNIT 5 Master Data Product

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UNIT OBJECTIVES

- Define Subscription Product
- Explain Product Modeling
- Describe Product Master Data Views
- Understand Subscription Specific Data
- Understand Cross Catalog Mapping (CCM)
- Explain SAP Convergent Charging (SAP CC) Tables
- Describe Advanced Variant Configuration (AVC)
- Explain Variant Configuration Modeling
- Detail Variant Pricing

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- Discuss Product Training Examples

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Unit 5
Lesson 1

Product Modelling Overview

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Define Subscription Product
- Explain Product Modeling
- Describe Product Master Data Views

Business Example

You are a product manager in Sales Organization 1010 Cloud4You. A new subscription cloud product for data storage is launched on 01.01.2020. You are required to design the new data storage product with name *SCR_DATA_STORAGE*. The product includes a recurring price component and a usage price component.

Explain Product Modeling

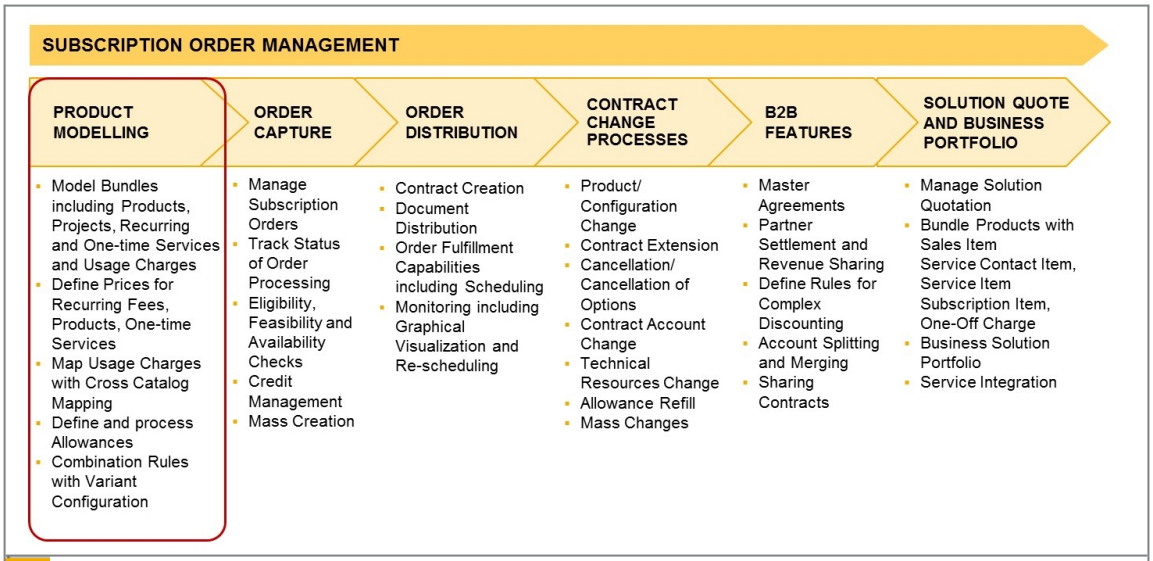

Figure 76: Overview Subscription Order Management

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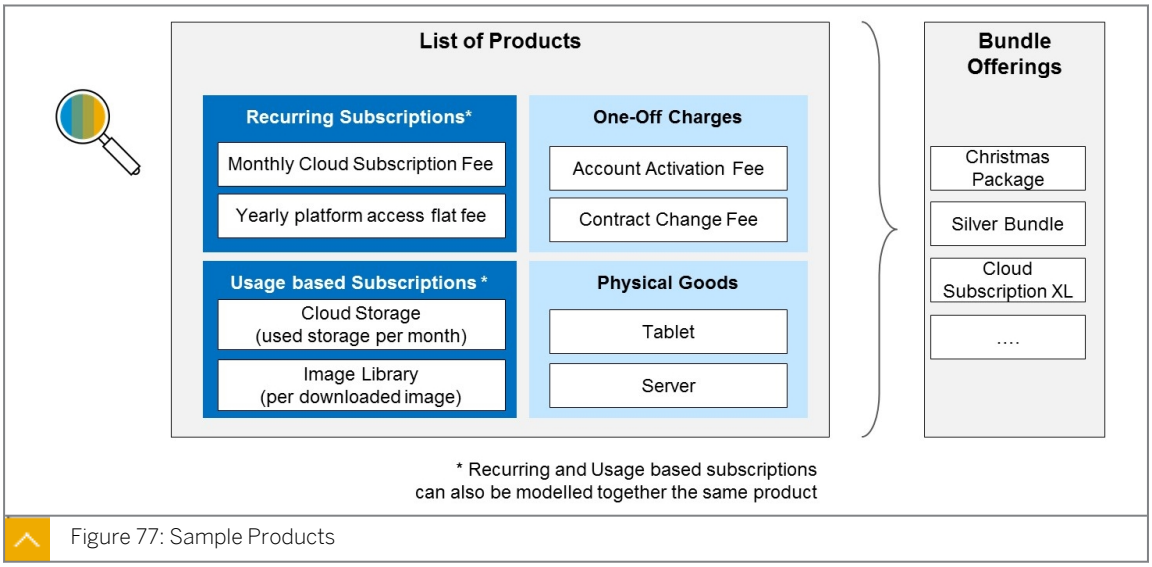

Figure 77: Sample Products

Subscription orders support different kind of products:

Subscription products

Subscription products can be used to subscribe to a service. Fees and charges can be managed in different ways.

- There are usage-based subscriptions for consumption, based pricing models for which the pricing is defined in SAP Convergent Charging.
- There can also be products, which shall be billed on a regular basis, for example, monthly, the recurring subscriptions
- There can also be one time fees, the one-off charges like an activation fee for a contract

All these subscription products are billed in SAP Convergent Invoicing as billable items. Billing documents are either being created directly in SAP Convergent Invoicing (recurring fees), being sent from Subscription Order Management (SOM) (one-off charges), or being sent from SAP Convergent Charging (usage based).

Physical goods

Physical goods can be treated as sales items, resulting in an SD sales order to support logistical processes, such as delivery or goods issue.

Bundles

Bundles can combine different kinds of products, to be sold in a bunch and to offer a specific process for the single bundle components. Bundles can be sales packages but also subscription products.

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A Material Master/Product…

- Has a Product type
- Can be configurable
- Can have Sales attributes….
- Has a Price…
- Can be Goods – inventory
- Needs different orders…
- Can be a Service….

Industry Examples:

1. Product One-OFF: Activation Fee for Cloud Selection Service
2. Product Sales Distribution: Sales of a Notebook
3. Recurring: Insurance Fee for a Notebook
4. Usage Charge: Rating and Charging of CPU Usage
5. Service Product: Installation of a Notebook

Figure 78: Definition Material Master

All kinds of Material Masters are maintained in the Fiori App *Manage Product Master Data*. The initial set up always requires a *Product* type to indicate the business scenario. In the meaning of SOM most Materials are Subscription Products.

Define Subscription Product

A Subscription Product…

- provides specific attributes, and settings required for capturing subscription orders and contracts resulting in provider contracts in FI-CA and SAP Convergent Charging
- is modelled as a S/4HANA product master
- specifies certain subscription-specific attributes within the product master like terms of duration, billing cycles, …
- defines the additional Subscription Product-Specific Data like cross catalog mapping, discounts and charges as an own „Manage Subscription Product-Specific Data“ Fiori application
- Is either relevant for recurring fee creation or/and usage fee rating.

Industry Examples:

1. Product REC_LICENSE: Recurring Product for License fee of Software
2. Product BSU_CLOUD_SEL_SERV: Cloud Selection Services with CPU, RAM and Bandwidth Storage including subscription fee and usage rating - B2B Process

Figure 79: Definition of a Subscription Product

A subscription product provides specific attributes and settings required for capturing subscription orders and contracts. These result in provider contracts in Contract Accounts Receivable and Payable (FI-CA) and SAP Convergent Charging. These include terms of duration, billing cycles, discounts, and charges, among other core attributes, using the subscription product.

The subscription product is modeled with S/4HANA product master and specifies certain subscription-specific attributes within the product master, such as terms of duration, billing cycles, and resource assignment schema.

In addition the product master is defined by its subscription product-specific data, like cross catalog mapping or discounts and charges.

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Due to this extension, you require 3 Apps while modeling subscription products

These applications are available from the SAP Fiori Launchpad.

Features:

- Display subscription specific product data in Fiori app *"Manage Subscription-Specific Data"*
- Navigate to Product Master from Search Result List
- Navigate to Mapping and Range Tables
- Enable a subscription product to support charge plan activation and external objects, discount / charge activation, and parameter values, based on characteristics
- Manage allowances

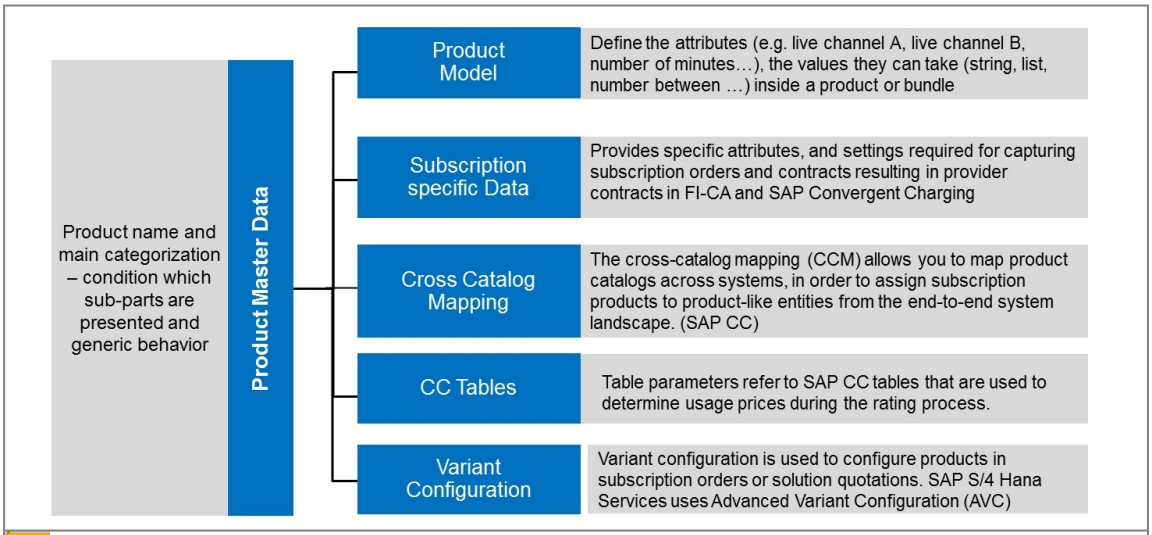

Figure 80: Product Elements Details: What is a Product in Subscription Order Management?

With Product Models it is possible to configure a product so that it allows the user/customer to choose from a number of options when ordering the product:

- The product contains a number of **characteristics** (attributes)
- The user / customer chooses a **value** for each characteristic
- Some characteristic values can also be selected automatically by predefined rules
- **Product components** are selected automatically by rules, based on the chosen characteristic values
- The product contains a number of **characteristics** (attributes)
- The user / customer chooses a **value** for each characteristic
- Some characteristic values can also be selected automatically by predefined rules
- **Product components** are selected automatically by rules, based on the chosen characteristic values

All products, but **subscription products with usage**, can have dependent components allocated to them:

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- **Less commercial relevance**, more driven by technical or process requirements
- Can have complex conditions (including **Business Rule Framework (BRF)**, allowing process and/or configuration dependencies) and generation rules are possible
- Often straightforward generation, but optional or alternative components are also possible.
- Dependent components usually do **not become part of the Subscription contract**.
- Usually appear on the lowest hierarchy level of a complex package

Technical Resources are the (typically unique) identifiers of a contract that are used in SAP CC to identify the contract for rating and charging.

In the product an *"assignment schema"* is assigned. The assignment schema contains one or more technical resource types. for example, one phone number and one SIM card ID.

Cross Catalog Mapping is the mechanism to integrate the pricing algorithm, defined in the rating catalog, with the commercial product.

Describe Product Master Data Views

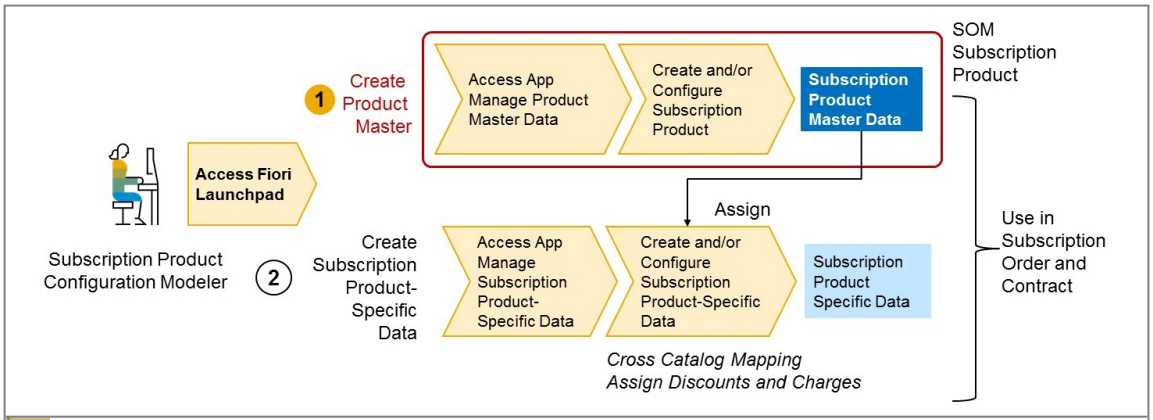

Figure 81: Create Subscription Product - Manage Product Master Data

To maintain a subscription product, it must be first created in the product master, using the *Manage Product Master Data* app. This subscription product is then extended by its subscription product-specific data, using the *Manage Subscription Product-Specific Data* app.

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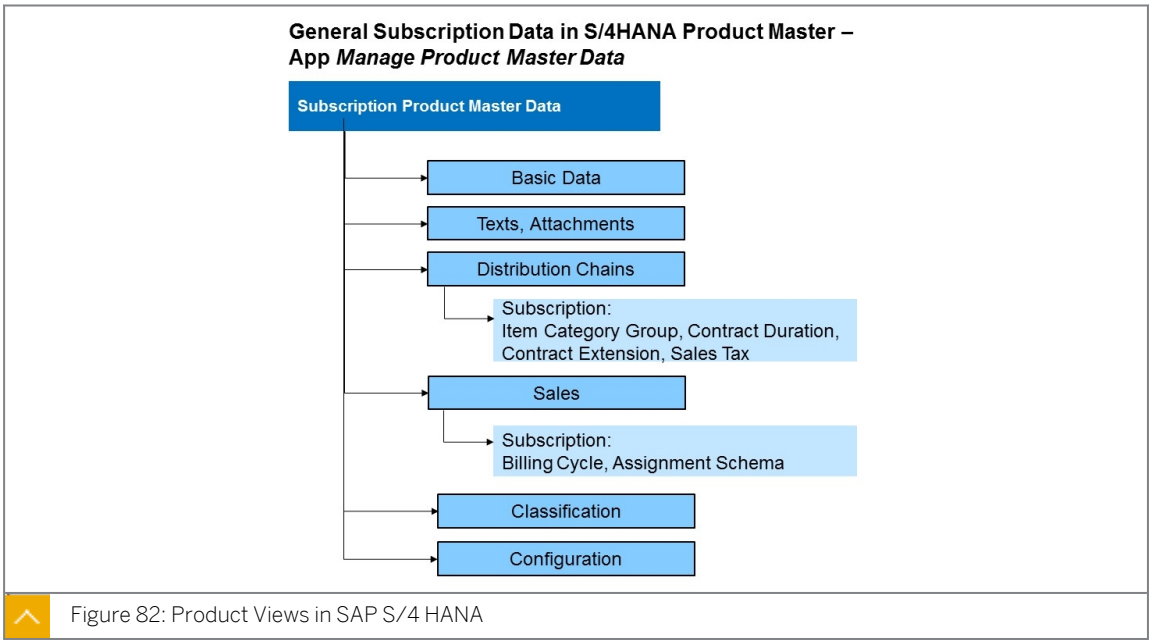

Figure 82: Product Views in SAP S/4 HANA

In S/4HANA only the product master, also known as the material master, is used to maintain central product data. In addition attributes like cross catalog mapping, or discounts and charges are maintained in their own application.

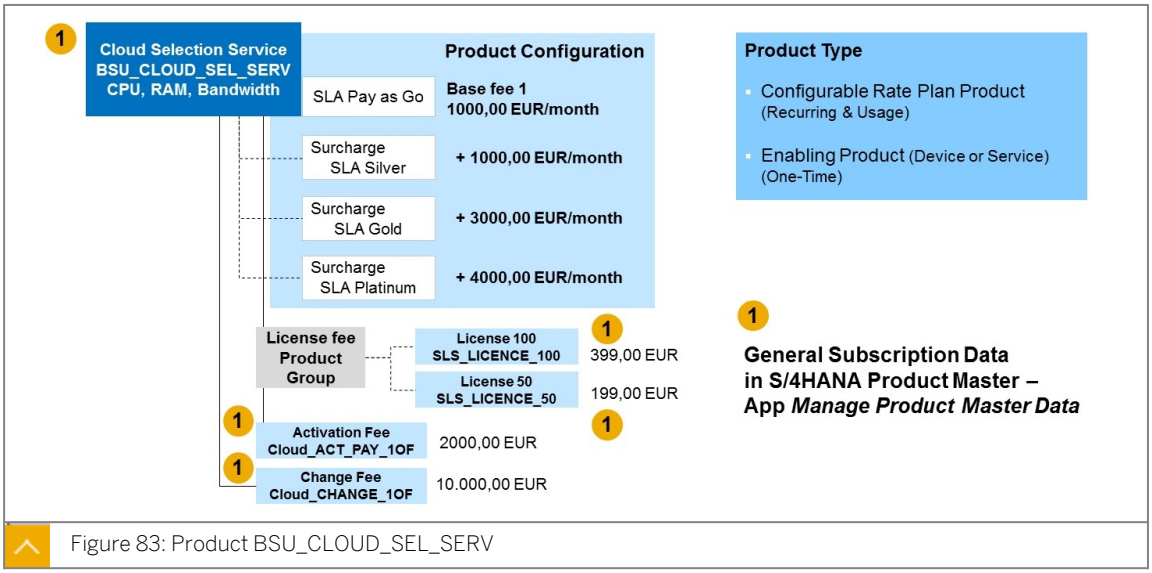

Figure 83: Product BSU_CLOUD_SEL_SERV

Example of a product structure:

In this case a bundle is the container for each product. The first step to *Create the Cloud Selection Service* is to maintain the material master with the App *Manage Product Master Data*. The same for the activation fee, the change fee, and the licenses.

LESSON SUMMARY

You should now be able to:

- Define Subscription Product
- Explain Product Modeling
- Describe Product Master Data Views

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Subscription Specific Data

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Understand Subscription Specific Data

Business Example

To set up the usage price component you need to add the subscription specific data to your storage product.

Understand Subscription Specific Data

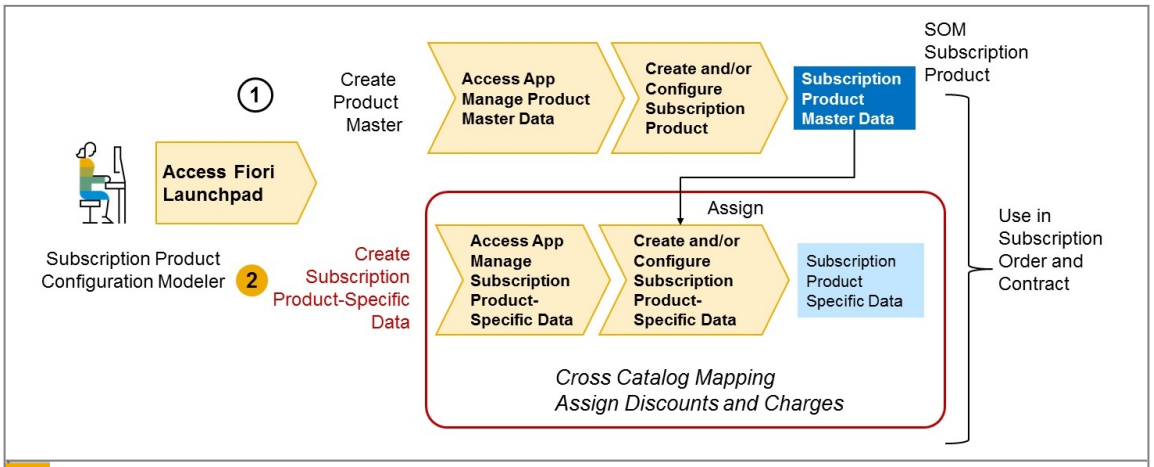

Figure 84: Create Subscription Product Specific Data

To maintain a subscription product, it must be first created in the product master using the *Manage Product Master Data* app. This subscription product is then extended by its subscription product-specific data using the *Manage Subscription Product-Specific Data* app.

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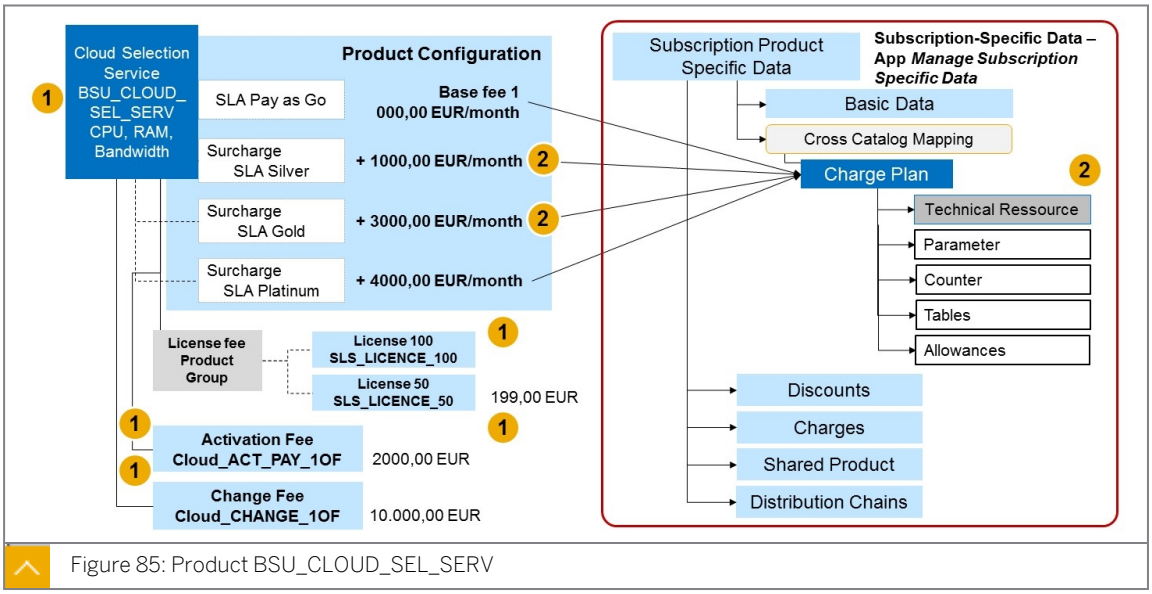

Figure 85: Product BSU_CLOUD_SEL_SERV

Example of a product structure

This figure shows the settings regarding the Cross Catalog Mapping for our Cloud Selection Service Product. The named *SLA Parameters* are linked to the *CCM - Parameters*.

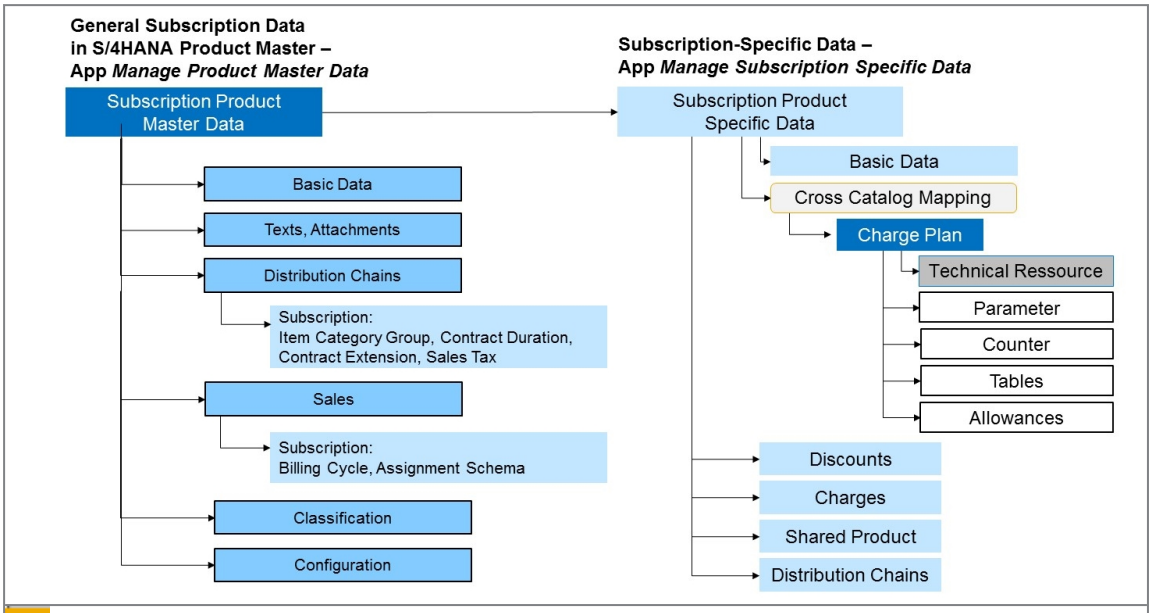

Figure 86: Product Views in SAP S/4 HANA

LESSON SUMMARY

You should now be able to:

- Understand Subscription Specific Data

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Unit 5
Lesson 3

Cross Catalog Mapping

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Understand Cross Catalog Mapping (CCM)

Business Example

Product Cloud Selection Service Product *BSU_CLOUD_SEL_SERV* has a cross catalog mapping with one chargeplan for usage. The chargeplan includes 3 Charges, CPU Usage Charge, RAM Usage Charge, and Bandwidth Usage Charge.

Cross Catalog Mapping (CCM)

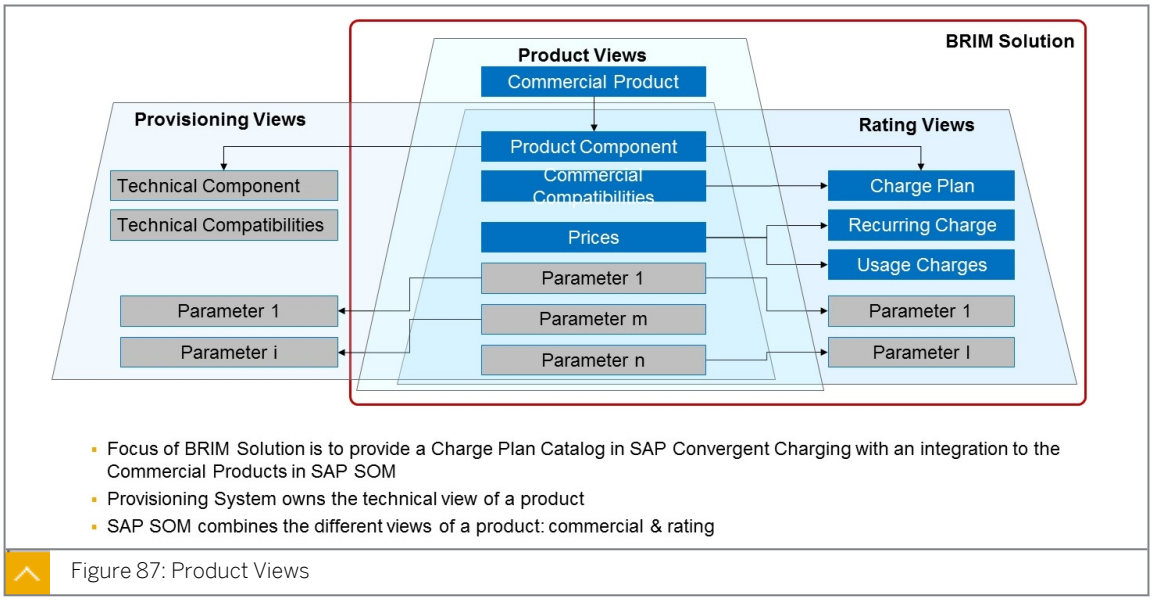

Figure 87: Product Views

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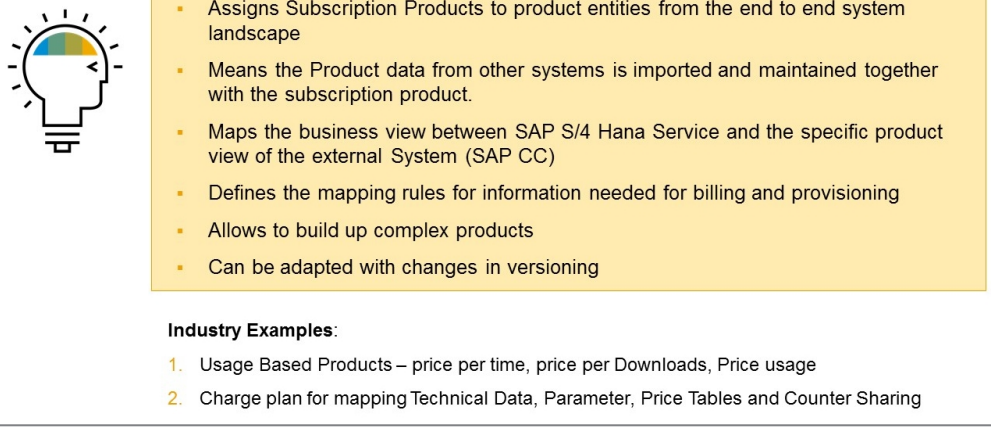

Figure 88: Definition Cross Catalog Mapping (CCM)

CCM is the mechanism to integrate the pricing algorithm defined in the rating catalog to the subscription product. It is possible to define tables, mapping, range tables, or parameters that can later be redefined in the Subscription Order.

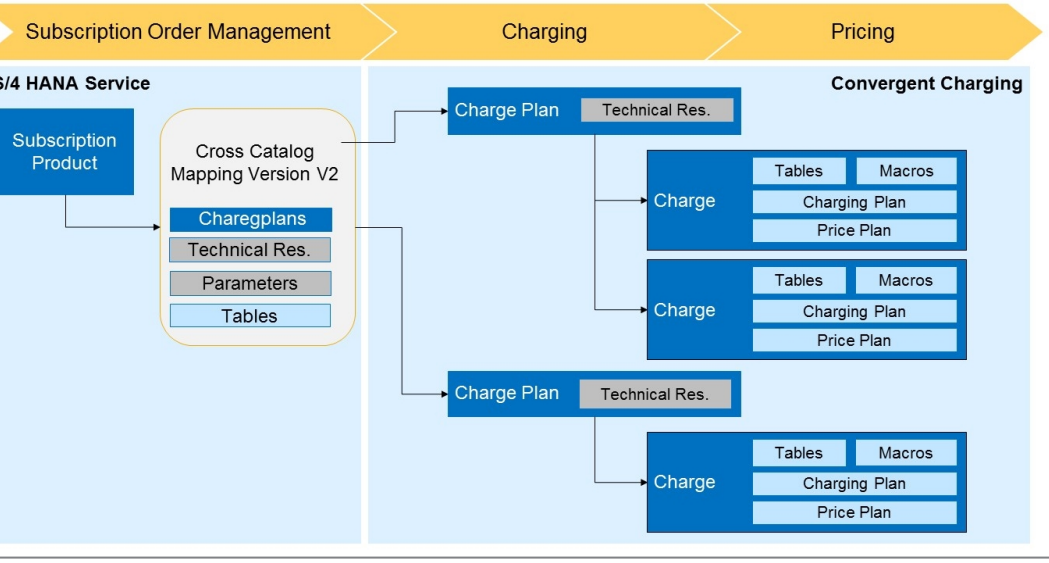

Figure 89: Cross Catalog Mapping: SOM Product with Charge Plan Mapping

CCM is also the mechanism to integrate the pricing algorithm defined in the rating catalog to the commercial product. The pricing engine consists of a graphical user interface to configure pricing trees for usage, recurring and one-time events. It is possible define tables (mapping or range tables) or parameters that can be later redefined.

Charging is the term which describes the process of determining the account of the customer / person, who is paying for the service consumption. The charging logic is stored in the so called *"charging plan"*, a decision tree based structure.

Rating is the term which describes the process of actually calculating what the usage of a certain service costs, such as how much has to be paid.

Pricing is the term describing the process of creating the logic that calculates the prices to be charged for the various services to the customers. The logic is created in so called *"Price*

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plans". These will be covered later in this presentation. The price is calculated, based on the consumption based parameters (Pay per Use).

- CPU (Times how long the user accessed the virtual box) 0,035 c / h

- RAM Usage (How much RAM the user requested) 0,03 c/ MB

- The monthly fee for the cloud select service is defined in the contract in Subscription Order Management (SOM) on a monthly base but triggered in Convergent Charging (CC).

Price plans and charging plans are stored in, so called, *"Charges"*. Many charges can be combined into a charge plan.

The charge plan is the key master data of the new model introduced in SAP CC.

A charge plan includes redefined and customized charges, redefined counters or parameters. It also includes the declarations of the elements that are expected in the provider contract for each activated charge plan, such as, account assignments, user technical identifiers, counters, and parameters.

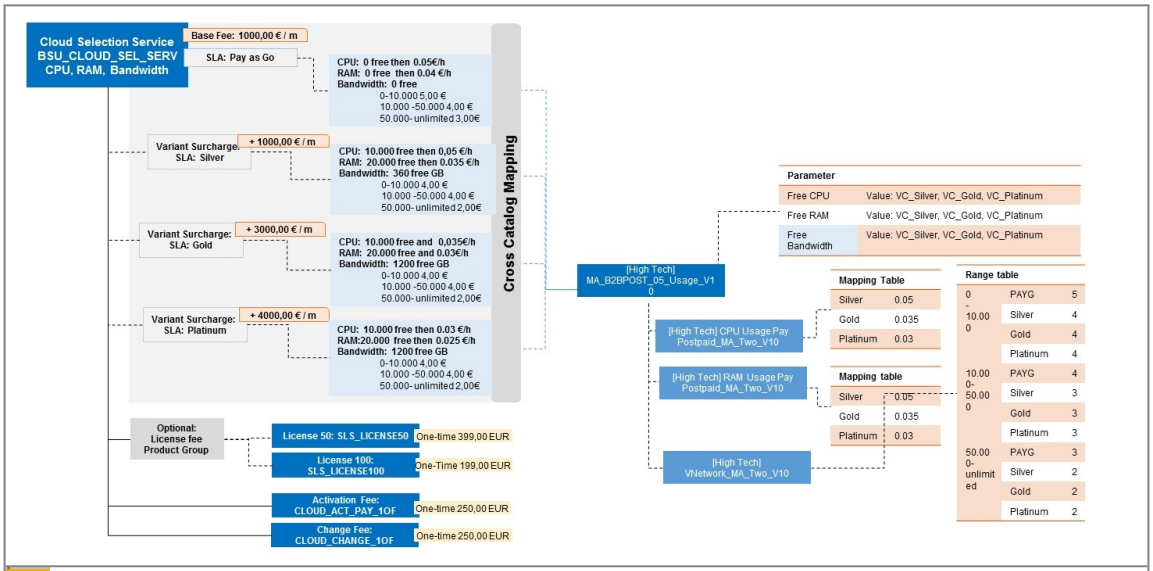

Figure 90: Example of Price configuration

This slide shows the price configuration of our product Cloud Selection Service.

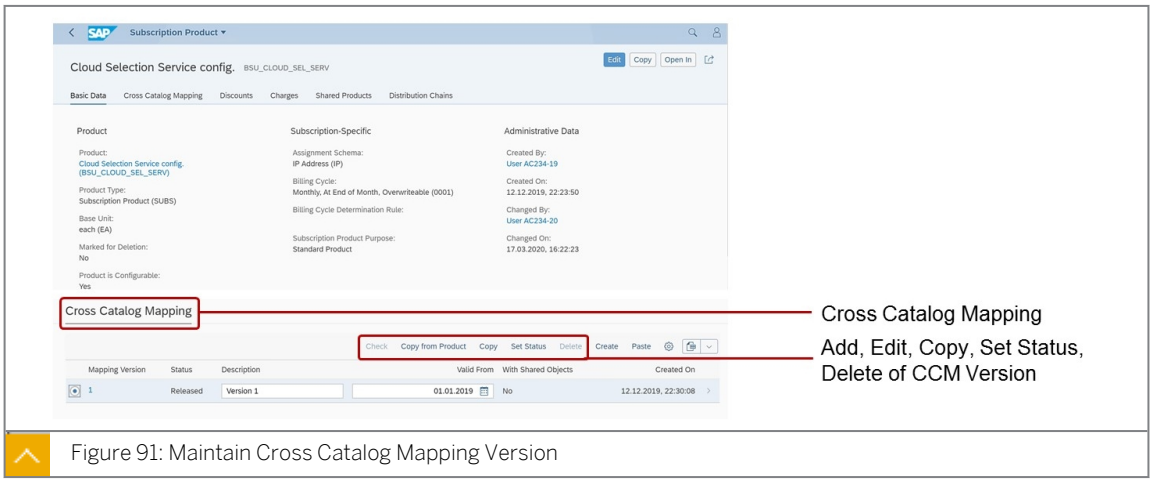

Figure 91: Maintain Cross Catalog Mapping Version

The CCM allows you to map product catalogs across systems in order to assign subscription products to product-like entities from the end-to-end system landscape. The product data

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from other systems is imported and maintained together with the subscription product. By enriching the imported data and defining mapping rules with these components and its sub-components a mapping is done. This mapping is between SAP S/4HANA Service business views, the subscription product, and the specific product view of the external product representation. Information needed in billing, provisioning, or other systems is gathered using the mapping rules defined in CCM.

Mapping versions allow maintenance of time dependent changes on the mapping of a product with an easy status management.

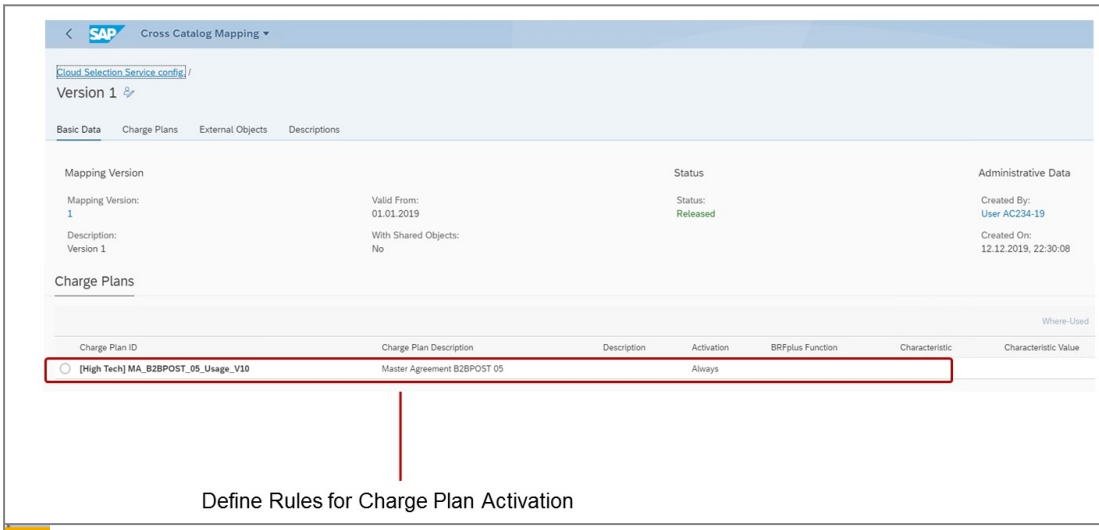

Figure 92: Cross Catalog Mapping Version Details - Charge Plans and External Objects

During charge plan assignment, the external parameters from that charge plan are taken over into the cross-catalog mapping and persisted. You can configure these parameters individually, for example, by redefining charge plan parameters or table parameters. By doing so, you can create individually configured products on the basis of charge plans. You can also further assign them as basic building blocks for complex product definitions. Adapting the charge plan parameters allows you to model individual products as per your specific requirements.

Exchange Charge Plan

You can change the charge plan in an existing assignment by selecting a new charge plan. In this case, the assignment still remains for the mapping version, but with a new charge plan. The assignment ID also remains unchanged. Settings of the previous charge plans, such as, parameter redefinitions are retained as far as parameter IDs are still identical. While new parameters are added and parameters that are not contained in the new charge plan are removed.

Table parameters are also retained as far as the table parameter ID and the table class ID of the new charge plan are identical.

Copy Charge Plan Assignment

The copied charge plan assignment has the same configuration as the source, except for the redefined tables, if any are used. Those are individual object-specific tables that are limited to a certain product, version, and charge plan assignment. In this case, these tables are created as new in SAP CC and then linked to the new charge plan assignment.

Check Charge Plan

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You can verify the accuracy of the charge plan by selecting the relevant charge plan and clicking on *Check*. In case of warnings and errors, the check result is displayed in the message section.

Where-Used

You can view the products, versions, and assignments where a charge plan is used.

Delete Charge Plan Assignment

You can delete a charge plan assignment.

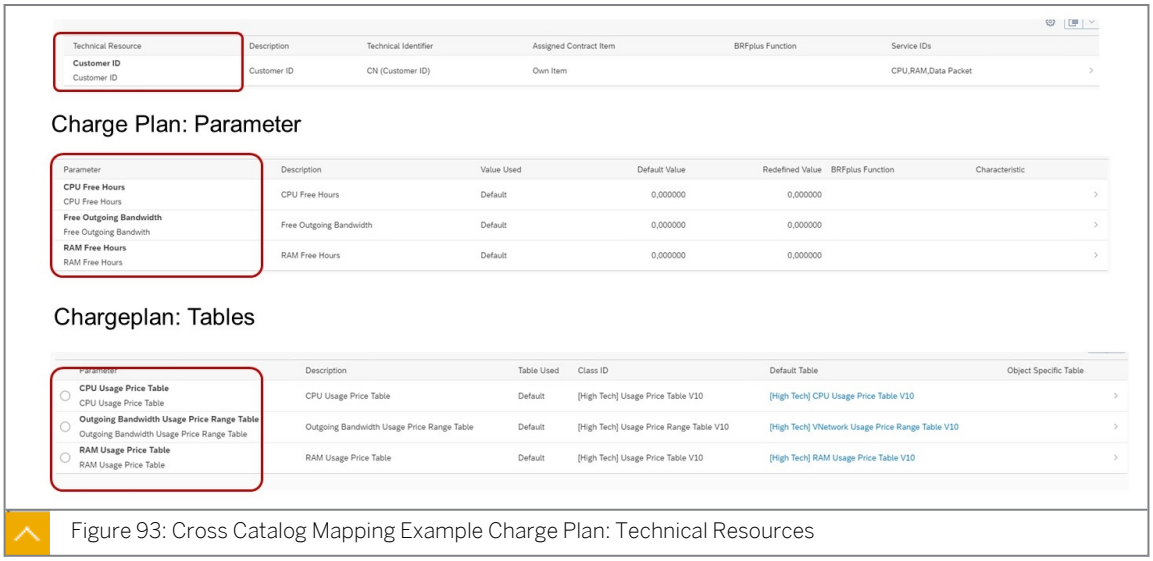

Figure 93: Cross Catalog Mapping Example Charge Plan: Technical Resources

Assignment of technical resources

For technical resources from SAP CC, you can specify the type of the identification by the technical identifier for the offered services. For example, an e-mail address, an IP address, just to name a few. Further, you can use technical resources to determine at which item level the user identification for the customer using this service is defined in the subscription contract.

To determine the technical identifier, use the value help to view all the types of identifier that are assigned to the resource assignment schema, maintained for this subscription product

Assign parameters from the charge plan

Set attributes for the parameters, default values or BRF+. Parameters are defined in the external object during its implementation in SAP CC. There are three data types for parameters, each represented by their individual details sections:

- String Parameters
- Numeric Parameters
- Date Parameters

You can activate parameters:

- *Default Value* (only available for optional parameters): The default value for the charge plan is used.
- *Redefined Value:* A constant value must be provided.
- *BRFplus Function:* The value is determined by the result of an executed BRFplus function.

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LESSON SUMMARY

You should now be able to:

- Understand Cross Catalog Mapping (CCM)

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Unit 5
Lesson 4

SAP Convergent Charging Tables

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explain SAP Convergent Charging (SAP CC) Tables

Business Example

Usage Prices for CPU and RAM is stored in a Mapping Table. Usage Prices for Bandwidth are stored in a Range Table.

Features Product and Cross Catalog Mapping

- Charge plan activation and external objects are based on characteristics. Charge plan parameter are also based on characteristics.
- Discount / charge activation are based on characteristics.

Features Subscription Order / Solution Quotation / Subscription Contract

- Configure subscription items with Advanced Variant Configurator (AVC) via UI and calculate variant surcharges (via SD pricing)
- Hand over configuration information to Contract Accounts Receivable and Payable (FI-CA) provider contract
- Hand over configuration information to Convergent Charging (CC)
- Change process change of configuration
- Display configuration change in the contract history

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SAP Convergent Charging Tables

You can use SAP CC Tables

- To add different parameters to the rating process
- To determine the usage per price during the rating process

SAP CC Tables are

- Stored only in SAP Convergent Charging in 3 different ways
  - Catalog Tables (Tables that are assigned to specific subscription products – must be created without assignment to a business partner)
  - Agreement Tables (any SAP CC Table created in the manage Mapping/Range Table application by assigning a business partner, or any SAP CC Table assigned to a Master Agreement or any SAP CC Table assigned to a subscription item that fulfills the prerequisites of Account split)
  - Subscriber Tables (An SAP CC table is stored as an subscriber table if it is assigned to a subscription item that does not fulfill the prerequisites for an account split)

Industry Examples:

1. RAM usage for Product BSU_CLOUD_SEL_SEV is stored in a Mapping Table
2. Bandwidth usage for Product BSU_CLOUD_SEL_SERV is stored in a Range Table

Figure 94: Definition SAP Convergent Charging Tables (SAP CC)

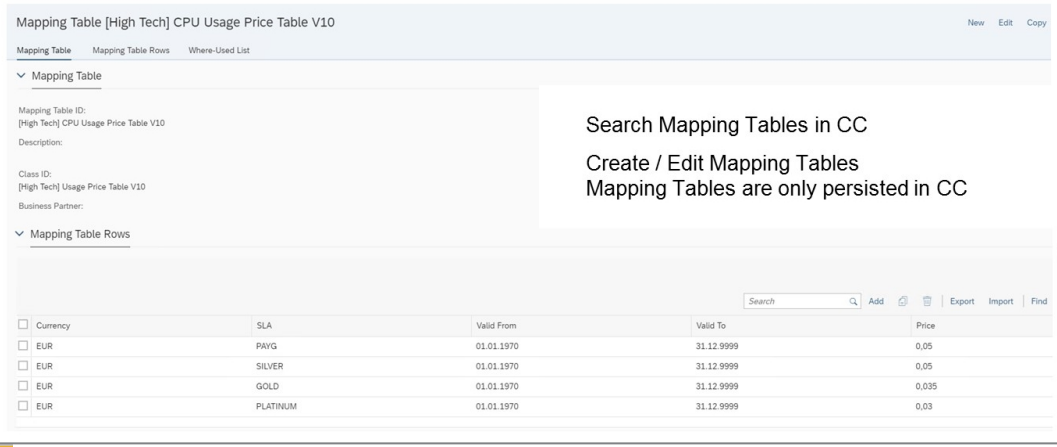

Figure 95: Manage Mapping Table: CPU Usage Table

You can use SAP Convergent Charging (SAP CC) tables to add different parameters to the rating process in SAP CC. You can, for example, determine the usage price from these tables during the rating process in SAP CC. SAP CC tables are stored in SAP CC and not in SAP S/4HANA Service Sales Order Management (SOM), which only stores references to the tables.

SAP S/4 HANA Service offers a search application to maintain the CC tables (mapping and range tables) in a standalone mode.

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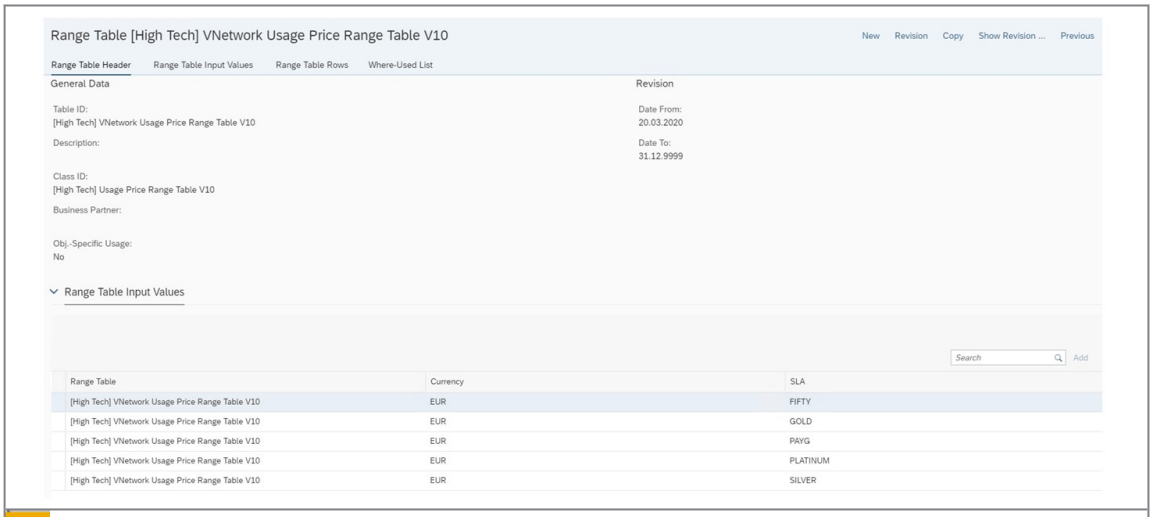

Figure 96: Manage Range Table: Bandwidth Usage Table

LESSON SUMMARY

You should now be able to:

- Explain SAP Convergent Charging (SAP CC) Tables

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Unit 5
Lesson 5

Advanced Variant Configuration (AVC Product Configuration)

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Describe Advanced Variant Configuration (AVC)
- Explain Variant Configuration Modeling
- Detail Variant Pricing

Business Example

A material, in general, is often used for goods/products that are subject of a business activity. A configurable material is a material, which has several variants, for example, Service Level Agreements, Service Combinations, Speed, and Free Volume.

You have recently started in your company and want to make yourself familiar with the product model for Cloud Selection Services. Therefore you check the master data and the configuration model of the product *BSU_CLOUD_SEL_SERV*. Depending on the configuration of the product a surcharge for the service level agreement is set.

In our Business Example Table, *SLA_INC_FREE_VOL* is read to set the Free Volume for CPU, RAM, and Bandwidth depending on the selected service level agreement.

Advanced Variant Configurator Solution Component:

- You can use the S4HANA Variant Configuration to create your Product Models. Variant Configuration via IPC using the Product Modeling Environment (PME) is no longer supported
- Product Modeling and Advanced Variant Configuration are not specific components to Subscription Order Management, but general S/4HANA features
- The following slides demonstrate a basic product modeling set up. For more details please check *help.sap.com -> S/4HANA Product Life Cycle Management (PLM)*
- SAP S/4HANA Service uses Advanced Variant Configuration (AVC)

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Describe Advanced Variant Configuration

Variant Configuration

- is used to configure products in subscription orders or solution quotations
- Is used in SAP S/4 HANA Service

With Variant Configuration

- it is not required to create a separate material for each variant of a product in your company, but it is possible to use one configurable material to cover all variants.
- depending on the characteristics and attributes values set for a product in a subscription item, several dependent features can be set or activated like the activation of charge plans in SAP Convergent Charging or the definition of parameters required for the price models in CC.

Industry Examples:

- BSU_CLOUD_SEL_SERV Cloud Selection Service has different configuration variants –select Service Level: Pay as you go, Silver, Gold, Platinum
- Depending on the Service Level the number of Free CPU, RAM and Bandwidth is set.
- Depending on the Service Level a surcharge is added to the basic fee
- Mobile Phone Package with different Tariffs – Volume of free sms and phone calls

Figure 97: Definition of Advanced Variant Configuration (AVC)

The Variant Configuration app is used to configure your products. Features of Product and Cross Catalog Mapping:

- Charge plan activation and external objects based on characteristics
- Charge Plan Parameter based on characteristics
- Discount / charge activation based on characteristics

Features of Subscription Order / Solution Quotation / Subscription Contract:

- Configure subscription items with Advanced Variant Configurator (AVC) via UI and calculate variant surcharges (via SD pricing)
- Hand over configuration information to Contract Accounts Receivable and Payable (FI-CA) provider contract
- Hand over configuration information to CC
- Change process change of configuration
- Display configuration change in contract history

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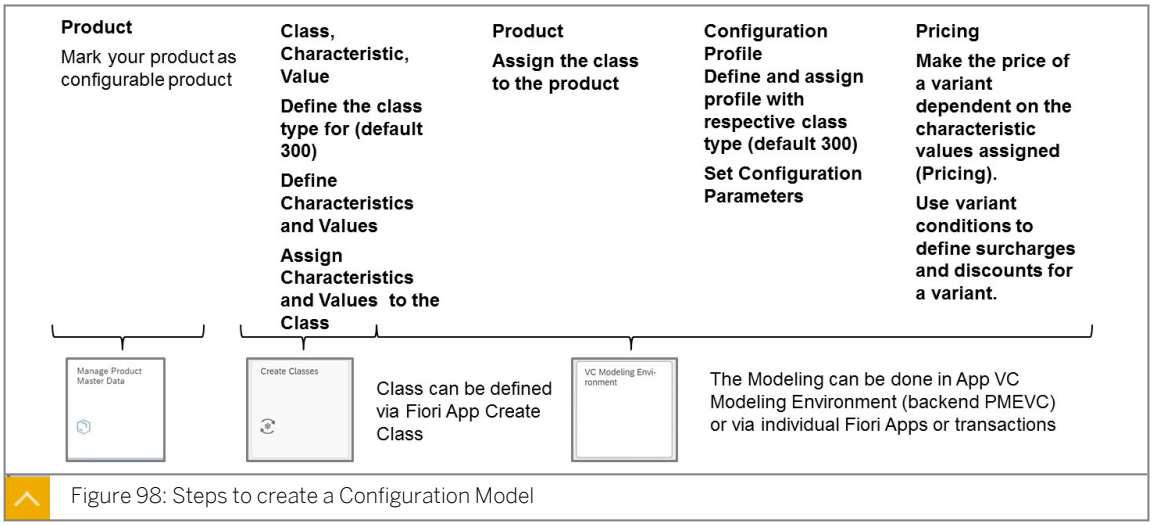

Figure 98: Steps to create a Configuration Model

The Steps to create a configuration Model:

1. In the first step the product needs to checked as a configurable product in the Manage Product Master Data app.
2. In the second step a class of the type *300* needs to be created with the Create Classes app.
3. The definition of characteristics and values is also done via the Created Classes app.
4. In the last Step the product needs to be assigned to the created class via the VC Modeling Environment app.

Explain Variant Configuration Modeling

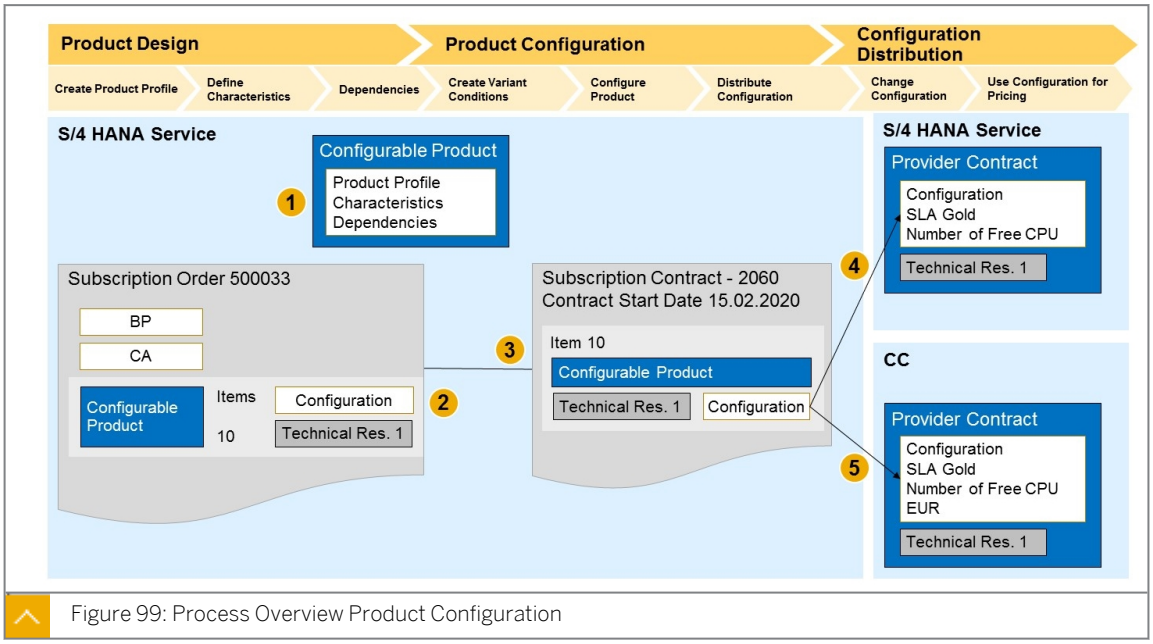

Figure 99: Process Overview Product Configuration

Step 1: Create a Product Profile

Step 2: Configure the Product in Subscription order

Step 3: Distribute the Configuration to Subscription Contract

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Step 4: Distribute the Configuration to Provider Contract in Contract Accounts Receivable and Payable (FI-CA)

Step 5: Distribute the Configuration to Provider Contract in Convergent Charging

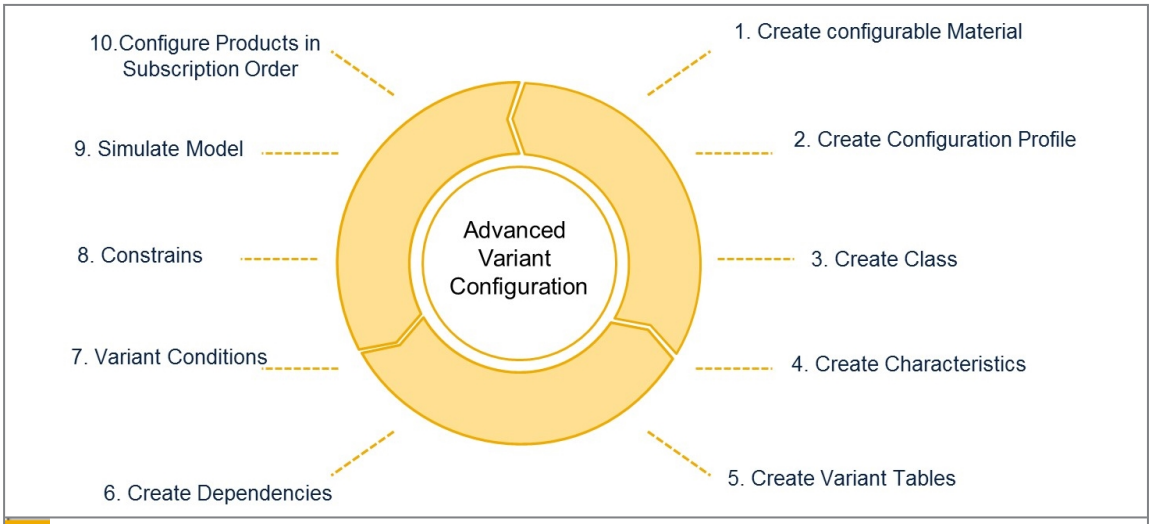

Figure 100: Variant Configuration Modelling

Product modeling steps for creation configurable product *BSU_CLOUD_SEL_SERV*:

1. **Create Configurable Material:**

   A material, in general, is often used for goods/products that are the subject of a business activity. A configurable material *BSU_CLOUD_SEL_SERV* is a material, which has several variants, such as Service Level Agreement, Free Volume, or Surcharges.

   In Material Master the setting, *BSU_CLOUD_SEL_SERV product is configurable* must be set to yes (X = active).

2. **Create Configuration Profile:**

   - Define the class type (default 300)
   - Assign a class which is valid for the respective class type
   - Set Configuration Parameters:
     - Processing Mode "Advanced variant Configuration" only AVC is supported with S/4 Service
     - BOM Explosion NONE, only NONE is supported
     - Process Plnd / Prod.Order

3. **Create Class:**

   Classes allow you to group objects together according to criteria that you define. You can consider classes as a pool, for example, the class "Cloud Service" may consist of the characteristics:

   - Service Level
   - Number of FREE RAM

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- Number of FREE Bandwidth

4. Create Characteristics:

   Characteristics describe the properties of objects. The values of a characteristic specify these properties. E.g. Characteristic "SLA" has the values 'PAY_AS_YOU_GO', 'SILVER', and 'GOLD'.Provisioning Parameter can be Speed.

5. Create Variant Tables:

   In variant tables, you can formulate dependencies between characteristics by listing value combinations row by row. In our business example a surcharge is calculated depending on the service level agreement.

6. Create Dependencies:

   Describe the interdependencies between characteristics and characteristic values. Control which components are selected from a bill of material and which operations are selected from a task list.

- Dependency - Precondition and selection condition

Preconditions are used to hide characteristics and characteristic values that are not allowed and thereby ensure that the configuration of an object is consistent. Selection conditions are used to ensure that all the objects relevant to a variant are selected.

- Dependency - Procedure

Procedures can overwrite default values that are set by other procedures. Procedures can set default values for a characteristic, which can be overwritten by the user.

- Dependency - Constraint and constraint net

Constraint and constraint net are used for declarative formulation of configuration rules, whose processing sequence is determined automatically. This dependency type is mainly for intensively interactive configuration tasks and for configuration tasks in which you need to take into account the dependencies between the characteristics of several objects. The main purpose of a constraint is to monitor the consistency of a configuration.

Constraints have the following distinguishing features:

- You can use constraints to describe the dependencies between completely different objects and their characteristics.
- Constraints are used to store information on which conditions must be fulfilled if the configuration is to be consistent.
- Constraints are not directly allocated to individual objects. They are grouped together to form dependency nets and allocated to a configurable material in the configuration profile.
- Constraints are declarative dependencies. The processing sequence of constraints and the point in time when constraints are processed are not relevant.
- Constraints are not processed in a specific order. You cannot determine when a specific constraint is used.
- In our business example table *SLA_INC_FREE_VOL* is read to set the Free Volume for CPU, RAM and Bandwidth depending on the selected service level agreement.

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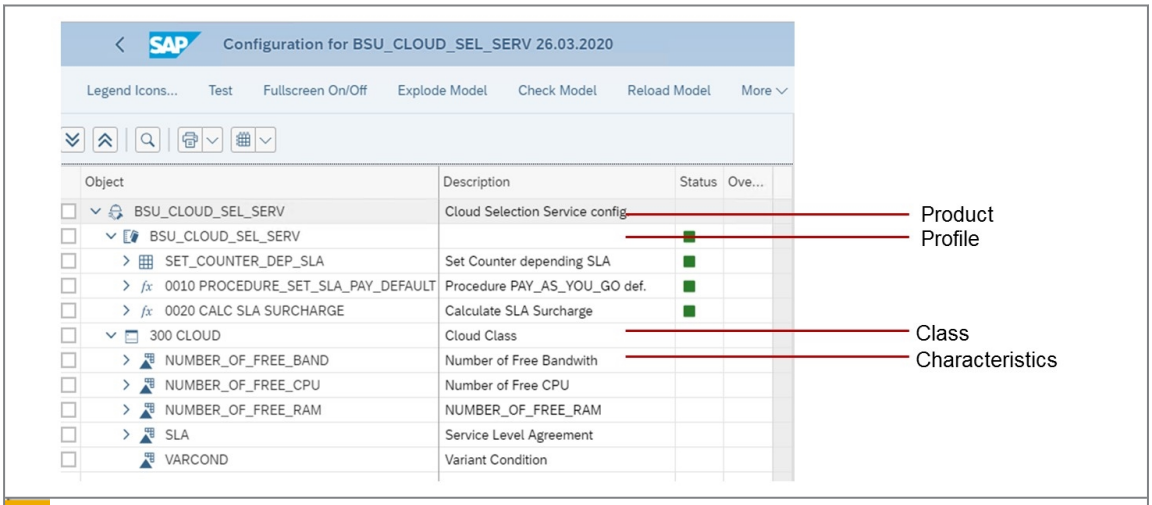

Figure 101: App VC Modeling Environment

Integration to CC and FI-CA:

Saving a subscription item in an order or a quote, or saving a subscription contract, the system persists configuration data and assigns the configuration key to the subscription item.

Convergent Charging:

During Contract Distribution to CC, the configuration data is read and the respective charge plan parameters are supplied with the values read from the configuration of the subscription item

FI-CA:

During Contract Distribution to FI-CA the configuration key is passed to FI-CA provider contract item. The FI-CA provider contract can access the configuration data via this key.

Detail Variant Pricing

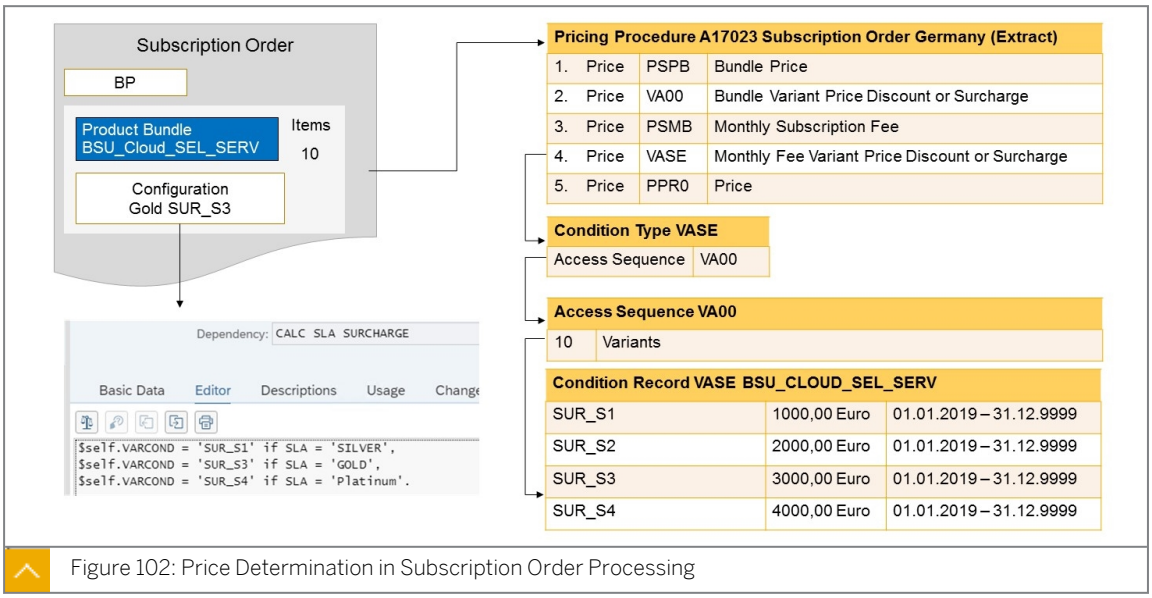

Figure 102: Price Determination in Subscription Order Processing

Create Variant Conditions:

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You can use variant conditions to influence the price of a configurable material depending on the characteristic values assigned. You can define variant conditions during product modeling.

Variant Prices:

Prices, surcharge or discount, can be defined based on a variant condition. Such variant conditions are linked to characteristics and values. The Variant Condition Key is part of the key of a Condition Record.

Pricing:

During configuration these variant conditions are derived, and the respective variant prices are determined and passed to the item. You can assign reference characteristics referring to *SDCOM-VKOND* using procedures and constraints.

The Variant Conditions Keys are collected in a reference characteristic. Typical names for this characteristic are, VARCOND, VKOND, or similar. In the procedure, you enter the characteristic with the table reference and the variant key as a value. In such a procedure you can:

- set the characteristic with the table reference and the variant key as a value
- set pricing factors for surcharges or discounts, which can depend partly on a specific characteristic value and partly on other factors, such as size

LESSON SUMMARY

You should now be able to:

- Describe Advanced Variant Configuration (AVC)
- Explain Variant Configuration Modeling
- Detail Variant Pricing

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Product Training Examples and Customizing

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Discuss Product Training Examples

Discuss Product Training Examples

Tabelle

Description	Product ID	MM Product Type	MM item Product category	Transaction Type	Order Item Type	CCM	Dis-count	Confi-gurable	Condition Type
Subscription Product without usage	**REC_INSURANCE**	SUBS	PRSR	PRVO	PROR	no	no	no	PSMB
Subscription Product with Recurring Fee and Usage	**SCR_CLOUD_UNLIMIT**	SUBS	PRSR	PRVO	PROR	yes	no	no	PSMB
Configurable Bundle Product for Cloud Services, including Licence and Activation Fee	**BSU_CLOUD_SEL_SERV**	SUBS	PRRP	PRVO	PROP	yes	no	yes	PSMB
Configurable Bundle Product for Cloud Services, including Licence and Activation Fee with Discount for RAM and Volume Usage	**BSU_CLOUD_SEL_SERD**	SUBS	PRPR	PRVO	PROP	yes	yes	yes	PSMB
Software License (License Inventory)	**SLS_LICENCE_100** **SLS_LICENCE_50**	HAWA	NORM	PRVO	PROS	no	no	no	PPR0
Activation Fee Change Fee	**CLOUD_ACT_PAY_1OF** **CLOUD_CHANGE_1OF**	SUBS	PRSV	PRVO	PROF	no	no	no	PSPB
Product Bundle for Subscription Contract Creation with Usage and Tablet	**BUN_CLOUD_USG**	SUBS	LUMF	PRVO	PRSP	no	no	no	PSPB
Sales Product: Hardware	**SLS_NB_HARDWARE**	HAWA	NORM	PRVO	PROS	no	no	no	PPR0
Product Bundle for Solution Quotation including Hardware, Installation Service, Support, Insurance and Recurring fee	**BSV_NOTEBOOK_C4U**	BUND	SRVP	SRVP	SRQP	no	no	no	PSPB
Sales Product: Hardware	**SLS_NB_HARDWARE**	HAWA	NORM	SRVP	SRQS	no	no	no	PPR0
Service Product: Installation	**SRV_INSTALL_NB**	NLAG	SCNP	SRVP	SRQM	no	no	no	PPR0
Service Product: Service Contract	**SVC_SUPPORT_CONTR**	NLAG	SRVP	SRVP	SRQM	no	no	no	PPR0

Figure 103: Product Example with Technical Settings (1/2)

This slide shows an overview of possible product setups with technical information. All named products are also examples shown during our training system. It is important here to highlight the needed Product Type in combination with the item category settings. The condition type is related to the standard scenarios, this can vary.

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Tabelle

Description	Product ID	MM Product Type	MM item Product category	Transaction Type	Order Item Type	CCM	Discount	Confi-gurable	Condition Type
Sharing Child Product with Counter	**BSU_CL_SHA_CHI**	SUBS	PRRP	PRVO	PROR	yes	no	no	n.a
Sharing Root Product with Counter	**BSU_CL_SHA_ROO**	SUBC	PRRP	PRVO	PROR	yes	no	no	n.a
Product for Partner Settlement without subscription with usage	**PNR_DOWNLOAD**	SUBP	PRRP	PRPO	PRPO	yes	no	no	n.a.
Configurable Bundle Product for Cloud Services, including Licence and Activation Fee with Allowance Chargeplan for Payment in Advance	**BSU_CLOUD_SEL_ALLO**	SUBS	PRRP	PRVO	PROP	yes	no	yes	PSMB
Master for Bundle Exercise	**SCR_DATA_MASTER**	SUBS	PRRP	PRVO	PROP	yes	no	no	PSMB

Figure 104: Product Example with Technical Settings (2/2)

This slide shows up an overview of possible product setups with technical information. All named products are also examples shown during our training system. It is important here to highlight the needed Product Type in combination with the item category settings. The condition type is related to the standard scenarios, this can vary.

Provide Overview of Product/Material Customizing

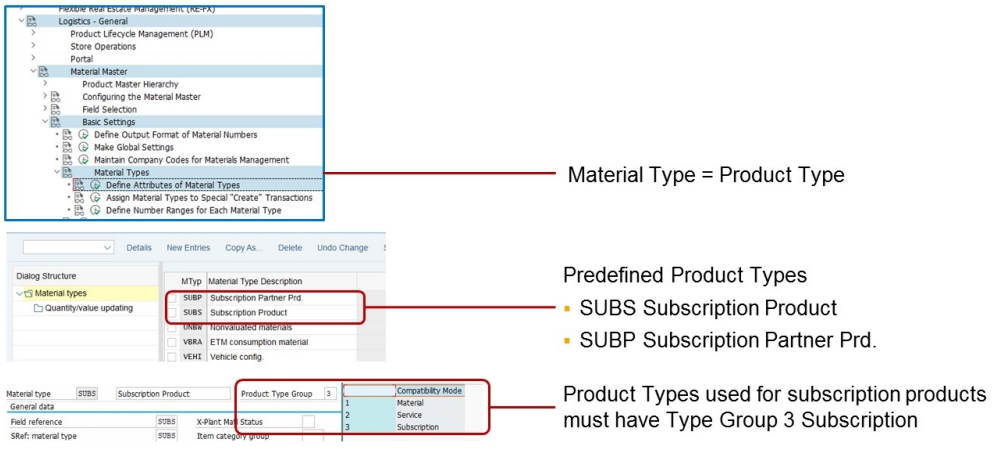

Figure 105: Product Master: Product Type Definition

You can define the attributes of each material type in the Implementation Guide (IMG).

SPRO → Logistics → General → Material Master → Basic Settings → Material Types → Define Attributes of Material Types.

For subscription orders and contracts, two product types are predefined:

- SUBS Subscription Product
- SUBP Subscription Partner Product

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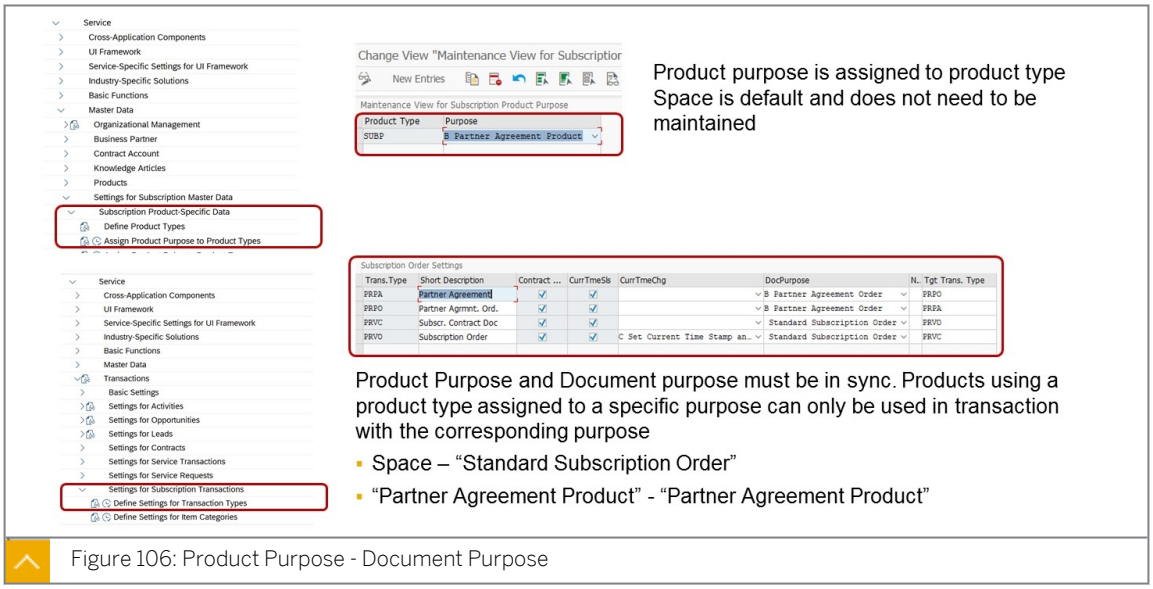

Figure 106: Product Purpose - Document Purpose

You can maintain the settings to assign product type to product purpose in IMG.

Service → Master Data → Settings for Subscription Master Data → Subscription Product-Specific Data → Assign Product Purpose to Product Types.

A product is classified using this product type and the assigned product purpose defines for which business processes this product can be used. The following product purposes are available:

- *Standard Product:* Standard product can be used in subscription orders or contracts. Products which are classified with a product type, for which no product purpose is defined, are always used as standard products.
- *Partner Agreement:* Product Partner agreement products define the rules for distribution of revenue for third-party companies or partners if customers buy or consume services that these partners are involved in providing. Partner agreement products can only be used in partner agreements or partner agreement orders.

If you use products in a manner that the purpose derived from the document does not conform to the purpose derived from the product type, the system displays an error during document processing.

You can maintain the settings for document purpose in IMG.

Service → Transactions → Settings for Subscription Transactions → Define Settings for Transaction Types.

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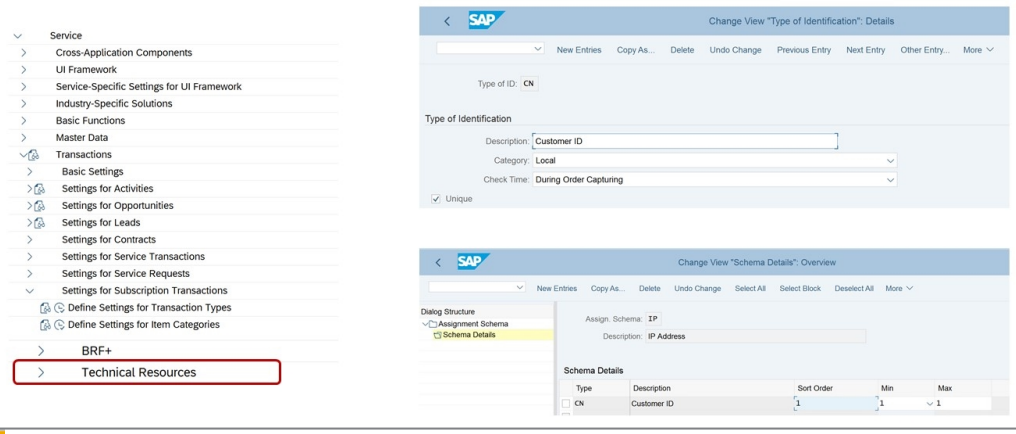

Figure 107: Product Master: Technical Resource

The resource type specifies the kind of technical resource, the source and uniqueness of the technical resource, and the stage in the subscription order or contract processing when it would be required.

The assignment schema defines which and how many resources of certain resource types can, or have to, be assigned to a contract using this schema.

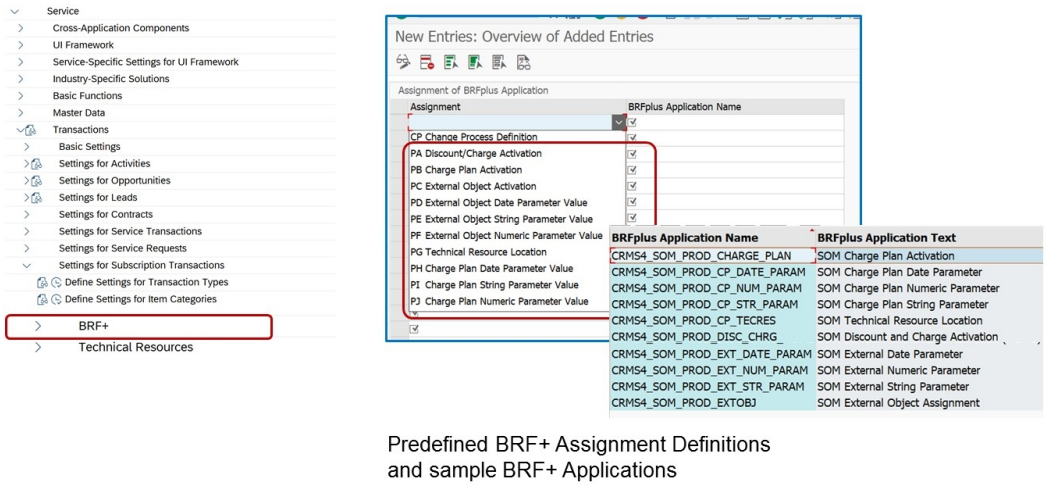

Figure 108: Subscription product-specific data: BRF+

There are assignments which are related to the maintenance of subscription product-specific data. All delivered applications are template BRFplus applications. If this template BRFplus application does not fit to your requirements, you can create and assign your own BRFplus application.

- **Discount/Charge Activation:** determine if a discount / charge is activated in a subscription order. *CRMS4_SOM_PROD_DISC_CHRG*
- **Charge Plan Activation:** determine if a charge plan is activated in a subscription contract. *CRMS4_SOM_PROD_CHARGE_PLAN*
- **External Object Activation:** determine if an external object is activated in a subscription order / contract. *CRMS4_SOM_PROD_EXTOBJ*
- **External Object Date / String / Numeric Parameter Value:** determine the value of a date / string / numeric parameter of an external object.

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CRMS4_SOM_PROD_EXT_DATE_PARAM / CRMS4_SOM_PROD_EXT_STR_PARAM / CRMS4_SOM_PROD_EXT_NUM_PARAM

- **Technical Resource Location:** determine the location in the contract where the technical resource can be found. *CRMS4_SOM_PROD_CP_TECRES*
- **Charge Plan Date / String / Numeric Parameter Value:** you can determine the value of a date / string / numeric parameter of a charge plan. *CRMS4_SOM_PROD_CP_DATE_PARAM / CRMS4_SOM_PROD_CP_STR_PARAM / CRMS4_SOM_PROD_CP_NUM_PARAM*

LESSON SUMMARY

You should now be able to:

- Discuss Product Training Examples

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Learning Assessment

1. Which of the following additional fields are required for subscription products?

   Choose the correct answers.

   - [ ] **A** Billing cycle determination rule
   - [ ] **B** Payment method
   - [ ] **C** Technical identifier
   - [ ] **D** Contract terms
   - [ ] **E** Extension rule

2. Which of the following components is necessary for product configuration installation?

   Choose the correct answer.

   - [ ] **A** AVC
   - [ ] **B** BOM
   - [ ] **C** BIT Class
   - [ ] **D** SOA Manager
   - [ ] **E** MW

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Learning Assessment - Answers

1. Which of the following additional fields are required for subscription products?

   Choose the correct answers.

   - [X] A Billing cycle determination rule
   - [ ] B Payment method
   - [ ] C Technical identifier
   - [X] D Contract terms
   - [X] E Extension rule

   That is correct! The payment method and technical identifier are not required as additional fields for subscription products.

2. Which of the following components is necessary for product configuration installation?

   Choose the correct answer.

   - [X] A AVC
   - [ ] B BOM
   - [ ] C BIT Class
   - [ ] D SOA Manager
   - [ ] E MW

   That is correct! The AVC component is required for product configuration installation.

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UNIT 6 Bundles Products

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UNIT OBJECTIVES

- Define Product Bundles
- List and explain Product Bundle Features
- Understand the Sales Bundle
- Understand the Contract Bundle
- Define Customizing for Bundles
- Understand Pricing and its determination
- Outline the Pricing Procedure Determination in SD
- Explain Condition Type PSPB
- Locate the Customizing for Pricing
- Undestand Discounts and Charges
- Perform the Customizing of Discounting

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Unit 6
Lesson 1

Overview Bundle

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Define Product Bundles
- List and explain Product Bundle Features
- Understand the Sales Bundle
- Understand the Contract Bundle
- Define Customizing for Bundles

Business Example

In cloud service business, new sales products are new combinations of basic products. These are typically bundles containing several products. For example, you are selling a product bundle containing Data Storage for Data Service Storage, with one time installation service fee, and a yearly recurring insurance fee. Another example is, you are selling a product bundle containing Hardware, installation service from technician, a Support Contract, and a recurring insurance fee.

Define Product Bundles

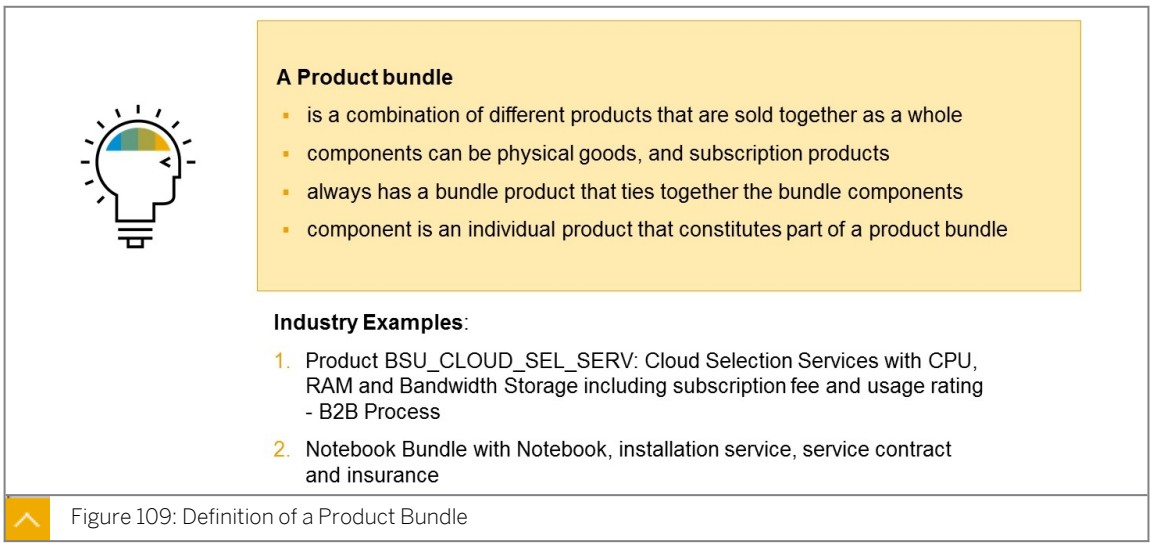

Figure 109: Definition of a Product Bundle

In our training, the main Product Bundles will be our Cloud Service Selection Product, *BSU_CLOUD_SEL_SERV* and our Notebook Care for you package, *BSV_NOTEBOOKS_C4U*

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List and explain Product Bundle Features

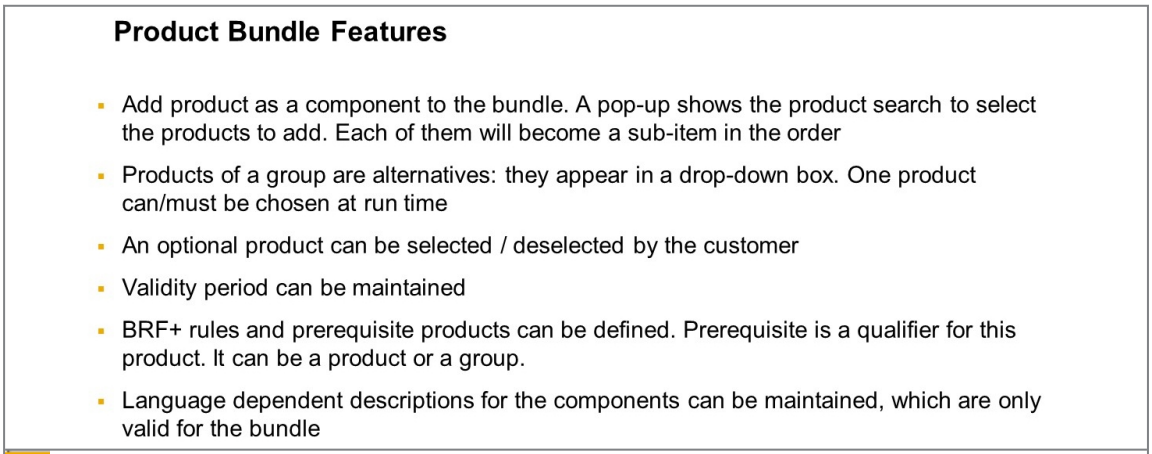

Figure 110: Product Bundle Features

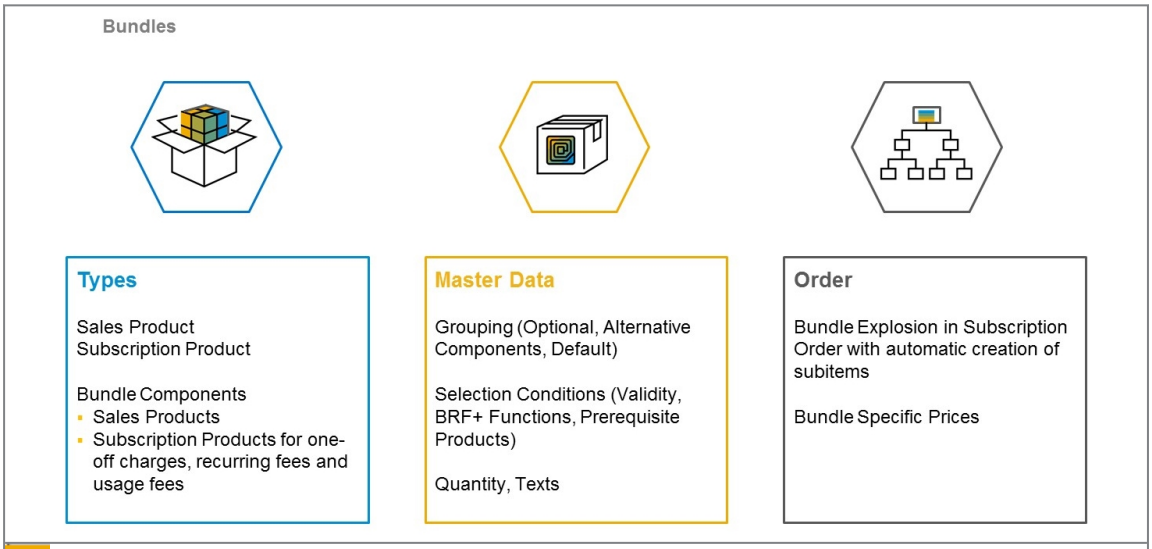

Figure 111: List and explain Product Bundle Features

The first step to creating a Bundle Product, is to define the Product Type, Sales Bundle or Subscription Bundle.

In the second step the Master Data is created with all relevant attributes and conditions.

The last step is the bundle explosion in the Subscription Order or in the Solution Quotation, with creation of all defined subitems within the Bundle.

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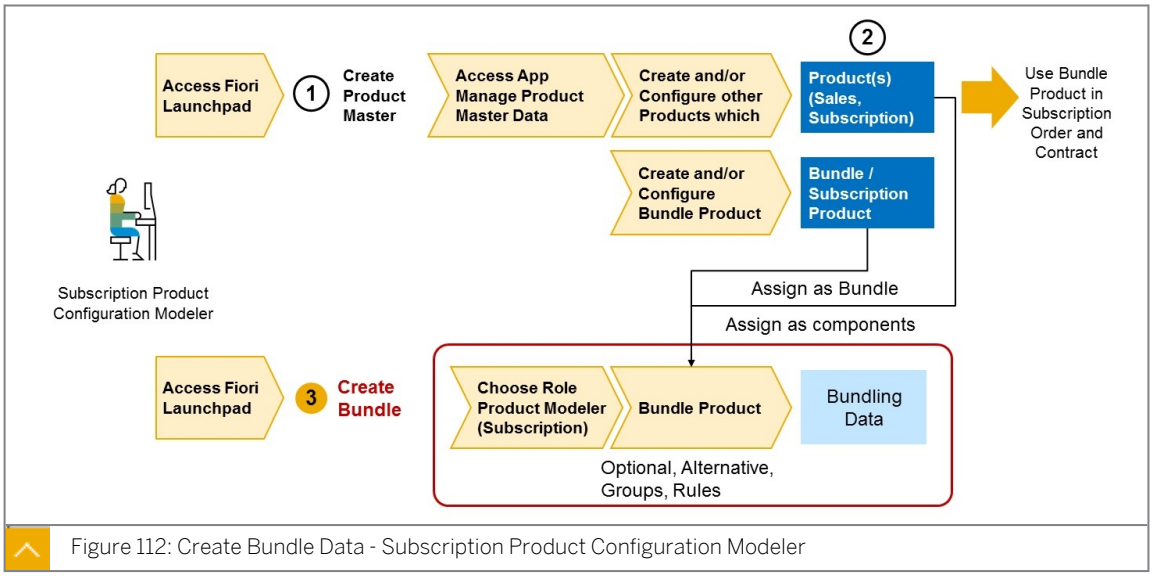

Figure 112: Create Bundle Data - Subscription Product Configuration Modeler

- Bundles are modeled in S/4HANA for Service. They are not specific for subscription order management, but are also used by service management.
- The single bundle components must already exist as product masters in S/4HANA product master.

The Sales Bundle

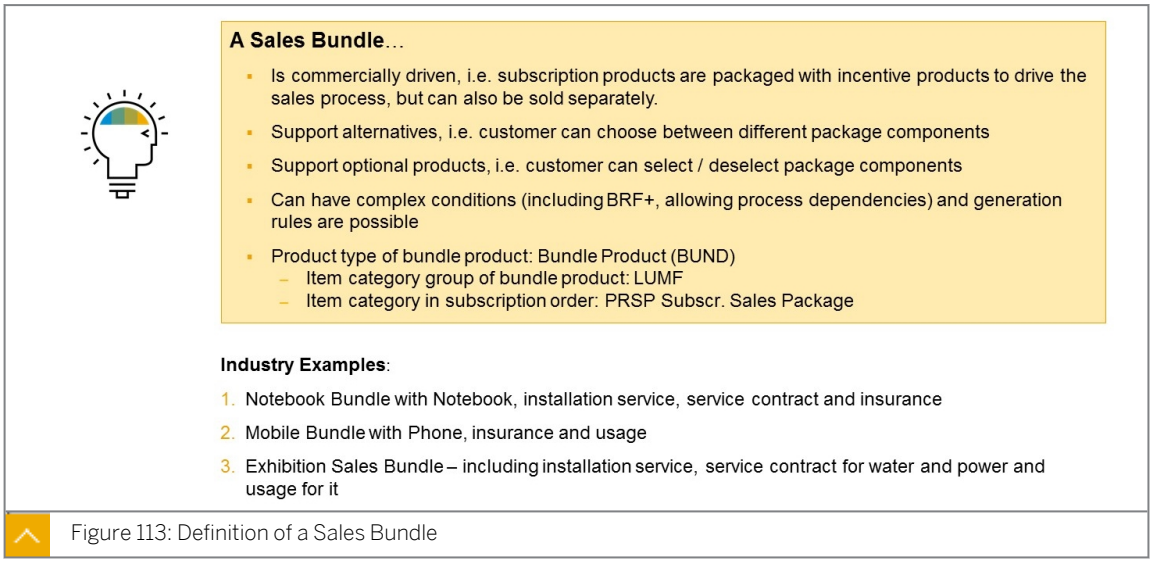

Figure 113: Definition of a Sales Bundle

In our training, our Sales Bundle is a Notebook Care for you package. *BSV_NOTEBOOKS_C4U*

Results:

- The bundle product is not taken over to the contract
- Each sub-item with a subscription product will become an independent subscription contract
- One-off items will not be taken over to the contract

To distinguish between Sales Bundle and Contract Bundle, the corresponding item category needs to be maintained.

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Our example has item category SRVP, the item category for Solution Quotation.

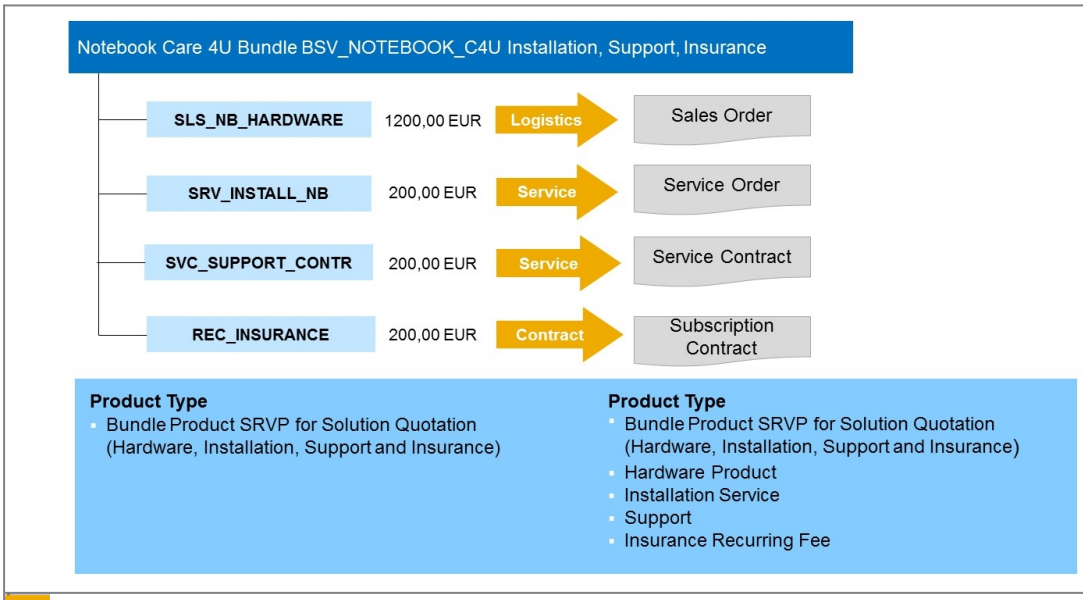

Figure 114: Sales Bundle Product BSV_NOTEBOOK_C4U

Example of a product structure:

In this case the bundle is the container for each product. You can either define a price on bundle product level or the price is being determined by the selected products. This product example is used for the Solution Quotation, where you can sell Hardware, Services, and Contract relevant items in one order. Resulting from the shown bundle, each item triggers a transaction type. Prices are defined for each product.

Understand the Contract Bundle

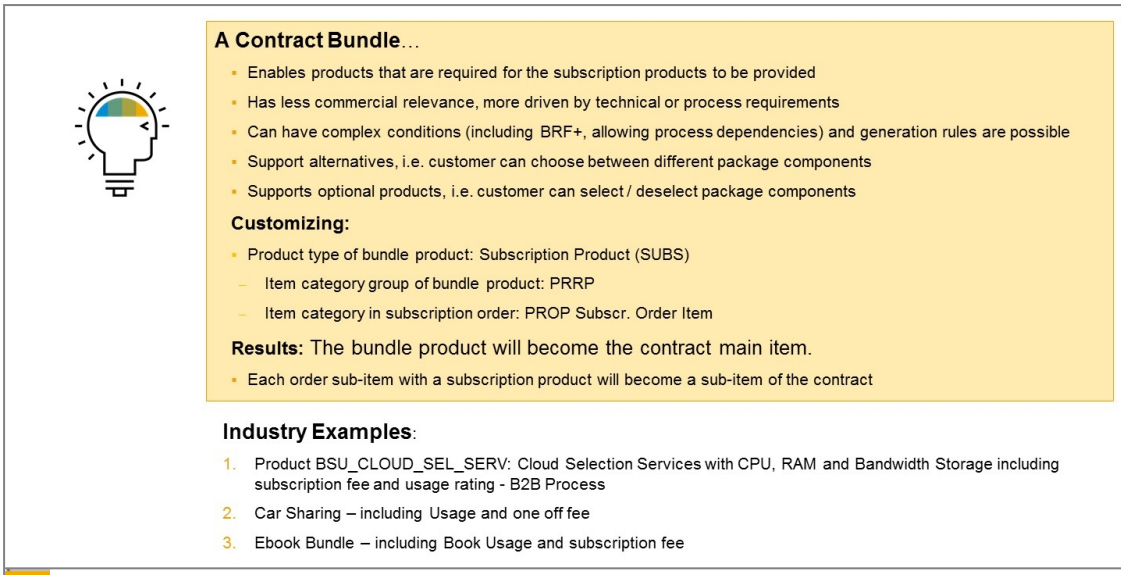

Figure 115: Definition of a Contract Bundle

In our training our Contract Bundle is a Cloud Selection Service, *BSU_CLOUD_SEL_SERV*.

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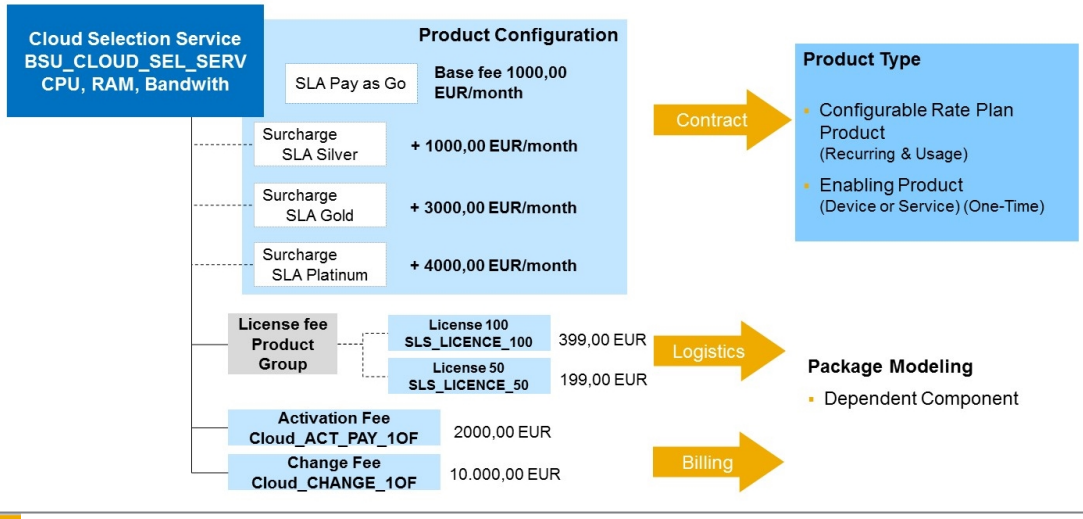

Figure 116: Contract Bundle Product BSU_CLOUD_SEL_SERV

Example of a product structure:

In this case a bundle is the container for each product. You can either define a price on bundle product level or the price is determined by the products selected.

Configurable products can be used for each type of product and on each level of granularity. This means, that it's also possible to have multiple configurable products within one configurable root product.

You can define the price for the configurable product, either in a fixed way on root level or by calculating it via the configuration of the product.

The interlinkages can be either used for bundling or for process products. Process products will automatically be recognized by the system, if all constraints for it are fulfilled. Accordingly, there is the possibility to map specific requirements just for certain transactions in the system, like activation fee for certain products and so on.

Define Customizing for Bundles

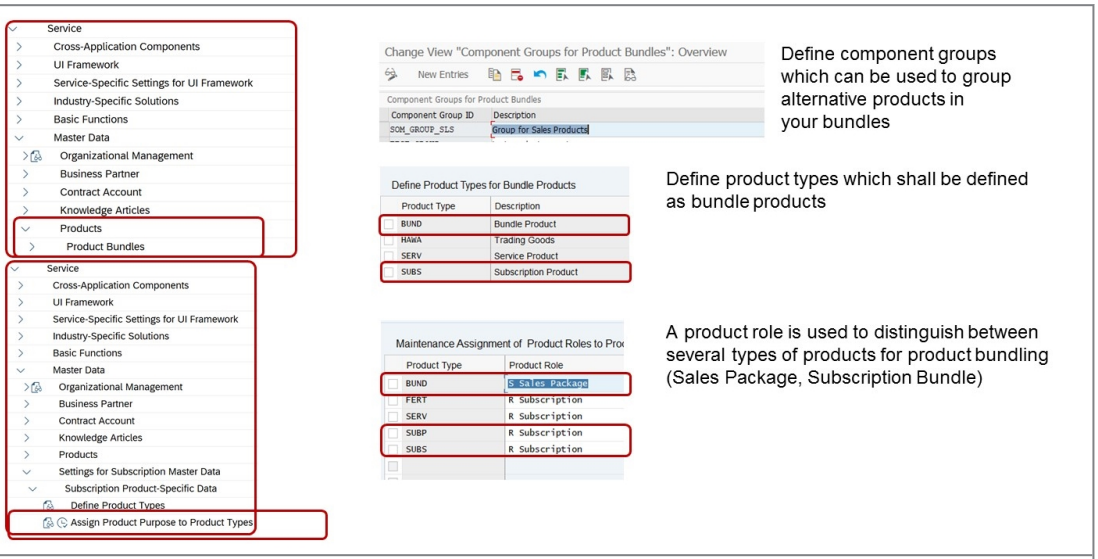

Figure 117: Customizing for Bundles

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Before similar sales components can be put together, to be used as alternative products in a bundle, the **component group** has to be created.

SPRO → Service → Master Data → Products → Product Bundles → Define Component Groups for Product Bundles.

Product type defines certain attributes of a product and has some control functions. Before products of a certain **product type can be used as a bundle**, the corresponding product type must be defined.

SPRO → Service → Master Data → Products → Product Bundles → Define Product Types for Bundle Products.

A product role is used to distinguish between several types of products for product bundling **(Sales Package, Subscription Bundle)**.

SPRO → Service → Master Data → Settings for Subscription Master Data → Subscription Product-Specific Data → Assign Product Roles to Product Types.

LESSON SUMMARY

You should now be able to:

- Define Product Bundles
- List and explain Product Bundle Features
- Understand the Sales Bundle
- Understand the Contract Bundle
- Define Customizing for Bundles

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Pricing (Calculation Scheme)

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Understand Pricing and its determination
- Outline the Pricing Procedure Determination in SD
- Explain Condition Type PSPB
- Locate the Customizing for Pricing

Understand Pricing and its determination

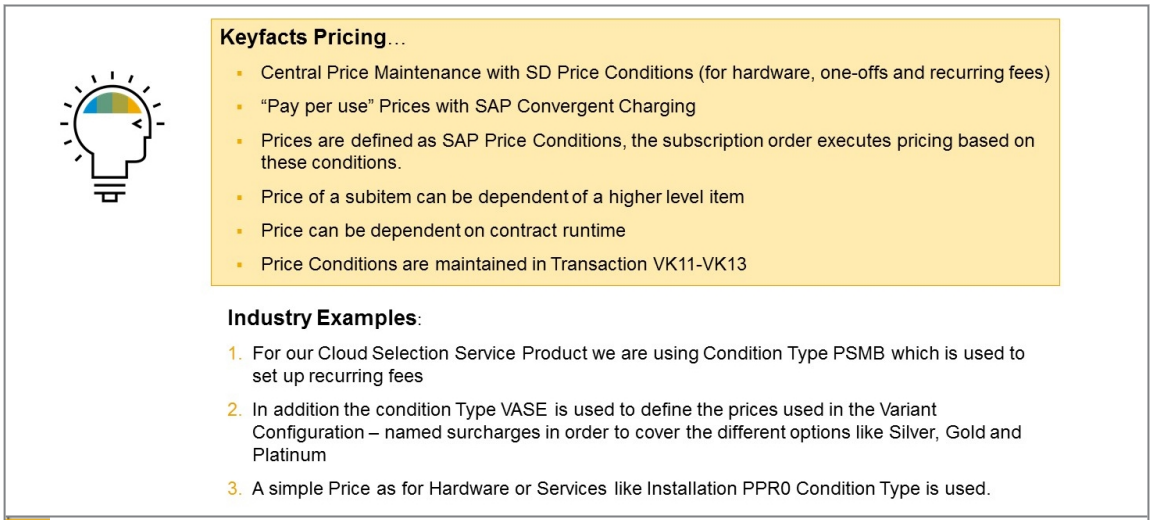

Figure 118: Pricing

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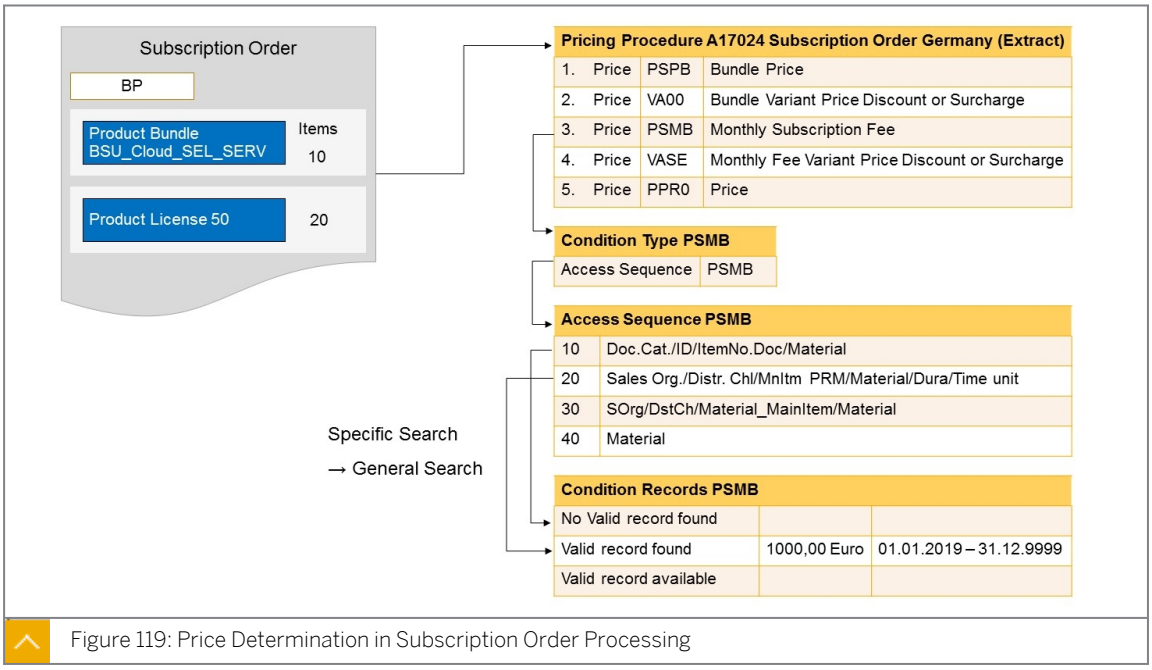

Figure 119: Price Determination in Subscription Order Processing

Pricing Procedure A17024 is a Standard Procedure, which is used for Subscription Transaction (Germany). It can be used as a template for any adaptations. The main Condition Type is PSPB - Bundle Price.

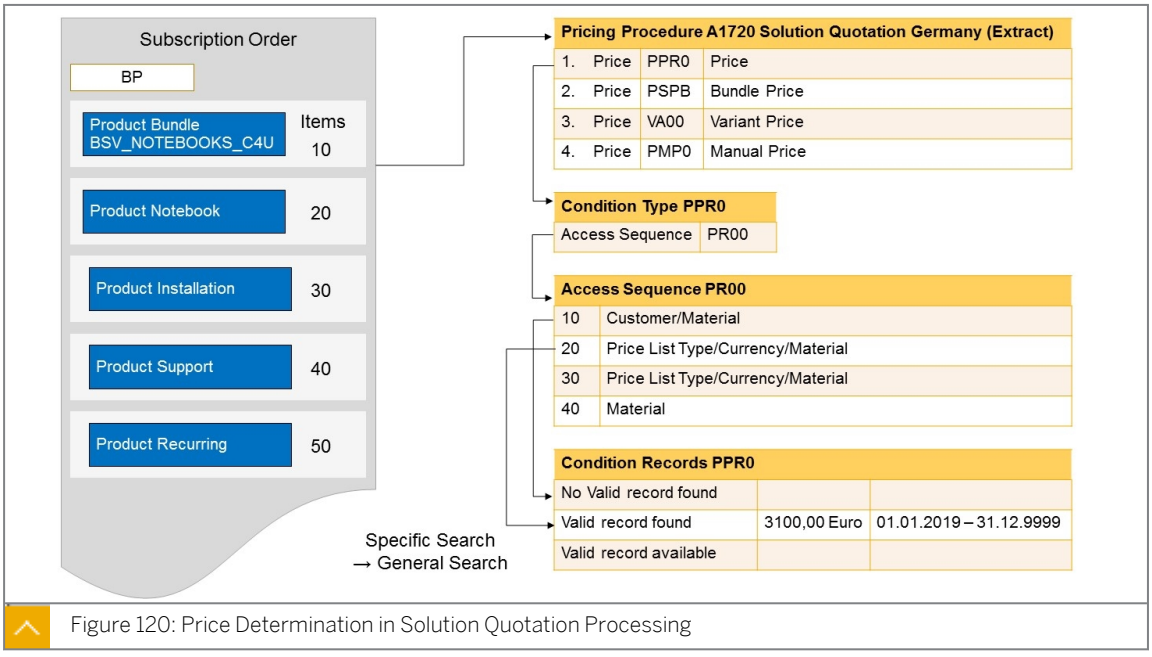

Figure 120: Price Determination in Solution Quotation Processing

Pricing Procedure A17020 is a Standard Procedure, which is used for Solution Quotation(Germany). It can be used as a template for any adaptations. The main Condition Type is PPR0 - Bundle Price.

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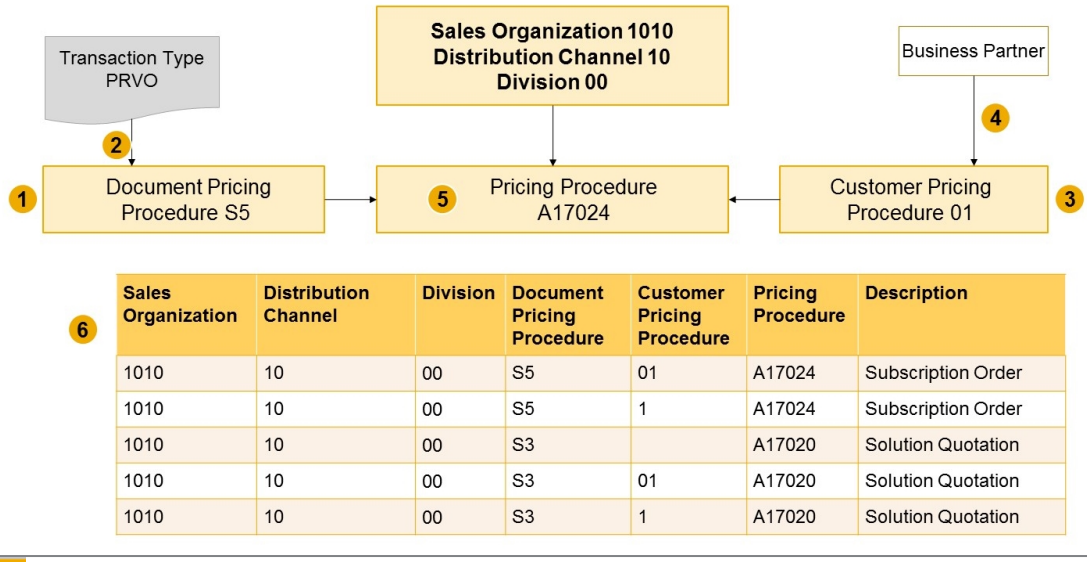
Outline the Pricing Procedure Determination in SD



6

Sales Organization	Distribution Channel	Division	Document Pricing Procedure	Customer Pricing Procedure	Pricing Procedure	Description
1010	10	00	S5	01	A17024	Subscription Order
1010	10	00	S5	1	A17024	Subscription Order
1010	10	00	S3		A17020	Solution Quotation
1010	10	00	S3	01	A17020	Solution Quotation
1010	10	00	S3	1	A17020	Solution Quotation

Figure 121: Pricing Procedure Determination in SD

Customizing Steps for Pricing of Conditions:

1. The **document pricing procedure** is defined in the Implementation Guide (IMG):

   SPRO → Sales and Distribution → Basic Functions → Pricing → Pricing Control → Define And Assign Pricing Procedures → Set Pricing Procedure Determination

   Or *Service → Basic Functions → Pricing → Configure Pricing*.

2. Document pricing procedure is assigned to the subscription transaction types and will be considered during the pricing procedure determination in IMG:

   Service → Transactions → Basic Settings → Define Transaction Types → Select a Transaction Type → Assignment of Business Transaction Categories → Sales → Customizing header.

3. 3. Settings for **customer pricing procedure** are maintained in IMG:

   SPRO → Sales and Distribution → Basic Functions → Pricing → Pricing Control → Define And Assign Pricing Procedures → Set Customer Pricing Procedures.

   Or *Service → Basic Functions → Pricing → Configure Pricing*.

4. You specify the customer pricing procedure in the Business Partner Customer master record for each sales area.

5. Settings for pricing procedure are maintained in IMG:

   SPRO → Sales and Distribution → Basic Functions → Pricing → Pricing Control → Define And Assign Pricing Procedures → Set Pricing Procedures.

   Or *Service → Basic Functions → Pricing → Configure Pricing*.

   In the pricing procedure, you define which condition types are to be taken into account and in which sequence. During pricing, the system automatically determines which pricing procedure is valid for a business transaction. The system then takes the condition types contained in the pricing procedure into account, one after the other.

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For subscription orders and contracts, pricing procedure A17023 is shipped as a sample pricing procedure, which can be adjusted to the customers' requirements. The condition PSMB is used for recurring subscription fees and PSPB for bundle prices / one off fees. Customers build their own pricing procedures.

6. Pricing Procedure Determination:

   The pricing procedure determination is defined in IMG:

   SPRO → Sales and Distribution → Basic Functions → Pricing → Pricing Control → Define And Assign Pricing Procedures → Set Pricing Procedures.

   Or *Service → Basic Functions → Pricing → Configure Pricing*.

You specify how the system determines pricing procedures for documents.

The document pricing procedure is derived from the transaction type. The customer pricing procedure is derived from the business partner.

Explain Condition Type PSPB

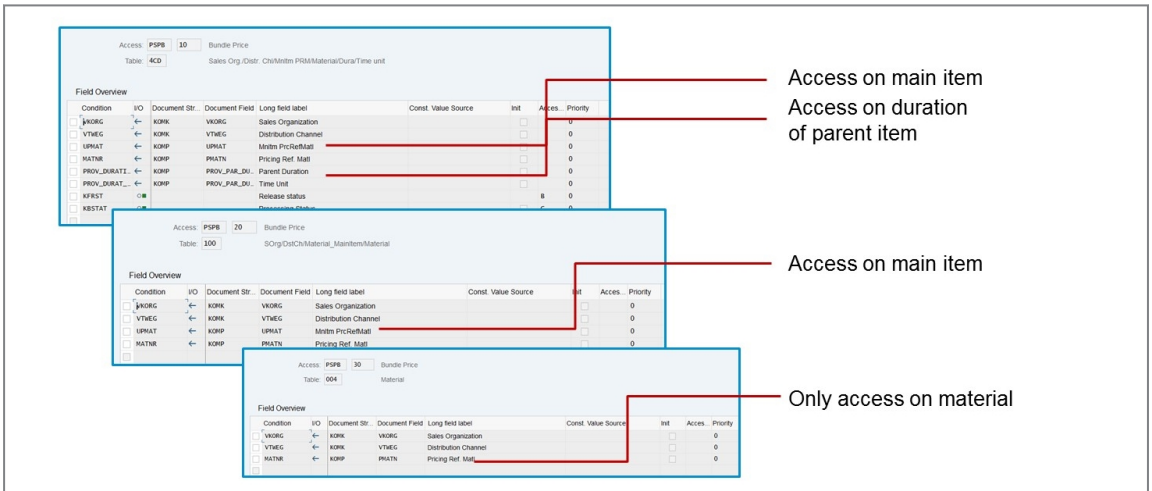

Figure 122: Access Sequence PSPB for Bundle Price - No. 10, 20, and 30

The Access Sequence shows, in detail, the condition used per table.

It is defined for each condition type in the Pricing Procedure. This search strategy defines the sequence in which the system reads the condition records for a condition type. Each access performed during the access sequence is made using a condition table. A condition table is a combination of fields, which form the key for a condition record. You can make an access dependent on certain requirements.

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Locate the Customizing for Pricing

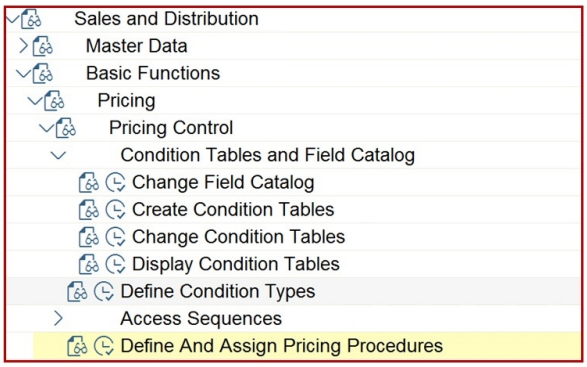

Figure 123: Pricing in IMG

Define the settings in Customizing for Sales and Distribution under *Basic Functions → Pricing → Pricing Control*:

- Define Condition Tables
- Define Condition Types
- Define Access Sequences
- Define And Assign Pricing Procedures

In the Customizing activity, *Define and Assign Pricing Procedures*,, only the following settings are relevant:

- Set Pricing Procedures
- Set Customer Pricing Procedures
- Set Document Pricing Procedures
- Set Pricing Procedure Determination

In addition, you must perform the following settings in *Customizing for S/4Hana Service*:

Under *Transactions → SPRO → Basic Settings*:

- Under *Define Transaction Types*, assign the pricing procedure to the relevant transactions.
- Under *Define Item Categories*, activate the pricing function for item categories

LESSON SUMMARY

You should now be able to:

- Understand Pricing and its determination
- Outline the Pricing Procedure Determination in SD
- Explain Condition Type PSPB
- Locate the Customizing for Pricing

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Unit 6
Lesson 3

Discounts and Charges

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Undestand Discounts and Charges
- Perform the Customizing of Discounting

Business Example

Cloud Product *BSU_CLOUD_Sel_Serv* has two discounts assigned. One Discount on RAM Usage and one yearly Discount for total volume amount of billable items.

Undestand Discounts and Charges

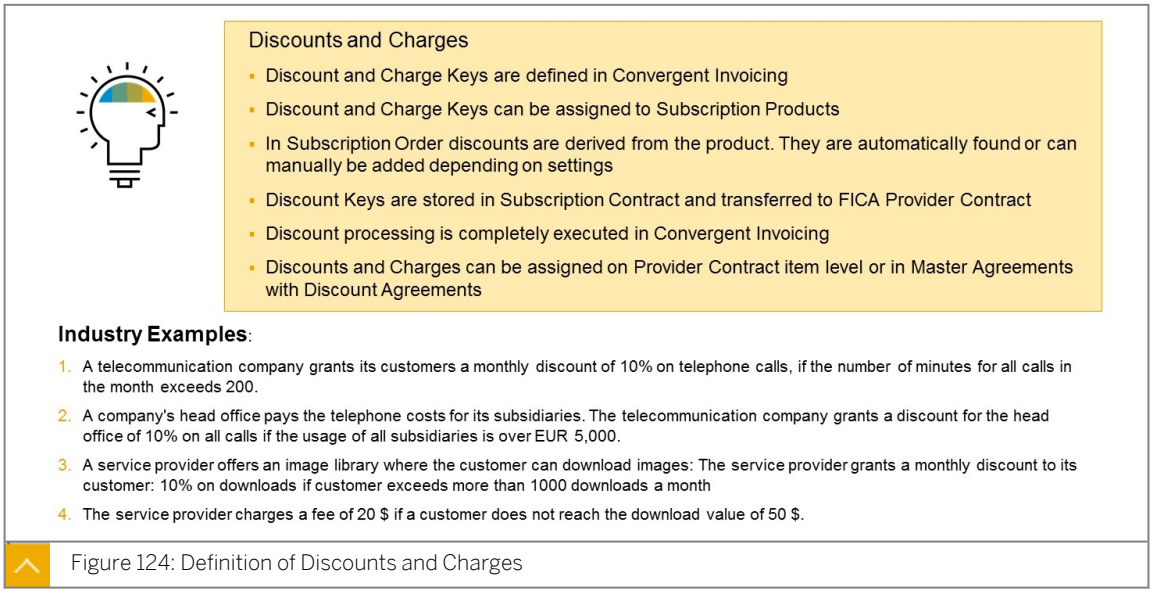

Figure 124: Definition of Discounts and Charges

Discount and Charge assignment during Product Modeling with the Fiori Launchpad app *"Manage Subscription Product-Specific Data"*.

Web Client UI to maintain Discount Keys in Subscription Order.

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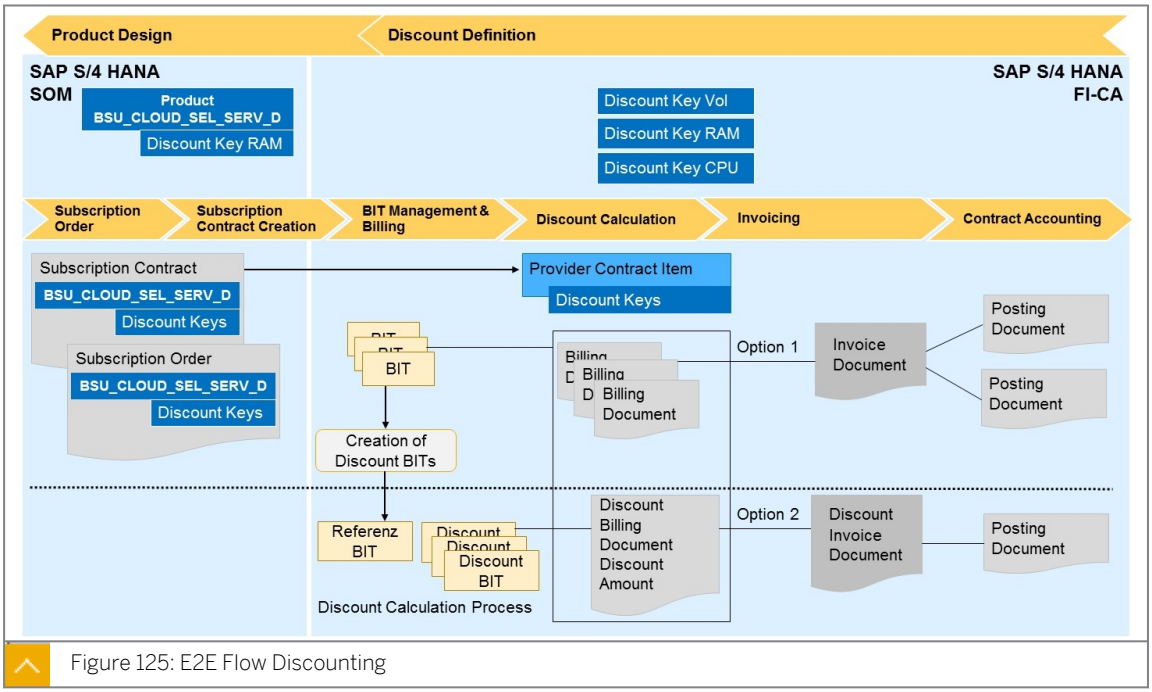

Figure 125: E2E Flow Discounting

You assign discounts and charges in master agreements,with business customers, at the level of the provider contract or at the level of discount agreements.

The system determines the discounts and charges during the billing of consumption, based on specifications entered in their master data. As part of this process, the system creates base items with the relevant discount or charge information. The system stores the base items similar to the way in which billable items are stored. The system calculates and processes discounts and charges during the processing of base items in a separate billing process. The billing document stores the results of the calculation.

Process Steps of Discounting and Charging of billable Items:

1. Configuration of Discount Key and assignment to Master Agreement or Provider Contract. In Standard Billing Process determination of Base Information and Creation of Base Bits
2. Calculation of Discounts and Charges in Discount Billing process
3. Create Discount / Charges Calculation Documents
4. Invoicing of Discount Billing Documents, either with Usage Billing Documents (Option1) or separate

Flexibility of new Discount and Charge functionality:

- Discount determination via discount agreement when usage BITs are billed (time independent processes possible, for example, quarterly sums for discount base amount
- Discount calculation on specified calculation level, Provider Contract Item, Provider Contract, Contract partner, or Agreement level
- Discount distribution (assignment to source of discount or assignment to, for example, headquarter)

The benefit of this solution are, that the same architecture is used in B2C and B2B. The reuse of discount key and discount definition, calculation modules on Provider Contract and Master Agreement level.

From Standard Billing and Invoicing processes document linkage and traceability is used.

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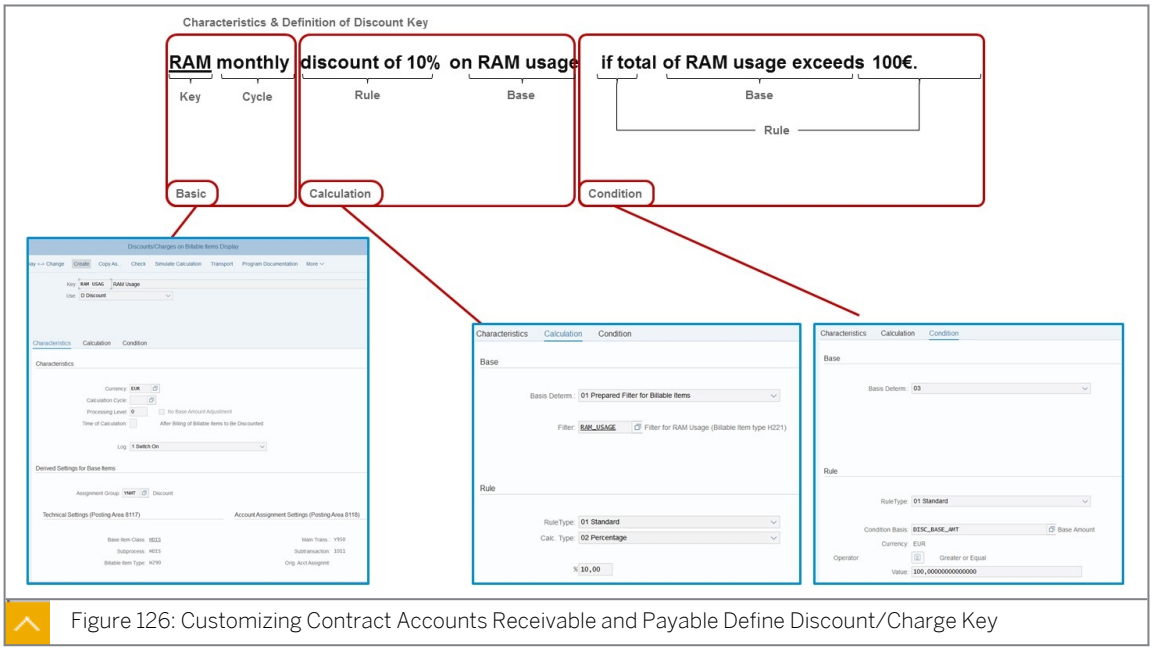

Figure 126: Customizing Contract Accounts Receivable and Payable Define Discount/Charge Key

Discounts and Charges on Billed Items

You configure discount/charge keys in three steps on the following three tabs:

Characteristics

On the upper part of the tab, you make general settings about the use, the discount or charge currency, the calculation cycle, and the processing level. In the lower part of the tab, the derived settings for the base items show you directly the currently valid derived values for the discount/charge key from posting areas 8117 and 8118 (as long as you have maintained the posting areas).

Calculation

On the upper part of the tab, you define the basis for the calculation, meaning which billed items form the basis for the discount or charge calculation. On the lower part of the tab, you define the calculation rule. That means that you specify here how the discount or charge is calculated for the billed items defined in the basis.

Condition (optional)

On the upper part of the tab, you define the basis for the condition. That means that you specify for which billed items the system applies a condition. On the lower part of the tab, you define the condition rule. That means that you specify which prerequisites have to be met in order for a discount to be granted or a charge levied.

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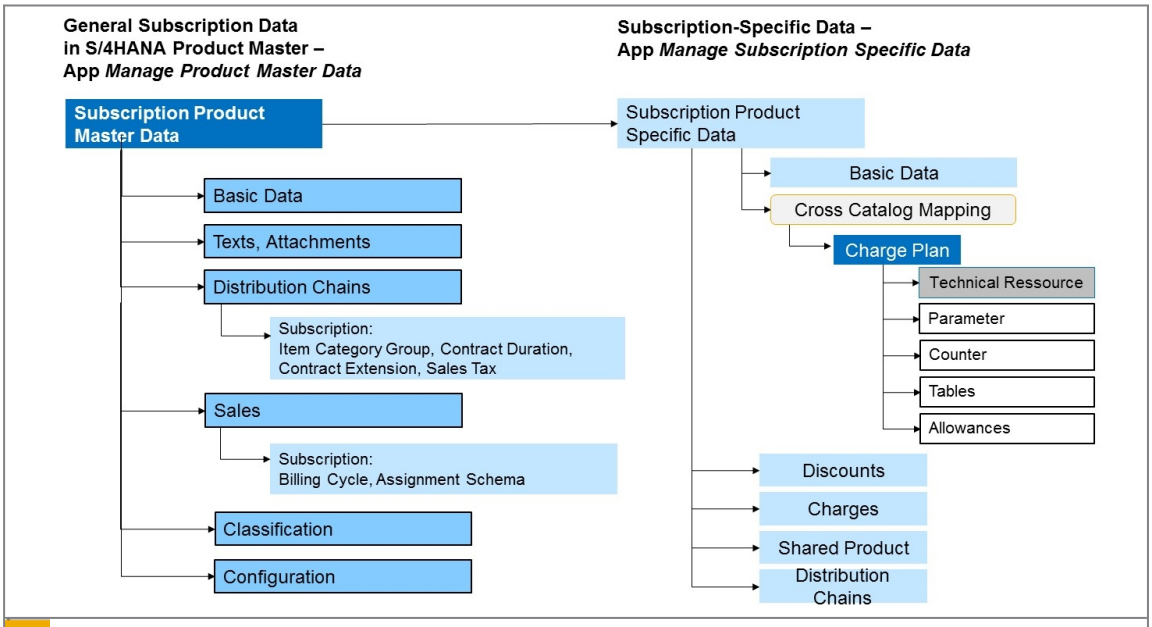

Figure 127: Product Views in SAP S/4 HANA

In S/4HANA only the product master (aka material master) is used to maintain central product data .

In addition, attributes like cross catalog mapping, or discounts and charges are maintained in an own app, *Manage Subscription Specific Data*.

Perform the Customizing of Discounting

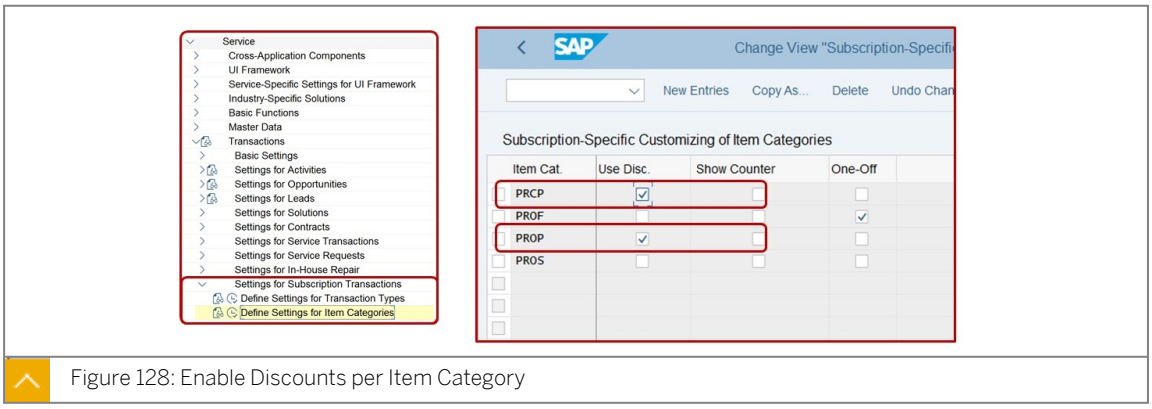

Figure 128: Enable Discounts per Item Category

You can enable discounts / charges for transactions on the item categories in Implementation Guide (IMG).

SPRO → Service → Transactions → Settings for Subscription Transactions → Define Settings for Item Categories.

Enter entries for the Item Categories for which you want to use discounts. Mark the checkbox *Use Disc.*

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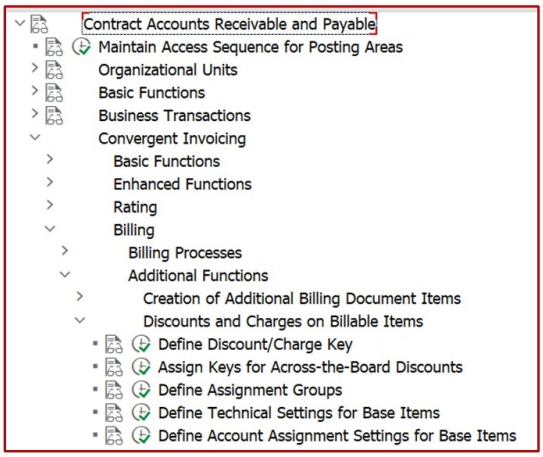
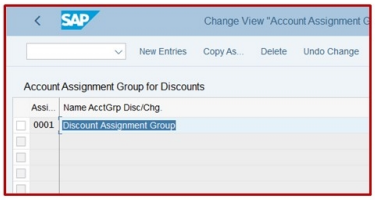
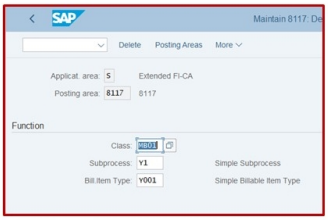


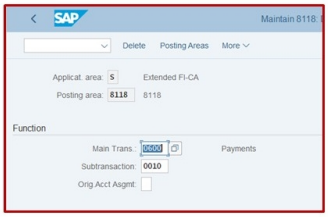
Figure 129: Customizing Contract Accounts Receivable and Payable



Define Assignment Groups



Define Base Item Class



Define Transactions for Base Items

Figure 130: Customizing Contract Accounts Receivable and Payable: Define Assignment Groups

Define Assignment Group:

SPRO → Financial Accounting → Contract Accounts Receivable and Payable → Convergent Invoicing → Billing → Additional Functions → Discounts and Charges on Billable Items → Define Assignment Groups.

Assignment groups are a grouping characteristic that you can use both in configuring discounts and charges and in posting areas 8117 and 8118.

Base-Item Class:

SPRO → Financial Accounting → Contract Accounts Receivable and Payable → Convergent Invoicing → Billing → Additional Functions → Discounts and Charges on Billable Items → Define Base Item Class.

Define the base item class settings for your assignment group/discount key combination.

In this activity, you make information available for discount/charge keys that is needed for creating base items.

Define Transactions for Base Items:

SPRO → Financial Accounting → Contract Accounts Receivable and Payable → Convergent Invoicing → Billing → Additional Functions → Discounts and Charges on Billable Items → Define Transactions for Base Items.

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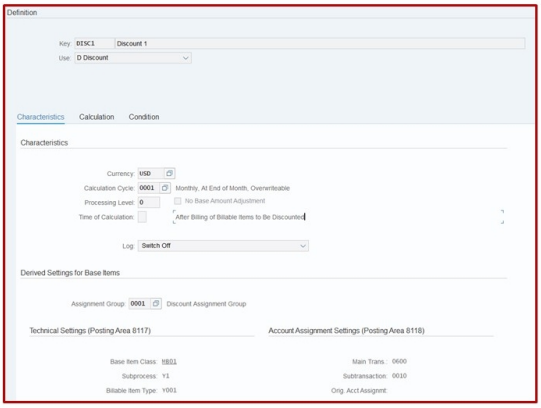
Define the transaction settings for your assignment group/discount key combination.

In this activity, you make information available for discount / charge keys that is needed for creating base items.

Dependent on the discount / charge key and the assignment group, you either specify a main transaction and sub-transaction, or you specify that the original account assignment should be used.



Discount/charge keys in provider contracts are used to grant discounts and enter charges on billed items.

Figure 131: Customizing Contract Accounts Receivable and Payable: Define Discount/Charge Key

SPRO → Financial Accounting → Contract Accounts Receivable and Payable → Convergent Invoicing → Billing → Additional Functions → Discounts and Charges on Billable Items → Define Discount/Charge Key.

LESSON SUMMARY

You should now be able to:

- Undestand Discounts and Charges
- Perform the Customizing of Discounting

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Learning Assessment

1. Which of the following can be included in a product bundle in SOM?

   Choose the correct answers.

   - ☐ **A** Hardware
   - ☐ **B** One offs
   - ☐ **C** Projects
   - ☐ **D** Subscriptions
   - ☐ **E** Service Items

2. Which of the following condition types are delivered in the Sales and Distribution component for subscription order management?

   Choose the correct answers.

   - ☐ **A** PSMB
   - ☐ **B** PSPB
   - ☐ **C** PPBS
   - ☐ **D** SOMP

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Learning Assessment - Answers

1. Which of the following can be included in a product bundle in SOM?

   Choose the correct answers.

   - [X] A Hardware
   - [X] B One offs
   - [ ] C Projects
   - [X] D Subscriptions
   - [X] E Service Items

   That is correct! Projects are not part of product bundles.

2. Which of the following condition types are delivered in the Sales and Distribution component for subscription order management?

   Choose the correct answers.

   - [X] A PSMB
   - [X] B PSPB
   - [ ] C PPBS
   - [ ] D SOMP

   That is correct! PSMB and PSPB are the correct condition types.

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UNIT 7 Subscription Contract Changes

Lesson 1

Subscription Contract Changes 141

UNIT OBJECTIVES

- Describe Subscription Contract Changes
- Browse Contract Change Customizing

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Unit 7
Lesson 1

Subscription Contract Changes

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Describe Subscription Contract Changes
- Browse Contract Change Customizing

Business Example

The customer decides to upgrade their service level agreement for the Cloud Selection Service, from Gold to Platinum. You are in the role of an agent who performs this change in the system.

Describe Subscription Contract Changes

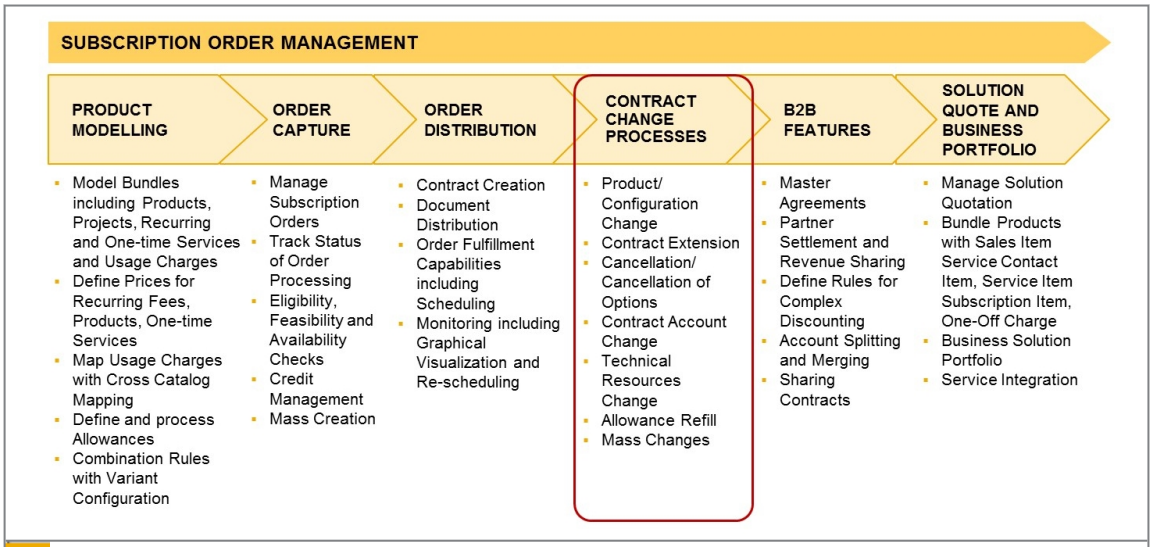

Figure 132: Overview Subscription Order Management

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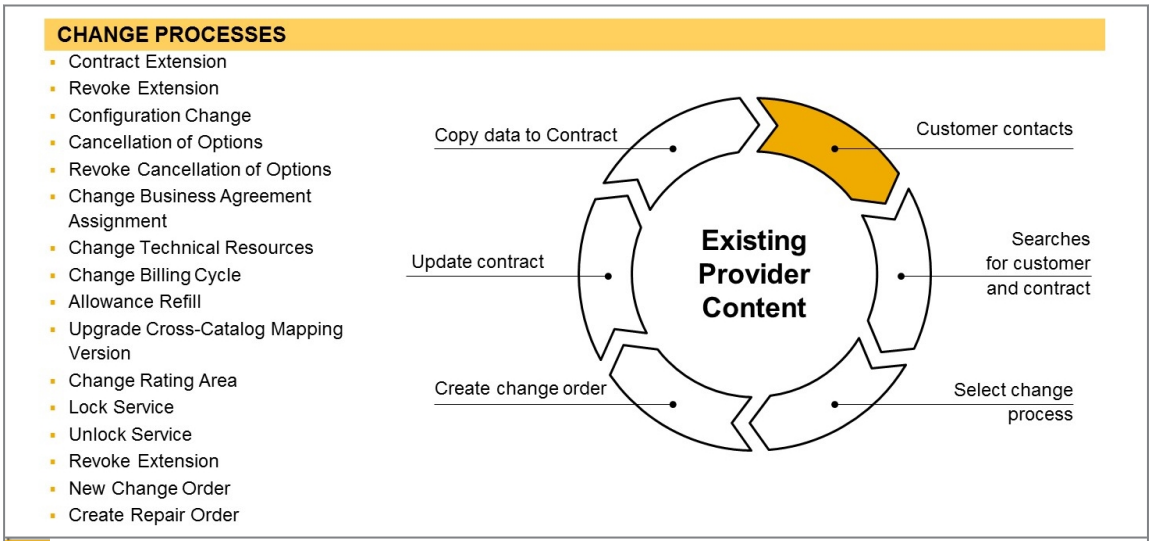

Figure 133: Contract Change Management

This slide outlines the contract changes that can be made in SAP Billing and Revenue Innovation Management (BRIM) and Subscription Order Management (SOM).

All contract change processes for existing customers are handled via change orders. This means that, if an agent triggers a change process, an order is created based on the contract. Within this order, the changes are being applied and afterward written back into the contract. Change orders can also be scheduled in the future and can include one or several of the following change processes:

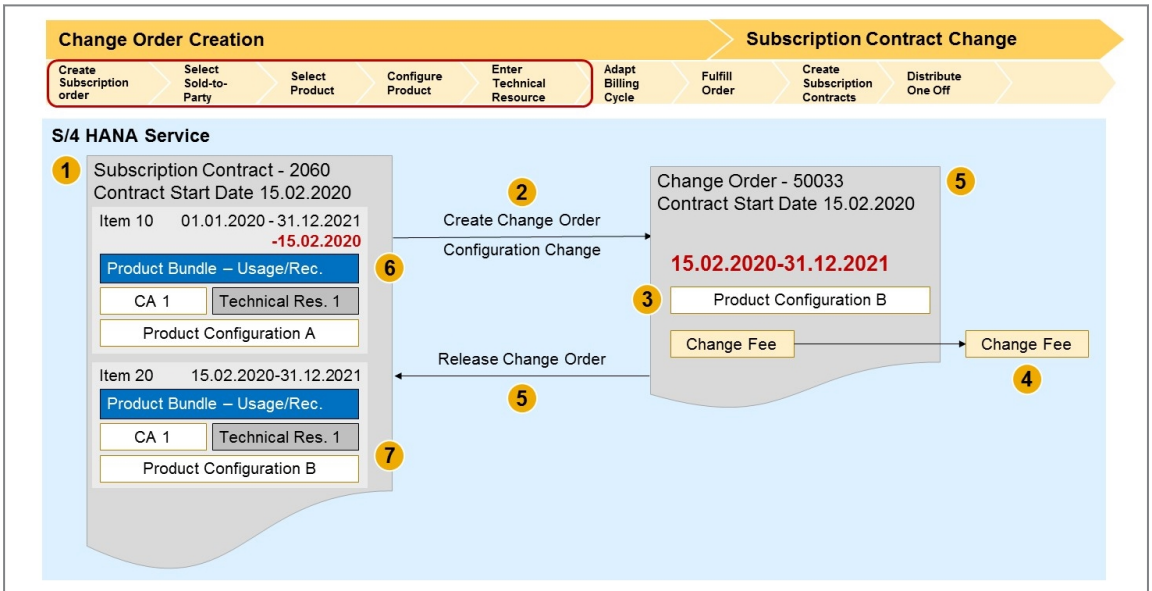

Figure 134: Process Overview Subscription Order Creation

From the contract, all relevant data is copied to the change order. The change order changes the validity dates of the existing contract item and copies the new data of the change order to a new contract item with relevant dates to the contract. The technical object is linked from the old contract item to the new contract item.

All contract changes are used to apply the changes to the contract. Therefore, the contract will always retain its previous ID and the related change processes are stored in detail as a contract history.

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Changes that should be performed in the future can accordingly also be scheduled in the change order for a future date. When the date is reached, the order will apply the changes to the contract.

1. Current Subscription Contract 2060. Contract Start Date 15.2.2020
2. Start Product Configuration Change for Product Configuration A
3. Product Configuration B is done. Time Slice beginning at 15.2.2020
4. Change Fee is created
5. Change Order is released and created
6. Changed item 10 is set to End Date 15.2.2020
7. Subscription Contract 2060 now has been changed, to item 20

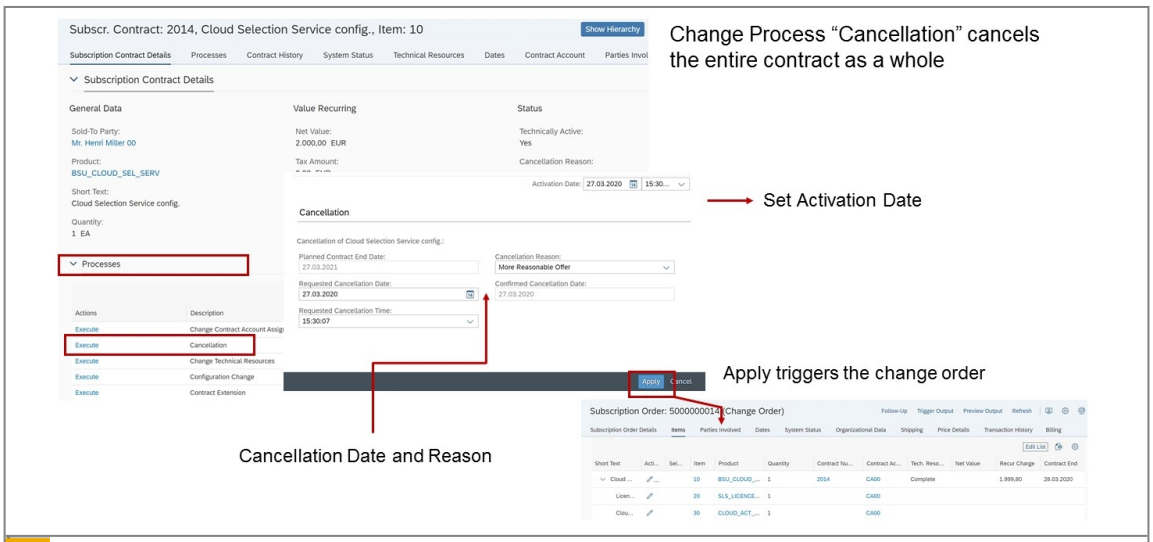

Figure 135: Contract Change - Cancellation

Cancellation

You can cancel an active contract by setting a specific cancellation date. In this case, the entire contract is canceled.

The following three types of cancellations exist, based on your entries in the Requested Cancellation Date and Requested Cancellation Time:

- **Immediate:** If you enter the current date and time, the contract is canceled immediately. You cannot revoke it.
- **Deferred:** If you enter a date or time in the future, the cancellation is deferred. You can revoke it if the requested cancellation date / time has not been reached.
- **Backdated:** If you enter a date or time in the past, the contract is canceled immediately with the entered requested cancellation date. You cannot revoke it.

Once the contract has been canceled, all sub items for the main item are also canceled. If you need to cancel only an option, such as a dependent contract item, you can use change process *"Cancellation of Options"*.

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Technical details of the standard change process for cancellation

Process Name: CANCEL
Process Class: CL_CRM_BTMF_TELCO_CANCEL
Process View: PROVDLPR/CancellationVS

Process Name: CANCEL_OPTION
Process Class: CL_CRM_BTMF_TELCO_CANCEL_OPT
Process View: PROVDLPR/CancelOptionViewSet

Process Name: REVOKE_CANCEL
Process Class: CL_CRM_BTMF_TELCO_CANCEL_REVOK

Process Name: REVOKE_CANCEL_OPTION
Process Class: CL_CRM_BTMF_TELCO_CANC_OPT_REV

Figure 136: Technical Details for Cancellation

Billing Cycle:A product change can lead to a change of billing cycle.

- CI Billing plans are assigned to the Contract Accounts Receivable and Payable (FI-CA) provider contracts, receive the new line item with the new cycle.
- If BITs have already been requested for the closed cycle, they must be reverted. The contract will stay inactive and must be redistributed

Price Change: A product change can lead to a price change.

- CI Billing plans assigned to FI-CA provider contracts, receive the new line item with the new price.

Switching between pure Recurring and Usage Based products:

- If a CI Billing plan needs to end as the product has no longer recurring fees involved, a new contract item time slice is created (new VTPID), the new item has no billing plan assigned.
- Same is valid vice versa, then the new time slice has a billing plan assigned.

Product Change can only be executed:

- From one Subscription Product to another Subscription Product.
- From one Subscription Product to another Subscription Product bundle.

Up / cross sell is not available.

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Define BRFplus

- BRFplus is a generic framework that can be used to define and maintain business rules for all kinds of business applications.
- BRFplus is enabled for change processes for:
  - Change Process Definition
  - Predefined templates are shipped
- BRFplus is also used for others purposes
  - Discount/Charge Activation
  - Charge Plan Activation
  - Charge Plan Date / String / Numeric Parameter Value
  - External Object Activation
  - External Object Date / String / Numeric Parameter Value
  - Technical Ressource Location

Industry Examples:

1. Matching responsible sales agents, or suitable products and locations to customers need
2. Additional cost calculation is necessary when product configuration changes for our Cloud Selection Product

Figure 137: Definition BRF+

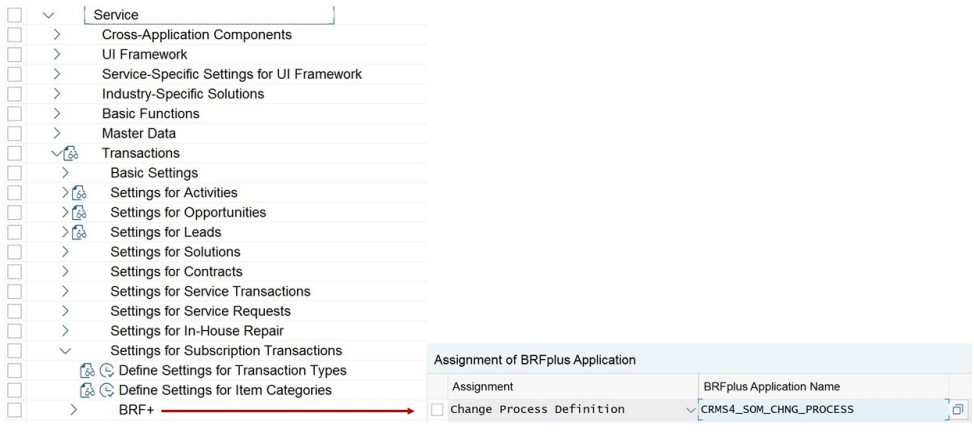

Figure 138: Defining BRFplus Applications

In Customizing, you can assign existing BRFplus applications. The assignment can service different purposes. For change processes the assignment purpose *"Change Processes Definition"* is predefined.

Service → Transactions → Settings for Subscription Transactions → BRF+

Detailed explanations can be found in the Implementation Guide (IMG) documentation and the F1 helps of the single fields.

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Detail Change Process Customizing

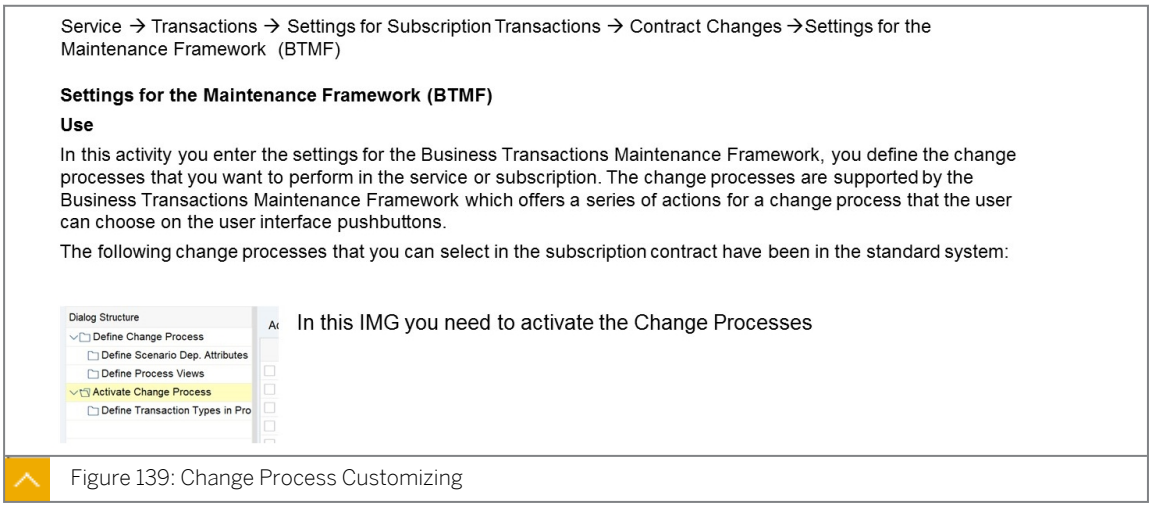

Figure 139: Change Process Customizing

In Customizing you can define your change processes you want to perform in the Business Transactions Maintenance Framework (BTMF).

SPRO → Service → Transactions → Settings for Subscription Transactions → Contract Changes.

Detailed explanations can be found in the Implementation Guide (IMG) documentation and the F1 helps of the single fields.

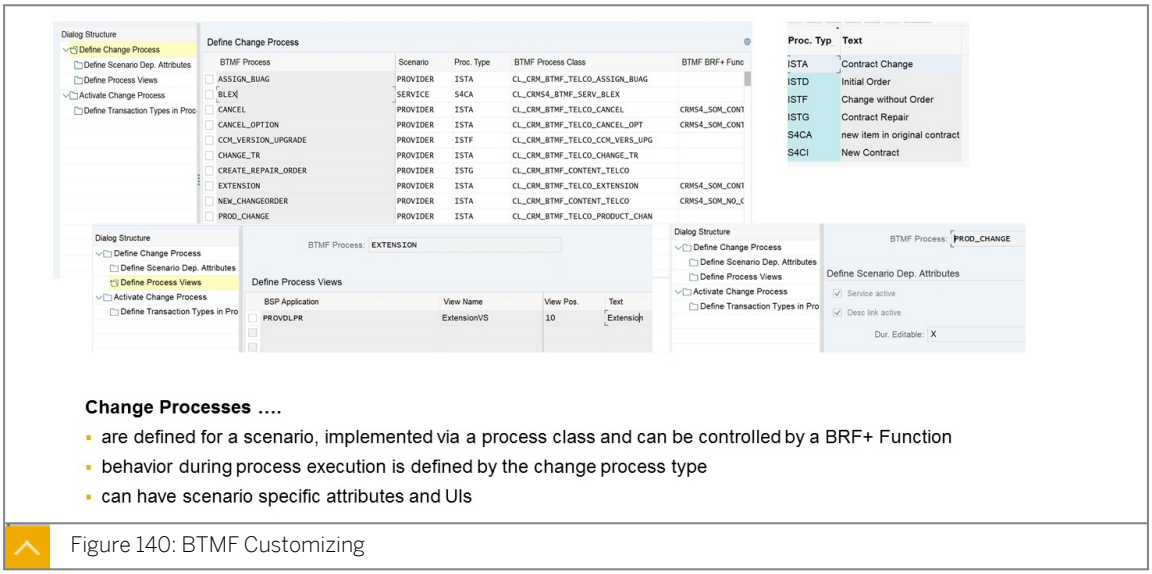

Figure 140: BTMF Customizing

Define Change Process:

- Predefined change processes are delivered. In addition own change processes can be defined.
- The BTMF scenario defines which transactions a change process is valid for. Scenario PROVIDER describes that a change process is valid for subscription contracts.
- The process type defines how the change process is executed, for example, ISTA: contract change with change order

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- The change process logic is implemented in the change process implementation class
- BRF+ functions can be used to control which processes can be executed

Define Scenario Specific Attributes:

You can specify additional attributes for the process according to the scenario selected. Views, which are to be displayed when the change process is performed, can be assigned for change processes with the PROVIDER scenario.

Define Process Views:

Views, which are to be displayed when the change process is performed, can be assigned for change processes with the PROVIDER scenario.

Activate Change Process:

Once the change processes have been defined, you activate them by entering them in the Activate Change Process section.

LESSON SUMMARY

You should now be able to:

- Describe Subscription Contract Changes
- Browse Contract Change Customizing

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Learning Assessment

1. For which of the following tasks do you use the Subscription Order Change process?

   Choose the correct answers.

   - ☐ **A** Create a new invoice
   - ☐ **B** Install discounts to a subscription fee
   - ☐ **C** Change a contract duration
   - ☐ **D** Correct a usage fee
   - ☐ **E** Change an assigned technical ID
   - ☐ **F** Change the sold-to-party

2. For each subscription contract change, a change order is created.

   Determine whether this statement is true or false.

   - ☐ True
   - ☐ False

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Learning Assessment - Answers

1. For which of the following tasks do you use the Subscription Order Change process?

   Choose the correct answers.

   - [ ] A Create a new invoice
   - [X] B Install discounts to a subscription fee
   - [X] C Change a contract duration
   - [ ] D Correct a usage fee
   - [X] E Change an assigned technical ID
   - [ ] F Change the sold-to-party

   That is correct! You use the Subscription Order Change process to install discounts to a subscription fee, change a contract duration, or change an assigned technical ID.

2. For each subscription contract change, a change order is created.

   Determine whether this statement is true or false.

   - [ ] True
   - [X] False

   That is correct! Each subscription contract change does not generate a change order.

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UNIT 8 Master Agreement

Lesson 1

Create Master Agreement 153

Lesson 2

Customer Hierarchy and Master Agreement 159

UNIT OBJECTIVES

- Explain Master Agreement
- Describe Master Agreement Creation in SOM
- Understand Authorized Partner and Partner Validation Rule
- Explain Customer Hierarchy with Master Agreement
- Understand Customizing for Master Agreement in SOM

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Unit 8
Lesson 1

Create Master Agreement

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explain Master Agreement
- Describe Master Agreement Creation in SOM
- Understand Authorized Partner and Partner Validation Rule

Business Example

Business Customer within an organizations structure, closes a master agreement for cloud products. The Customer has negotiated customer prices for usage and monthly fees and a volume discount for total revenue of 10% the year. The customer hierarchy sales area and development areas are assigned to the master agreement.

Explain Master Agreement

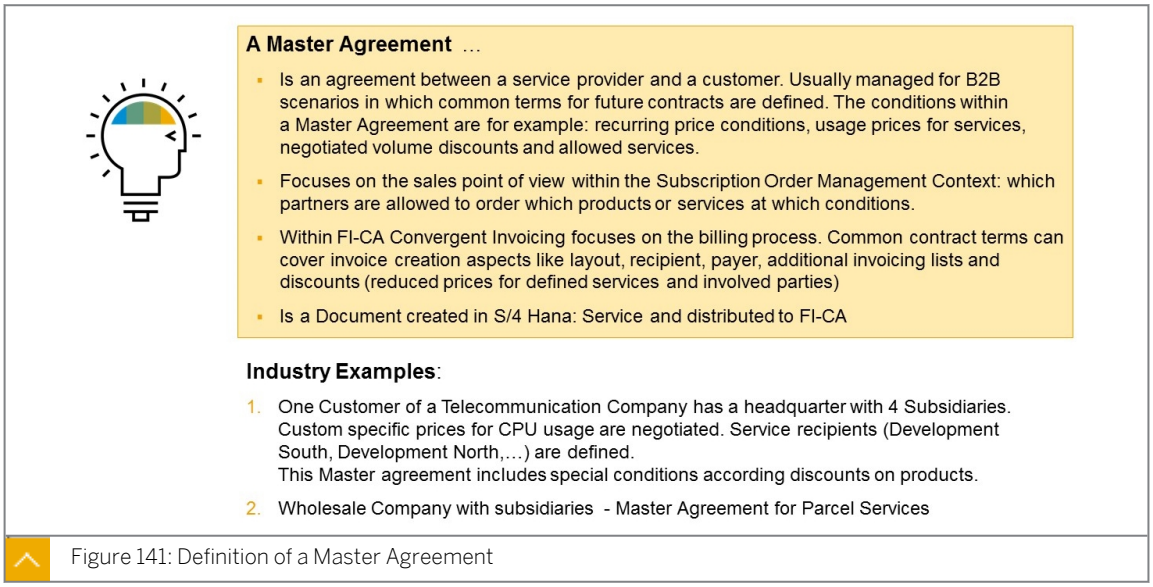

Figure 141: Definition of a Master Agreement

Master agreements are created in S/4HANA Service via the Manage Master Agreement app . Sales Order Management (SOM) focuses on the ordering side of the master agreement.

The master agreement is used as a "product catalog", in which you can define individual products (configuration) for a customer. Every product is represented as an item of the master agreement. The product redefinition is maintained on the item level. This means that you don't have to create new specific products for every new customer. Those products are based on standard products, but in the context of the master agreement, they can have a master agreement specific product number, as well as a specific product description or different prices.

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In addition to the product list, the business partners that can participate in the master agreement are also captured. These can either be contract partners later on or they can be service recipients.

While SOM focuses on the ordering side of a master agreement, S/4Hana Convergent Invoicing (CI) focuses on the billing side.

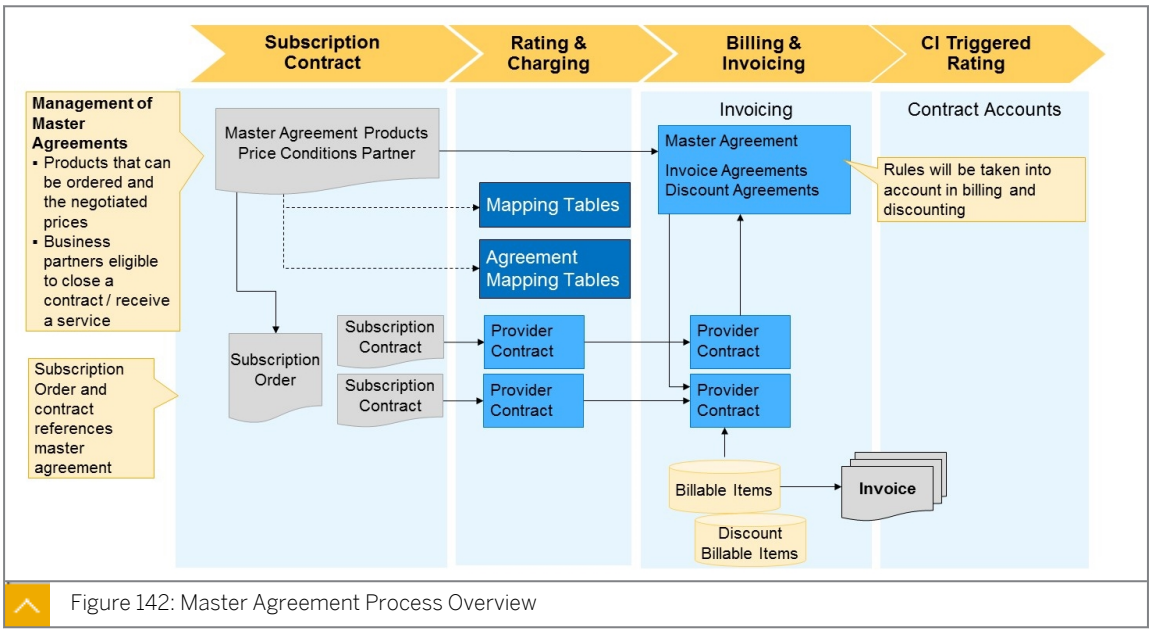

Figure 142: Master Agreement Process Overview

This business scenario shows the creation of a Master Agreement:

The master agreement data is distributed between SAP S/4HANA Service, SOM and SAP S/4HANA, Contract Accounts Receivable and Payable (FI-CA) and each component stores the relevant data for its core processes.

Those products are based on standard products but in the context of the master agreement can have a master agreement specific product number as well as a specific product description or different prices.

You can manage various business partner roles in the master agreement:

- Master Agreement (Contract) Partner: The master agreement partner is the partner with whom you have negotiated the master agreement. A master agreement only has one master agreement partner that is assigned on the Master Agreement header. The sold-to party category is assigned to the master agreement partner.
- Authorized Contract Partner: Authorized contract partners can perform releases and act as the contract partner (payer), sold-to party, and service recipient in the provider contract as a default setting.
- Authorized Sold-to Party: Authorized sold-to parties can perform releases and act as the sold-to party and service recipient in the provider contract as a default setting. However they cannot act as the contract partner (payer) in the provider contract. The default contract partner is used here.
- Authorized Ship-To/Service Recipient: Authorized service recipients cannot perform releases or act as the contract partner in the provider contract. They can be entered as an alternative service recipient in the provider order.

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It is possible to redefine the standard products due to customer or master agreement purposes.

Master Agreement Functionality

- Create Master Agreement
- Assign Products, assign Business Partners with hierarchies, maintain Configuration for Products, Redefine sub-items, and Assign / Redefine Cross Catalog Mapping Tables and Parameters
- Master agreements can be offered to an individual partner or group of partners represented by a customer hierarchy
- Call-offs can be requested for partners assigned to the hierarchy
- The customer hierarchy is also stored with the Contract Accounts Receivable and Payable (FI-CA) provider contract
- Define Master Agreement specific prices
- Release Master Agreement Items
- Replicate Master Agreement in FI-CA
- Display Invoice Agreements, Discount Agreements, and Invoice List Agreements
- Display related subscription orders, contracts, and solution quotations (related objects)
- Create Subscription Order / Solution Quotation / Sharing Contract from Master Agreement
- Auto determine a Master Agreement during creation of Subscription Order / Solution Quotation /Sharing Contract
- Search for Master Agreement

Describe Master Agreement Creation in SOM

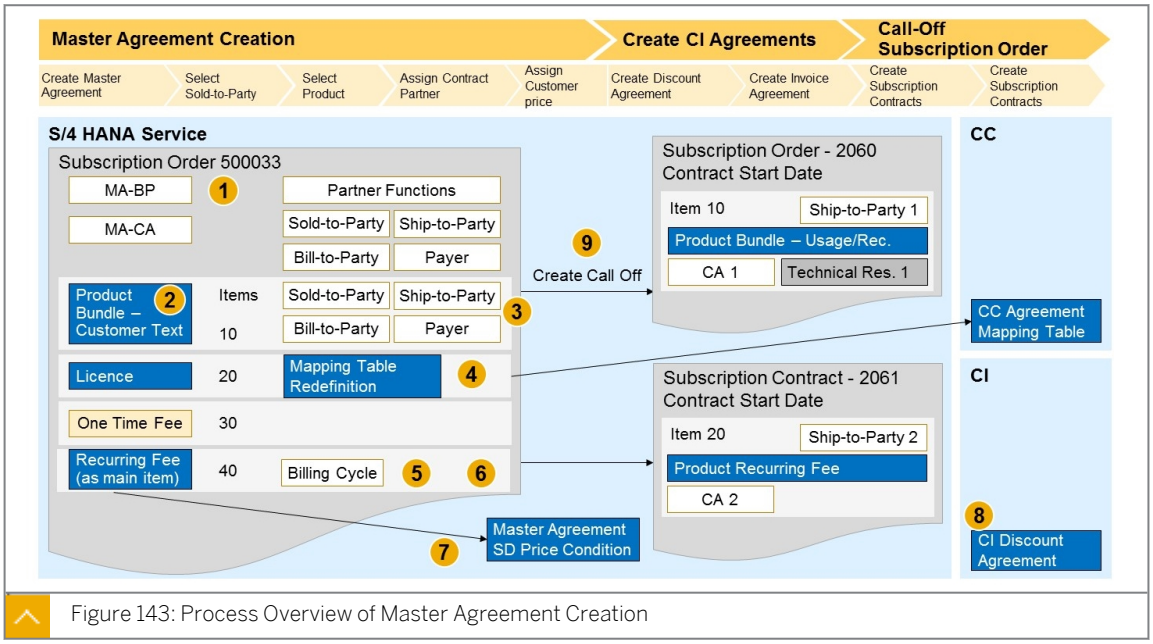

Figure 143: Process Overview of Master Agreement Creation

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The following process steps are done during master agreement creation:

1. Access the FIORI App *Manage Master Agreement* and create a new Master Agreement.
2. Assign a master agreement contract partner.

   The master agreement contract partner needs to have a contract account for technical reasons as well. The master agreement contract account is used in Convergent invoicing in order to assign billable items.
3. Create a master agreement item as a reference for subscription order call offs:
   - Select a product
   - Define Custom Product IDs and Custom Text. Configure the product and fix the configuration if required
   - Find subitems of bundle in the product list and redefined subitems
4. Partner function in Master Agreement:

   The contract partner is the partner with whom you have negotiated the master agreement. A master agreement only has one contract partner and it is assigned on the Master Agreement header. Authorized contract partners act as the contract partner, sold-to party and service recipient in the call-off object as a default setting. Authorized sold-to parties act as the sold-to party and service recipient in the call-off object as a default setting. They cannot however act as the contract partner in the provider contract. The default contract partner is used here. Authorized service recipients cannot act as the contract partner in the call-off object. They can be entered as an alternative service recipient in the call-off object.

   The Master Agreement Partner (Contract Partner) is the partner that you negotiated the master agreement with. This partner is also the legal partner. In addition, it is possible to maintain a contact person and a description.

   On Master Agreement item level, you have to assign Authorized Partners. Authorized Partners can perform call offs (such as provider orders) for the master agreement or can receive goods and services. If the standard behavior is not sufficient, it is possible to define your own rule via the BAdI *"Check Authorized Partners for Provider Master Agreement"*.

   The authorized partners can be added as business partners and the hierarchy structure itself can be added or a hierarchy node. In this case, a hierarchy node is added.
5. Convergent Charging Tables:
   - Depending of the Master Agreement negotiation, it might be necessary to change the entries of a standard Mapping or Range Table, only for this specific Master Agreement.
   - In case of redefining, the system automatically copies the default table and creates a new table. Mapping or Range Tables will be created as Agreement Tables with relation to Master Agreement.
   - It is also possible to assign another exiting table.
   - Via the hyperlink you can navigate to the CC table.
6. Master Agreement - Convergent Charging Parameters

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The concept of redefining Convergent Charging Parameter is similar to the tables. The system reads the parameter from the standard product (cross catalog mapping) and display them in assignment block "Convergent Charging Parameters". The following options are possible to redefine a parameter:

- *Redefined Value:* The column "Redefined Value" is editable and you can enter an individual value.
- *Characteristic:* The column "Characteristic" is editable and you can select a characteristic from the product configuration Attribute: The column "Characteristic" is editable and you can select a attribute from the product configuration.
- *BRF Event:* The column "BRF plus function" is editable and you can select an existing BRF Event

7. Redefine Product Billing Cycle / Billing Cycle determination Rule

   You can set a master agreement specific billing cycle / billing cycle determination rule, which will then be copied to the subscription item of the call-off object

8. Manage the status: Release master agreement item.
9. Define recurring price conditions for master agreement items.
10. The master agreement is replicated to Contract Accounts Receivable and Payable (FI-CA). Discounts, invoice agreements and invoice list agreements are assigned to the master agreement and shown in the Subscription Order Management (SOM) side of master agreement.
11. The master agreement item is ready for subscription offs. All call-off document are referenced to master agreement for discount, invoicing agreements
12. Create a subscription order for the sold-to party and the master agreement is set as reference for the subscription order.
13. Master agreement conditions are transferred to subscription call offs.

Detail Authorized Partner and Partner Validation Rule

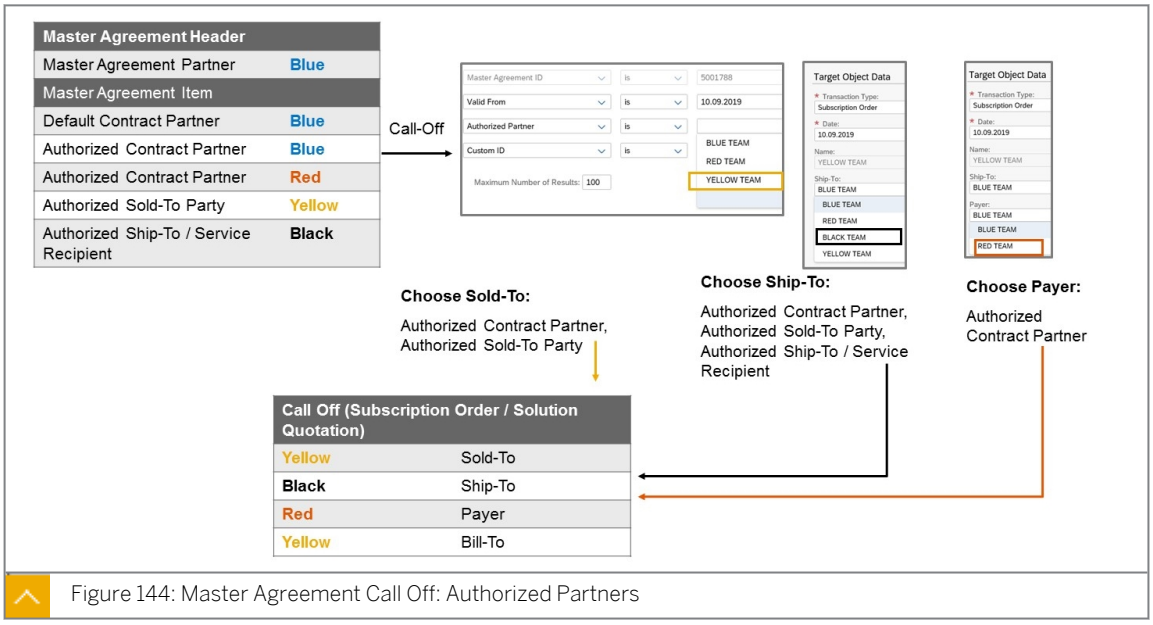

Figure 144: Master Agreement Call Off: Authorized Partners

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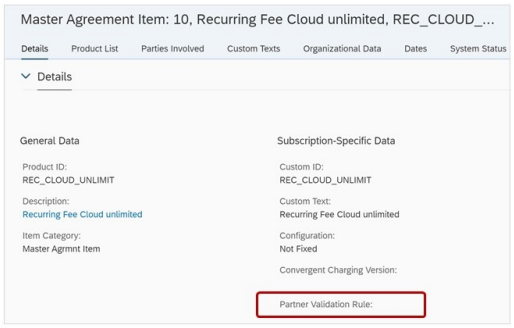
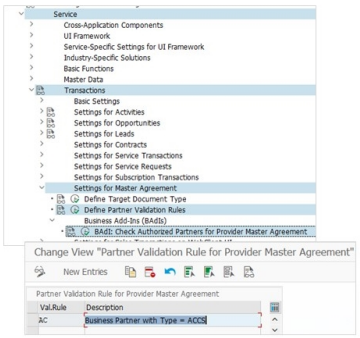
After releasing the items within the Master Agreement, the Call off can be done to create the Subscription Orders or the Solution Quotations for the Authorized Partners. Press *Call off* and search for the Authorized Partner to proceed a Call off. Afterward, select which Transaction Type should be created: Subscription Order or Solution Quotation. In addition you can manage your Partners as Ship To or Payer for the creation.





On Master Agreement item level, you have to assign Authorized Partners.

Authorized partners act as partners in the call-off objects.

Partner validation Rules are predefined in customizing and used as a Filter for a BAdI Implementation. The default implementation is using filter SPACE.

You can find the corresponding setting in IMG: Service → Transactions → Settings for Master Agreement

Figure 145: Master Agreement - Partner Validation Rule

Additional Partner Validation rules can be used to determine required Partners for the Master Agreement.

LESSON SUMMARY

You should now be able to:

- Explain Master Agreement
- Describe Master Agreement Creation in SOM
- Understand Authorized Partner and Partner Validation Rule

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Unit 8
Lesson 2

Customer Hierarchy and Master Agreement

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explain Customer Hierarchy with Master Agreement
- Understand Customizing for Master Agreement in SOM

Explain Customer Hierarchy with Master Agreement

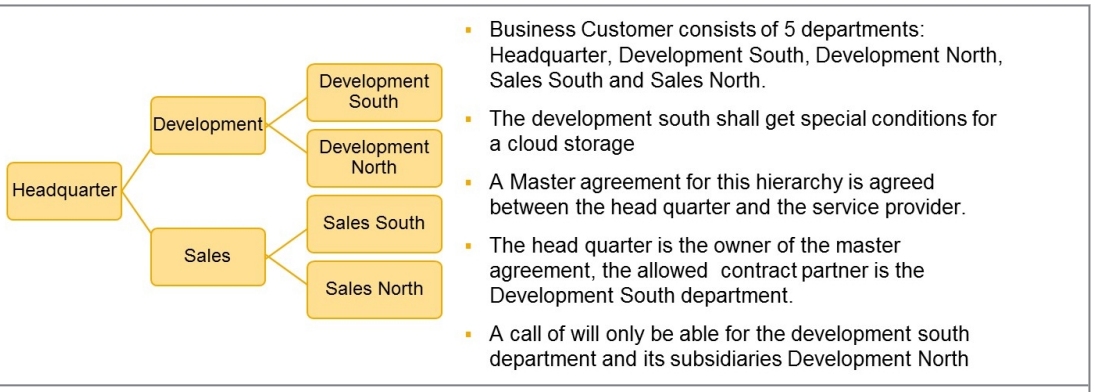

Figure 146: Sample Use Case

Customer Hierarchy Key:

A prerequisite to use a customer hierarchy in a master agreement is that the node ID which represents the customer hierarchy key is existing.

The key *'Node ID'* is a short representation of the full key of a customer hierarchy, which consists of the customer node:

- Sales Organization
- Distribution Channel
- Division
- Start of Validity

Create Master Agreement with Customer Hierarchy:

1. Assign a Customer Hierarchy as Authorized Partner to your Master Agreement
2. Search for your Customer Hierarchy by selecting the search help in Authorized Partner
3. Select your set up Customer Hierarchy
4. Hierarchy is assigned as authorized Partner

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5. After saving your Master Agreement the Hierarchy is copied to the Contract Accounts Receivable and Payable (FI-CA) Master Agreement

Create a Call off:

1. As authorized contract partners you can choose the hierarchy and the assigned partners
2. The selected partners are copied to the order / quotation

Understand Customizing for Master Agreement in SOM

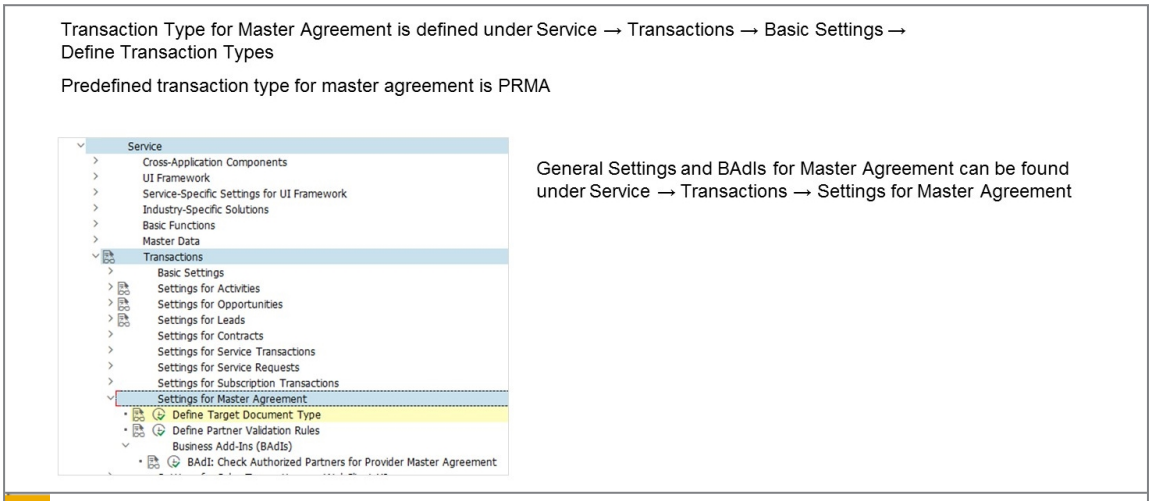

Figure 147: Customizing for a Master Agreement

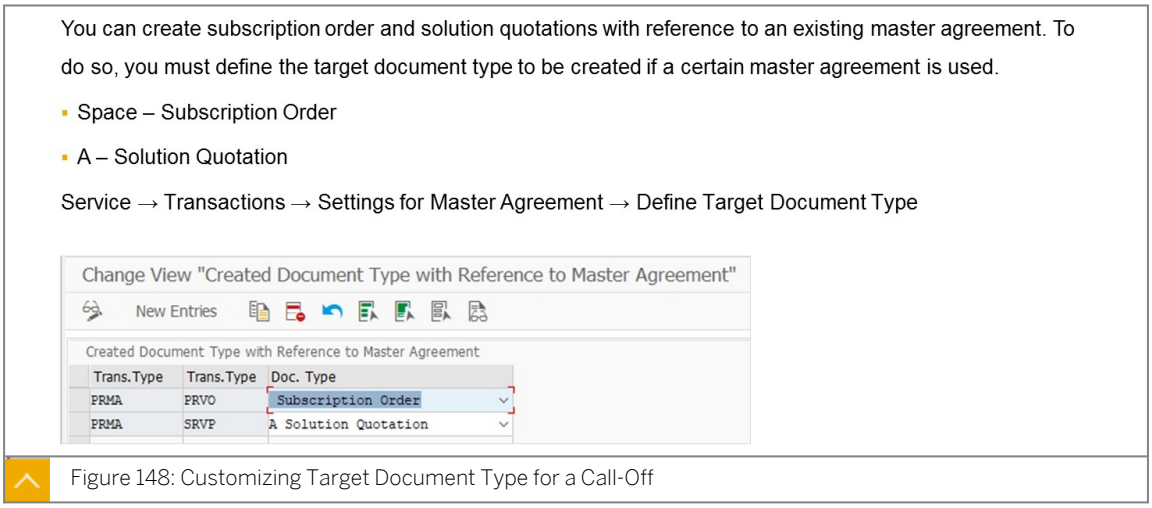

Figure 148: Customizing Target Document Type for a Call-Off

Customizing Target Document Type for a Call-Off:

You can create a subscription order and solution quotations with reference to an existing master agreement. To do so, you must define the target document type to be created if a certain master agreement is used.

- Space - Subscription Order
- A - Solution Quotation

Service → Transactions → Settings for Master Agreement → Define Target Document Type

Authorized Partner Validation Rule:

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On Master Agreement item level, you have to assign Authorized Partners. Authorized partners act as partners in the call-off objects.

Partner validation Rules are predefined in customizing and used as a Filter for a BAdI Implementation. The default implementation is using filter SPACE.

You can find the corresponding setting in Implementation Guide (IMG): *Service → Transactions → Settings for Master Agreement*.

BADI for Date Determination:The default implementation is using the contract start date for the check.

You can use Business Add-In (BAdI) *CRM_PRVMA_DATE_DETERMINE* to check which date shall be used for checking the validity of a subscription order / solution quotation with reference to a master agreement item.

The date must be within the validity period defined in the master agreement item referenced. If the check fails, the system creates an error message.

Enhancement Spot *ES_CRM_PRVMA_API*

BAdI Definition *CRM_PRVMA_DATE_DETERMINE*

Default Implementation *CL_CRM_PRVMA_DATE_DETERM_DFL*

Replication to Contract Accounts Receivable and Payable (FI-CA): IMG: *Service → Transaction → Data Exchange in Service*.

Replication of Master Agreement between Service and FI-CA is done via Data Exchange. Custom specific logic can be implemented. In addition you can use BADI Data Exchange *CRMS4_MAGR_DATA_EXCH* to enrich data during exchange.

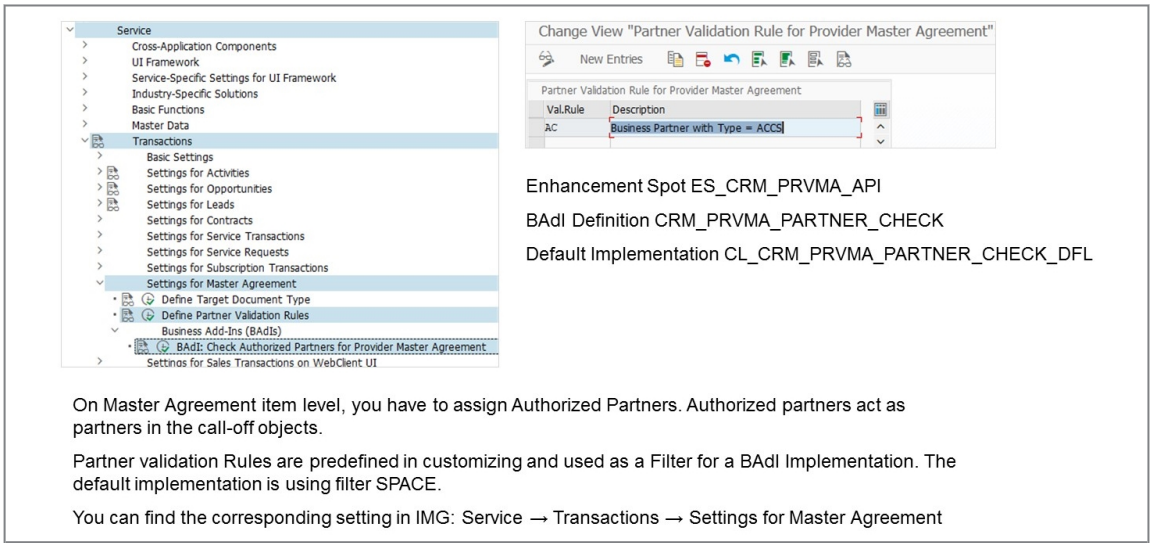

Figure 149: Master Agreement - Partner Validation Rule

Master Agreement - BAdI Date Determination

You can use BAdI *CRM_PRVMA_DATE_DETERMINE* to check which date shall be used for checking the validity of a subscription order / solution quotation with reference to a master agreement item. The date must be within the validity period defined in the master agreement item referenced. If the check fails, the system creates an error message.

Enhancement Spot *ES_CRM_PRVMA_API*

BAdI Definition *CRM_PRVMA_DATE_DETERMINE*

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Default Implementation *CL_CRM_PRVMA_DATE_DETERM_DFL*

The default implementation is using the contract start date for the check.

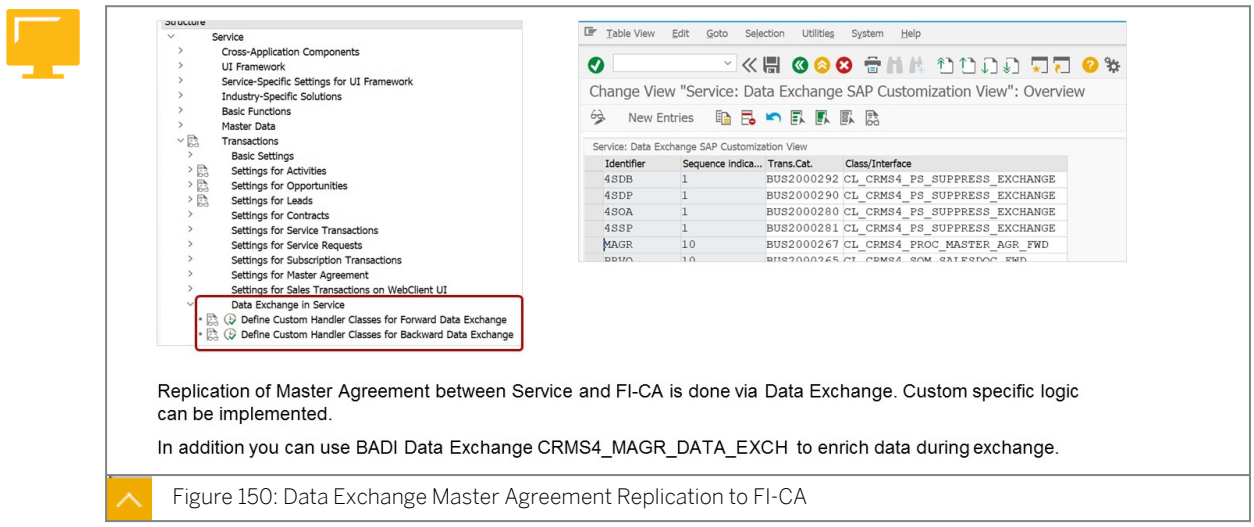

Figure 150: Data Exchange Master Agreement Replication to FI-CA

LESSON SUMMARY

You should now be able to:

- Explain Customer Hierarchy with Master Agreement
- Understand Customizing for Master Agreement in SOM

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Learning Assessment

1. Master Agreement is used to assign sales conditions for services to a group of solution quotations or subscription contracts.

   Determine whether this statement is true or false.

   - ☐ True
   - ☐ False

2. Which of the following can the Master Agreement define?

   Choose the correct answers.

   - ☐ **A** Who is allowed to order the service
   - ☐ **B** Which price is defined for a service
   - ☐ **C** The amount of the bill
   - ☐ **D** What can be ordered

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Learning Assessment - Answers

1. Master Agreement is used to assign sales conditions for services to a group of solution quotations or subscription contracts.

   Determine whether this statement is true or false.

   [X] True

   [ ] False

   That is correct!

2. Which of the following can the Master Agreement define?

   Choose the correct answers.

   [X] A Who is allowed to order the service

   [X] B Which price is defined for a service

   [ ] C The amount of the bill

   [X] D What can be ordered

   That is correct! The amount of the bill is not defined in the Master Agreement.

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UNIT 9

Solution Quote (Sales, Subscription and Service)

Lesson 1

Solution Quotation 167

UNIT OBJECTIVES

- Understand the Solution Quotation Process
- Provide a Solution Quotation Product Example
- Create a Solution Quotation
- Understand Master Agreement with Solution Quotation
- Describe Solution Quotation Order Distribution and Customizing

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Unit 9
Lesson 1

Solution Quotation

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Understand the Solution Quotation Process
- Provide a Solution Quotation Product Example
- Create a Solution Quotation
- Understand Master Agreement with Solution Quotation
- Describe Solution Quotation Order Distribution and Customizing

Business Example

As a sales representative you want to sell a tablet (good), including the service for installation, a service contract, and a subscription contract for data storage service to your customer. You have designed a product, Notebook Care, for your package, which includes all requirements. You set up a Solution Quotation to see your Notebook bundle.

Solution Quotation Definition

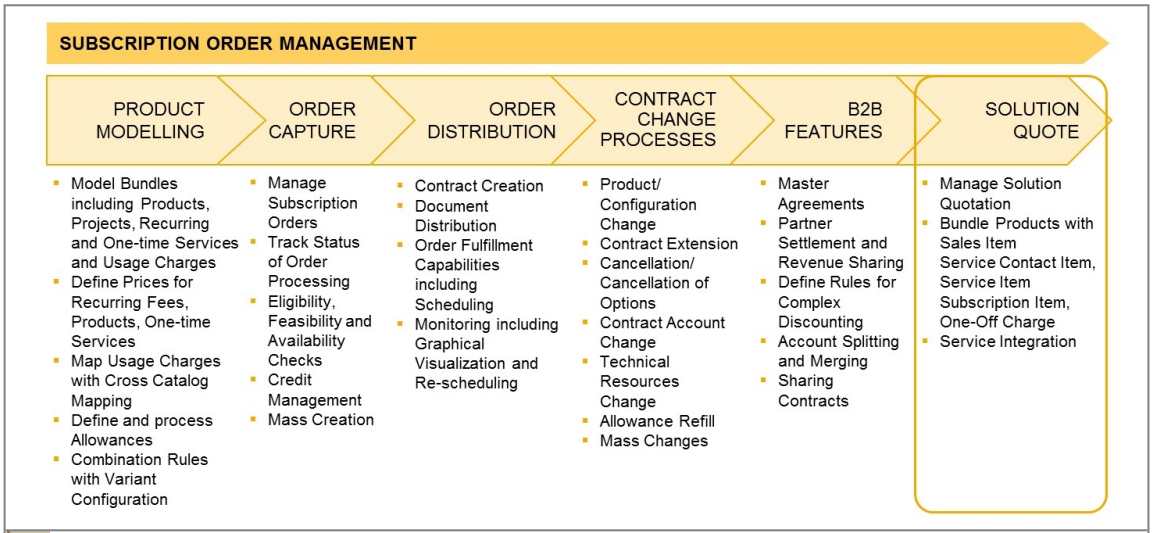

Figure 151: Overview of Subscription Order Management

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A Solution Quotation …

- Is a legally binding offer with fixed conditions made to a customer for the performance of different kinds of services, subscriptions and for the delivery of physical goods.
- Allows ordering Service, Sales and Subscription products in one SAP order document
- Perform and tracks the creation of all follow up documents.
- can be related to a Master Agreement.

Industry Example:

1. Buying a Notebook including installation service, support contract and monthly subscription fee.
2. Buying a Car including mobile services (navigation services, mobility insurance, service contract)

Figure 152: Definition of Solution Quotation

The Solution Quotation can be used to sell physical goods in combination with Service Transactions and with Subscription Orders. The outcome is triggered automatically via the Action Profile set on the Transaction Type SRVP.

Solution Quotation Process Overview

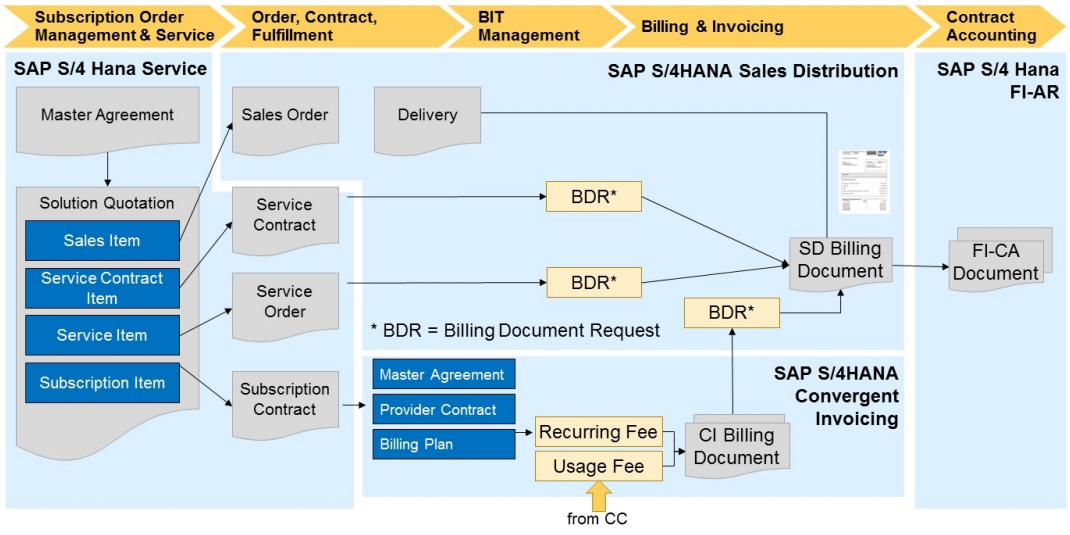

Figure 153: Overview of Solution Quotation Process

This Integration overview shows the Solution Quotation in FI-AR context:

1. Step 1 the Solution Quotation is created and accepted
2. Step 2 Sales Order, Service Contract, Service Order, and Subscription Order are distributed
3. Step 3 monthly billing plan items are requested (CI) --> BIT creation
4. Step 4 Convergent Charging Rating (Usage)
5. Step 5 Billing in Convergent Invoicing (CI)
6. Step 6 Transfer Billing Document as BDR to FI-AR

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7. Step 7 Invoicing in FI-AR

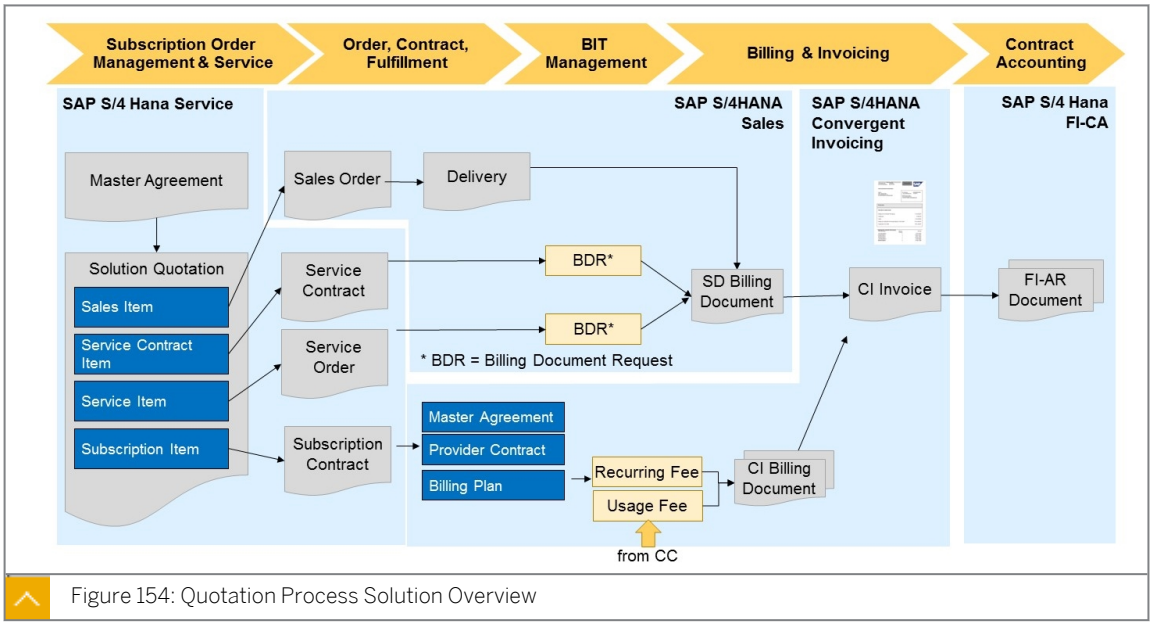

Figure 154: Quotation Process Solution Overview

This Integration overview shows the Solution Quotation Scenario in Contract Accounts Receivable and Payable (FI-CA) context. After accepting the Solution Quotation (Order Fulfillment), the Billing Document Requests and Billing Plan items are created and transmitted to Billing in FI-CA → CI Billing.

All changes regarding the solution quotation are done in outgoing orders. Changes to a subscription contract are made via subscription contract change processes. There are no change processes available in solution quotation.

Solution Quotation Product Example

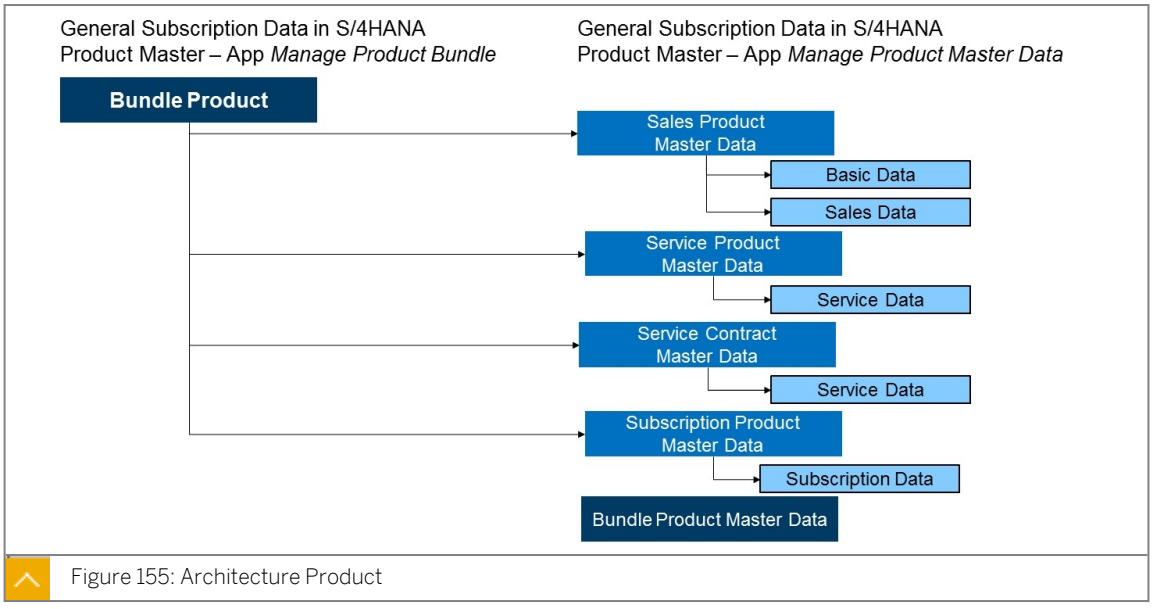

Figure 155: Architecture Product

In S/4HANA only the product master, also known as the material master, is used to maintain the central product data.

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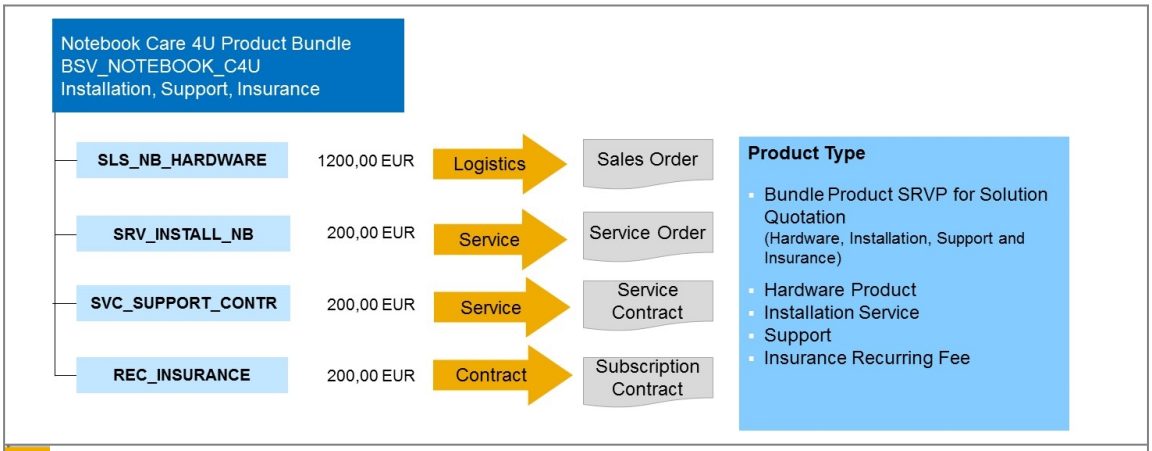

Figure 156: Product Bundle BSV_NOTEBOOK_C4U

Example of a product structure:

In this case, the bundle is the container for each product. You can either define a price on bundle product level or the price is being determined by the selected products. This product example is used for the Solution Quotation, where you can sell hardware, services, and contract relevant items in one order. Resulting from the shown bundle, each item triggers a transaction type.

Solution Quotation Creation

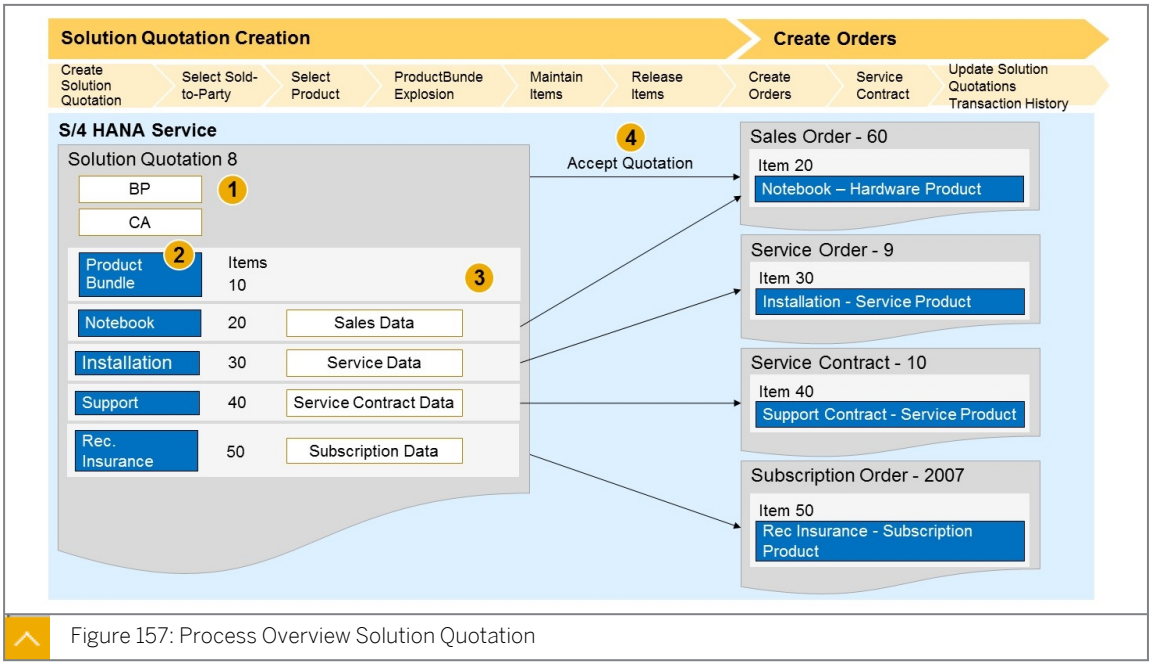

Figure 157: Process Overview Solution Quotation

Access the SAP Fiori app, Manage Solution Quotation and create a new Solution Quotation. This business scenario shows the creation of a Solution Quotation.

The Solution Quotation is a special service order that can be used to sell product bundles that consist of physical products and services, as well as contract relevant items. When the Solution Quotation has been accepted, a Sales Order, a Service Order, a Service Contract, and a Subscription Contract are automatically created.

In the first step you log on to the SAP Fiori Launchpad, go to the Subscription Order Management section, and select the App, Manage Solution Quotations.

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Solution Order Creation

Selection of the transaction type SRVP Solution Quotation.

Step 1: At header level you need to select following:

- Customer Data:

  Select an existing Sold to Party, the sold to party entered is also used for the required partner functions: Ship-to Party, Bill-to Party, and Payer are determined via the Partner Determination within Customizing.

- Organizational Data:

  The Organizational Data is automatically determined according to the entered business partner. This is done via customizing of the organizational profile for the Solution Quotation. This also includes the needed Service Organization.

Step 2: At item level you need to select following:

- Product Data:

  The products entered in the Solution Quotation are represented as items. Products bundle can have complex models that produce additional sub items .

- Contract Account:

  This contains the information about the payment and financial data for long term business. The data contained in the contract account influences processes within invoicing, taxation, and correspondence processing. If Contract Accounts Receivable and Payable (FI-CA) is activated, a contract account needs to be maintained.

Step 3: Release all items

- After the Solution Quotation is captured you need to release all items to finalize the process.

Step 4: Accept Quotation

- As last step the Solution Quotation needs to be accepted to get the order fulfillment. After accepting it, a Sales Order, a Service Order, a Service Contract, and a Subscription Contract are automatically created.
- Update the transaction history with the follow up documents.

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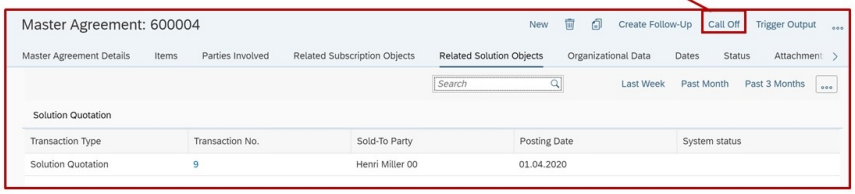
Master Agreement with Solution Quotation

Direct Assignment of a Master Agreement

"Call Off" button on the Master Agreement allows direct creation of a solution quotation based on this Master Agreement



Master Agreement Determination

On creation of a new Solution Quotation, once Sold-to and product have been entered the system checks for existing master agreements and proposes to assign one of them in a new pop-up.

Figure 158: Master Agreement with Solution Quotations

In SPRO, the setup in the node *Service → Transactions → Settings for Master Agreement → Define Target Document Type*, you define the transaction types of the solution quotation used in this process.

Solution Quotation Order Distribution and Customizing

Solution Quotation is now able to directly trigger the ODI, based on the respective ODI schema definition.

This is needed to handle one-off charges, as those would not become part of a subscription contract and therefore would not be handled by the respective ODI triggers.

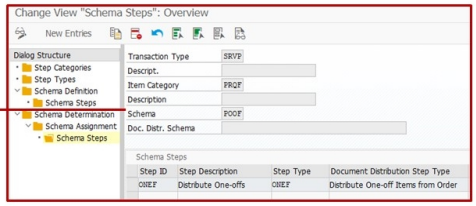


Figure 159: Object Distribution Infrastructure (ODI) in the Solution Quotation

With the ODI schema POOF, the Solution Quotation is now able to distribute one-offs to Convergent Invoicing (SAP CI)

With S4HANA for customer management 1809 FSP01 you were already able to use solution quotation but subscription order was not included.

Starting with S4HANA Service 1909 Subscription Items were enables for Solution Quotation.

In order to activate subscription business in the solution quotation the respective transaction type of the solution quotation needs the assignment of the subscription order transaction category.

This assignment is part of delivery customizing for the solution quotations, but may need to be added to customer specific transaction types.

SPRO
Service -> Transactions -> Basic Settings -> Define Transaction Types

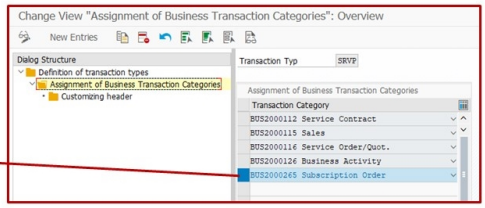


Figure 160: Solution Quotation With Or Without Subscriptions

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This does not have an immediate effect on the UI of the Solution Quotation header overview page however, there are no subscription specific UI configurations delivered for the solution quotation details section and no dynamic field display based on this criterion.

The actual effect this setting has is only taking place in the background. If the assignment of the subscription order bus type is not done. due to a customer not using subscription business in the solution quote, then certain coding is not run through, resulting in an improved performance.

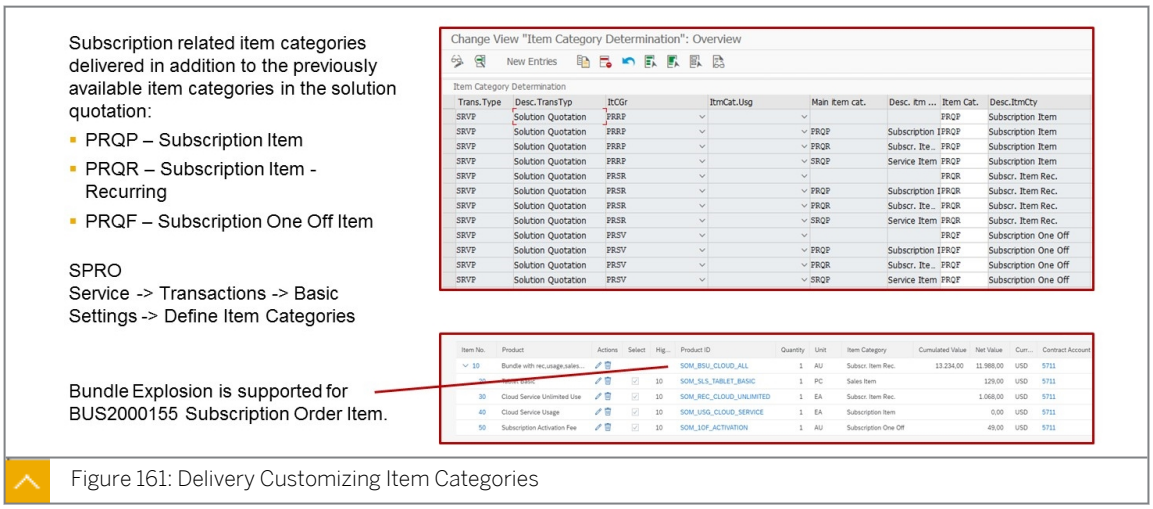

Figure 161: Delivery Customizing Item Categories

In addition to this, copy control customizing was defined for Solution Quotation as a source and Subscription Contract as a target.

LESSON SUMMARY

You should now be able to:

- Understand the Solution Quotation Process
- Provide a Solution Quotation Product Example
- Create a Solution Quotation
- Understand Master Agreement with Solution Quotation
- Describe Solution Quotation Order Distribution and Customizing

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Learning Assessment

1. Which of the following items does a solution quotation create?

   Choose the correct answers.

   - [ ] **A** Project Items
   - [ ] **B** Sales Items
   - [ ] **C** Service Items
   - [ ] **D** Subscription Items
   - [ ] **E** Service Contract Items

2. A solution quotation is integrated with which of the following?

   Choose the correct answers.

   - [ ] **A** Sales and Distribution
   - [ ] **B** Convergent Invoicing
   - [ ] **C** Convergent Charging
   - [ ] **D** Material Management
   - [ ] **E** Service Management

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Learning Assessment - Answers

1. Which of the following items does a solution quotation create?

   Choose the correct answers.

   - [ ] A Project Items
   - [X] B Sales Items
   - [X] C Service Items
   - [X] D Subscription Items
   - [X] E Service Contract Items

   That is correct! A solution quotation creates all of the listed items with the exception of project items.

2. A solution quotation is integrated with which of the following?

   Choose the correct answers.

   - [X] A Sales and Distribution
   - [ ] B Convergent Invoicing
   - [ ] C Convergent Charging
   - [ ] D Material Management
   - [X] E Service Management

   That is correct! A solution quotation is integrated with Sales and Distribution, and Service Management.

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UNIT 10 Partner Agreement

Lesson 1

Partner Agreement 179

UNIT OBJECTIVES

- Describe Partner Agreement
- Explain Partner Product
- Create a Partner Agreement
- Define Partner Agreement Customizing

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Unit 10
Lesson 1

Partner Agreement

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Describe Partner Agreement
- Explain Partner Product
- Create a Partner Agreement
- Define Partner Agreement Customizing

Business Example

In cloud selection service bandwidth data transfer is sold. All customer usage of bandwidth service impacts a gateway fee for a partner, who provisions the gateway service for data transfer. A partner agreement for cloud selection service partner GATEWAY-To-WORLD is captured, in order to share, for all bandwidth usages, 3% of the customer revenue with this partner .

Understand Partner Agreement

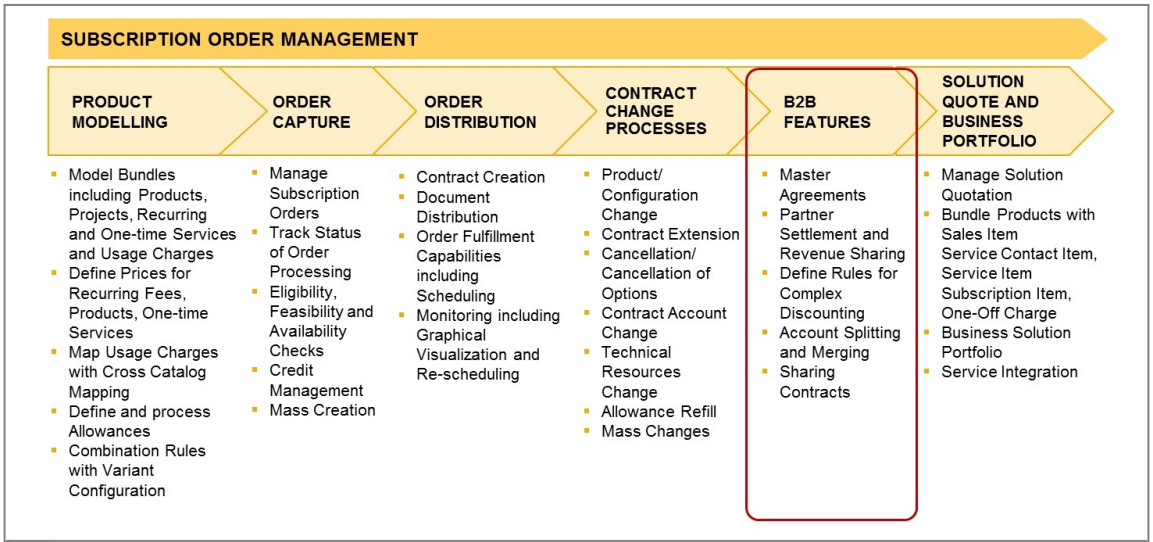

Figure 162: Overview of a Subscription Order Management

Partner Settlement Features in Subsciption Order Management (SOM):

- Subscription Partner Orders and Subscription Partner Contracts are created in SOM
- Partner processing itself is only done in Convergent Invoicing and Convergent Charging.

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- Subscription Partner Contracts, create a partner agreement (provider contracts for a partner) in Contract Accounts Receivable and Payable (FI-CA)
- Subscription Partner Orders can only be created for products of a material type which is customized for partner processing, in addition, a settlement rule needs to be assigned.

Figure 163: Definition Partner Agreement

As traditional industries increasingly change their business model to a service-based model, a revenue sharing business model becomes more and more important. In such a model, the service provider, who has the contractual agreement with the end customer, shares parts of the revenue with their content providers or has to pay some royalties.

When an end customer is buying a service, the content partner should get part of the revenue that the services provider makes with the end customer.

For this reason, a service provider closes a partner agreement with the content provider. The partner agreement contains the conditions that were negotiated between the service provider and the content partner, such as the service and the revenue split.

The provider that acts as a billing operator, including invoicing and collections towards the end consumer, must forward a certain amount of money to the corresponding partners or third-parties.

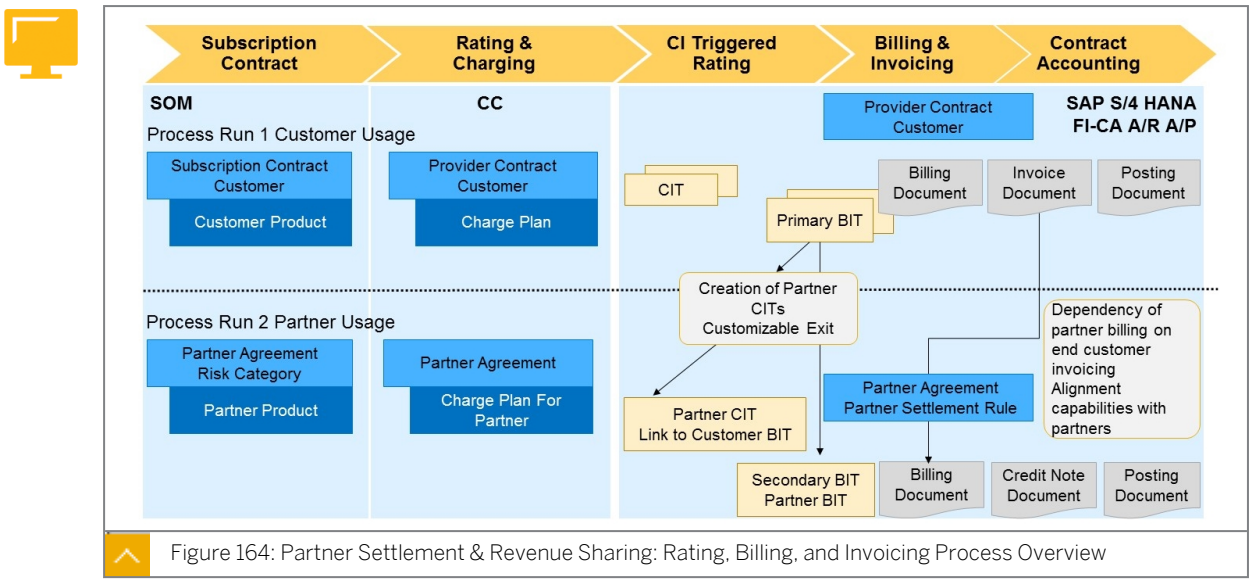

Figure 164: Partner Settlement & Revenue Sharing: Rating, Billing, and Invoicing Process Overview

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Process Steps in Partner Settlement & Revenue Sharing:

1. Define Risk Category for the partner in the partner settlement rule assigned in the partner agreement.
2. Link Customer BITs with the Partner BITs for payment control.
3. Partner credits can be adjusted for the concerned partner items when an end customer receivable is subject to dunning or written off.
4. 4. The generated adjustment BITs are tracked and linked to the original partner and end customer BITs

Explain Partner Product

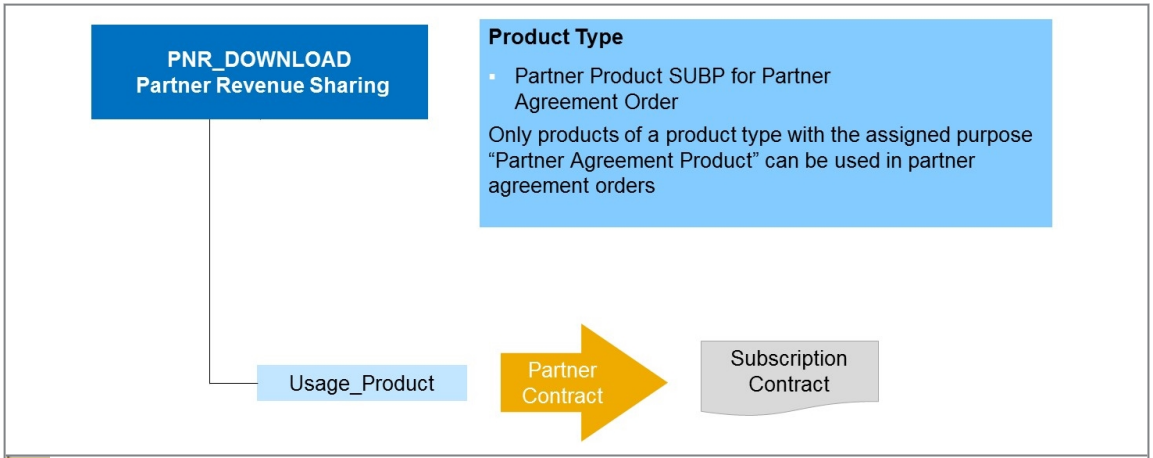

Figure 165: Product PNR_DOWNLOAD

The PNR_DOWNLOAD Product shows up the scenario to create a Partner Agreement and includes the Settlement Rules for Convergent Invoicing.

Describe Partner Agreement Creation

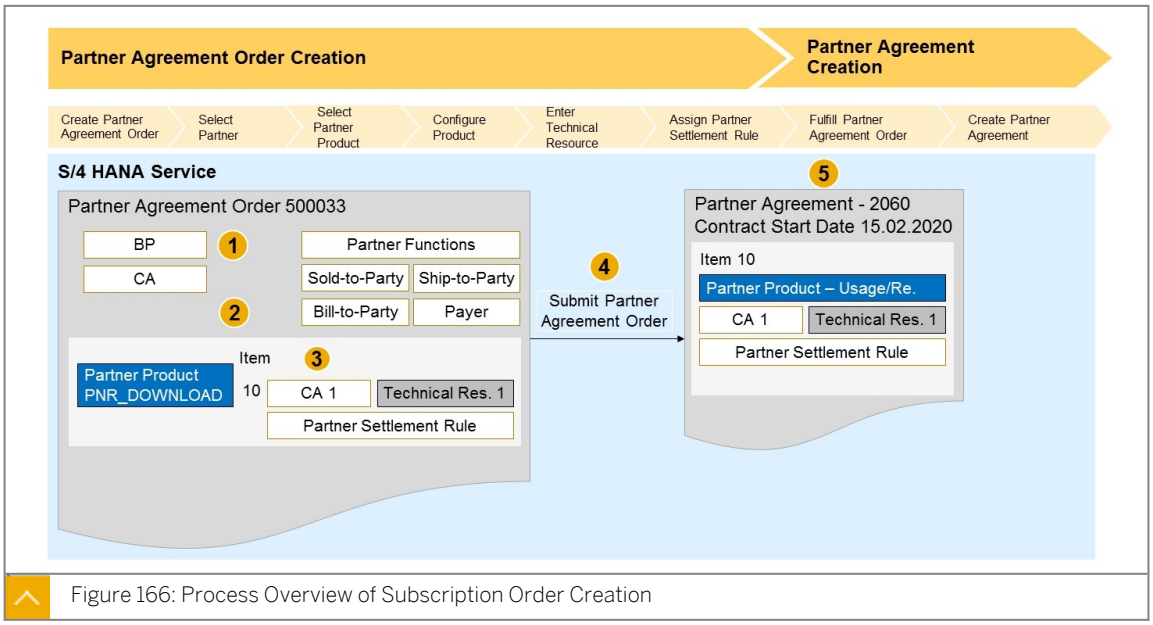

Figure 166: Process Overview of Subscription Order Creation

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This business scenario shows the creation of a Partner Agreement:

1. Step 1. Partner Agreement Order Creation

   In the first step you log on to the Fiori Launchpad, go to the *Subscription Order Management* section and select the *Manage Subscription Order* app.

   Selection of the transaction type *PRPO Partner Agreement Order*.

2. Step 2. At header level you need to select the following:

   Partner Data:

   Select an existing Partner as Sold to Party, the sold to party entered is also used for the required partner functions. Ship-to Party, Bill-to Party, and Payer are determined via the Partner Determination PROV0001 within Customizing.

   Organizational Data:

   The Organizational Data is automatically determined according to the entered business partner. This is done via customizing of the organizational profile for the subscription order with *PROVHEADER*.

   Partner Contract Account:

   This contains the information about the partner credit payment and financial data for long term business. The data contained in the contract account influences processes within invoicing, taxation, and correspondence processing. If Contract Accounts Receivable and Payable (FI-CA) is activated a contract account need to be maintained.

3. Step 3. At item level you need to select the following:

   Product Data:

   Partner Products have the product type *SUBP*. Only Partner Products can be entered in Partner Agreement orders. Partner Products have a partner settlement rule assigned. This is used for risk categorization in Credit Payment control.

   Enter the product or item related data:

   Contract start date, product configuration, technical resources, and choose the contract accounts.

   Technical Resources:

   If the Partner consumption items are rated in Convergent Charging, a technical resource is needed for Charging and Rating of Partner consumption items.

4. Step 4. Submit Partner Agreement Order

   After the Partner Agreement Order is captured you need to submit it to trigger the order fulfillment. The result of the order fulfillment is the Partner Agreement Order creation.

5. Step 5. Partner Agreement distribution in Billing and Revenue Innovation Management (BRIM) landscape.

   Depending on the Partner Product modeling as a recurring and / or usage product, the partner agreements are distributed in the BRIM landscape, same as other subscription products.

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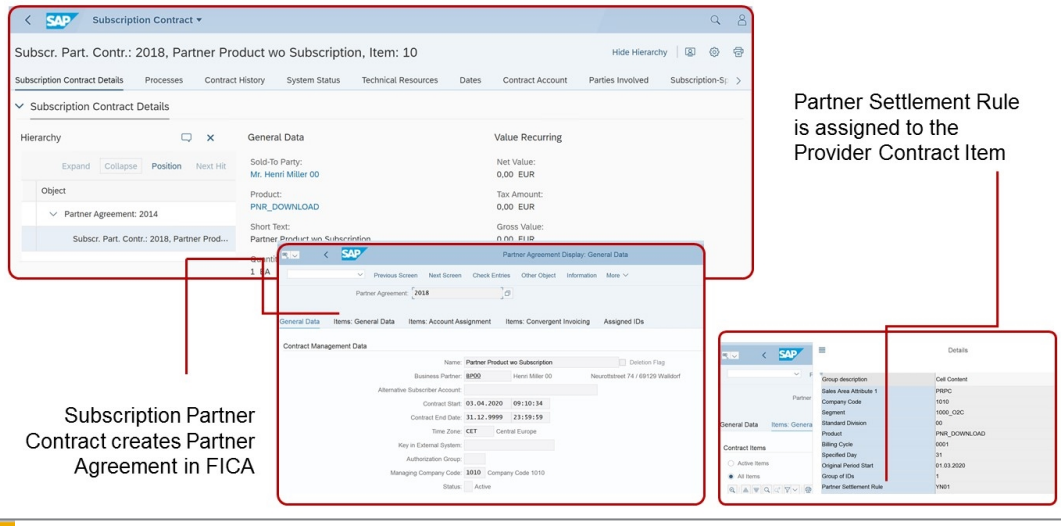

Figure 167: Partner Agreement

The partner agreement is replicated to a partner agreement in FI-CA. Partner agreements are a special kind of provider contract. The settlement rule is copied to the partner agreement in FI-CA. To display partner agreements in FI-CA, you can use transaction *FP_PV3*.

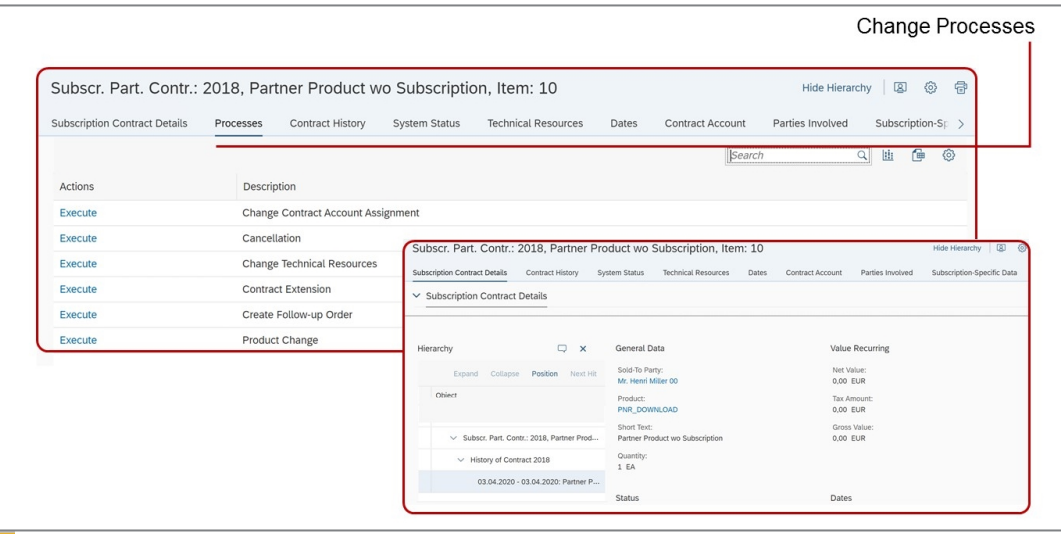

Figure 168: Change Processes

A change process for the Subscription Partner Contract is the same as for a Subscription Contract. Unit 7 covers the Contract Changes.

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Define Partner Agreement Customizing

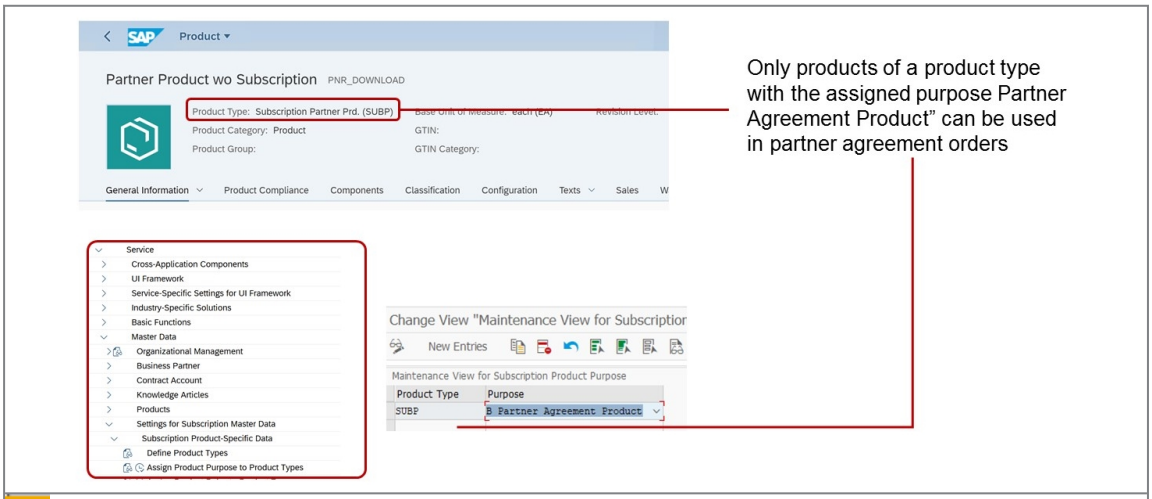

Figure 169: Product Type for Partner Products

You can maintain the settings in Implementation Guide (IMG):

→ *Service* → *Master Data* → *Settings for Subscription Master Data* → *Subscription Product-Specific Data* - → *Assign Product Purpose to Product Types*

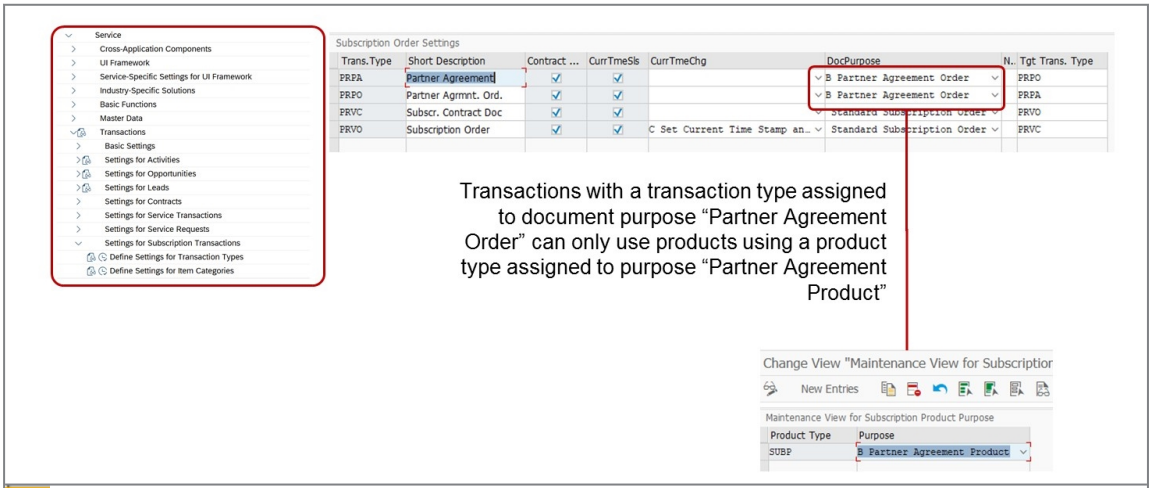

Figure 170: Figure: Customizing Partner Agreement

You can maintain the settings in IMG:

Service → *Transactions* → *Settings for Subscription Transactions* → *Define Settings for Transaction Types*

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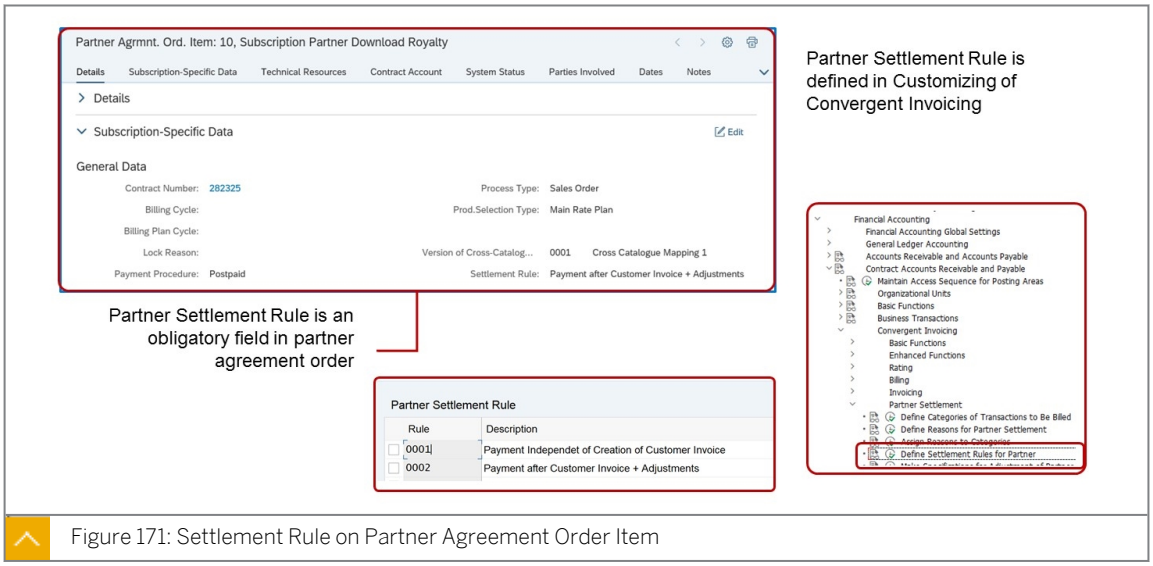

Figure 171: Settlement Rule on Partner Agreement Order Item

A partner settlement rule in Convergent Invoicing specifies:

- The point in time at which the partner shares are settled
- Whether a preliminary invoice is created
- Whether the invoice document is posted in Contract Accounts Receivable and Payable
- The payment mode
- Whether the partner bears the default risk if the end customer fails to pay

It is assigned to the partner agreement order.

LESSON SUMMARY

You should now be able to:

- Describe Partner Agreement
- Explain Partner Product
- Create a Partner Agreement
- Define Partner Agreement Customizing

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Learning Assessment

1. Partner agreement differs from subscription contract with an additional field for risk management of partner revenue settlement. Which of the following is the additional field in partner contract?

   Choose the correct answer.

   - ☐ A Partner Settlement Category
   - ☐ B Partner Settlement Rule
   - ☐ C Partner Risk Category
   - ☐ D Partner Subscription Contract

2. In an ordering process for a partner agreement, you need a different product than in an ordering process for a customer contract.

   Determine whether this statement is true or false.

   - ☐ True
   - ☐ False

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Learning Assessment - Answers

1. Partner agreement differs from subscription contract with an additional field for risk management of partner revenue settlement. Which of the following is the additional field in partner contract?

   Choose the correct answer.

   - [ ] A Partner Settlement Category
   - [x] B Partner Settlement Rule
   - [ ] C Partner Risk Category
   - [ ] D Partner Subscription Contract

   That is correct! The additional field in partner contract is the *Partner Settlement Rule* field.

2. In an ordering process for a partner agreement, you need a different product than in an ordering process for a customer contract.

   Determine whether this statement is true or false.

   - [x] True
   - [ ] False

   That is correct!

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UNIT 11

Integration with SAP Convergent Charging

Lesson 1

SOM Integration with Convergent Charging 191

Lesson 2

Account Splitting 207

UNIT OBJECTIVES

- Explain SOM Integration with Convergent Charging
- Detail the Convergent Charging Data Model
- Understand Process Flow for usage based fees
- Explain Table Definition and Redefinition
- Show Integration with CITs and BITs into Convergent Invoicing
- Describe Technical Integration with Convergent Charging
- Recurring Fees in SAP Convergent Charging
- Illustrate the Account Splitting process
- Use Account Splitting customizing

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Unit 11
Lesson 1

SOM Integration with Convergent Charging

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explain SOM Integration with Convergent Charging
- Detail the Convergent Charging Data Model
- Understand Process Flow for usage based fees
- Explain Table Definition and Redefinition
- Show Integration with CITs and BITs into Convergent Invoicing
- Describe Technical Integration with Convergent Charging
- Recurring Fees in SAP Convergent Charging

Business Example

The O2C company offers cloud infrastructure to their customer.

The product Storage Service offers storage due to Data Volume Storage:

- 1 GB of Storage costs 2 EUR
- 4 GB of Storage costs 3 EUR

Explain SOM Integration with Convergent Charging

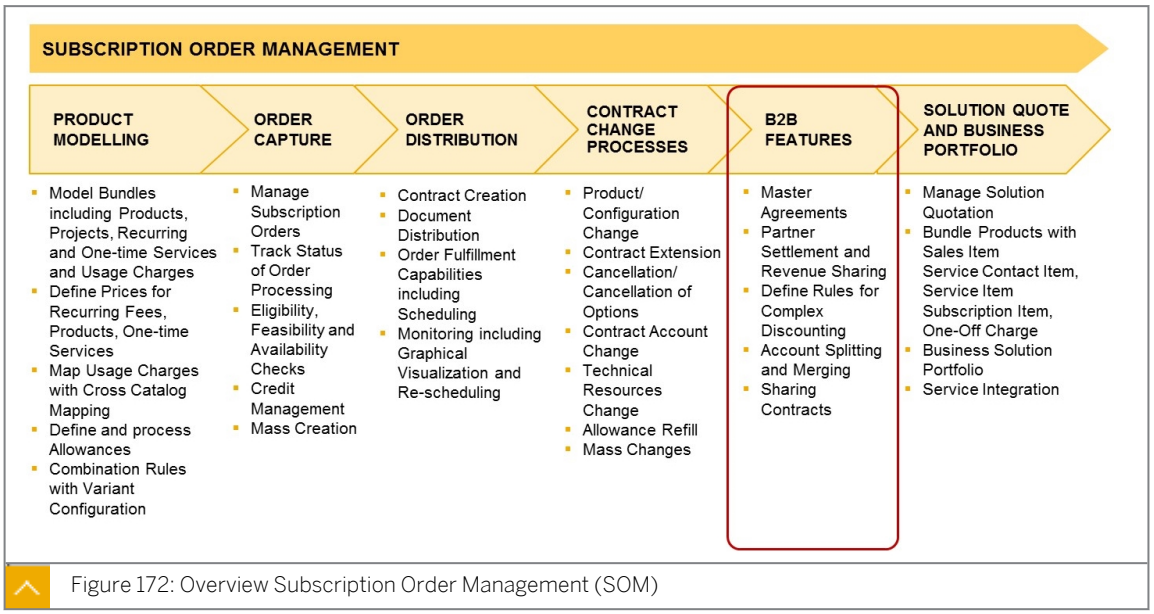

Figure 172: Overview Subscription Order Management (SOM)

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Convergent Charging is a Billing and Revenue Innovation Management (BRIM) solution component for service pricing and modeling, for usages.

Understand the process definition for Pricing, Rating ,and Charging:

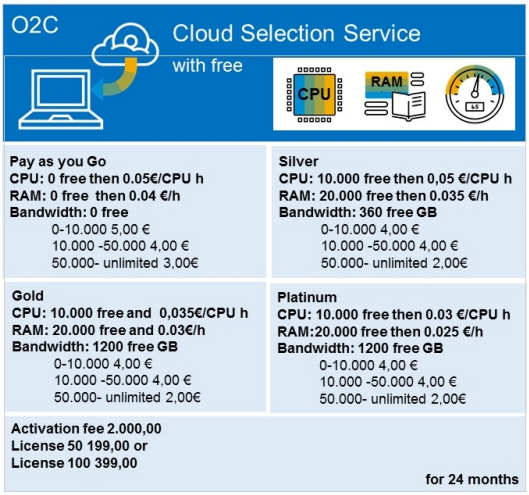
Process Definitions:

- **Pricing** in Convergent Charging is the term describing the process of creating the logic that calculates the prices to be charged for the various services to the customers. The logic is created in so called **Price Plans** which are stored in so called **Charges**.
- **Rating** in Convergent Charging is the term which describes the process of actually calculating what the usage of a certain service costs, i.e. how much has to be paid.
- **Charging** in Convergent Charging is the term which describes the process of determining the account of the customer / person, who is paying for the service consumption. The charging logic is stored in the so-called **Charging Plan**, a decision tree-based structure similar to price plans.

Industry Examples:

1. Car Sharing: license plate B-KN-123 driven for 25 km starting at 01.02.2020 9:15 up to 11:35. The usage costs 5,00 Euro.
2. Call with mobile +49 170 12345 of Mr. Miller from Germany to Switzerland to mobile +41 160 34567. Mr Miller has to pay 3,36 Euro according to his mobile tariff.
3. SAP fellow DB Transportation travel ticket from Hamburg to Amsterdam. The ticket costs 118,20 Euro.

Figure 173: Process Definition Pricing, Rating, Charging



The O2C company offers cloud infrastructure to their customer. The product Cloud Selection Service is a subscription and is paid monthly.
The monthly fee depends on the selected service level agreement.

Additionally the product is bundled with an activation fee and a license fee for delivering the license key.

The product has a minimum duration for the subscription of 24 months.

The company Video Cutting is a large company that deals with videos in the cloud. The company receives the cloud platform and physical infrastructure for their work from the O2C company.

The IT Department of the company decides to order the product Cloud Selection Service in the variant Gold.

Figure 174: Business Example

The O2C company offers cloud infrastructure to their customer. The product Cloud Selection Service is a subscription and is paid monthly.

The monthly fee depends on the selected service level agreement. Four options are possible: Pay as you go, Silver, Gold, and Platinum. Depending on the selected service level the customer can consume a number of CPU hours, RAM hours, and GB for Bandwidth for free.

Additionally the price for usage consumption for RAM, CPU, and Bandwidth depends on the service level agreement. Pricing, Rating, and Charging of usage consumption. according to the product model. is done in Convergent Charging.

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Detail the Convergent Charging Data Model

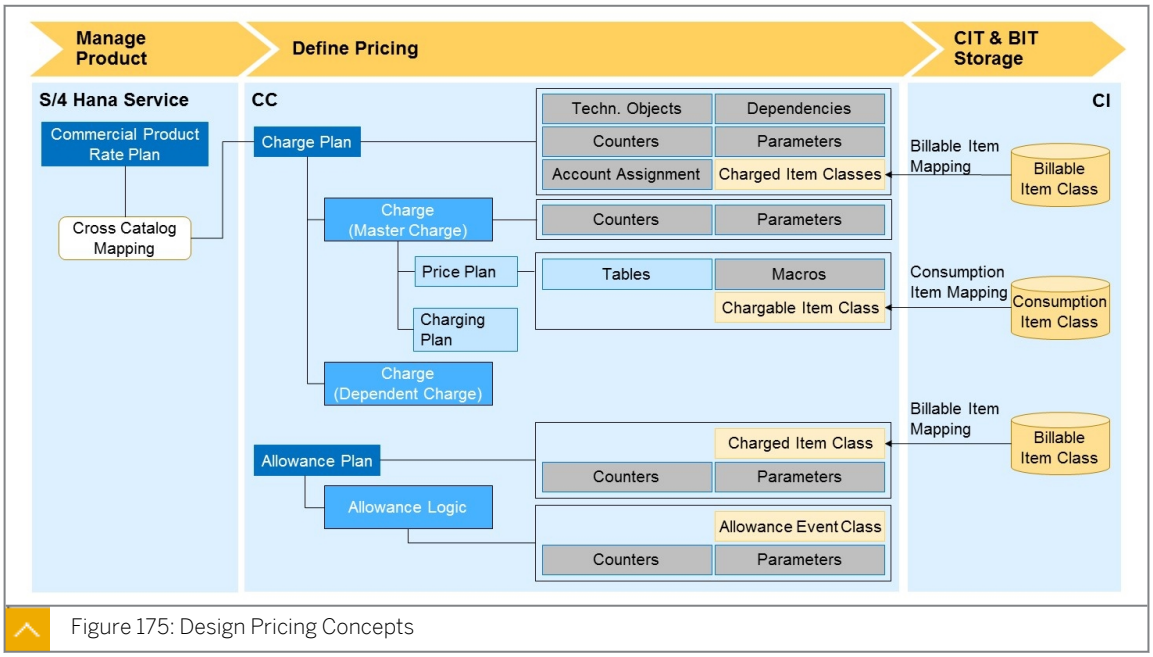

Figure 175: Design Pricing Concepts

Explanation of Objects in Convergent Charging (CC):

- Charge Plans are the building blocks of a commercial offer for the customer and combine charges to meaningful packages. Charge plans transfer counter, parameter, and technical objects to charges.
- Tables contain the price or other information. There are different table types in place: Mapping and Range tables. Both are adaptable to subscriber Mapping tables and Subscriber Range tables.
- Macros contain reusable logic of price plans.
- The charge is the representation of an amount, which is charged to someone.
- CC uses three types of charges: one-time, recurring, and usage charges.
- A charge includes two plans:
  1. *Charging Plans* describe how to determine the account to credit / debit.
  2. *Price Plans* describe how to calculate the price.
- The Chargeable Item Class is responsible for the format of data provided by a mediation system, which is to be processed by CC.
- In the integrated Billing and Revenue Innovation Management (BRIM) scenario, the chargeable item class will be automatically derived from the consumption item class in SAP Convergent Invoicing (CI). This occurs during the import of the consumption item mapping. Consumption Items store the unrated incoming data events and allow deferred rating or re-rating.
- The Charged Item Class describes the output data format. Billable Items are the rated events, which will be processed by SAP CI into invoices. Billable Items class and Charged Items class are mapped via Billable Items mapping.

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- An allowance allows the customer to consume a defined volume (allowance counter) of a service during a defined time.

Therefore the following objects in CC are used:

Allowance plan:

The allowance plan defines the behavior of an allowance when the charging and refilling processes are executed. The allowance plan consists of:

- One, or several, allowance logic objects
- A set of counters linked to the allowance logic objects
- A set of parameters optionally linked to the allowance logic objects
- A status, which can be open or released.

Allowance logic:

The allowance logic performs calculations only when the allowance plan is triggered by:

- Any allowance event
- A periodic allowance event
- A one-shot allowance event

From the financial view allowances, create deferred revenues in S/4HANA CI.

Understand Process Flow for usage based fees

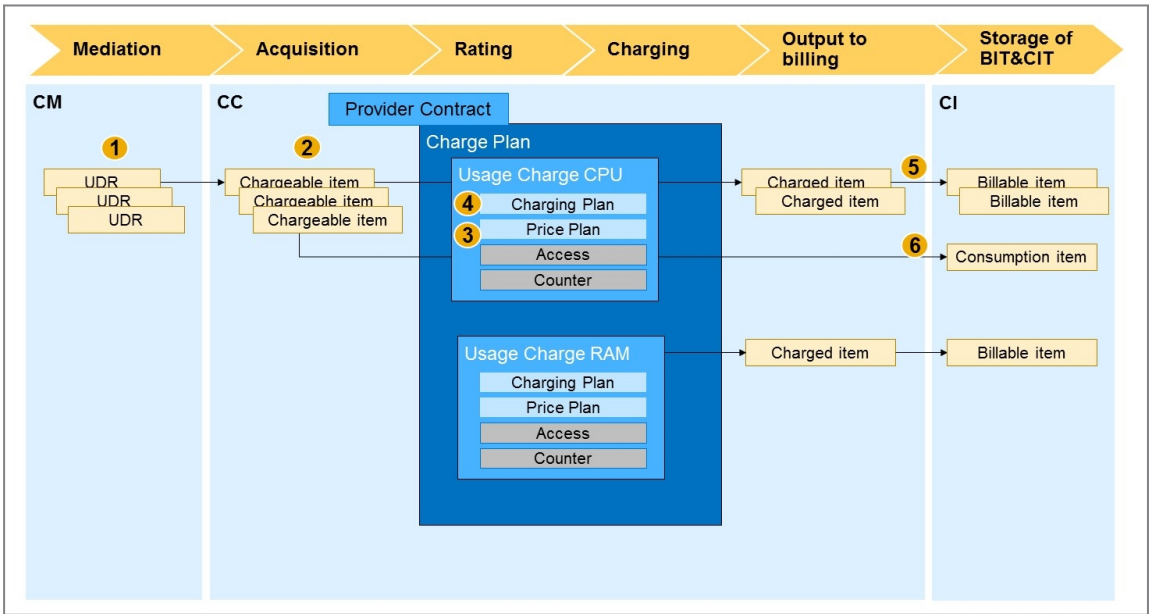

Figure 176: Usage Acquisition, Charging and Rating

Convergent Charging Rating & Charging Scenario with creation of consumption items:

Step 1: Usage data Mediation

The mediation engine is responsible for transforming the data sent by the network infrastructure into records which can be handled and processed by the rating engine. These records need to contain all data necessary for rating and charging:

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- User ID pointing to the correct (technical) user using the service
- Service ID pointing to the correct service which was used
- Consumption date in the right common format
- Any user-defined properties which are necessary for the rating and charging process
- Convert to chargeable item and sent to Convergent Charging (CC)

Step 2: Acquisition

A Provider Contract is made up of at least one or more contract items. Each contract item corresponds to a Charge Plan, which includes the charges. How many Contract Items (or Charge Plans) are activated inside a provider contract is defined in the Product Model and the contained Cross Catalog Mapping in SOM.

- Acquisition of the chargeable item by CC
- An access object will only be created if the charge contains a usage rate. Charges with only recurring rates do not have an access
- The UserID is a unique identifier of the subscriber account / provider contract
- The ServiceID is a unique technical identifier of the service (charge)
- The access is the combination of UserID and Service ID
- The usage events can be assigned to the usage charge, with help of the access

Step 3: Rating

Apply pricing to events based on:

- Pricing logic in the price plan for CPU usage
- Update counters, for example, CPU usage counter for free CPU usage

Step 4: Charging

- Charging in the Charging Plan means assignment to the external account.
- Determine the account that will be charged based on the charging plan.

Step 5 and 6: CC processes the chargeable items and generates the corresponding charged items.

- Charged Items are mapped to billable items with charged item mapping.
- Map Charged Items to billing format and send to the Billing System.
- Convergent Invoicing receives the usage data with a price and is assigned to a specific customer account.

Step 6: Update status of CIT

- Loading in Convergent Invoicing of the corresponding rated consumption items.

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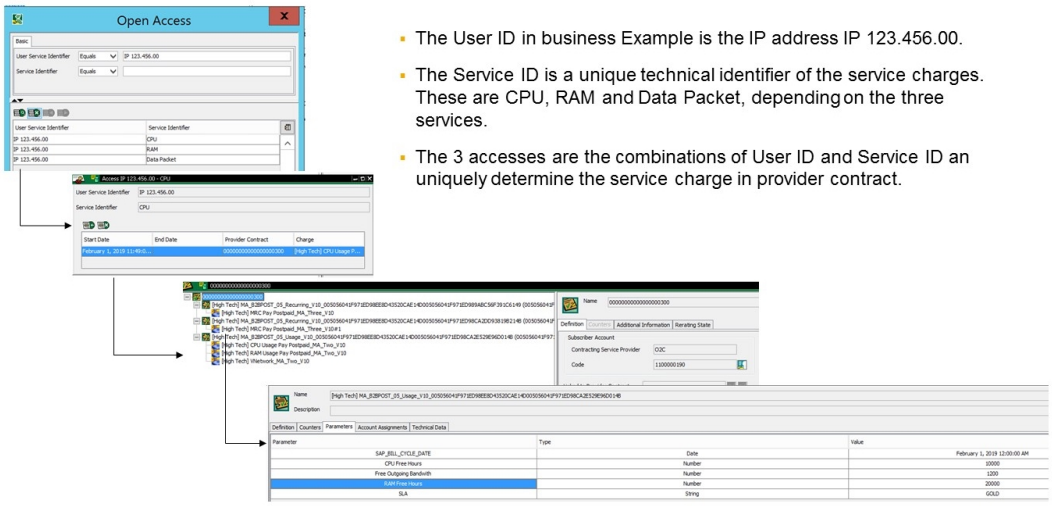

Figure 177: Access in Business Example Provider Contract Cloud Selection Service

The slides show the accesses and parameters of our business example provider contract 300.

The User ID in our cloud example is an IP address *IP 123.456.00*. Three services are assigned to the cloud product, Service IDs: CPU, RAM, and Data Packet.

The combination of user Id and service Id defines the accesses. CPU, RAM, and Bandwidth Usage can be assigned to the charge with help of User ID and Service ID in the consumption data record.

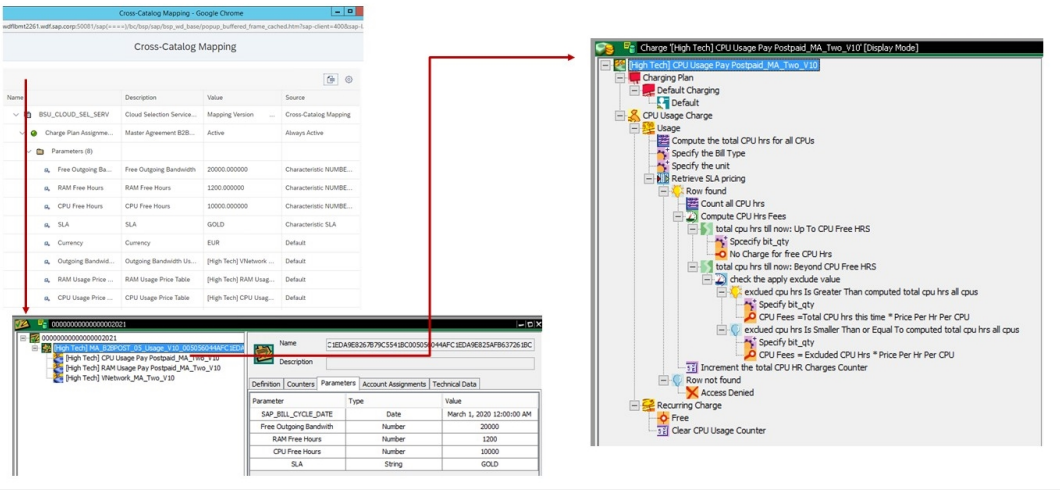

Figure 178: Use of Charge Plans and Usage Charge in Business Example BSU_CLOUD_SEL_SERV

Steps in calculate usage charge for CPU:

Free Volume is 10000 according to the provider contract conditions. First CDR CPU; IP 123.456.03; 2019/02/06; CPUs 35; Hours 100

1. Compute the total CPU hours for all CPU = 35*100 = 3500
2. Specify the BIT Type out Product Bill Type, Bill Type =BIT Type is H211.
3. Specify the unit is H.

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4. Retrieve SLA Pricing (read the range table)
5. Count all CPU hours. Start with Counter = 0, the counter is set to 3500.
6. With the first usage the counter is up to 10.000.No charge for free 3500 CPU. Create an output of the leaf on BIT, with amount 0 for the first usage.

Every "leaf" of the price plan of the Charge CPU Usage Pay Postpaid_MA_Two_V7 is ending with a function.

Explain Table Definition and Redefinition

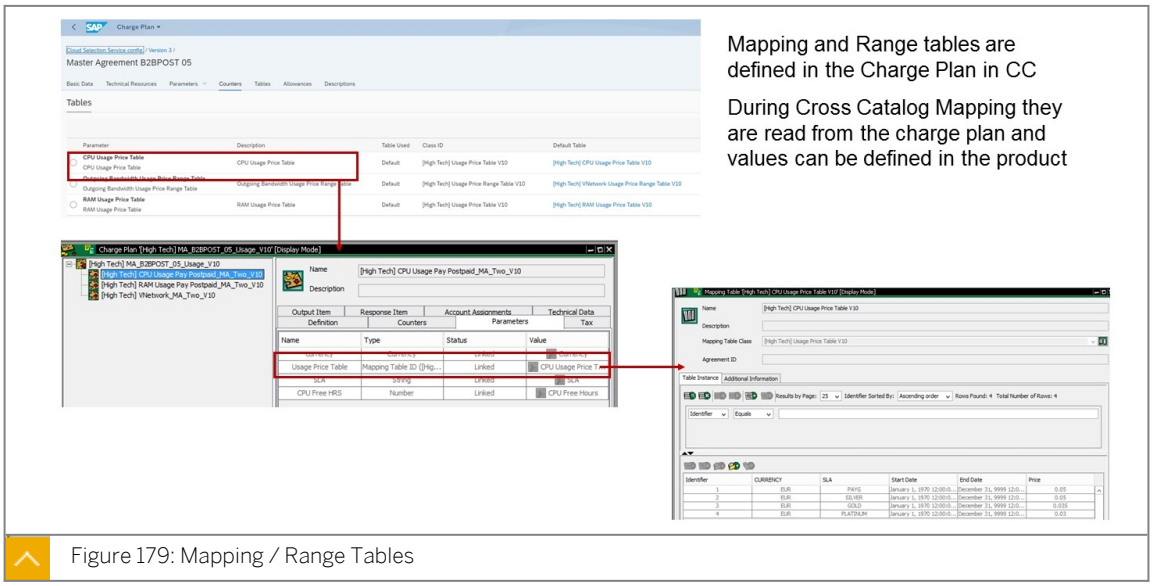

Figure 179: Mapping / Range Tables

Table parameters refer to SAP Convergent Charging (CC) tables that are used to determine usage prices during the rating process.

Table parameters are special string parameters. During contract creation, SAP CC expects the ID of a specific table type. The charging table type is defined by the table class that is displayed together with the table parameter. SAP CC allows you to use different tables for each contract and therefore, the integration with cross-catalog mapping (CCM) allows you to overwrite the default tables from SAP CC, at subscription product and contract level. You are able to leverage mapping and range tables here.

- **Default Table**
- *Redefined Table:* You can redefine a table. A copy of the default table with a specific ID will be created in SAP CC. The ID of the redefined table is shown in the Object Specific Table column. The redefined table must be maintained in the app, manage mapping / range tables
- *Assigned Table:* You can assign a table. Assigned tables are defined independently from a specific product. If the table class ID of the table parameter matches, the available table can be assigned. Changes to the content of an assigned table are visible to all products, mapping versions, and assigned charge plans, using this assigned table. Thus, it is possible to make personalized changes (centrally) to the configuration of all subscription products, using this assigned table.

Convergent charging tables (SAP CC tables - mapping or range tables) are tables in an SAP CC system that are used during the charging and rating process in SAP CC. The table can be

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referenced in a charge plan using a parameter. This parameter value becomes the default value.

Charge plan parameters can be overwritten in the CCM of a subscription product. For parameters of the type, mapping or range table, you can create a new table and assign its identifier to a parameter value.

Subscription Order:

- The default table that is defined in the CCM of the product is copied as the default table in the subscription order item. If an object-specific table (redefined or assigned) is defined in the product, it is copied as a default table into the subscription order item.
- An assigned table can be a table from the CC standard catalog or a table from the subscriber account assigned in the subscription order.
- If a redefined table is created in the subscription order, it is stored in the subscriber account for the CC system. You can change the contents of a redefined table.

Show Integration with CITs and BITs into Convergent Invoicing

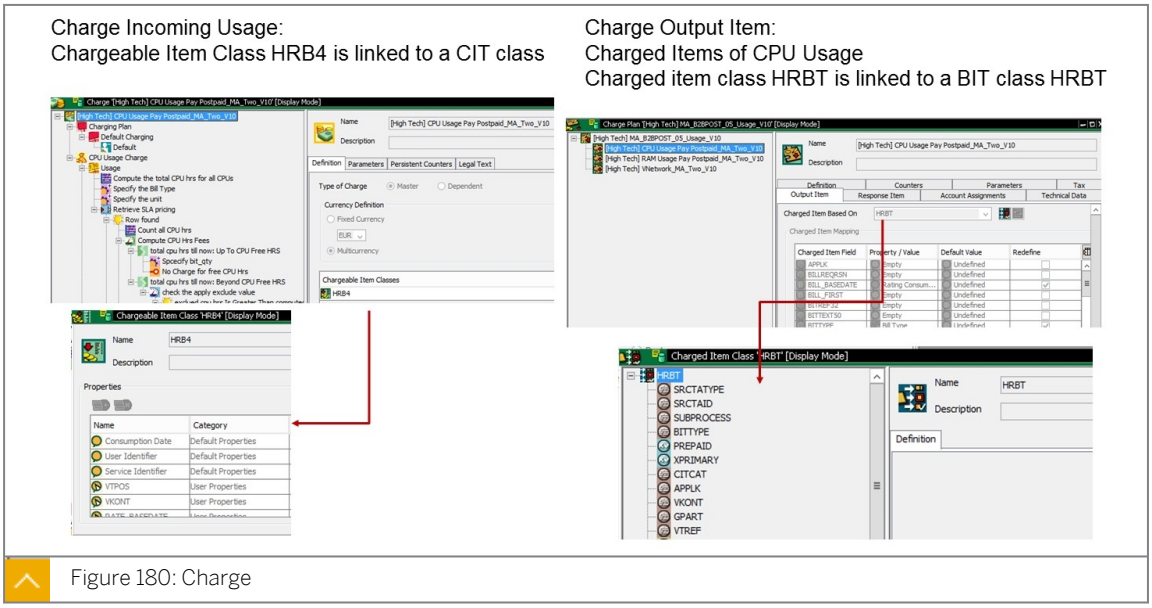

Figure 180: Charge

SAP Convergent Charging (CC) handles the rating (pricing) and charging (determination of invoice recipient) of services. As a result billable items are created that you transfer to billing and that, as an example, in the case of re-rating, you can also reverse.

Through the integration of SAP Convergent Invoicing (CI), Contract Accounts Receivable and Payable (FI-CA), SAP CC, and S/4Hana Service (SOM), SAP ensures the distribution and synchronization of master data across all involved systems (see replication from an FI-CA system to a CC system).

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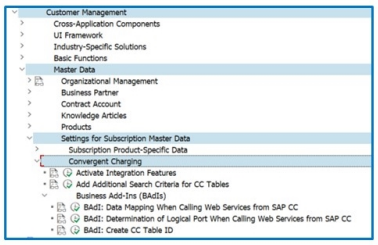
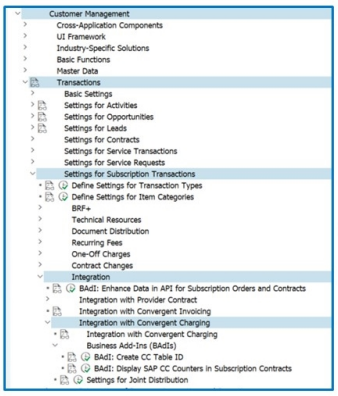
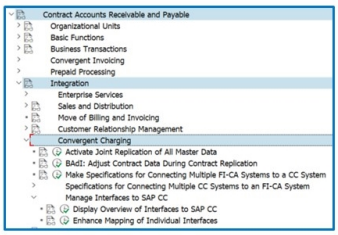


Settings for Mapping Tables



Settings for Counters, Joint Distribution



FI-CA Integration with Convergent Charging

Figure 181: Customizing for Integration with Convergent Charging

Settings for Mapping Tables:

SPRO → Service → Master Data → Settings for Subscription Master Data → Convergent Charging

Settings for Counters, Joint Distribution:

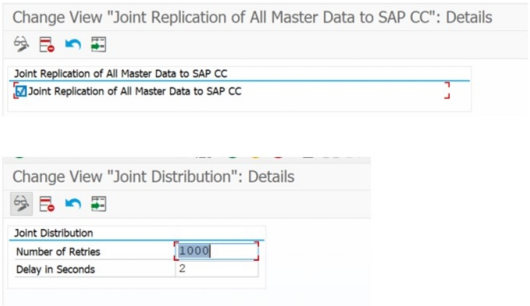
SPRO → Service → Transactions → Settings for Subscription Transactions → Integration

FI-CA Integration with Convergent Charging:

SPRO → Financial Accounting → Contract Accounts Receivable and Payable → Integration

Convergent Charging

Describe Technical Integration with Convergent Charging



Joint Replication must be activated when working with CC

The system replicates the following master data objects jointly to SAP Convergent Charging:

- Business partner
- Contract account
- Prepaid account
- Provider contract

Multiple Events can influence the processing

- Event 5707 is used to update the ODI distribution step status. In this event a wait and retry mechanism is implemented which's factors can be customized

Figure 182: Joint Replication / Joint Distribution

Activation of Joint Replication / Joint Distribution

When working with Convergent Charging (CC), joint replication must be activated:

SPRO → Financial Accounting → Contract → Accounts Receivable and Payable → Integration → Convergent Charging → Activate Joint Replication of All Master Data

Settings for Joint Distribution in S/4Hana Service

Since the master data and contract replication are confirmed asynchronously in joint distribution, the subscription contract might still be locked when the status update is triggered from the provider contract.

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Define the number of retries to lock the subscription contract for the status update from the provider contract and the waiting interval between those retries.

SPRO → Service → Transactions → Define Settings for Item Categories → Integration → Integration with Convergent Charging → Settings for Joint Distribution

Integration of S/4Hana with SAP CC is realized via Web Services calls:

Webservices	
CO_CRM_ISX_CC_ACTIVATION_SRV	ActivationServices
CO_CRM_ISX_CC_CATALOG_SERVICES	CatalogServices
CO_CRM_ISX_CC_CHARGEABLE_ITEM	ChargeableItemChargingServices
CO_CRM_ISX_CC_RATING_SERVICES	RatingServices
CO_CRM_ISX_CC_SUBSCRIBER_MAPPI	SubscriberMappingTableManagementService
CO_CRM_ISX_CC_SUBSCRIBER_RANGE	SubscriberRangeTableManagementService
CO_ISX_CC_CONTRACT_STATE_MANAG	ContractStateManagementService
CO_ISX_CC_CTR_PROV_SERVICES	ContractProvisioningServices
CO_ISX_ALLOWANCE_MANAGEMENT_SE	AllowanceManagementServices

Figure 183: CC Web Services

You can use Transaction *SOAMANAGER* to configure Web Services:

Web Services used for S/4Hana Service follow the name pattern **FKKCC**

Web Services called from Contract Accounts Receivable and Payable (FI-CA) follow the name pattern **ISX**

Recurring Fees in SAP Convergent Charging

Recurring fees can be managed either via CI Billing Plans or via SAP Convergent Charging. Recurring fees in CC is especially required for complex recurring fee calculation logic.

Features:

- Transfer price data for SAP CC using Data Replication Framework (DRF)
- Create and update price tables in SAP CC
- Create subscription orders / solution quotations with recurring fees
- If required maintain manual recurring fees
- Derive CC price table or create CC price table (entry) in case of manual fees during contract distribution
- SAP CC can now create recurring fees and pass them to Convergent Invoicing for further processing

Figure 184: Recurring Fees in SAP Convergent Charging

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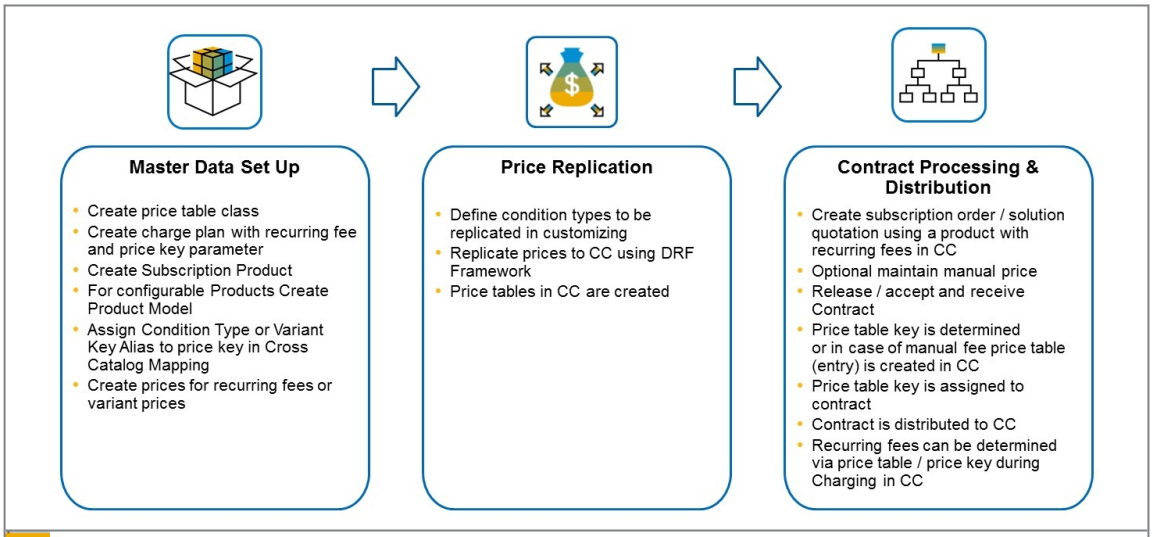

Figure 185: Process Steps for Recurring Fees in SAP CC

In addition to charging recurring fees via billing plans, you can also charge recurring fees via SAP Convergent Charging (CC). A mapping version for charge plans, to provide the configuration for recurring fees can be created under *Cross Catalog Mapping in Subscription Product-Specific Data*. Here you can assign *Charge Plans* from SAP CC containing one or more parameters from the type *Price Key* (this is a charge plan parameter from type *Mapping Table Key*) and one from type *Price Table* (this is a charge plan parameter from type *Mapping Table ID* using mapping table class *Price Table*) that are configured in SAP CC. This will be used for storing information needed for the recurring charge. The price key and price table will be interpreted and sent to SAP CC during subscription contract activation where it is stored in the provider contract.

As long as there is no manual price in the subscription contract, the price is sent to SAP CC when the corresponding condition table is saved and stored in SAP CC in the price table related to the used subscription product. Manual contract prices will be sent to SAP CC together with the contract distribution and stored in agreement tables used for manual prices.

A recurring fee can be configured in the assigned *Charge Plan* of the *Cross Catalog Mapping Version* either by a *Condition Type* or by a *Variant Condition Key Alias*. When using condition types, only condition types are allowed that are not used for variant configuration.

If you want to use a condition type relevant for variant configuration, you have to use the *Variant Condition Key Alias*. In this case where you do not enter the condition type itself, instead you have to enter either the *Variant Condition Key* directly or only a part of the key with a wild card. If you use a wild card you must ensure that during product configuration in the order only one variant condition key is found. In the value help only variant condition keys are displayed, which are maintained in the *Variant Configuration Model* under the section *Variant Prices*.

By default the flag *Price is Mandatory* is set, that is, during contract distribution a price must be found for this price key. You can remove this flag if the price is not mandatory. In this case a value 0 is set as parameter value if no price is found.

Recurring fees dependent on specific subscription product relevant characteristics can be defined by the *Variant Condition Key Alias*.

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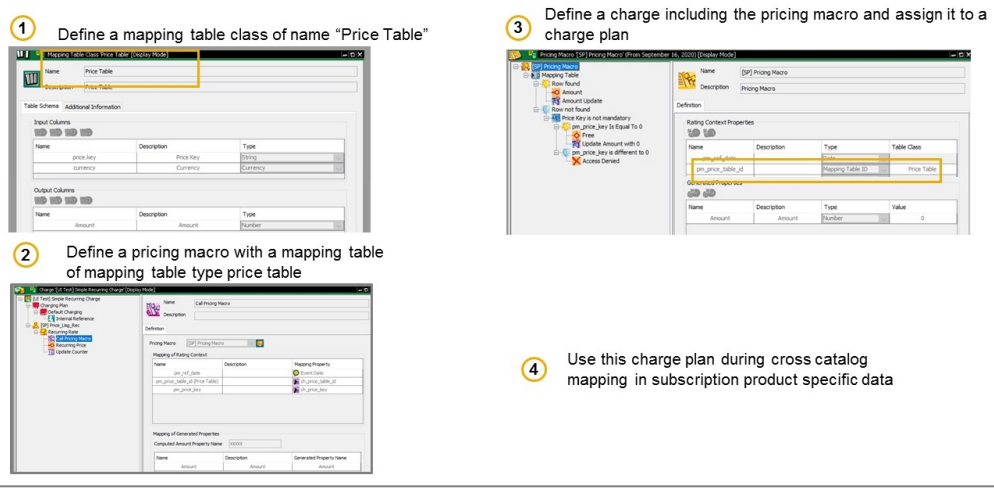

Figure 186: Master Data Set Up SAP CC: Sample for Defining a Charge Plan for Recurring Fees in CC

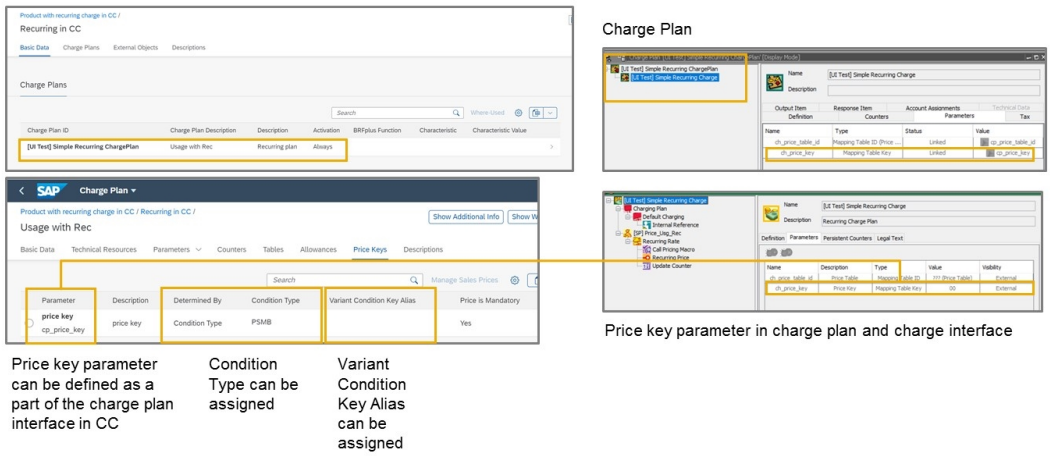

Figure 187: Master Data Set Up: Subscription Product Specific Data - Price Key Assignment in Charge Plan Mapping

The definition of recurring fees is also part of the subscription-specific data of subscription product. When you assign a charge plan with recurring fees, a price key parameter is available and can be linked to the condition types or variant condition key aliases that define the condition record referenced during contract creation.

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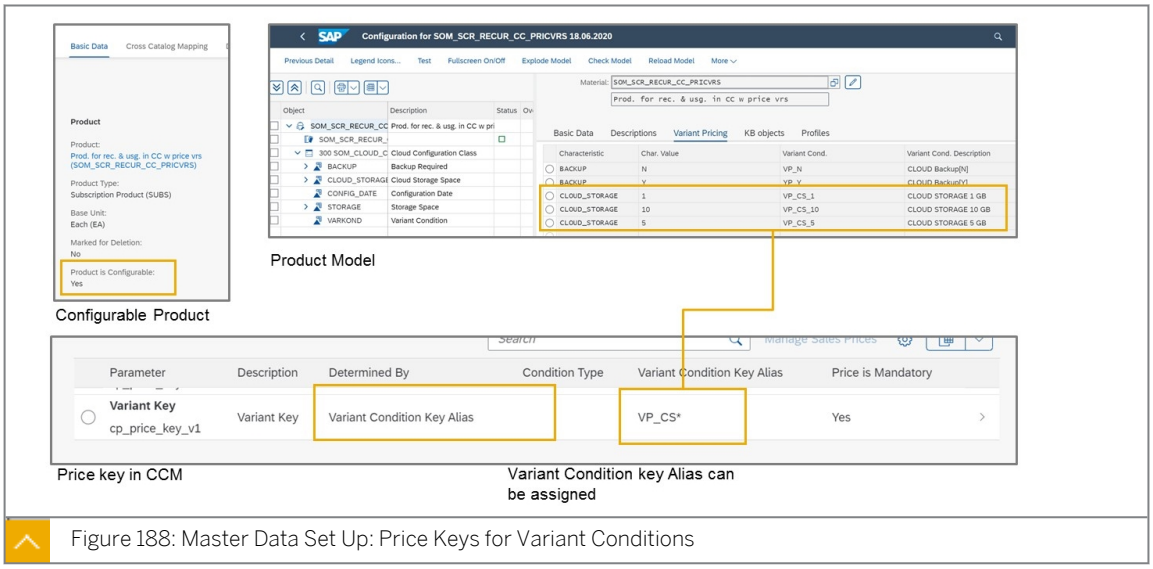

Figure 188: Master Data Set Up: Price Keys for Variant Conditions

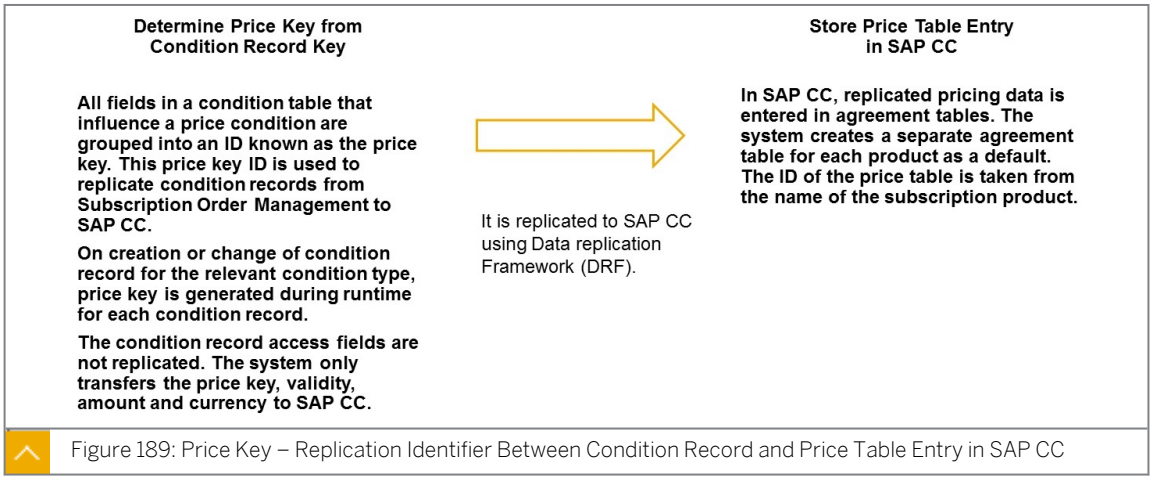

Figure 189: Price Key – Replication Identifier Between Condition Record and Price Table Entry in SAP CC

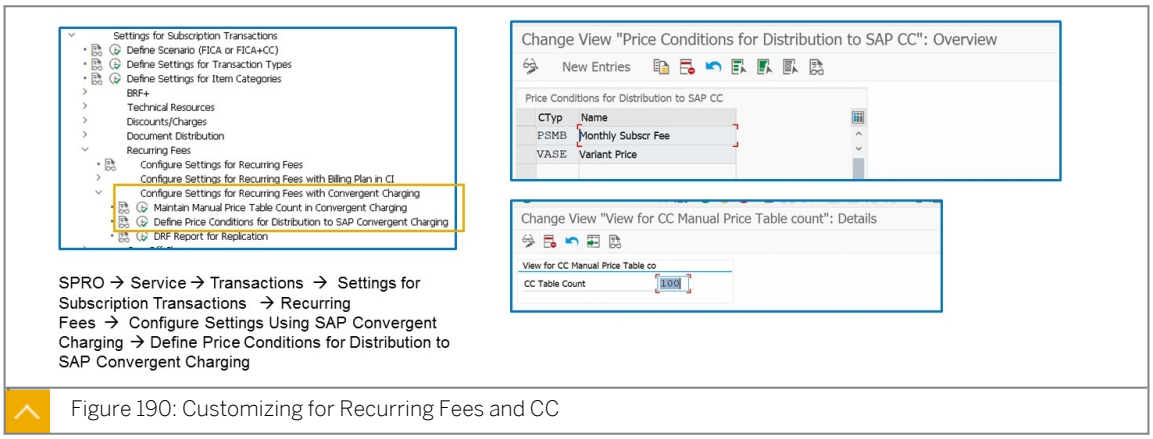

Figure 190: Customizing for Recurring Fees and CC

Manual Prices

Define number of price tables to be created in SAP Convergent Charging to handle manual prices.

Based on the expected number of subscription contracts with manual prices, define number of tables that can be created in SAP Convergent Charges to hold manual prices.

Also consider SAP Convergent Charging recommendation on number of agreement tables and number of table lines per agreement table for performance efficiency.

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Pricing Conditions for Recurring Fees

Define pricing conditions for recurring fees that are relevant for replication of prices to SAP Convergent Charging.

When recurring fees is maintained for products using, the standard price maintenance apps or transactions for these price conditions, the price data can be replicated to SAP Convergent Charging using DRF. Only the condition types mentioned in the customizing table are taken in account for replication to SAP Convergent Charging.

Additionally, you can use transaction `DRFLOG` for error monitoring.

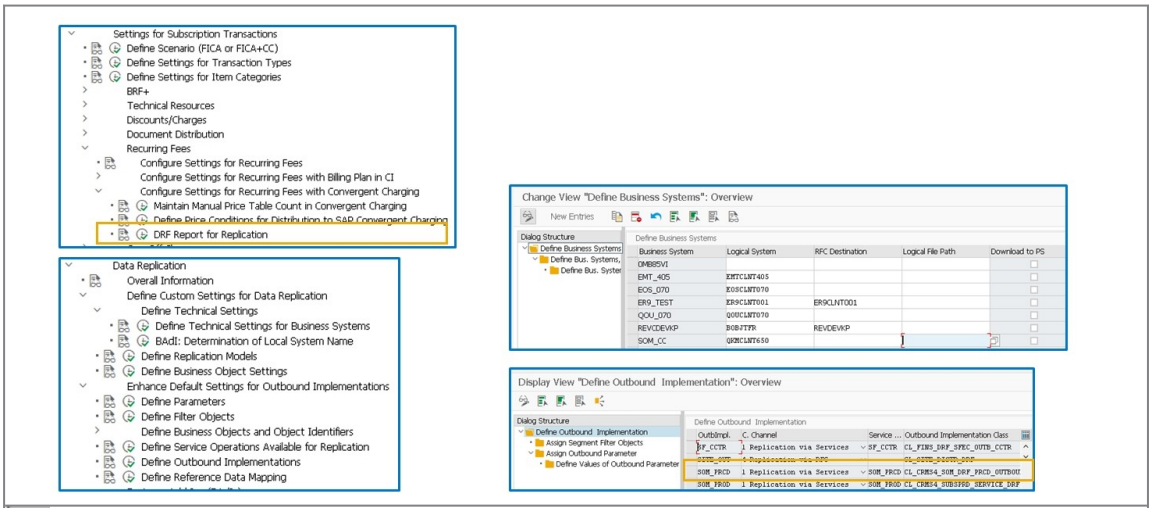

Figure 191: Customizing for Recurring Fee and DRF Settings

This report can be used to create one or multiple outbound messages that are sent using an outbound implementation that is defined for a replication model. You can create the outbound messages directly or in the background.

Activities

The Data Replication Framework is used to replicate price condition records to SAP Convergent Charging and to replicate the subscription product-specific data to external systems. You can define the replication model using the following steps:

1. Go to transaction `DRFIMG`.
2. Navigate to *Data Replication → Define Custom Settings for Data Replication → Define Replication Model*.
3. The default replication model for price replication to SAP Convergent Charging is *SOM_PRCD*. For the subscription product-specific data, the default replication model is *SOM_PROD*.

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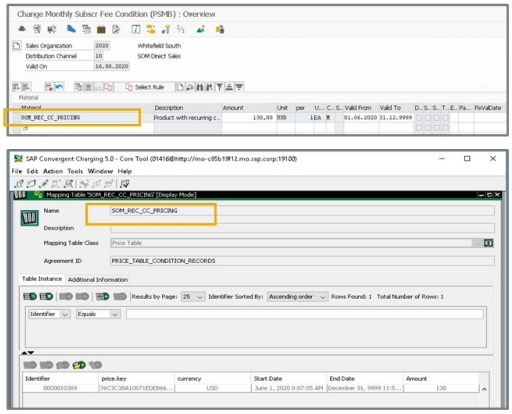
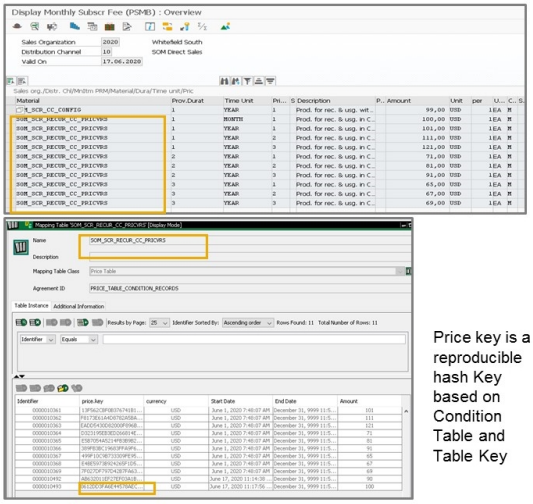
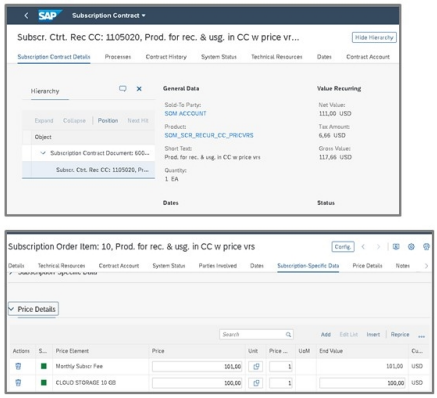


One Agreement Table per Product in CC



Price key is a reproducible hash Key based on Condition Table and Table Key

Figure 192: Prices Replicated as Agreement Tables to SAP CC



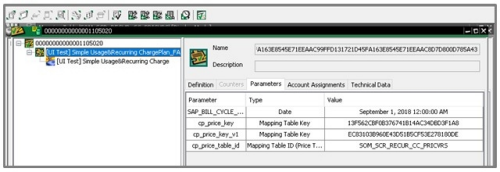
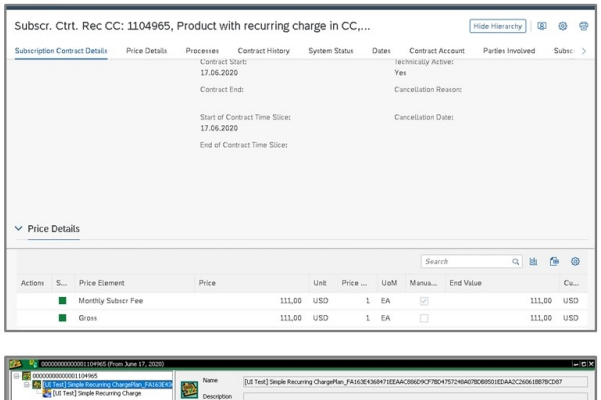
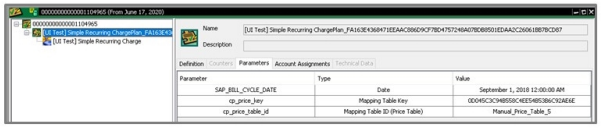


Figure 193: Contract with Base Fee and Variant Price





Note:
When working with manual prices in combination with list prices, you need to model the different recurring fees in two charge plan mappings in CCM. Nevertheless, you can use the same charge plan.

Manual prices are stored in generic agreement tables, which are determined by an algorithm which agreement table to use for which contract. The number of required agreement tables can be defined in customizing.

Figure 194: Contract with Manual Price

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LESSON SUMMARY

You should now be able to:

- Explain SOM Integration with Convergent Charging
- Detail the Convergent Charging Data Model
- Understand Process Flow for usage based fees
- Explain Table Definition and Redefinition
- Show Integration with CITs and BITs into Convergent Invoicing
- Describe Technical Integration with Convergent Charging
- Recurring Fees in SAP Convergent Charging

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Unit 11
Lesson 2

Account Splitting

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Illustrate the Account Splitting process
- Use Account Splitting customizing

Illustrate the Account Splitting Process

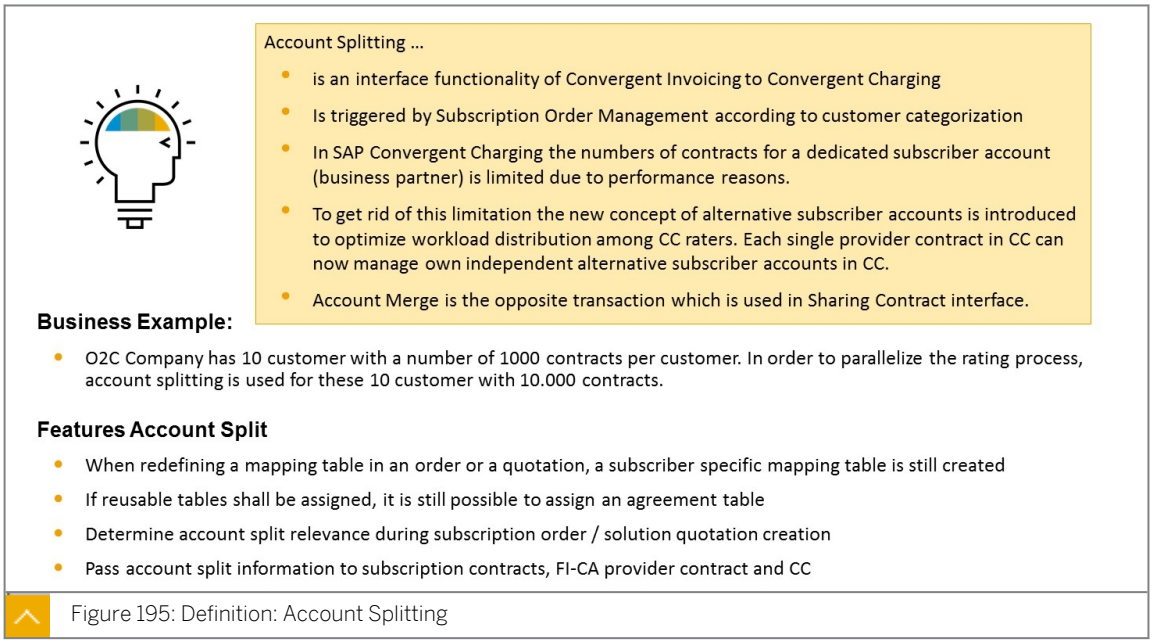

Figure 195: Definition: Account Splitting

In SAP Convergent Charging, the numbers of contracts for a dedicated subscriber account (business partner) is limited due to performance reasons. To get rid of this limitation, the new concept of alternative subscriber accounts is introduced when distributing the provider contract to SAP CC to optimize workload distribution among SAP CC raters. Each single provider contract in SAP CC can now manage their own independent alternative subscriber accounts in SAP CC.

Definition Shared CC Tables - Agreement Tables:

- As a consequence, agreement tables are introduced as a new variant of SAP CC tables. Agreement tables are mapping or range tables that are shared between multiple objects with a specific relation.
- This relation can be a subscriber account identifier, so that all provider contracts, with different alternative subscriber accounts belonging to one specific subscriber account, can share the same table. Or, alternatively, the relation can be a master agreement ID, so that

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all provider contracts. with reference to this master agreement, can use the same agreement table.

As of S/4 HANA 1909, agreement tables are introduced as new variants of SAP CC tables. Agreement tables are mapping or range tables which are shared between multiple objects with a specific relation. This relation can be either:

- A subscriber account identifier, so that all provider contracts with different alternative subscriber accounts belonging to one specific subscriber account can share the same table.
- Or the relation can be a master agreement ID, so that all provider contracts with reference to this master agreement can use the same agreement table.

Definition Shared CC Tables – Agreement Tables with 1909

- As a consequence of account split, agreement tables are introduced as new variant of CC tables. Agreement tables are mapping or range tables that are shared between multiple objects with a specific relation.
- This relation can be either a subscriber account identifier, so that all provider contacts with different alternative subscriber accounts belonging to one specific subscriber account can share the same table
- Or the relation can be a master agreement ID, so that all provider contracts with reference to this master agreement can use the same agreement table.

Features Shared CC Tables – Agreement Tables with 1909

- Manage CC Tables via Application “Manage Mapping / Range Table"
- Create an agreement table for a specific business partner, the mapping table is created as agreement table.
- Search for CC tables (searching with a subscriber account finds agreement and subscriber tables; searching without subscriber account finds catalog tables and agreement tables for master agreements)
- Use CC Tables in Subscription Orders / Solution Quotations
- Assign catalog or agreement tables for the specific partner to subscription items
- Create a redefined CC table in a subscription order depending on the scenario (account split) as a subscriber or as an agreement table
- Use CC Tables in Master Agreement
- Create a redefined CC agreement table in a master agreement

Features Account Split and Shared CC Tables with 1909

- Identify Account Split
- Offer the customer the flexibility to define whether a mapping / range table is created as an agreement table (Customizing + BAdI)
- Determine account split relevance during subscription order / solution quotation creation
- Pass account split information to subscription contracts, FI-CA provider contract and CC
- Manage CC Tables via the application *Manage Mapping / Range Table*.

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- Create an agreement table for a specific business partner; the mapping table is created as an agreement table
- Search for CC tables (searching with a subscriber account finds agreement and subscriber tables; searching without a subscriber account finds catalog tables and agreement tables for master agreements)
- Use CC Tables in Subscription Orders / Solution Quotations
- Assign catalog or agreement tables for the specific partner to subscription items
- Create a redefined CC table in a subscription order depending on the scenario (account split) as subscriber or as agreement table
- Use CC Tables in Master Agreement
- Create a redefined CC agreement table in a master agreement

As of S/4 HANA 2020, agreements tables are nor used, because this was an object-specific agreement table, which could not be reused for other documents. Especially in a B2C or B2B2C case, this would lead to a huge amount of agreement tables, which has a negative impact on the performance in SAP CC. Therefore CC Tables, which are created as redefined tables from a subscription item during order or quotation creation, are created as subscriber mapping tables instead of object-specific agreement tables even if account split is active.

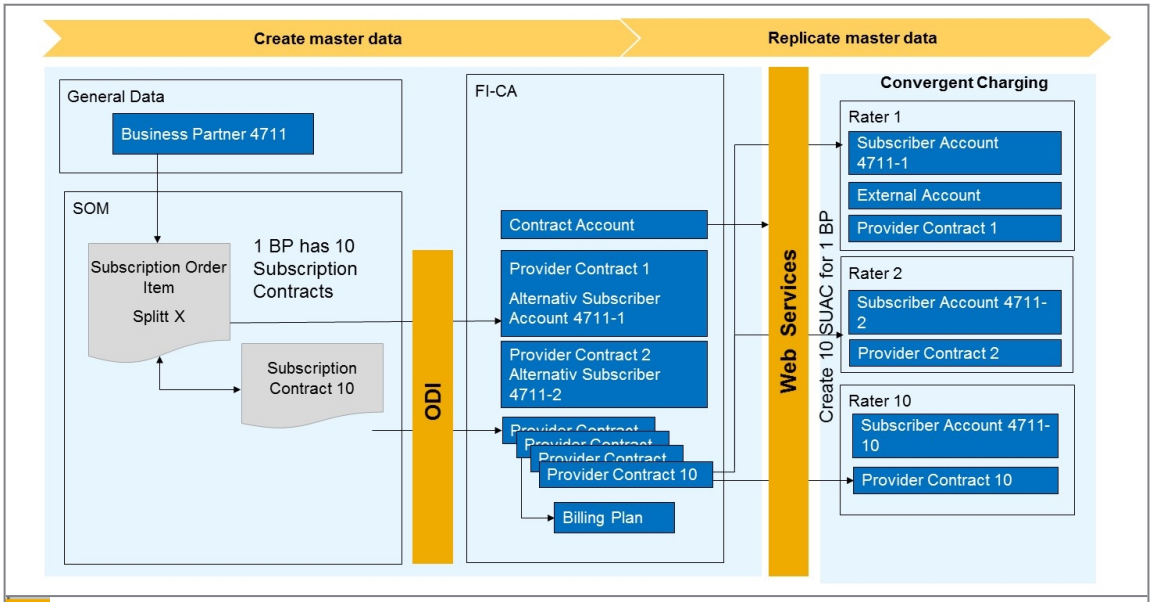

Figure 196: Data Distribution in BRIM Account Splitting

Process Steps in Account Splitting

Account splitting is a performance optimization method for the rating of B2B business partner with a high volume of subscription contracts and a high volume of usage.

Preparation: Create a subscription order with a split in the subscription order item category. Alternatively, a BADI can be implemented with customer-defined rules for splitting.

1. Provider contract creation concatinates an additional field alternative subscriber account that consists of a business partner number and a Provider Contracts number.
2. The Web Service interface to Convergent Charging creates an own subscriber account for each provider contract with an alternative Subscriber Account Number.

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3. Create 10 Subscriber Accounts with alternative Subscriber Account Numbers via the FI-CA Joint Master Data Distribution.
4. Subscriber accounts are distributed to different raters for performance optimization in parallel processing.

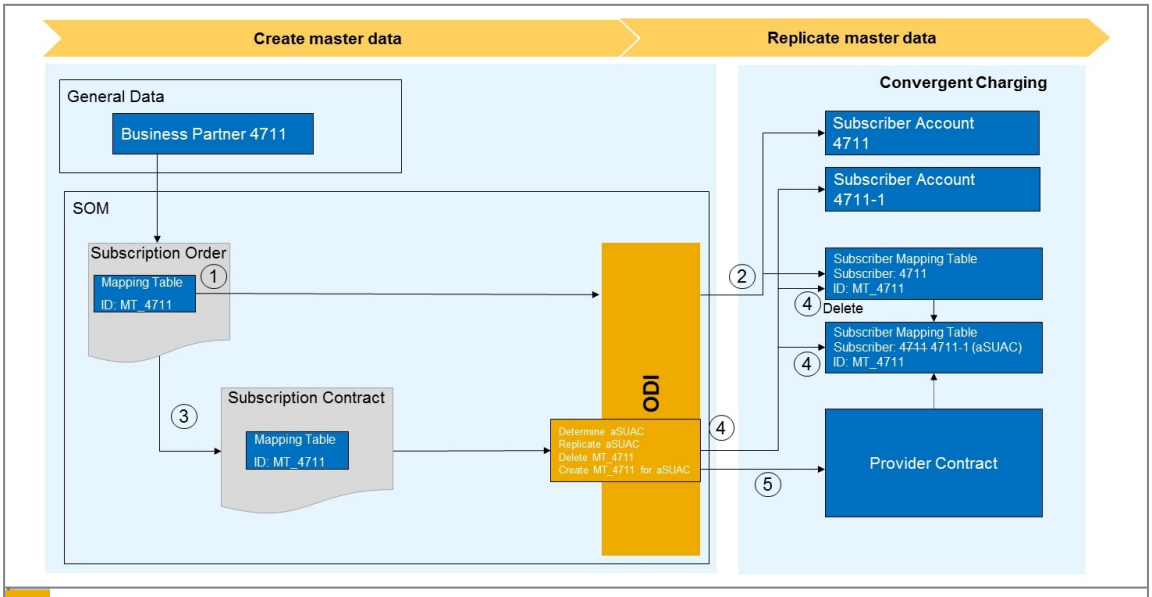

Figure 197: Subscriber Table Creation Flow for Account Splitting

Process Steps for Subscriber Mapping Table in Account Splitting

1. Redefining a table in the order / quotation, will always create a subscriber specific table.
2. Replicate Subscriber 4711 and create MT_4711. This table is created for the business partner (= subscriber account), as it is already the case if no aSUAC is in the game.
3. Subscription Contract is created, which is using the mapping table.
4. A new ODI step is introduced before the contract is distributed to SAP CC. In this ODI step, the existing subscriber table will be deleted and re-created with the same ID using the aSUAC as subscriber account. Deletion is necessary, since the table IDs (even the subscriber table IDs) are unique within a CC system. In a change order, when the aSUAC is already known, we can create the subscriber table directly for the aSUAC.
5. The Provider Contract is replicated to CC pointing to the subscriber mapping table of the aSUAC.

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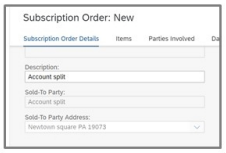
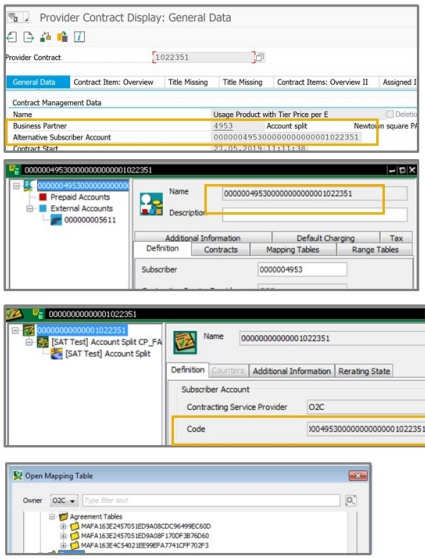


Subscription Order is identified for Account Split

- either by customizing settings
- or by BAdI Implementation

Optional Mapping Table is assigned to Subscription Order

Subscription Contract is created and distributed



FI-CA Contract with persisted Alternative Subscriber Account ID

Subscriber Account (Alternative)

CC Provider Contract for Alternative Provider Account

Mapping Tables

Figure 198: Account Split in Convergent Charging

Use Account Splitting Customizing

Customizing / BAdI Activate Account Split

First level
- Second level
  - Third level
    - Fourth level
      - Fifth level

Alternative you can maintain your custom specifc logic via BAdI

Transactions -> Settings for Subscription Transactions -> Integration -> Integration with Convergent Charging -> Business Add-Ins (BAdIs) -> BAdI: Determine Split for Subscriber Account in SAP CC

Enhancement Spot ES_CRMS4_SOM_1O_EXTENSION

BAdI Definition CRMS4_SOM_BADI_CC_ACC_SPLIT

Sample Class CRMS4_SOM_ACC_SPLIT_BP_EXMP

Figure 199: Customizing / BAdI Activate Account Split

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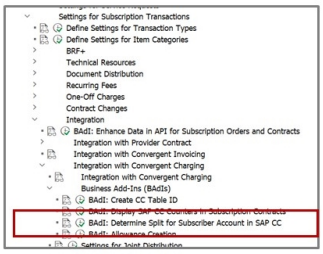
BAdIs Account Split



Description	BAdI Name / Enhancement Spot
BAdI: Determine Split for Subscriber Account in SAP CC	CRMS4_SOM_BADI_CC_ACC_SPLIT

You can change the indicator for subscriber account split in SAP Convergent Charging (SAP CC).

The indicator for subscriber account split is determined by IMG activity Define Settings for Item Categories. If this customizing is not sufficient for your requirements you can implement method DETERMINE_CC_ACCOUNT_SPLIT.

Example:
Depending on the settings in your business partner you want to decide if a subscriber account split is relevant or not. In this case you can read the business partner data in your BAdI implementation and set the indicator accordingly. You can have a look at example implementation CRMS4_SOM_ACC_SPLIT_BP_EXMP which set the indicator for account split if the business partner is an organization.

Figure 200: BAdIs Account Split

LESSON SUMMARY

You should now be able to:

- Illustrate the Account Splitting process
- Use Account Splitting customizing

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Learning Assessment

1. In the Billing and Revenue Innovation Management (BRIM) architecture, Convergent Charging (CC) is which of the following?

   Choose the correct answers.

   - ☐ A The charging and rating engine for usage
   - ☐ B The billing and invoicing engine
   - ☐ C The pricing engine for consumption

2. Technical integration of SAP S/4HANA Sales Order Management (SOM) with SAP CC is done with which one of the following?

   Choose the correct answer.

   - ☐ A Charge Plans
   - ☐ B Cross Catalog Mapping
   - ☐ C Convergent Mediation
   - ☐ D Technical Identifier

3. Which one of the following technologies is used to replicate conditions for recurring fees to SAP CC?

   Choose the correct answer.

   - ☐ A ODI
   - ☐ B DRF
   - ☐ C Middleware
   - ☐ D CPI

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Learning Assessment - Answers

1. In the Billing and Revenue Innovation Management (BRIM) architecture, Convergent Charging (CC) is which of the following?

   Choose the correct answers.

   - [X] A The charging and rating engine for usage
   - [ ] B The billing and invoicing engine
   - [X] C The pricing engine for consumption

   That is correct! Convergent Charging is the charging and rating engine for usage, and the pricing engine for consumption.

2. Technical integration of SAP S/4HANA Sales Order Management (SOM) with SAP CC is done with which one of the following?

   Choose the correct answer.

   - [ ] A Charge Plans
   - [X] B Cross Catalog Mapping
   - [ ] C Convergent Mediation
   - [ ] D Technical Identifier

   That is correct! The technical integration is done with Cross Catalog Mapping.

3. Which one of the following technologies is used to replicate conditions for recurring fees to SAP CC?

   Choose the correct answer.

   - [ ] A ODI
   - [X] B DRF
   - [ ] C Middleware
   - [ ] D CPI

   That is correct! DRF is the technology used to replicate conditions for recurring fees to SAP CC.

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UNIT 12

Integration with SAP Convergent Invoicing

Lesson 1

Process Overview 217

Lesson 2

Discounts and Charges 219

Lesson 3

Billing Plan 223

UNIT OBJECTIVES

- Explain the BRIM Invoicing Process Overview
- Discounts and Charges
- Define a Billing Plan
- Provide the Billing Plan Process Overview
- Perform the Customizing of the Billing Plan

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Unit 12
Lesson 1

Process Overview

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explain the BRIM Invoicing Process Overview

Business Example

Customer Headquarter 00 has a subscription contract for cloud selection service level Gold. The contract was closed at 01.01.2020.

The first invoice will be created at end of January. The first invoice will include the activation fee of 250 EURO, the basic subscription fee of 1000 Euro, and the recurring subscription fee of 3000 Euro for service level Gold. Depending on the usage, there are usage fees for CPU, RAM, and Bandwidth. All these billable items are collected in Convergent Invoicing (CI). The billable items are enriched with accounting information. Billing is a pre-aggregation for invoicing, in order to reduce the volume of data.

Explain the BRIM Invoicing Process Overview

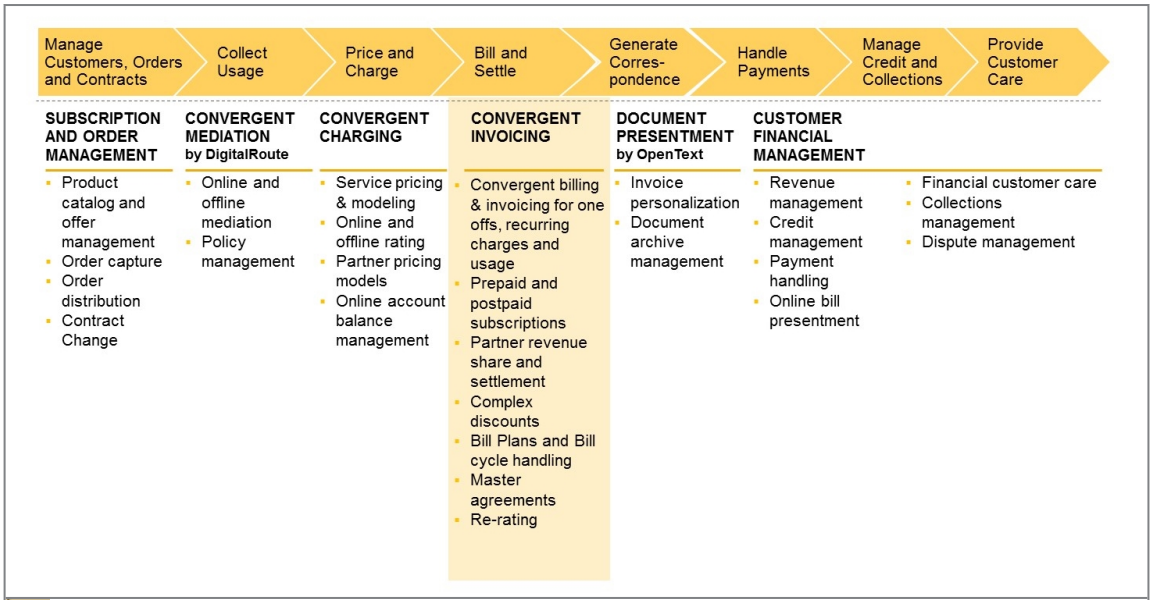

Figure 201: SAP a Billing and Revenue Innovation Management (BRIM) Solution Landscape

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A Billable Item (BIT) …

- is a priced event transaction record received by SAP Convergent Invoicing from a source system which has to be applied to a customer or partner statement of invoice.
- is stored and processed by SAP Convergent Invoicing.
- is the process starting point in S/4 HANA Convergent Invoicing for billing and invoicing
- includes all accounting information

Results by usage based transactions, which occur when a customer uses a service and the usage generates a transaction which should be charged to customer and/or provider:

1. CPU or RAM usage (0,035 per CPU hours)
2. Voice calls (0,2 Euro per minute international)
3. Parcel Shipment (per size and weight)

Results by recurring charge:

1. Monthly fees for Cloud Selection Service (Basic Fee 1.000 Euro, SLA Gold 3.000 Euro)

Results by a one time transactions:

1. One time Account Set Up: Activation Fee 399 Euro
2. One time Change Fee: 1000 Euro

Figure 202: Definition of a Billable Item (BIT)

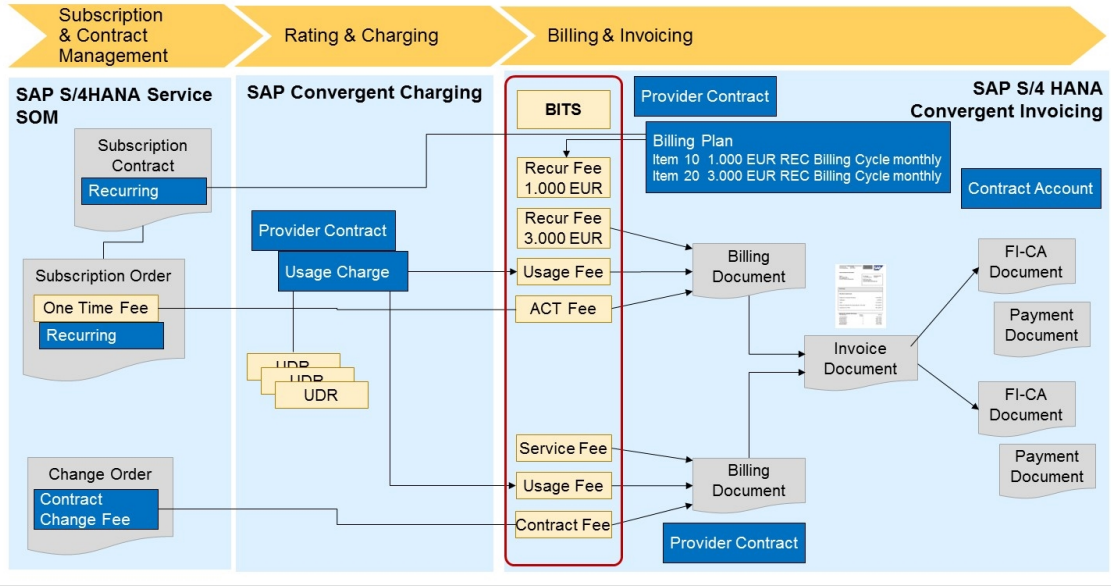

Figure 203: Process Overview of Billable Item Management, Billing and Invoicing

Process Steps in Convergent Invoicing

1. **Step 1:** Billable Item Management: One-Off, Usage BITs, and Recurring BITs are stored and enriched for accounting in Convergent Invoicing (CI).
2. **Step 2:** Billing Process is pre-aggregation for invoice preparation.
3. **Step 3:** Invoice Creation and Creation of Contract Accounts Receivable and Payable (FI-CA) Documents for accounting.

LESSON SUMMARY

You should now be able to:

- Explain the BRIM Invoicing Process Overview

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Unit 12
Lesson 2

Discounts and Charges

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Discounts and Charges

Business Example

In Subscription Business there are a lot of price reductions possible.

1. Cumulative prices for usage.
2. Contract closure start up.
3. Sell a volume for a special price, for example, 1000 CPU for 500 Euro.
4. With discount and charges in this chapter we handle discounts and charges on base of billed items.

   Revenue related discounts:

   Define a discount of 10% for a provider contract item, if the total revenue of the year exceeds the amount of...

Discounts and Charges

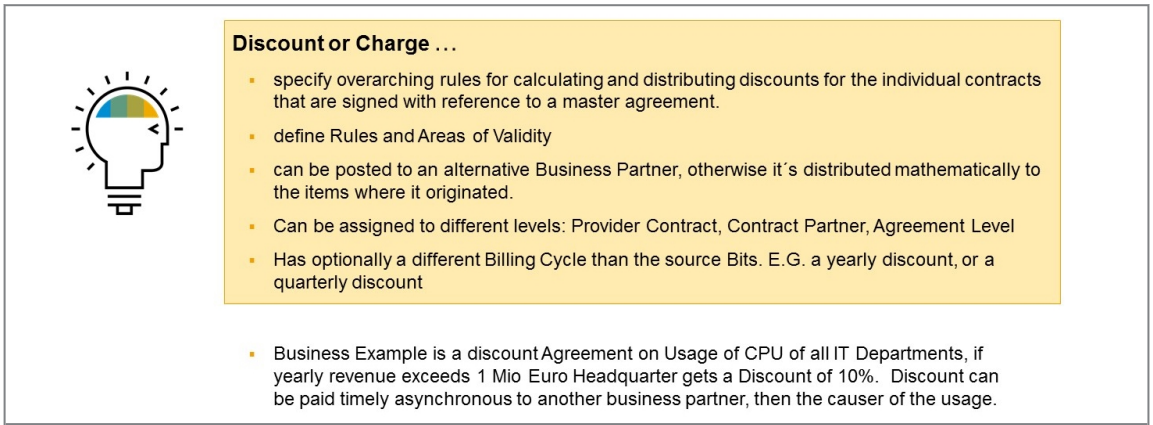

Figure 204: Definition of Discounts and Charges

Discount agreement:

Dependent on business partners, products, and further selections, it can be decided what kind of discounts will be applied to the billable items.

For a more convenient maintenance of the agreements' area of validity, it is possible to define business partner and product groups. By those groups the single records are bundled together and can be assigned to an agreement more easily (instead of entering each entry separately, which can become quite cumbersome).

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The agreements are built up out of a rules part and an area of validity part. The rules part is specific for each type of agreement. It tells what has to be done when a billable qualifies for an agreement, for example, the rules part of an invoice agreement tells who the invoice recipient is. The area of validity specifies the conditions under which the rules are applied. The conditions can be set up based on business partners, products, and further selections, based on filters and function modules.

Alternative Business Partner:

The Alternative Business Partner for posting of discounts is the business partner, to which the credit memo for the discount from the current agreement is posted.

If no Alternative Business Partner is given, then the system credits the discount according to where it originated. The system distributes the discount mathematically to the items where it originated and proportionately to the business partners and contract accounts belonging to the items.

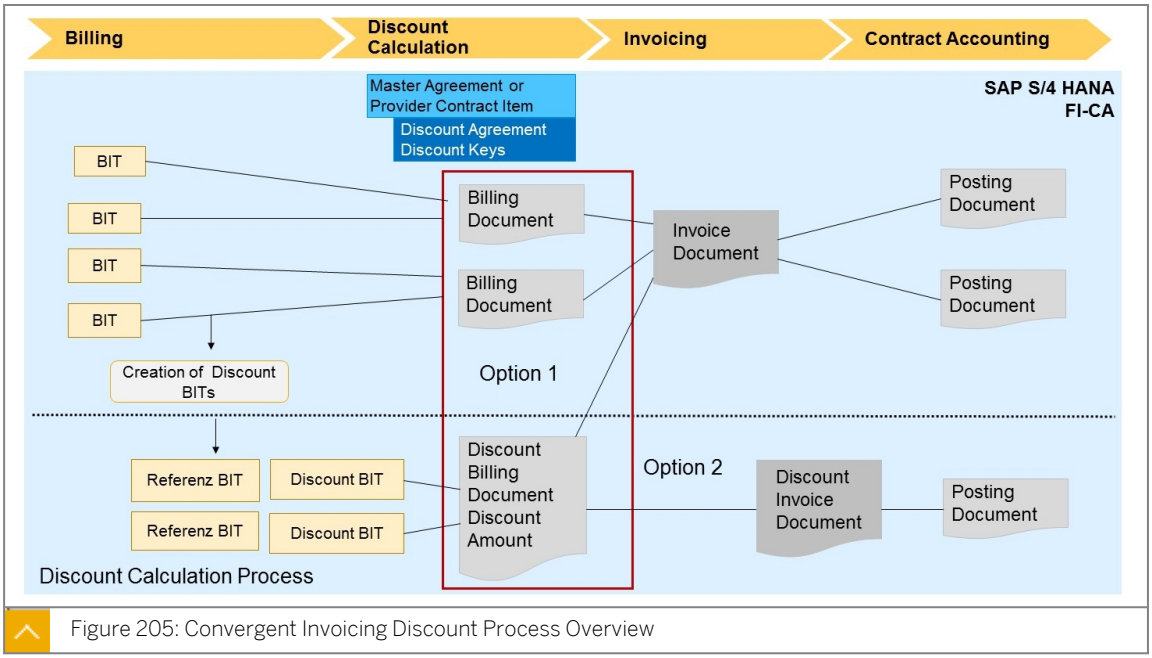

Figure 205: Convergent Invoicing Discount Process Overview

Process Steps of Discounting and Charging of Billable Items:

1. Configuration of a discount key and assignment to the master agreement or provider contract
2. In the standard billing process, determination of base information and creation of base BITs
3. Calculation of discounts and charges in a discount billing process
4. Create discount/charges calculation documents
5. Invoicing of discount billing documents either with usage billing documents (Option1) or separate

Flexibility of new Discount and Charge functionality:

- Discount determination via discount agreement when usage BITs are billed. Time independent processes is possible, for example, quarterly sums for discount base amount

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- Discount calculation on specified calculation level, such as provider contract Item, provider contract, contract partner, or agreement level
- Discount distribution. The assignment to source of discount or assignment to, for example, headquarter

The benefit of this solution is that the same architecture is used in B2C and B2B. The reuse of discount key and discount definition, or calculation modules on provider contract and master agreement level.

From standard billing and invoicing processes, document linkage and traceability is used.

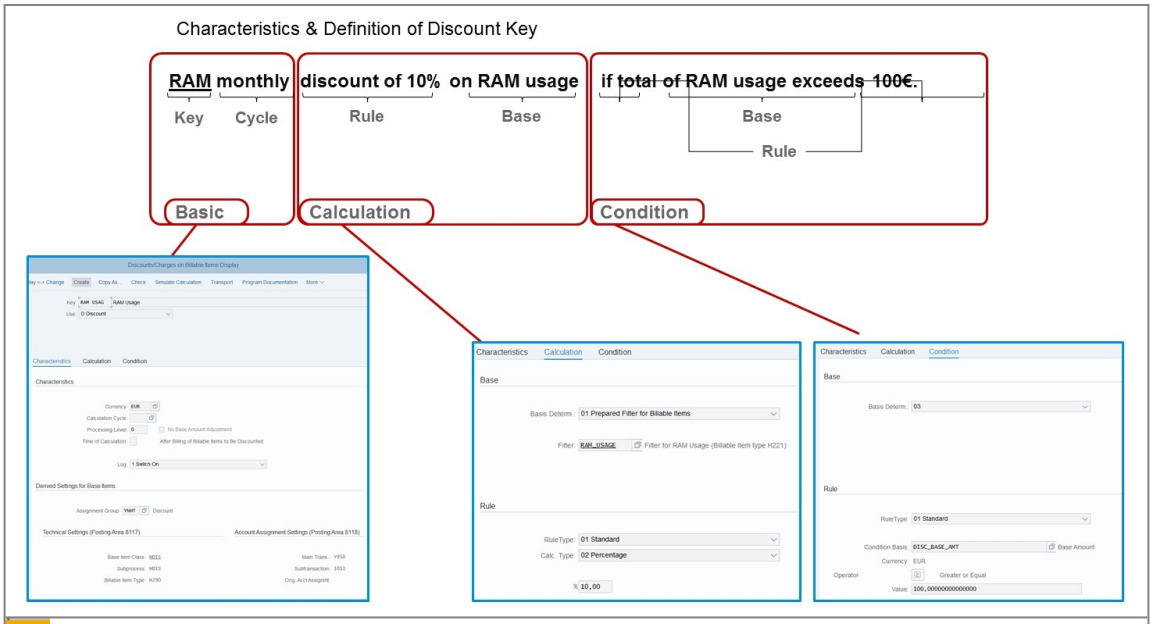

Figure 206: Customizing Contract Accounts Receivable and Payable to Define Discount/Charge Key

Discounts and Charges on Billed Items

Configuration

You configure discount/charge keys in three steps on the following three tabs:

Characteristics

On the upper part of the tab, you make general settings about the use, the discount or charge currency, the calculation cycle, and the processing level. In the lower part of the tab, the derived settings for the base items show you directly the currently valid derived values for the discount/charge key from posting areas 8117 and 8118, as long as you have maintained the posting areas.

Calculation

On the upper part of the tab, you define the basis for the calculation, meaning which billed items form the basis for the discount or charge calculation. On the lower part of the tab, you define the calculation rule. That means that you specify here how the discount or charge is calculated for the billed items defined in the basis.

Condition (optional)

On the upper part of the tab, you define the basis for the condition. That means that you specify for which billed items the system applies a condition. On the lower part of the tab, you define the condition rule. That means that you specify which prerequisites have to be met in order for a discount to be granted or a charge levied.

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LESSON SUMMARY

You should now be able to:

- Discounts and Charges

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Unit 12
Lesson 3

Billing Plan

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Define a Billing Plan
- Provide the Billing Plan Process Overview
- Perform the Customizing of the Billing Plan

Business Example

Customer Headquarter 00 has a subscription contract for cloud selection service level gold. The contract was closed at 01.01.2020. For each month a recurring Fee of 1000 Euro and a recurring fee of 3000 Euro for service level agreement gold are created.

Define a Billing Plan

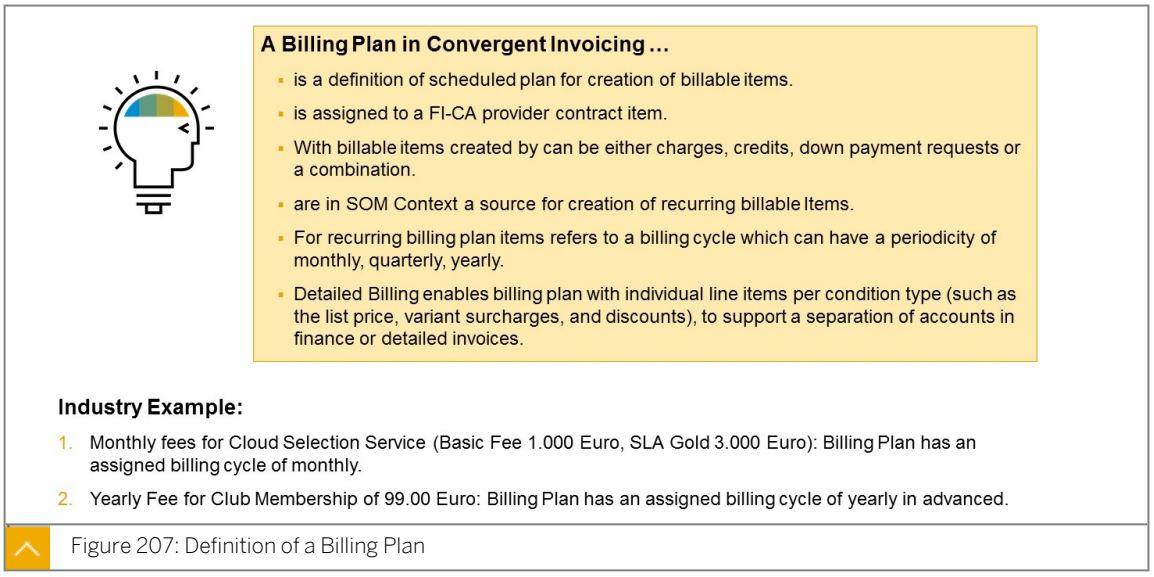

Figure 207: Definition of a Billing Plan

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Provide the Billing Plan Process Overview

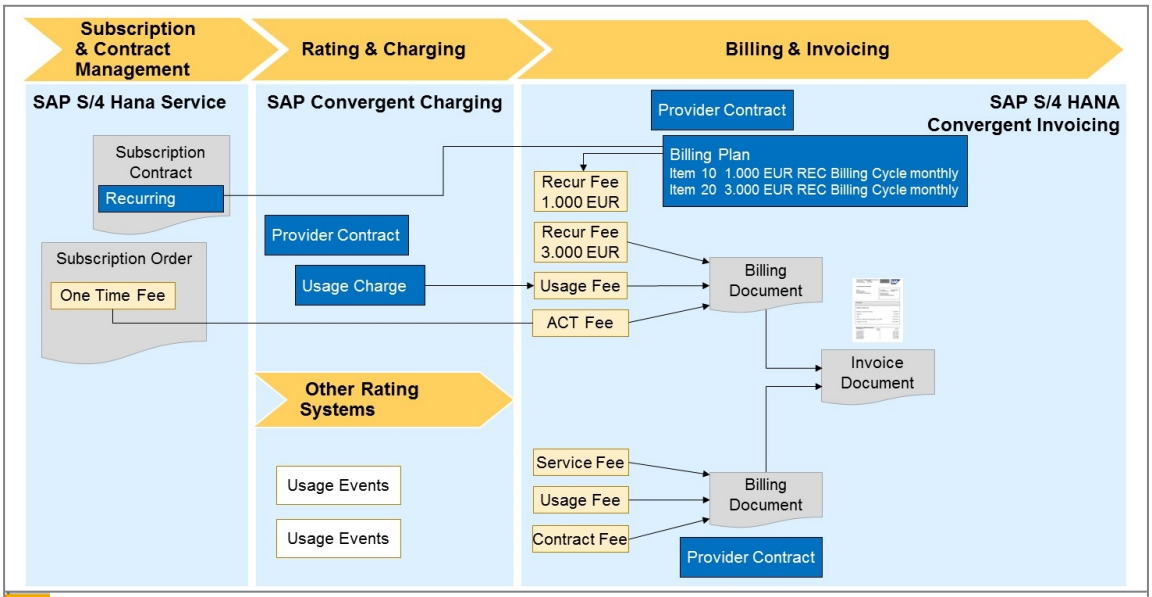

Figure 208: Process Request for a Billing Plan Item

Process Steps in Creation of Recurring Fees:

1. **Step 1:** Create a provider contract item, with an assigned billing plan. The billing plan is released with creation.
2. **Step 2:** The billing plan items include price, billing cycle, and accounting information. The price is a manual price in subscription item PSMB Position.
3. **Step 3:** Regularly request billing plan items. Created billable items have a reference to the billing plan.

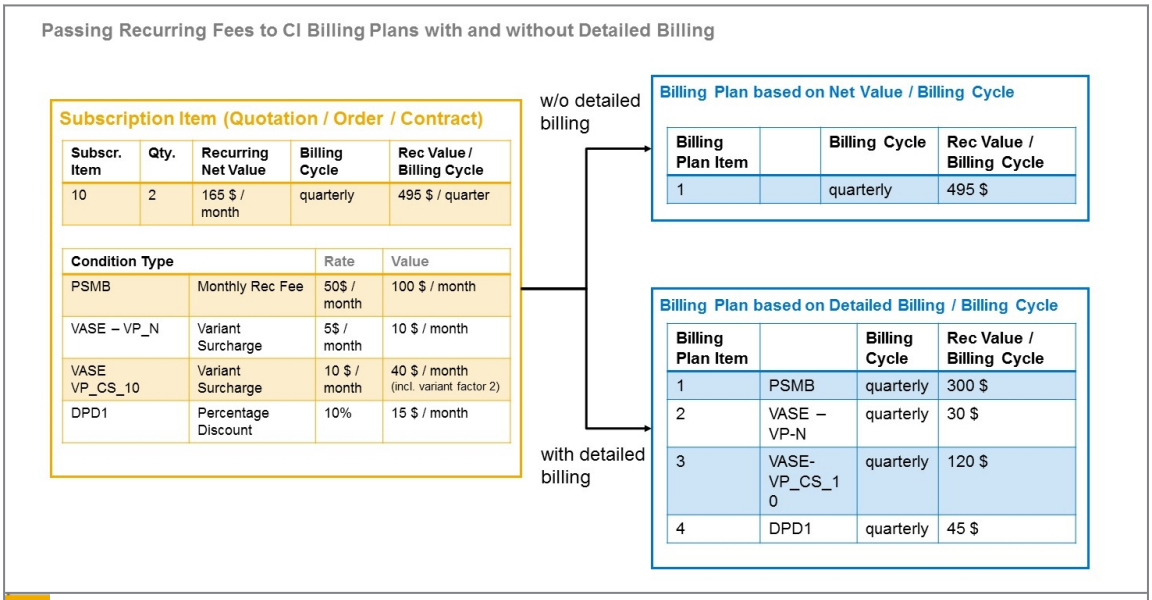

Figure 209: Passing Recurring Fees to CI Billing Plans with and without Detailed Billing

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It is now possible to create a billing plan with one detailed billing plan item for a recurring condition type in Convergent Invoicing. This enables generation of a detailed invoice to customers with details of the recurring fee, discounts, and surcharges applied.

This feature further supports accounting reconciliation.

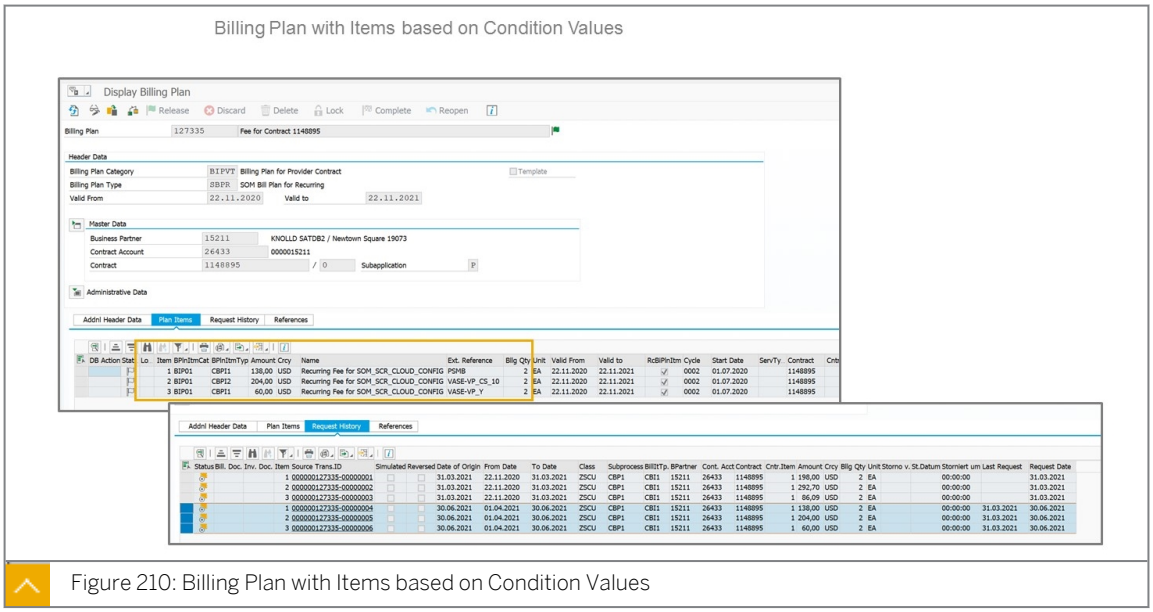

Figure 210: Billing Plan with Items based on Condition Values

Perform the Customizing of the Billing Plan

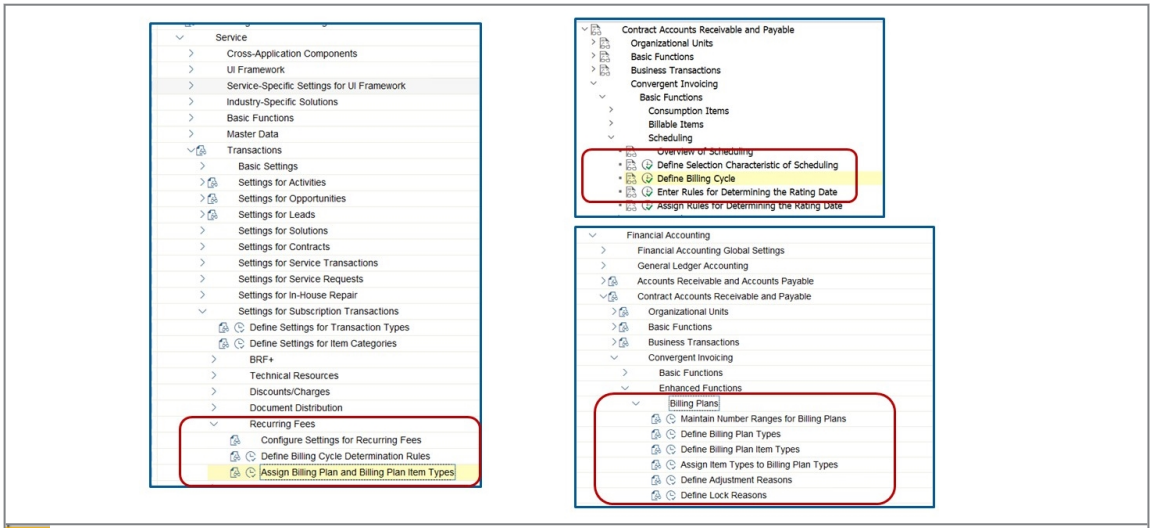

Figure 211: Product Master: Customizing for Recurring Fees

You can maintain the settings for Recurring Fees in the Implementation Guide (IMG):

Service → Transactions → Settings for Subscription Transactions → Recurring Fees

You can maintain the settings for the Billing Cycle in :

SPRO → Financial Accounting → Contract Accounts Receivable and Payable → Convergent Invoicing → Scheduling → Define Billing Cycle

You can define the billing plan in IMG:

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SPRO → Financial Accounting → Contract Accounts Receivable and Payable → Convergent Invoicing → Enhanced Functions → Define Billing Plan Types

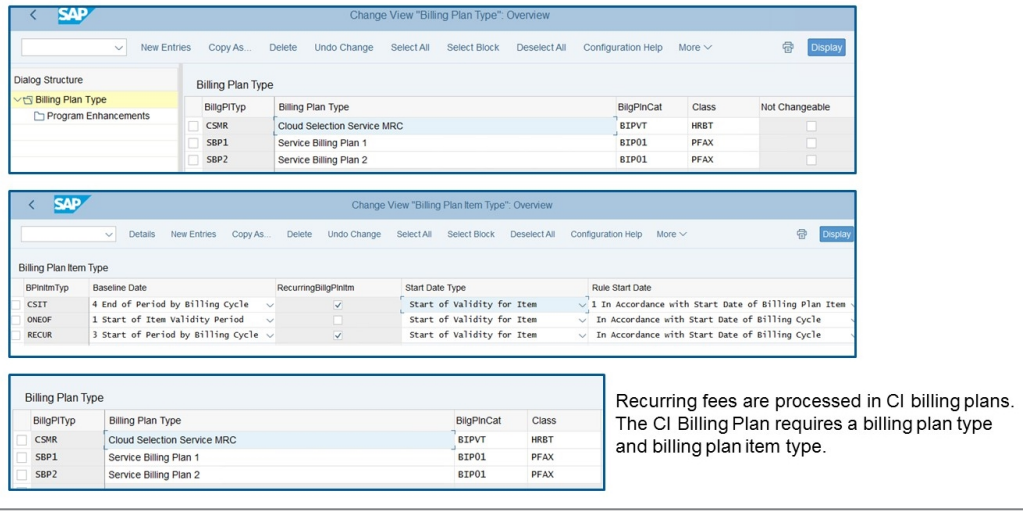

Figure 212: CI Customizing for Billing Plan Types and Billing Plan Item Types

You can define the billing plan type in IMG:

SPRO → Financial Accounting → Contract Accounts Receivable and Payable → Convergent Invoicing → Enhanced Functions → Define Billing Plan Types.

It is mandatory to specify the billing plan type, when billing plans are being created.

You can define the billing plan item type in IMG:

SPRO → Financial Accounting → Contract Accounts Receivable and Payable → Convergent Invoicing → Enhanced Functions → Define Billing Plan Item Types

It is mandatory to specify the item types when creating billing plan items.

You can assign billing plan item type to the billing plan type in IMG:

SPRO → Financial Accounting → Contract Accounts Receivable and Payable → Convergent Invoicing → Enhanced Functions → Assign Item Types to Billing Plan Types.

Specify which item types are supported in which particular billing plan type.

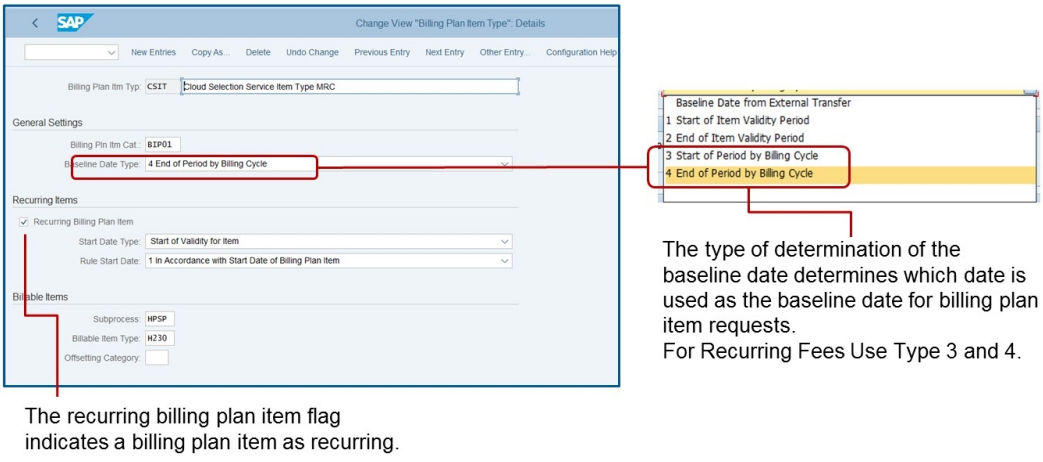

Figure 213: CI Customizing for CI Billing Plans with Recurring Fees

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Type of Determination of the Baseline Date

The type of determination of the baseline date determines which date is used as the baseline date for billing plan item requests. For Recurring Fees, Use *Type 3 or 4.*

- **'3'** Start of period in accordance with the billing cycle.

  Start of billing cycle period is used as the baseline date.

- **'4'** End of period in accordance with the billing cycle.

  End of billing cycle period is used as the baseline date.

Recurring Billing Plan Item

The recurring billing plan item flag indicates a billing plan item as recurring. The billing plan item can be requested multiple times within a validity period. The frequency of the request can be determined by using the billing cycle defined in the item, for example, monthly at the end of the month.

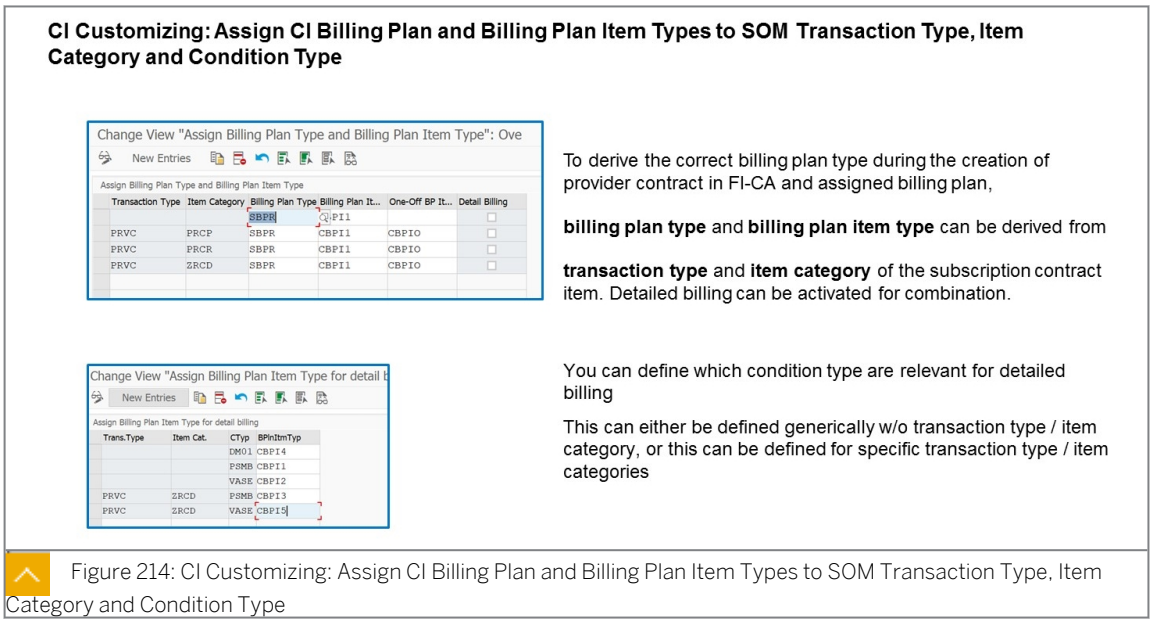

Figure 214: CI Customizing: Assign CI Billing Plan and Billing Plan Item Types to SOM Transaction Type, Item Category and Condition Type

You can assign Billing Plan Types and Billing Plan Items Types to Transaction Types and Item Categories in IMG

SPRO → Service → Transactions → Settings for Subscription Transactions → Recurring Fees → Configure settings using Billing Plans with Convergent Invoicing → Assign Billing Plan and Billing Plan Item Types

Recurring fees are processed in CI billing plans. The CI Billing Plan requires a billing plan type and billing plan item type.

To derive the correct billing plan type during the creation of provider contract and assigned billing plan, billing plan type and billing plan item type can be derived from transaction type and item category of the subscription contract item.

For Detailed Billing customizing in IMG:

SPRO → Service → Transactions → Settings for Subscription Transactions → Recurring Fees → Configure settings using Billing Plans with Convergent Invoicing → Define Price Conditions for Distribution to SAP Convergent Invoicing

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LESSON SUMMARY

You should now be able to:

- Define a Billing Plan
- Provide the Billing Plan Process Overview
- Perform the Customizing of the Billing Plan

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Learning Assessment

1. Recurring charges are created by billing plans in Convergent Invoicing (CI). Which of the following billing cycles are possible in the standard?

   Choose the correct answers.

   - ☐ **A** Yearly Billing Cycle
   - ☐ **B** Daily Billing Cycle
   - ☐ **C** Monthly Billing Cycle
   - ☐ **D** Quarterly Billing Cycle

2. Which of the following is the master data object for creating recurring fees in the Sales Order Management (SOM) process?

   Choose the correct answer.

   - ☐ **A** Billing Document
   - ☐ **B** BOM
   - ☐ **C** Consumption Item
   - ☐ **D** Billable Item
   - ☐ **E** Billing Plan
   - ☐ **F** Billing Request

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Learning Assessment - Answers

1. Recurring charges are created by billing plans in Convergent Invoicing (CI). Which of the following billing cycles are possible in the standard?

   Choose the correct answers.

   - [X] A Yearly Billing Cycle
   - [X] B Daily Billing Cycle
   - [X] C Monthly Billing Cycle
   - [X] D Quarterly Billing Cycle

   That is correct! All of the listed billing cycles are possible in the standard.

2. Which of the following is the master data object for creating recurring fees in the Sales Order Management (SOM) process?

   Choose the correct answer.

   - [ ] A Billing Document
   - [ ] B BOM
   - [ ] C Consumption Item
   - [ ] D Billable Item
   - [X] E Billing Plan
   - [ ] F Billing Request

   That is correct! The Billing Plan is the master data object in this context.

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UNIT 13 Counter Concepts

Lesson 1

Persistent Counter and Shared Counter 233

Lesson 2

Allowance Counter 239

Lesson 3

Shared Allowances 245

UNIT OBJECTIVES

- Explain the Shared Counter Process Overview
- Provide the Process Overview of Counter Sharing
- Describe the Customizing for Counter Sharing
- Define Allowances
- Outline the Allowance Creation Process
- Outline the Allowance Use Process
- Describe the Allowance Subscription Product Design
- Explain the Subscription Contract Change Process for Allowance Creation
- Describe Shared Allowances

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Unit 13
Lesson 1

Persistent Counter and Shared Counter

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explain the Shared Counter Process Overview
- Provide the Process Overview of Counter Sharing
- Describe the Customizing for Counter Sharing

Business Example

Training product cloud selection service *BSU_CLOUD_SEL_SERV* , uses counter in Convergent Charging (CC) for free volume of CPU, RAM, and Bandwidth usage. For product definition in Sales Order Management (SOM), the counter are defined in SOM as Counter as well. Two subsidiaries of a customer, with two provider contracts, do not share the volume counter of CPU, RAM, or bandwidth. In case they should share the volume, a sharing counter is necessary.

Explain the Shared Counter Process Overview

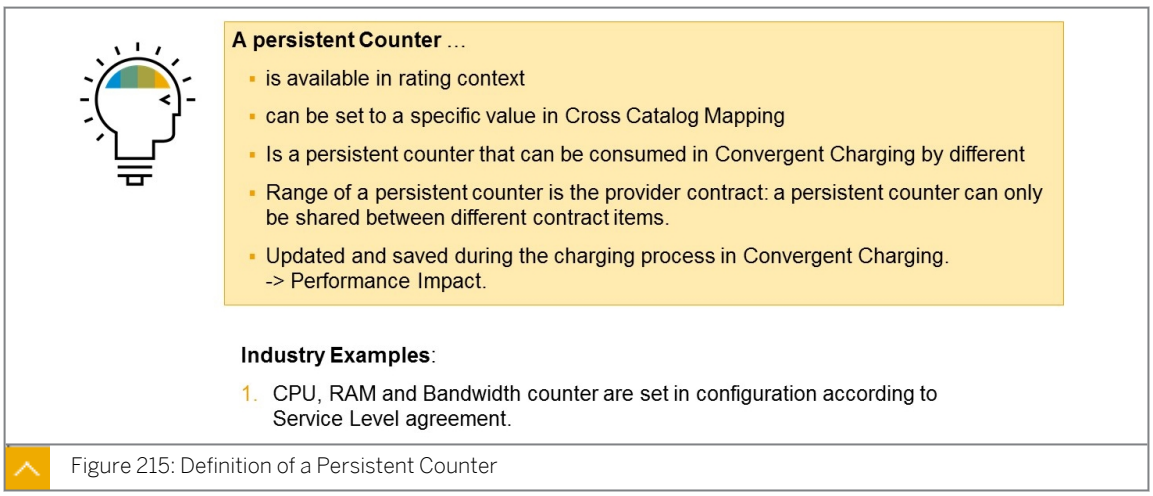

Figure 215: Definition of a Persistent Counter

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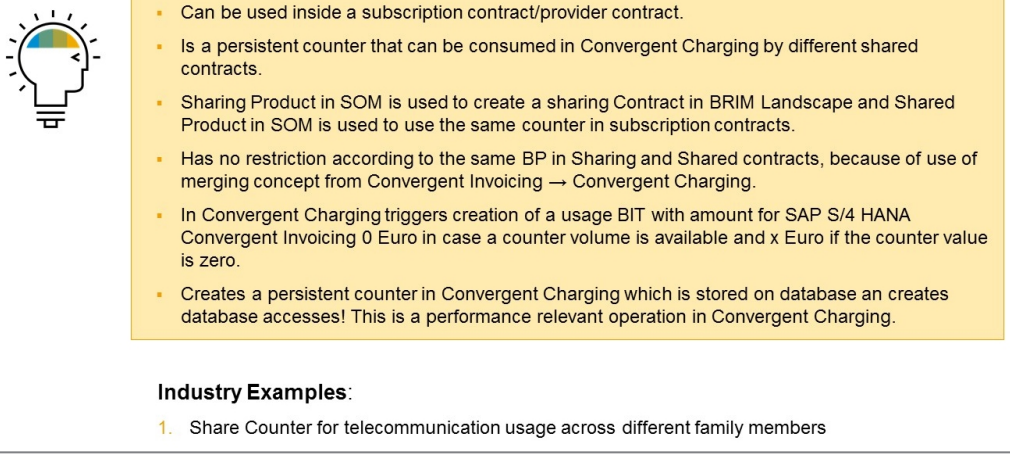

Figure 216: Definition of a Sharing Counter

The Definition

Sharing allows you to share counters or allowances during the rating process in CC, within one contract or across multiple contracts.

Sharing Options:

- Share counters and allowances within one contract
- Share objects, counters and allowances, across a sharing contract and its shared contracts
- Share allowances between different contracts

To enable sharing, specific settings to the subscription product must be provided.

Sharing contracts and shared allowances cannot be used together. A product which supports shared allowances cannot be used in a sharing or a shared contract and vice versa.

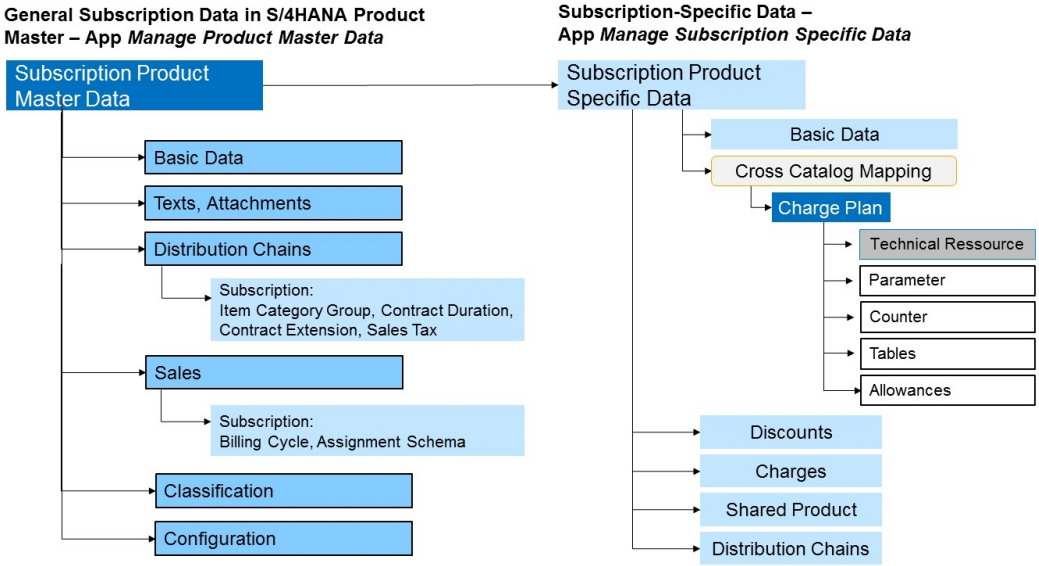

Figure 217: Product Views in SAP S/4HANA

Product modeling in SOM, according to CC Counter, is done in Cross Catalog Mapping.

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Additionally shared products are assigned in a section of subscription specific data.

Provide the Process Overview of Counter Sharing

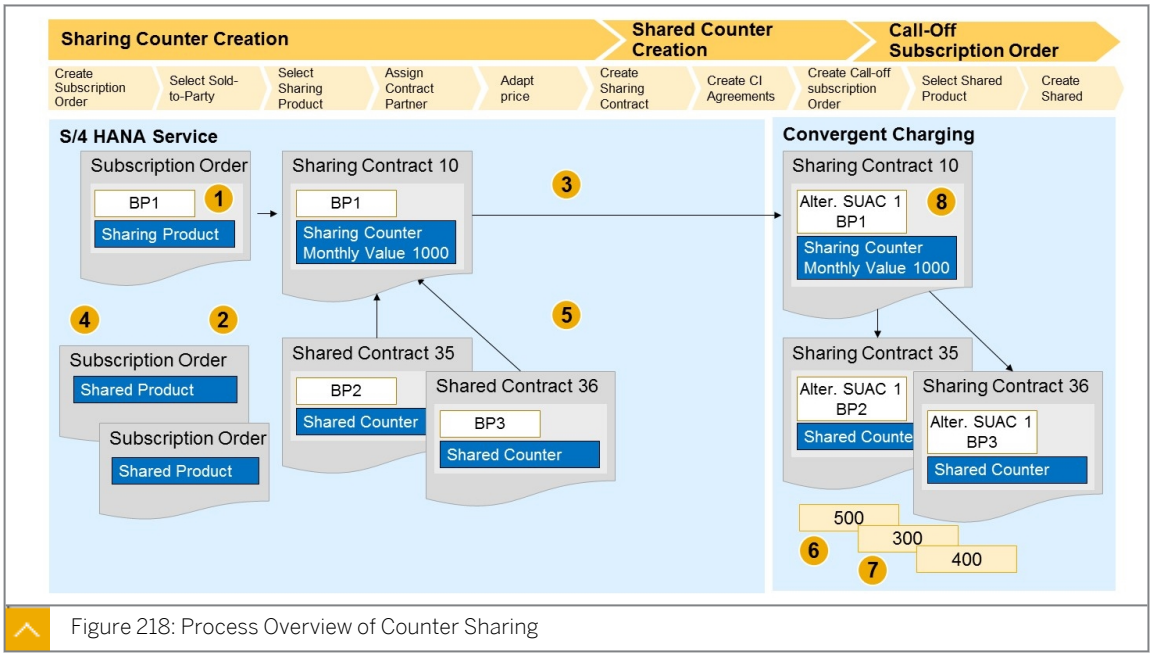

Figure 218: Process Overview of Counter Sharing

The following process steps are done during Shared Counter Creation:

1. Access the Fiori App *Subscription Order Management* and create a new subscription order with transaction type.
2. Assign a BP1 as a sold-to-party contract partner.
3. Create a contract hierarchy: *Root and Leaf*.
4. Create the sharing contract.
5. Create a subscription order item, which creates the counter for shared contracts.
6. Select the sharing product, which creates the counter for the contract.
7. Create a subscription order (shared product) and assign the sharing contract number to the subscription order item. The dependent contract is created.
8. Merge the subscriber account.
9. Create shared contracts with shared counter:
   - Usage rating with counter reduction
   - Usage rating with counter to zero
   - Monthly counter reset

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Describe the Customizing for Counter Sharing

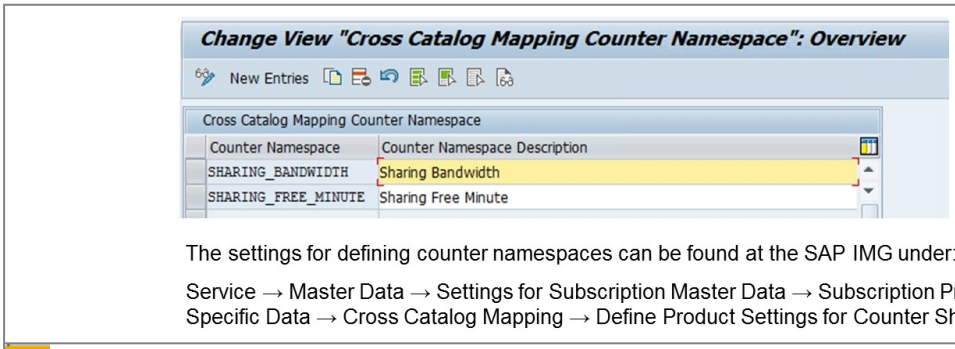

Figure 219: Sharing - Define Counter Namespaces for Counter Sharing

Define Product Settings for Counter Sharing Use

In this customizing activity, you can create and edit namespaces for SAP S/4HANA Service. It is required, so these namespaces can be used in Cross Catalog Mapping (CCM). These namespaces are required for sharing counters that are used by different charge plans.

Namespaces must be defined in SAP S/4HANA Service. This is in addition to the names that you have already defined for counters in Convergent Charging (CC). You can only share counters that have been defined with the same name in CC. Counters can be shared between different charge plans of the same contract or between charge plans of a sharing contract and its shared contracts.

Requirements:

- You have defined counters in SAP CC with the same name.

Activities:

1. On the change view Cross Catalog Mapping Counter Namespace, choose *New Entries*.
2. Enter a counter namespace.
3. Enter a counter namespace description.
4. Save your entries.

Business Example for Counter Sharing:

- A charge plan contains a counter with the name *SHARING_BANDWIDTH*. This counter counts the bandwidth that have been sent since the beginning of the month.
- A different charge plan contains a basic charge that also has a counter with the name *SHARING_BANDWIDTH*. This counter also counts the bandwidth that has been used. Once 1000 bandwidth has been used, the basic charge is waived.
- Both charge plans have to use the same counter. To share the counter, you must define a name in CC and assign the same namespace in CCM.

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Figure 220: Customizing Counter Display

You can activate the display of counters in Implementation Guide (IMG):

SPRO → Service → Transactions → Settings for Subscription Transactions → Define Settings for Item Categories

You can define which counters are displayed on subscription / provider contracts:

SPRO → Contract Accounts Receivable and Payable → Basic Functions → Provider Contract → Counter on Provider Contract

These views can then be used in BAdI *CRM_ISX_PROV_CC_COUNTER*.

LESSON SUMMARY

You should now be able to:

- Explain the Shared Counter Process Overview
- Provide the Process Overview of Counter Sharing
- Describe the Customizing for Counter Sharing

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Unit 13
Lesson 2

Allowance Counter

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Define Allowances
- Outline the Allowance Creation Process
- Outline the Allowance Use Process
- Describe the Allowance Subscription Product Design
- Explain the Subscription Contract Change Process for Allowance Creation

Define Allowances

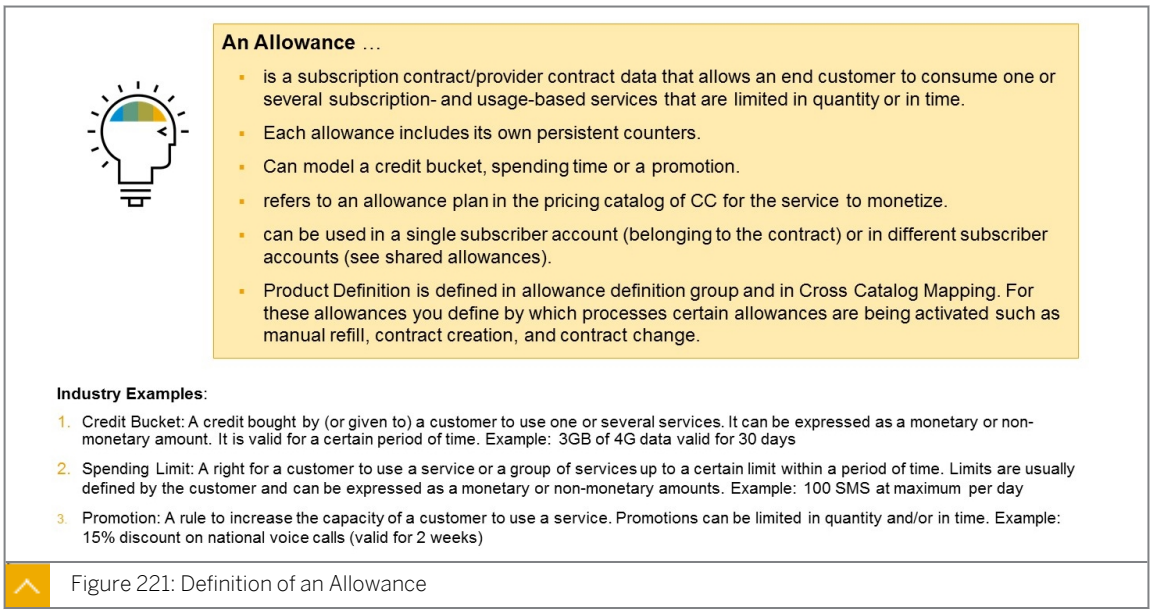

Figure 221: Definition of an Allowance

Features Product Modeling:

- Manage the allowance definition group via a Fiori App
- During product modeling in Cross Catalog Mapping (CCM), you can define the availability and activation behavior of allowances, by assigning allowance definition groups to the charge plan

Features Order / Contract :

- Simulate determination of allowance during order capture and display allowance on subscription order

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- Activate allowances during contract distribution for new and changed contracts
- Process manual allowance refill
- Display the current balance of allowances on a subscription contract

Allowance Definition

- An **Allowance** is used by a customer to consume services. An allowance can be limited in quantity and/or limited in time.
- An allowance can model a credit bucket, a spending limit or a promotion.
- Each allowance includes its own persistent counters.

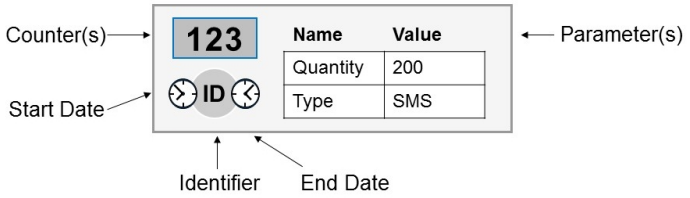

Figure 222: Allowance Definition

Allowance Lifecycle

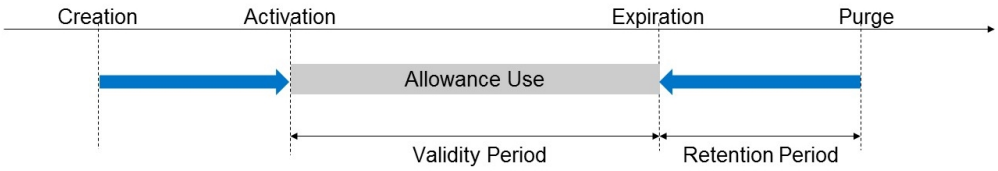

Figure 223: Allowance Lifecycle

Allowance Example Payment in Advance Mapping Table in Convergent Charging

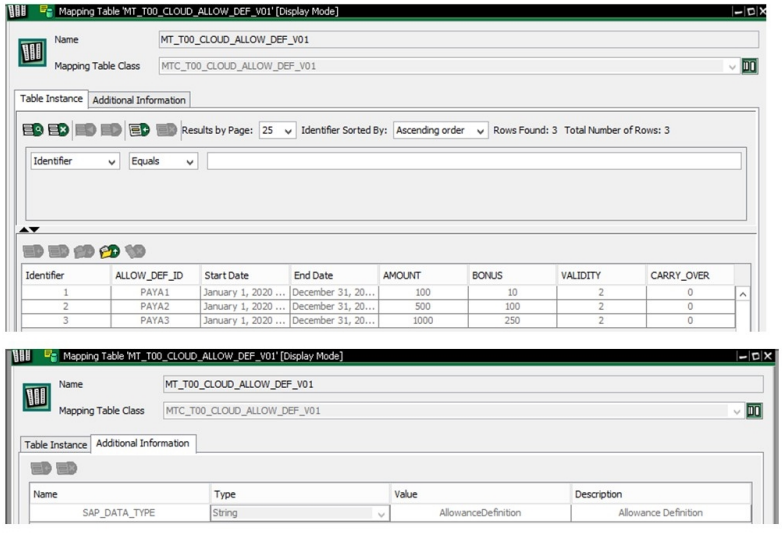

Figure 224: Define Allowances

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Outline the Allowance Creation Process

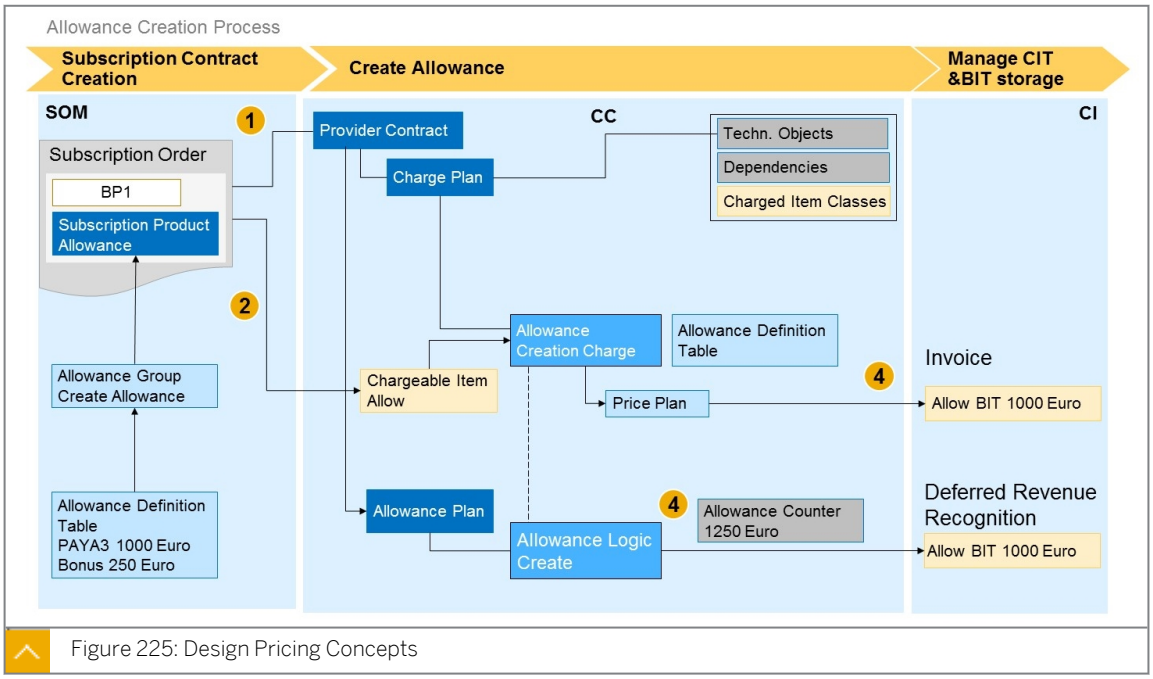

Figure 225: Design Pricing Concepts

In your SAP Solution, the Cross Catalog Mapping (CCM) in Sales Order Management (SOM) defines which allowances must be created, when end customers sign contracts for a subscription product, or when changes are made to the customers' contract. SAP SOM creates allowances in the SAP Convergent Charging (CC) system, on demand, or automatically, when the CC provider contract is activated or changed.

1. **Step 1:** Create a subscription Order with an allowance subscription product.

   SAP SOM sends a charging operation request to SAP CC with an ALLO chargeable item (1). This specialized chargeable item contains the identifier of the allowance definition to be created and the date/time at which the allowance must be activated.

2. **Step 2:** In SOM a webservice request is sent to CC for an allowance creation request. Allowance creation is an additional step in the Order Distribution Infrastructure (ODI) scheme.

   SAP CC processes this charging request, identifies and executes the relating contracted charge (2) in this contract.

3. **Step 3:** Create an allowance.

   This charge retrieves the corresponding allowance definition from the allowance definition table, creates the corresponding allowance, and initializes the allowance according to the allowance definition (3).

4. **Step 4:** The charge can generate billable items for reporting purposes (4) and the allowance can generate billable items for deferred revenue recognition purposes (4).

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Outline the Allowance Use Process

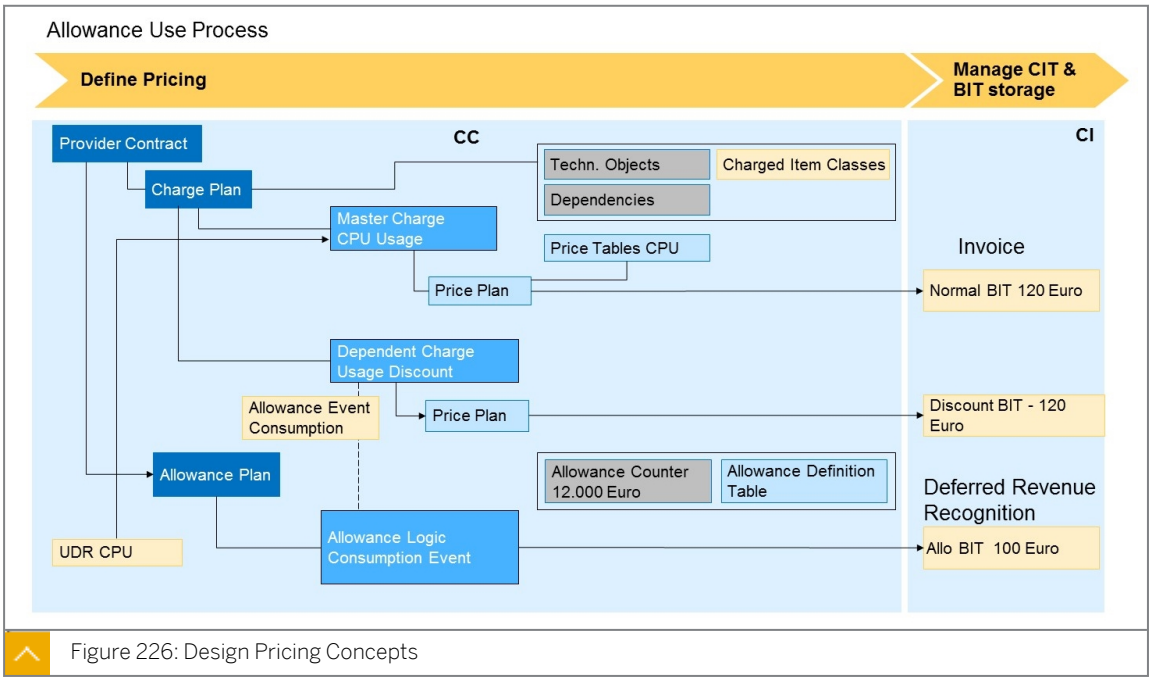

Figure 226: Design Pricing Concepts

In your SAP Solution, the Cross Catalog Mapping (CCM) in the SAP Sales Order Management (SOM) system defines which allowances must be created when end customers sign contracts for a subscription product or when changes are made to the customers' contract. SAP SOM creates allowances in the SAP Convergent Charging (CC) system, on demand or automatically, when the CC provider contract is activated or changed.

SAP SOM sends a charging operation request to SAP CC with an ALLO chargeable item (1). This specialized chargeable item contains the identifier of the allowance definition to be created and the date/time at which the allowance must be activated.

SAP CC processes this charging request, identifies and executes the relating contracted charge (2) in this contract.

This charge retrieves the corresponding allowance definition from the allowance definition table, creates the corresponding allowance, and initializes the allowance according to the allowance definition (3).

The charge can generate billable items for reporting purposes (4) and the allowance can generate billable items for deferred revenue recognition purposes (4).

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Describe the Allowance Subscription Product Design

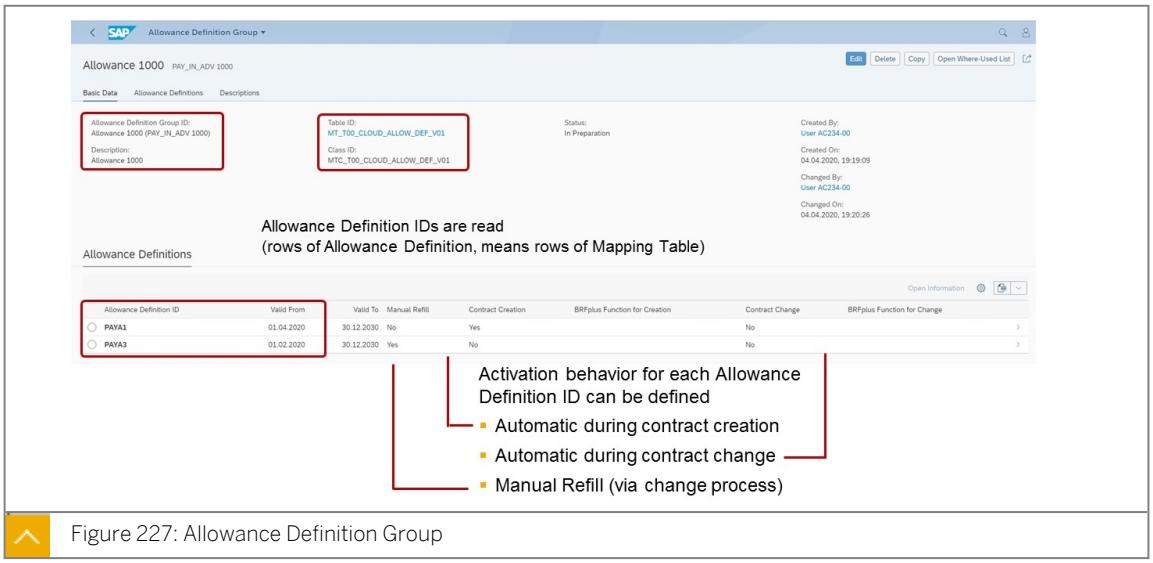

Figure 227: Allowance Definition Group

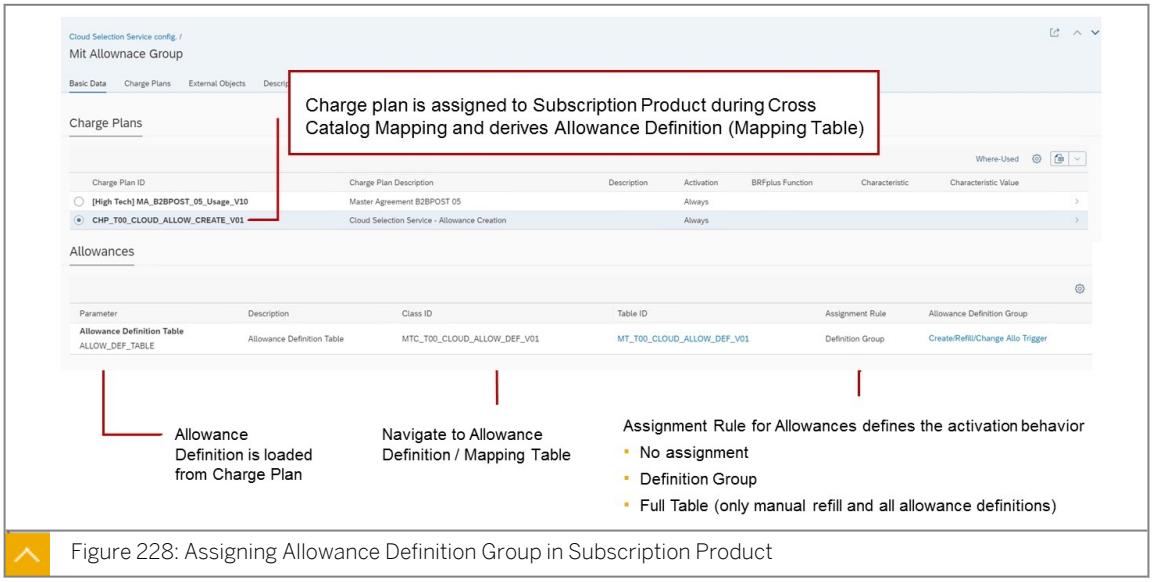

Figure 228: Assigning Allowance Definition Group in Subscription Product

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Explain the Subscription Contract Change Process for Allowance Creation

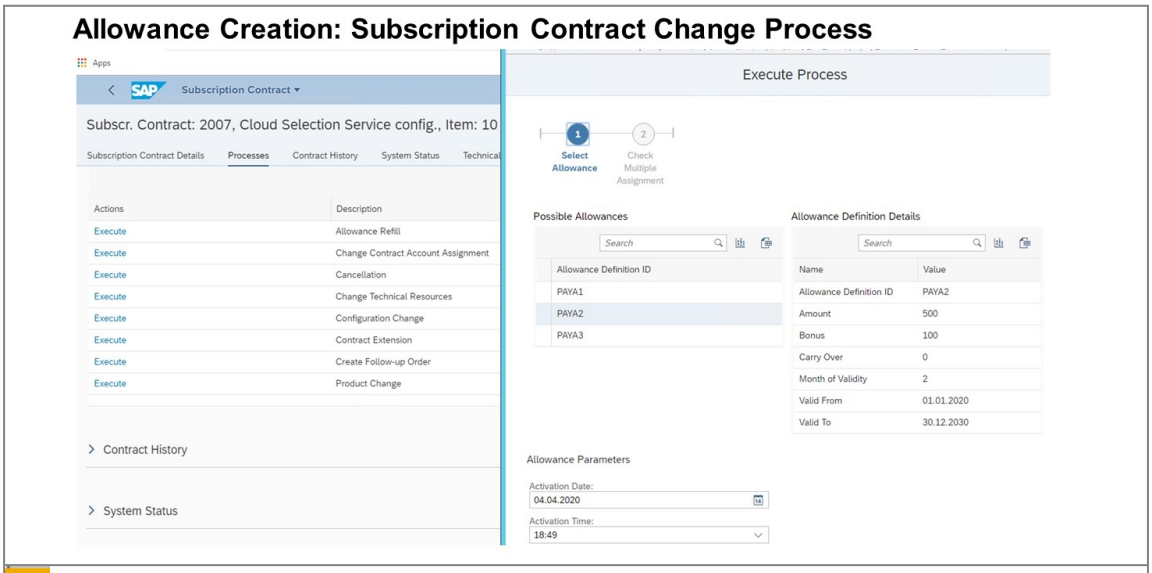

Figure 229: Explain the Subscription Contract Change Process for Allowance Creation

LESSON SUMMARY

You should now be able to:

- Define Allowances
- Outline the Allowance Creation Process
- Outline the Allowance Use Process
- Describe the Allowance Subscription Product Design
- Explain the Subscription Contract Change Process for Allowance Creation

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Unit 13
Lesson 3

Shared Allowances

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Describe Shared Allowances

Business Example

Subscription Product *BSU_CLOUD_SEL_ALLO* includes a Payment in Advance model, which offers a prepaid scenario with a discount for the payment type. Customers can pay 100 Euro, 500 Euro, or 1000 Euro in advance and get a bonus of 10% , 20%, or 25%. The amount paid in advance is in Convergent Charging (CC), calculated with an allowance counter.

Describe Shared Allowances

A Shared Allowance …

- Is an allowance that can be shared between independent contracts using the same shared ID: Shared allowances are identified during rating process via a share ID.
- The share ID is a combination of a share ID parameter defined in cross catalog mapping during product modeling, and a sharing group defined in the subscription contract.
- The 'Manage Sharing Groups' app (F4432) enables an Internal Sales Representative to define sharing groups, which are then used for shared allowances in the subscription order, solution quotation and the master agreement.
- Sharing contracts and shared allowances cannot be used together. A product which supports shared allowances cannot be used in a sharing or a shared contract and vice versa.
- The sharing group must be assigned during order capturing if shared objects is enabled for the subscription product
- **Rerating is not possible with Shared Allowances.**

Industry Examples:

1. Share a Payment in Advance between different Provider Contracts in Convergent Charging.

Figure 230: Definition of a Shared Allowance

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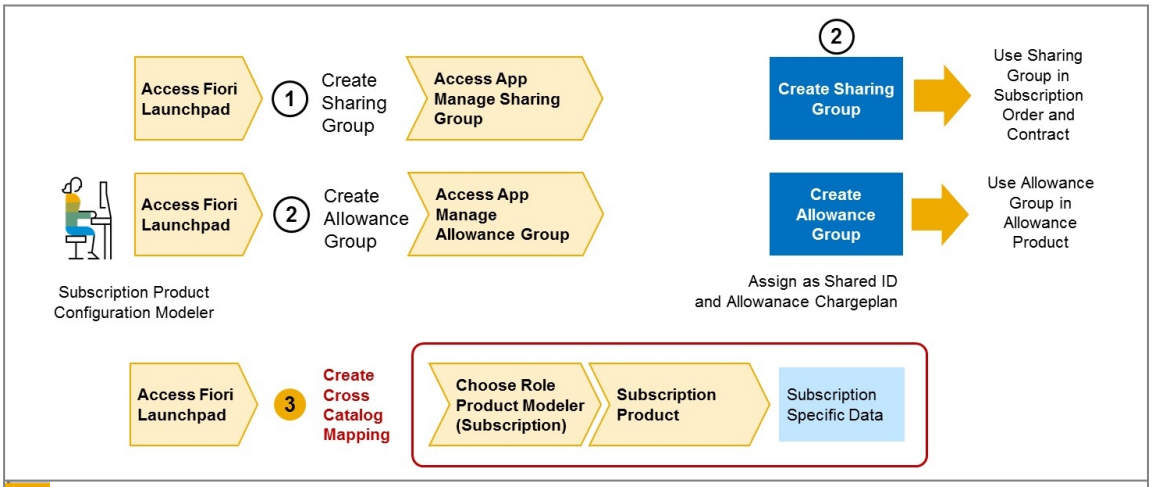

Figure 231: Create Shared Allowance Data - Subscription Product Configuration Modeler

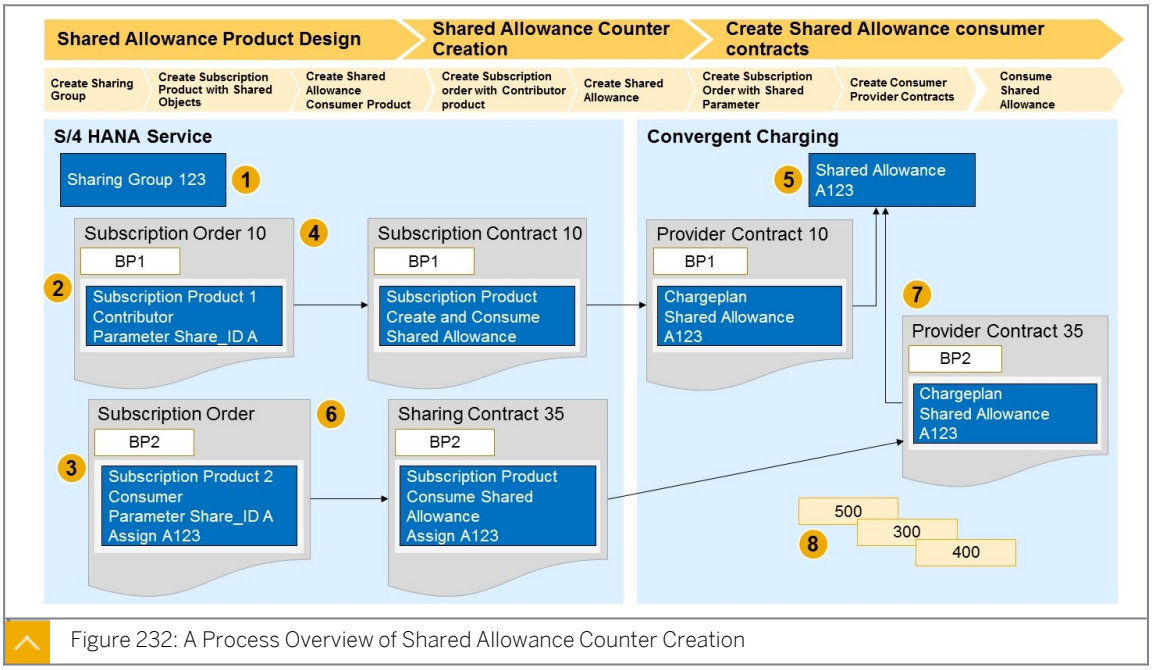

Figure 232: A Process Overview of Shared Allowance Counter Creation

Product Design for Shared Allowance:

1. **Step 1:** Sharing group creation. Create a sharing group as customer master data, for example, for *Headquarter 02*, if all allowances should be shared in this group.
2. **Step 2:** Create a subscription product with an allowance as a contributor product with Cross Catalog Mapping (CCM) assignment of shared objects.

   A. Set Shared Objects = YES.

   B. Assign a Shared ID as a Parameter.

   The chargeplan in Convergent Charging (CC) is a sharing chargeplan.
3. **Step 3.** (Optional). Create a subscription product with an allowance as a consumption product of the shared allowance. The consumption product leaves out the shared allowance creation.

   It´s also possible to create and consume with the contributor product runtime.

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4. **Step 4:** Create a subscription order and a subscription contract as a contributor for the shared allowance in Sales Order Management (SOM).
5. **Step 5:** The shared allowance is created in CC.

   The shared allowance id is a concatenation of the sharing group assigned in the subscription order and a created shared ID. In our process example it´s shared *ID A123*.
6. **Step 6:** Create a second subscription order and assign the shared ID in this subscription order.
7. **Step 7:** A provider contract in CC is as a consumer provider contract is for the shared allowance.
8. **Step 8:** Consumption of all assigned provider contracts reduce the shared allowance.

	Sharing Contracts	Shared Allowances
Shared Objects	Focus is more on counters	Focus is more on allowances Currently Allowances can only count downwards
Coupling	Tight Coupling during product modelling between sharing and shared product	Loose Coupling via Sharing ID, Assignment of Sharing Group ID during order / quotation creation
Account Split / B2B	In CC the subscriber account for sharing and shared contracts must be unique. In Subscription Order Management we allow different contract partners (payers). They are merged artificially to the same subscriber account ID in CC (account merge).	Account Split is supported as the allowance is only coupled via the share ID. The contracts are completely independent.
Rerating	Rerating is supported	Support of Rerating not available yet

Figure 233: Compare Scenarios

LESSON SUMMARY

You should now be able to:

- Describe Shared Allowances

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Learning Assessment

1. Allowances in Billing and Revenue Innovation Management (BRIM) can be used to do which of the following?

   Choose the correct answer.

   - [ ] **A** Grant Sales Order Management (SOM) user access rights.
   - [ ] **B** Give a customer a service credit.
   - [ ] **C** Check customers' credit worthiness.
   - [ ] **D** Allow a developer to do customizing.

2. Where are allowances in BRIM modeled?

   Choose the correct answers.

   - [ ] **A** In a Convergent Charging (CC) charge plan
   - [ ] **B** In Convergent Invoicing (CI) Billing
   - [ ] **C** In a Sales Distribution Sales Order
   - [ ] **D** in a Service Management Product
   - [ ] **E** In a SOM product

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Learning Assessment - Answers

1. Allowances in Billing and Revenue Innovation Management (BRIM) can be used to do which of the following?

   Choose the correct answer.

   - [ ] A Grant Sales Order Management (SOM) user access rights.
   - [x] B Give a customer a service credit.
   - [ ] C Check customers' credit worthiness.
   - [ ] D Allow a developer to do customizing.

   That is correct! Allowances can be used to give a customer a service credit.

2. Where are allowances in BRIM modeled?

   Choose the correct answers.

   - [x] A In a Convergent Charging (CC) charge plan
   - [x] B In Convergent Invoicing (CI) Billing
   - [ ] C In a Sales Distribution Sales Order
   - [ ] D in a Service Management Product
   - [x] E In a SOM product

   That is correct! Allowances are modeled in a CC charge plan, CI Billing, and in a SOM product.

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UNIT 14 SOM Fioris

Lesson 1

Fioris in SOM 253

UNIT OBJECTIVES

- Understand Fioris in SOM
- Identify Fiori SOM in Business Catalogs

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Unit 14
Lesson 1

Fioris in SOM

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Understand Fioris in SOM
- Identify Fiori SOM in Business Catalogs

Business Example

Get a detailed overview of the Fiori Launchpad apps used during the training.

Understanding Fiori in Sales Order Management

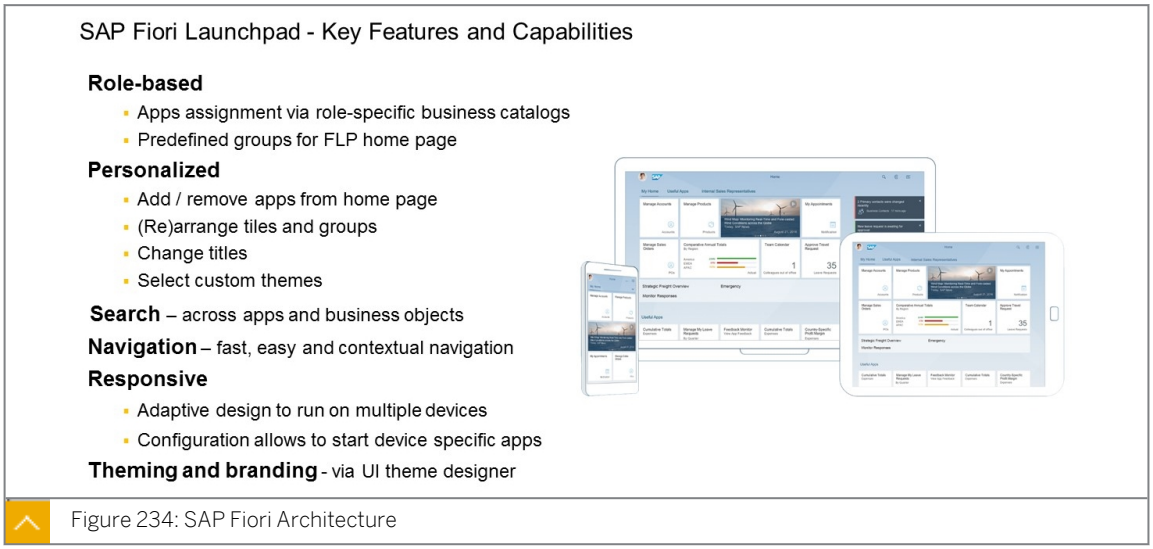

Figure 234: SAP Fiori Architecture

The SAP Fiori architecture introduces the high-level architecture of SAP Fiori.

The SAP Fiori apps are built with SAPUI5 technology for web applications, are iOS native and so can be used on smartphones, tablets, and desktop computers. The SAP Fiori Launchpad is used to launch and run the apps. The business data that is consumed in the apps is retrieved at runtime from the back-end systems by using OData services.

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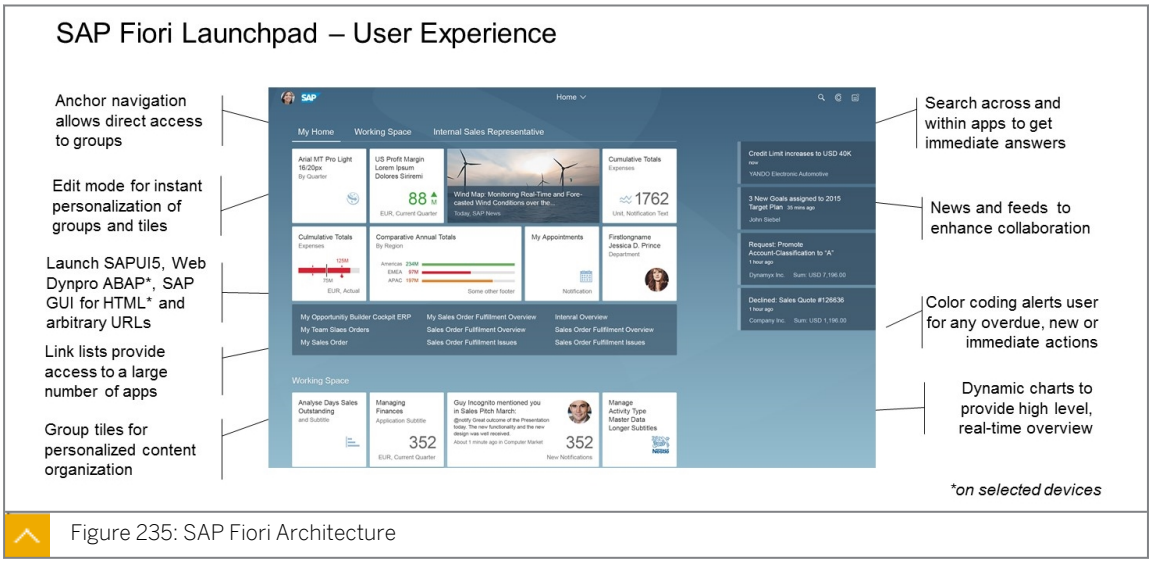

Figure 235: SAP Fiori Architecture

SAP Fiori is a consistent and holistic user experience for business users for all lines of business and across devices and deployment options.

The SAP Fiori concept is fundamentally a simplification of the user experience relying on modern design principles such as role-based, adaptive, coherent, simple, and delightful:

- *Role-based* means that it is designed for you, your needs, and how you work.
- *Adaptive* means that it adapts to multiple use cases and devices.
- *Coherent* means that it provides one fluid, intuitive experience.
- *Simple* means that it includes only what is necessary.
- *Delightful* means that it makes an emotional connection.

The SAP Fiori design is the visual design, information architecture, colors, and interaction patterns that are defined in the SAP Fiori design guidelines. Based on user roles and business processes, SAP Fiori simplifies doing business. SAP Fiori sets the standard for enterprise user experience by removing unnecessary complexity. We put users in control of their business tasks by giving them only what they really need. This is reflected in the five design principles already mentioned.

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Tiles / Apps

visual representation of applications **contain Intent**

(=combination of semantic object + action)

Catalogs

collections of apps available for user in home page or App Finder

Groups

subsets of apps displayed on home page as tiles or link

Roles

business roles including catalogs and groups

Target Mappings

map intent to application to be launched

(only required in FLP on ABAP)

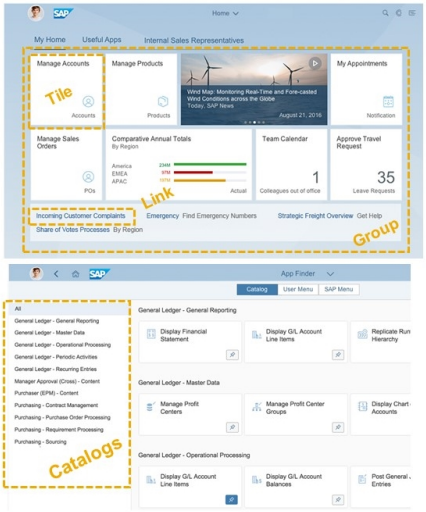

Figure 236: SAP Fiori Launchpad Content Entities

SAP Fiori Launchpad is the access point to apps on mobile or desktop devices.

Catalog: A set of apps you want to make available for one role. Depending on the role and the catalog assigned to the role, the user can browse through the catalog, choose apps from this catalog, and add them to the entry page of the SAP Fiori Launchpad.

Technically, apps are represented by the following:

- KPI tiles to launch the app
- App launcher tiles to launch the app.

Group: A subset of catalog that contains the apps visible on the SAP Fiori launchpad entry page. Which tiles are displayed on a user's entry page depends on the group assigned to the user's role. In addition, the user can personalize the entry page by adding or removing apps to pre-delivered groups or self-defined groups.

Roles (PFCG): Contain references to catalogs and groups, and provides users with access to the apps in these groups and catalogs.

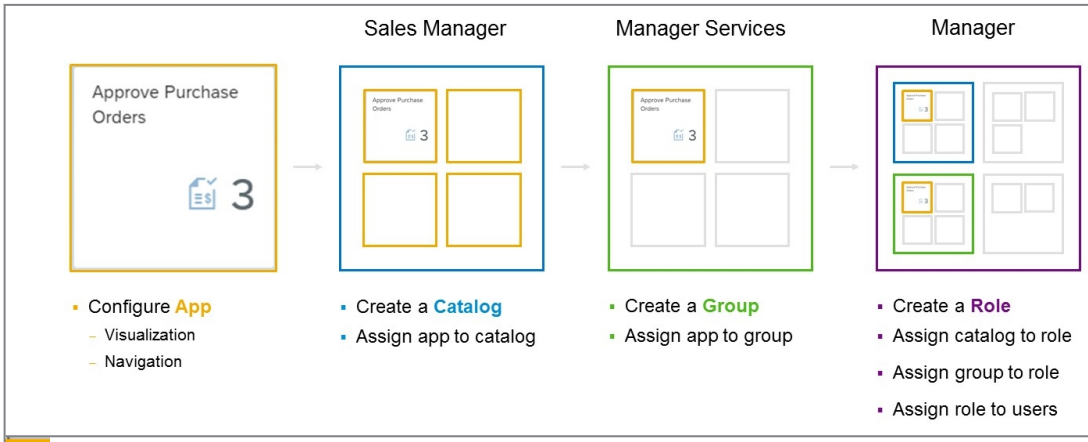

Figure 237: Managing Launchpad Content: Workflow

Steps to create the Fiori Launchpad content

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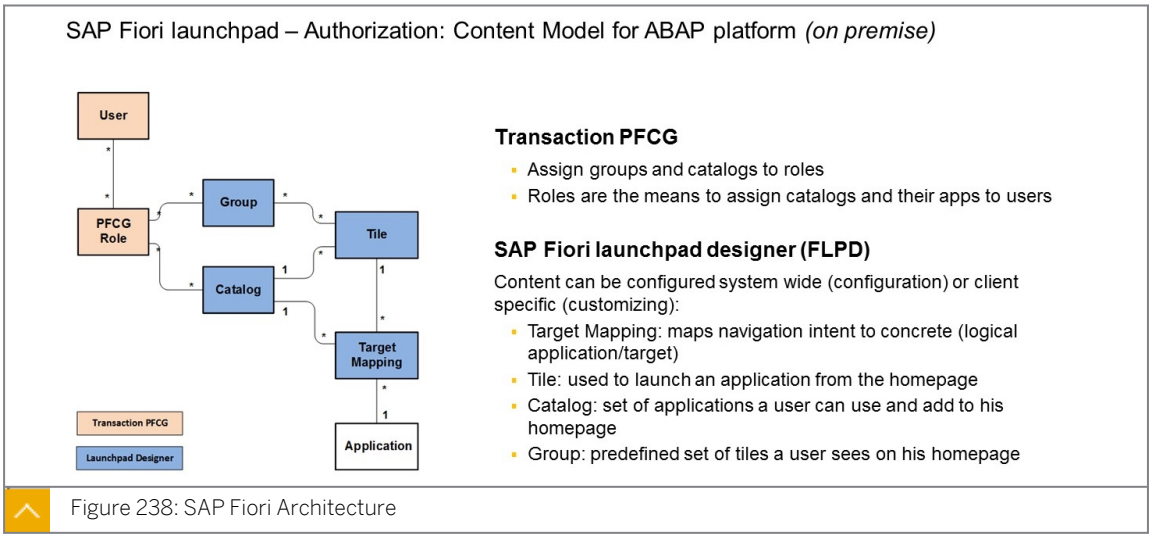

Figure 238: SAP Fiori Architecture

Use the technical catalogs delivered by SAP as repositories to create your own catalogs as an administrator. To display specific apps to a user by default in the entry page of the SAP Fiori Launchpad, create a group. Then add apps from your catalog to the group.

Both the catalog and the group must be assigned to users. Do this by assigning groups and catalogs to users' PFCG roles.

The business PFCG roles with the assigned catalogs and groups serve as examples, based on which, you can create your own business catalogs, groups, and roles.

Identify Fiori Sales Order Management in Business Catalogs

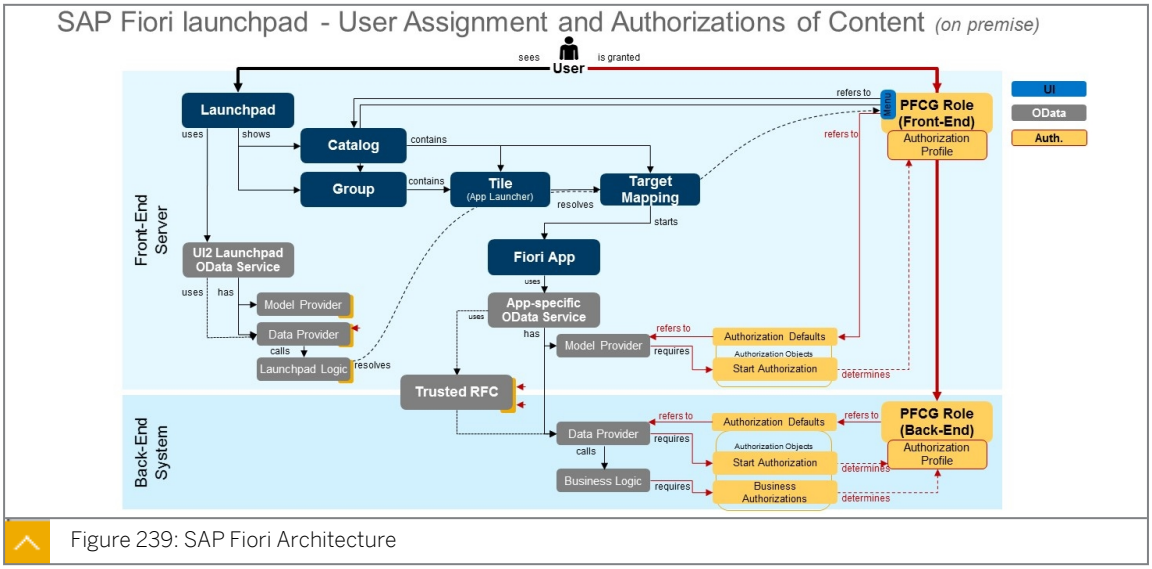

Figure 239: SAP Fiori Architecture

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Catalog:
SAP_SOM_BC_SUBSCR_ORDER

Fiori Tiles	Description
Manage Subscription Order	Create Subscription Orders
Manage Subscription Contract	Search for Subscription Contracts
Manage Solution Quotation	Create Solution Quotation
Mass Run	Execute Mass Runs
Manage Master Agreement	Create Master Agreement
Manage Sharing Groups	Manage Sharing Groups

Catalog:
SAP_SOM_BC_SUBSCR_PROD

Fiori Tiles	Description
Manage Product Master Data	Create Material Master
Manage Subscription Product specific Data	Create Subscription Specific Data
Manage Mapping Tables	Manage Mapping Tables
Manage Range Tables	Manage Range Tables
Manage Product Bundles	Create Product Bundles
Manage Allowance Definition Groups	Allowances Groups Definition
VC Modeling Environment	AVC Product Modeling

Figure 240: Standard SOM Fiori business Catalog

The shown business catalogs are delivered in the Fiori Launchpad designer as standard for Sales Order Management (SOM).

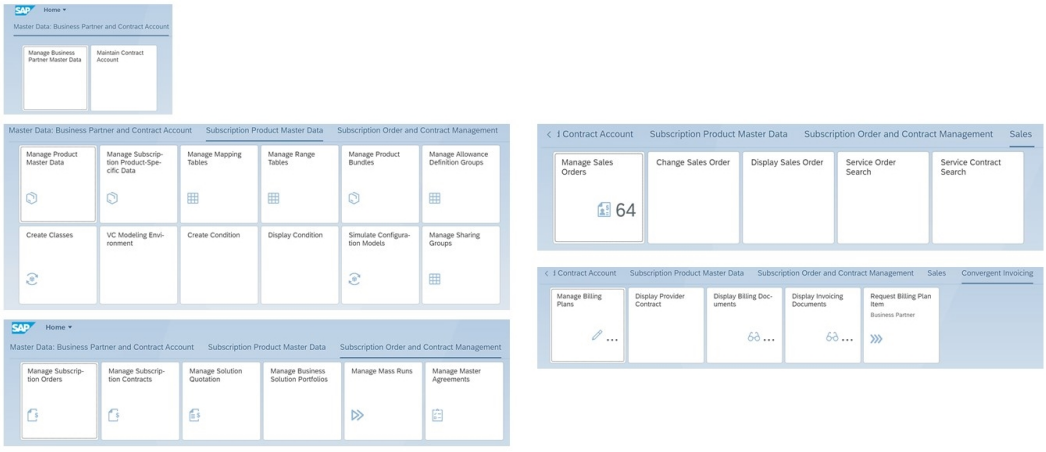

Figure 241: Custom Fiori Launchpad - Custom Fiori Catalog

For our training environment, we have set up a custom specific Fiori catalog, to group all the relevant Fiori apps necessary for the Course.

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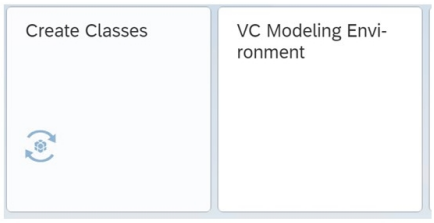
Section	Fiori Tiles	Section	Fiori Tiles	Section	Fiori Tiles
Master Data: Business Partner and Contract Account	Manage Business Partner Master Data	**Subscription Product Master Data**	Display Condition	**Sales**	Manage Sales Orders
	Maintain Contract Account		Simulate Configuration Model		Change Sales Order
Subscription Product Master Data	Manage Product Master Data		Manage Sharing Groups		Display Sales Order
	Manage Subscription Product Specific Data	**Subscription Order and Contract Management**	Manage Subscription Order		Service Order Search
	Manage Mapping Tables		Manage Subscription Contract		Service Contract Search
	Manage Range Tables		Manage Solution Quotation	**Convergent Invoicing**	Manage Billing Plans
	Manage Product Bundles		Manage Business Solution Portfolio		Display Provider Contract
	Manage Allowances Definition Groups		Mass Runs		Display Biling Documents
	Create Classes		Manage Master Agreement		Display Invoicing Documents
	VC Modeling Environment				Request Billing Plan items
	Create Condition				

Figure 242: Custom Fiori Launchpad - A Detailed Overview

On this slide you find structural detail of the Fiori Launchpad. The section titles can be defined custom wise, all Fiori apps with sequence add to the section are located in one catalog, custom defined catalog.



Product Modelling can be managed with Fiori Apps provided by Fiori Role
"*Product Configuration Modeler*"
"*SAP_BR_PRODUCT_CONFIG_MODELER*"

Instead of VC Modelling Environment App you can also use stand alone apps for the single modeling steps which are assigned to the role

To launch the Advanced Variant Configurator you need to include target mapping

sematic object VariantConfiguration
Action configureLegacy

into your Fiori Launchpad Catalog which is assigned to your business role.

To reference the semantic object you can use technical catalog SAP_TC_LO_VC_COMMON

The Fiori business role "SAP_BR_INTERNAL_SALES_REP" contains the respective target mapping.

Figure 243: Roles

LESSON SUMMARY

You should now be able to:

- Understand Fioris in SOM
- Identify Fiori SOM in Business Catalogs

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Learning Assessment

1. In which of the following tiles is the product modeling of Sales Order Management (SOM) products done?

   Choose the correct answers.

   - [ ] **A** Manage Solution Quotation
   - [ ] **B** Manage Product Bundles
   - [ ] **C** Manage Product Master Data
   - [ ] **D** Manage Range Tables

2. Subscription product modeling is only possible in an SAP S/4HANA backend system and not in a tile.

   Determine whether this statement is true or false.

   - [ ] True
   - [ ] False

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Learning Assessment - Answers

1. In which of the following tiles is the product modeling of Sales Order Management (SOM) products done?

   Choose the correct answers.

   - [ ] A Manage Solution Quotation
   - [X] B Manage Product Bundles
   - [X] C Manage Product Master Data
   - [X] D Manage Range Tables

   That is correct! Product modeling for SOM products is done in all of the listed tiles with the exception of the Manage Solution Quotation tile.

2. Subscription product modeling is only possible in an SAP S/4HANA backend system and not in a tile.

   Determine whether this statement is true or false.

   - [ ] True
   - [X] False

   That is correct!

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UNIT 15 Mass Runs

Lesson 1

Mass Runs in SOM 263

Lesson 2

APIs for Mass Run 269

UNIT OBJECTIVES

- Detail Mass Runs in SOM
- Know the API for Mass Run

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Unit 15
Lesson 1

Mass Runs in SOM

LESSON OBJECTIVES
After completing this lesson, you will be able to:

- Detail Mass Runs in SOM

Business Examples

Business Example Mass Run: Create with Master Agreement

A customer's headquarters and the service provider both agree on a master agreement. Development South and Development North have closed 100 subscription contracts.

The customer collects the relevant master data for the service recipients, for example, business partner number and contract duration. The collected data is uploaded to the mass run. If the data validation was successful then the subscription order and contract can be created and distributed. If an error occurs, the subscription orders will not be created.

Business Example Mass Run: Create without Master Agreement

The service provider needs to run a mass activation of contracts for the same product, with identical contract start date, but for different sold-to parties. The service provider collects the relevant sold-to and ship-to parties together with the technical resource IDs.

The collected data is uploaded to the mass run. If the data validation was successful then the subscription order and contract can be created and distributed. If an error occurs the subscription orders will not be created.

Business Example Mass Run: Change

A service provider needs to extend a set of contracts. The service provider preselects the contracts to be extended. The system processes the contract extension change process. Depending on the settings in order data distribution, the subscription contracts are updated and distributed at the given activation date.

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Detail Mass Run in Subscription Order Management

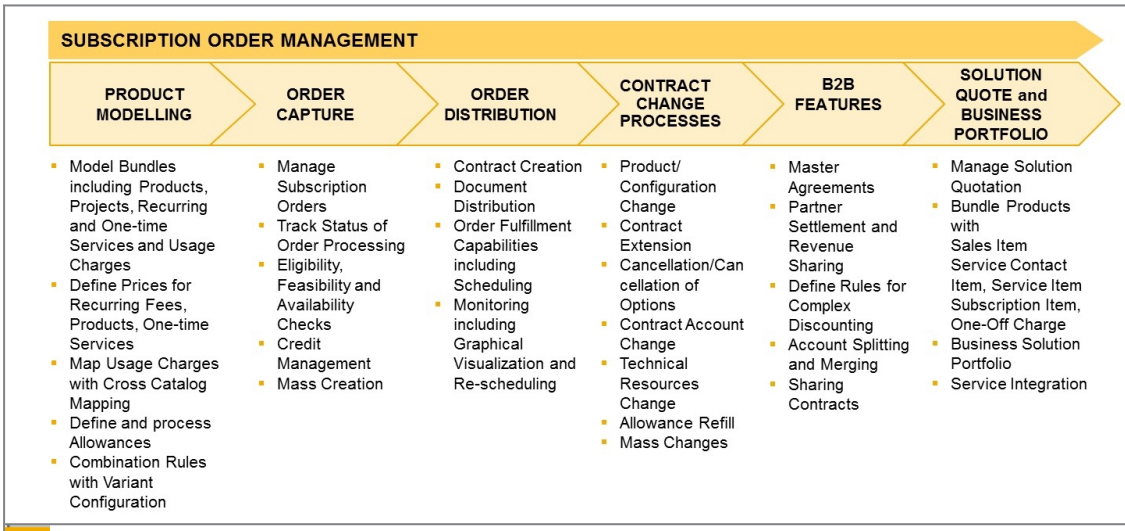

Figure 244: Overview Subscription Order Management

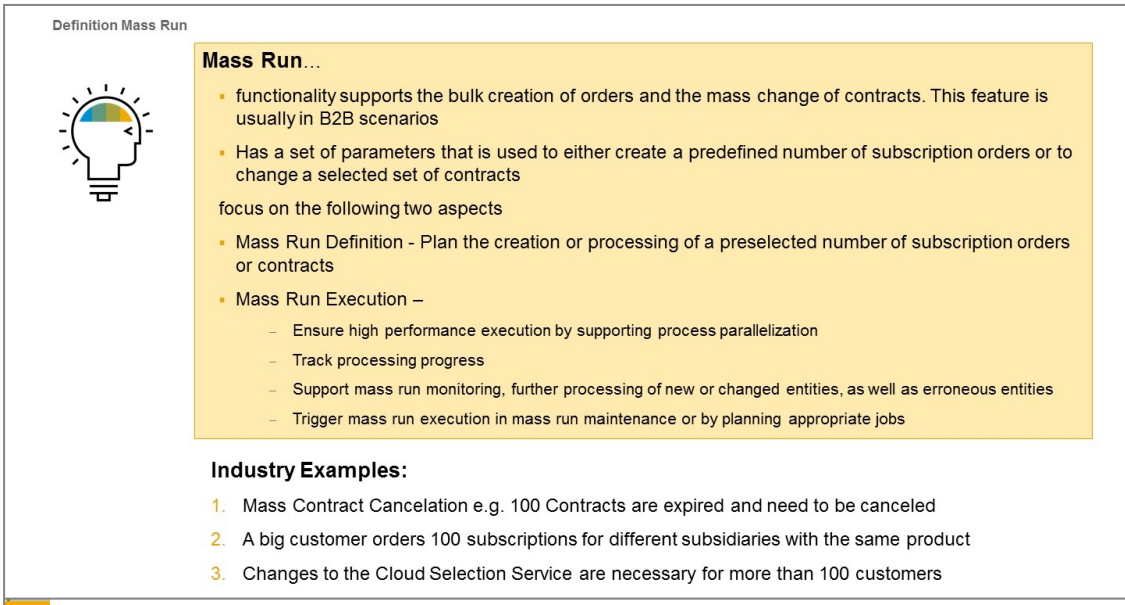

Figure 245: Definition of a Mass Run

You can find predelivered sample mass run types such as ...

- Create subscription orders with reference to a master agreement
- Create subscription orders for a specific product and individual sold to parties and technical IDs
- Run a mass change for change of cross catalog mapping version
- Run a mass cancellation of contracts

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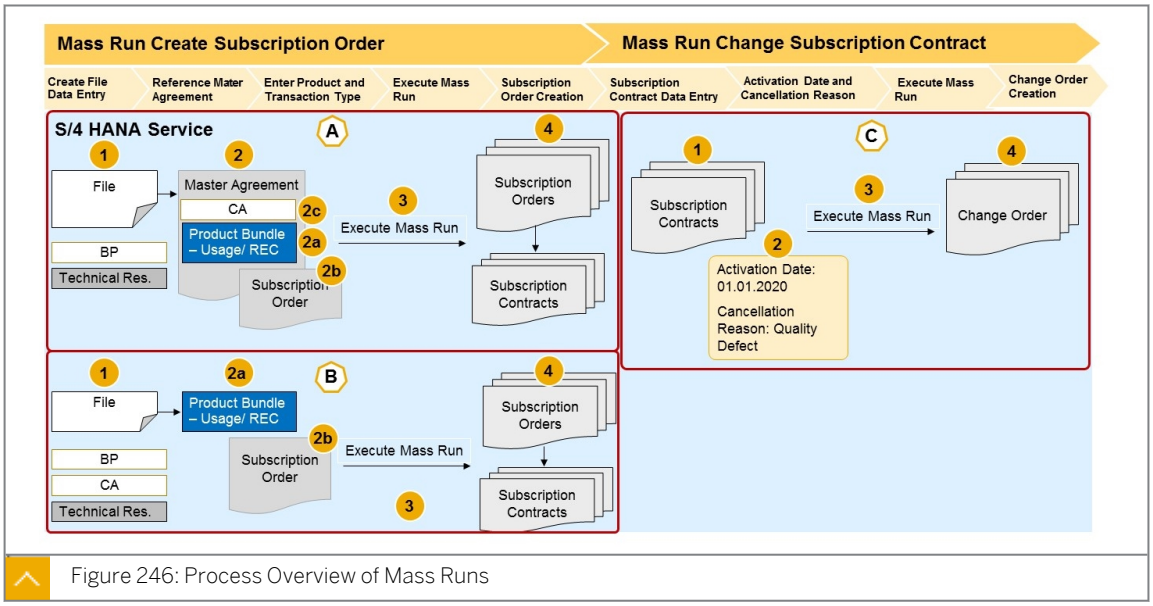

Figure 246: Process Overview of Mass Runs

This business scenario shows the Mass Run Process Flow

In scenario A a subscription order/contract is created with reference to a master agreement:

1. The data is entered as a CSV File and contains business partner data, such as sold to party and ship to party, as well as the corresponding technical resource
2. Choose the master agreement as reference for the order creation and add following data:
   a. Product Data - The field is called Custom ID
   b. Transaction Type- Subscription order PRVO
   c. Contract Account
3. Execute a mass run
4. Check subscription order and subscription contract creation

In the scenario B subscription order / contract is created without reference to a master agreement.

1. The data is entered as a CSV File and contains business partner data, such as sold to party and ship to party and the corresponding contract account and corresponding technical resource
2. Add the following data: *Enter Product Data*
   a. Enter the transaction type - *Subscription Order PRVO Execute Mass Run*
   b. Check the subscription order and subscription contract creation
3. Execute a mass run
4. Check the subscription order and subscription contract creation

In scenario C, a mass run for contract change, a cancellation is shown

1. The data entry consists in subscription contract IDs
2. Enter an activation date and cancellation reason

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3. Execute a mass run
4. Check change order creation

Please consider that step 1, data entry from scenario A and B can be done without a CSV File, manually.

Header Status	Description	Allowed Activities
Open	Initial status	▪ Can be edited ▪ Not observed by execution jobs ▪ Cannot be started ▪ Can be manually switched to *Released for Processing*.
Released for Processing	Can only be set manually	▪ Can no longer be edited ▪ Status can be switched back to *Open* ▪ Execution be started by choosing *Run* ▪ Mass run is observed by execution jobs
In Process	Set automatically once the elements are processed manually or by an automatic job	▪ Mass run cannot be changed ▪ Transient status: Status switches to *Processed Successfully* or *Processed with Errors* after processing ▪ Mass run display has to be refreshed by choosing *Update* after processing
Processed Successfully	Set automatically after processing Prerequisite: All elements in status Processed successfully or Processed Manually.	▪ Final status
Processed with Errors	Set automatically after processing Prerequisite: At least one element in status Processed with Errors	▪ Elements with errors can be manually adjusted or transferred to a follow-up mass run ▪ Once all elements have been manually adjusted to *Processed Successfully* or taken over into a new mass run, the status switches to *Processed Successfully*.

Figure 247: Mass Run Data - Header Status

- Mass runs consist of a general header data, a mass run category and a type of specific header and element data
- Mass run ID, identifies a mass run and is taken from a number range in the customizing
- Earliest execution date for this mass run, observed by the report if an automatic job is used. It is not evaluated if the execution is started by choosing run
- Mass run type defines the behavior of the mass run (activity type)

Mass Run Process Data:

- Create without a master agreement, supports, product, employee, contract start, and transaction type.
- Create with master agreement, in addition it also supports, master agreement id, custom product ID, and sharing contracts, together with several partners.
- Correctness of entered process data can be checked. On process header level all fields are maintained, which are shared between all orders / contracts within a mass run.

Process data for mass runs depends on the mass run category and type:

- Create without master agreement support product, employee, contract start and transaction type
- Create with master agreement, in addition, support master agreement id, custom product ID and sharing contracts, together with several partners

Correctness of entered process data can be checked. On process header level all field are maintained which are shared between all orders / contracts within a mass run.

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Element Status	Description	Allowed Activities
Open	Initial status of an element	▪ Element can be edited (according to the field settings), but the status cannot be changed. ▪ Element can be deleted. ▪ Element can be checked. ▪ Element will be processed when mass run is processed.
Successfully Processed	Element has been successfully processed without any errors being reported	Final status.
Processed with Errors	Errors occurred during processing	▪ Messages can be displayed choosing Show Messages in the Actions column ▪ Element can be reprocessed, transferred to a new mass run or status can be set to Processed Manually.
Processed Manually	Final status that can only be set manually	▪ Can be set if the contract of an element, processed with errors, has been corrected manually

Figure 248: Mass Run Process Data: Elements - Status

- The element information can either be entered via search, .csv file upload, or using a predefined selection
- Details are displayed on a separate page. Click details.
- To avoid performance problems and enable high-volume processing, you will find another search screen to limit the number of displayed contracts.
- To verify the entered information choose *Check All*

Use Background Processing

Mass runs can be executed in two ways:

1. Execute a mass run in dialog mode on the UI
2. Schedule a mass run to be executed at a certain point of time.

Mass run scheduling requires the report *CRM_ISX_MR_PROCESSING* to be scheduled periodically as a background job (transaction SM36).

General selection criteria:

- Planned execution date is earlier or the same as the time at which the report is executed
- Current mass run status is *Released for Processing*

Additional optional selection criteria (to be specified during report execution):

- Mass run type priority:
- For execution order the selected set of mass runs is sorted descending by priority from the mass run header data

Mass Run Error Monitoring

In the elements table (*Create*) or in details section (*Change*)you can find details about the mass run results. For **successfully** processed elements, order / contract ID are displayed. For **not successfully** processed elements, you can check the details by choosing *Show Messages*. A pop-up window opens and provides details about the error.

Mass Creation – Processing in Dialog Mode

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- After data entry the mass run status can be set to *Released for Processing*
- Executing the mass run in dialog mode, by choosing *Run Processing*, will start to observe the current processing status. Choose *Update*
- The summary provides an overview about the status of the elements
- The header status is set as summary of the elements' status
- The results for each elements are shown in the *Elements* table in column *Status*

LESSON SUMMARY
You should now be able to:

- Detail Mass Runs in SOM

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APIs for Mass Run

LESSON OBJECTIVES
After completing this lesson, you will be able to:

- Know the API for Mass Run

Know the API for Mass Run

SAP delivers an API (Function Module) to create mass runs:

CRMS4_SOM_MR_CREATE

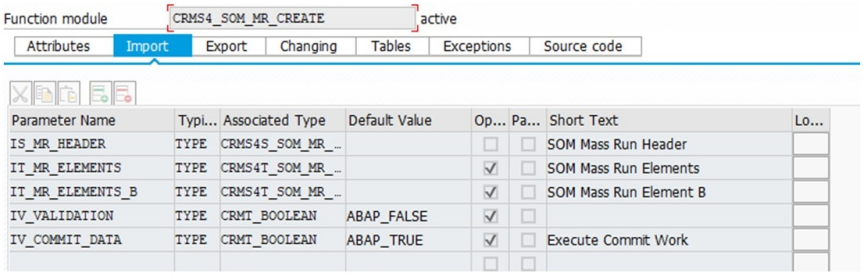

Figure 249: API for Mass Run

With the API *CRMS4_SOM_MR_CREATE* you are able to create Mass Runs, the same way as using the API for order and contract creation.

LESSON SUMMARY
You should now be able to:

- Know the API for Mass Run

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Learning Assessment

1. Which of the following tasks can mass runs be used for in Subscription Order Management (SOM)?

   Choose the correct answers.

   - [ ] A Creating a subscription order.
   - [ ] B Creating a subscription contract.
   - [ ] C Creating an invoice.
   - [ ] D Canceling a subscription contract.

2. Mass runs in SOM are created with the help of which of the following Fiori apps?

   Choose the correct answer.

   - [ ] A *Manage Master Data*
   - [ ] B *Manage Mass Runs*
   - [ ] C *Manage Subscription Order*
   - [ ] D *Manage Subscription Contract*

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Learning Assessment - Answers

1. Which of the following tasks can mass runs be used for in Subscription Order Management (SOM)?

   Choose the correct answers.

   - [X] A Creating a subscription order.
   - [X] B Creating a subscription contract.
   - [ ] C Creating an invoice.
   - [X] D Canceling a subscription contract.

   That is correct! Mass runs can be used for all of the listed tasks with the exception of creating an invoice.

2. Mass runs in SOM are created with the help of which of the following Fiori apps?

   Choose the correct answer.

   - [ ] A *Manage Master Data*
   - [X] B *Manage Mass Runs*
   - [ ] C *Manage Subscription Order*
   - [ ] D *Manage Subscription Contract*

   That is correct! Mass runs in SOM are created with the help of the *Manage Mass Runs* app.

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UNIT 16 APIs

Lesson 1

API 275

UNIT OBJECTIVES

- Understand the Order and Contracts API
- Understand the Subscription Contract API
- Understand the Product Master API
- Understand the Solution Quotation API
- Understand how the contract lifecycle can be managed using the API
- Understand how contact changes are done using the API

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Unit 16
Lesson 1

API

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Understand the Order and Contracts API
- Understand the Subscription Contract API
- Understand the Product Master API
- Understand the Solution Quotation API
- Understand how the contract lifecycle can be managed using the API
- Understand how contact changes are done using the API

Order and Contract APIs

Integration Architecture

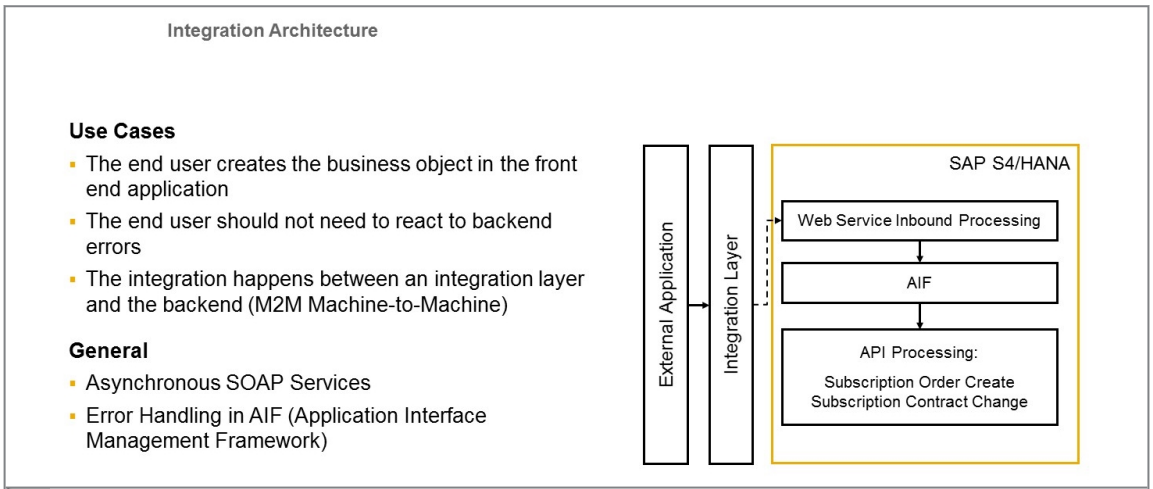

Figure 250: Integration Architecture

The subscription order Application Programming Interface (API) provides another option in subscription order and contracts maintenance, in addition to the existing Web-based UIs (Back Office and Fiori Launchpad).

The SAP Application Interface Framework is used to monitor interfaces as well as execute error handling. Under the /CMSOM namespace, the SOAP APIs can be monitored. You can find more information about AIF in the SAP Help Portal.

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SubscriptionOrderRequest_In

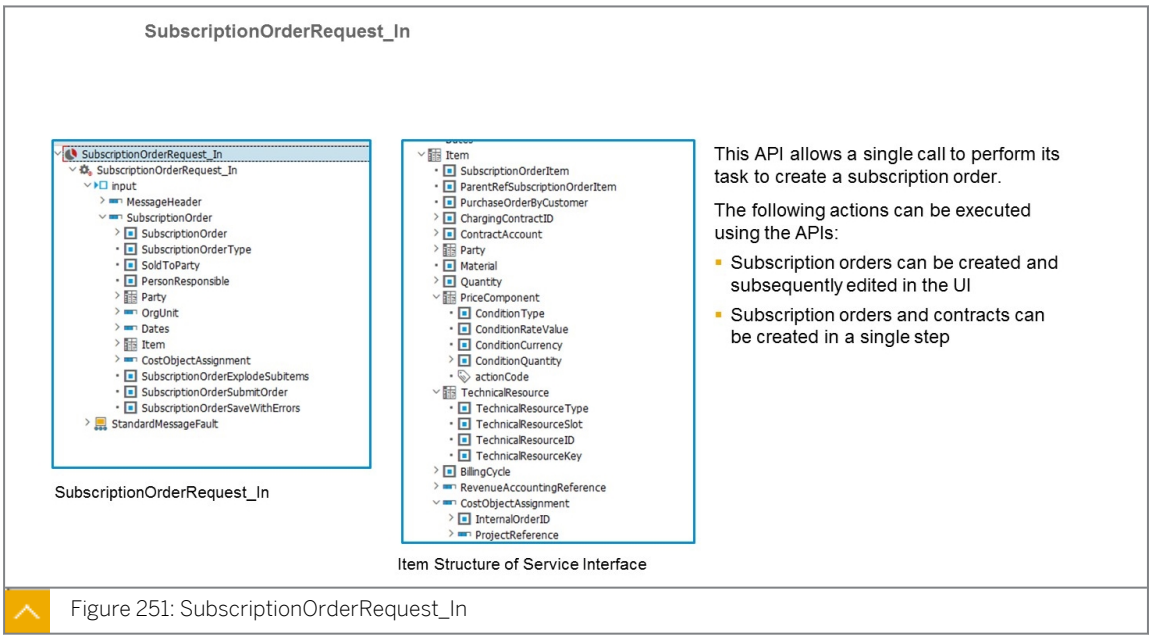

Figure 251: SubscriptionOrderRequest_In

You can open the service with transaction `SPROXY`.

To create subscription orders (with contracts), the following SOAP web services can be used:

- Create a subscription order and contract with a referenced order:

  Service Provider: SubscriptionOrderRequest_In

  Namespace: http://sap.com/xi/S4CRM/

The SOAP Interface has the following basic components:

- General data: This is primarily the header data for the order or contract, or contract accounts.
- Alternative partner data: You can define alternative partners at the item level, such as the partner or the sales organization.
- Item data: All data required to create an item, for example, products and dates (UTC).
- Cost Account Assignment: You can define the accounting objects at the header or item level.
- Control Flags for Provider Order API: Control indicators are used to control the creation process. For example, if the order must be saved to the database, you must set the Save and Commit indicators.
- The interface of the Web service delivered with S4HANA 1809 FSP01 was extended with some additional attributes:
  - Variant configuration
  - Master agreement
  - Sharing of contracts and groups

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- Partner settlement rule

  The interface is still compatible with the 1809 FSP01 delivery, that is, there are no changes to the existing interface structure, just new attributes in addition to the existing ones.

The SOAP Service has no response. Errors are tracked in AIF.

Control Parameters:

- SubscriptionOrderRequest_In
  - SubscriptionOrderExplodeSubitems

    Specifies if the receiving backend performs a package explosion on bundle items or not. This allows the suppression of the explosion in cases where the sending system is sending the full package structure.
  - SubscriptionOrderSubmitOrder

    Specifies if the created subscription order is submitted immediately, resulting in a subscription contract.
  - SubscriptionOrderSaveWithErrors

    Specifies if the subscription order is persisted even if it contains technical or functional errors. This depends on the intended error handling behavior. If set to false, the issues from this asynchronous webservice call are only visible in the AIF monitoring transaction.
- SubscriptionContractChangeProcessReleaseChange
  - SubscriptionContractChangeProcessReleaseChange

    Specifies whether the "release" step of the BTMF change process framework is triggered.

SubscriptionContractChangeProcessRequest_In

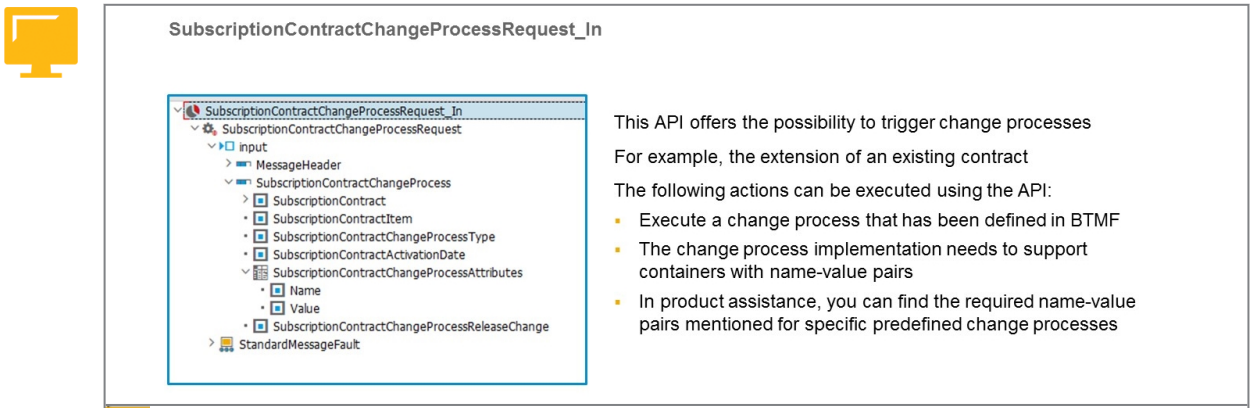

Figure 252: SubscriptionContractChangeProcessRequest_In

You can open the service with transaction `SPROXY`.

To change subscription contracts using change processes, the following SOAP web service can be used:

- Service Provider: SubscriptionOrderRequest_InAIF

  Namespace: /CMSOMAIF

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Interface Name: SBRO_R_IN

- Service Provider: SubscriptionContractChangeProcessRequest_InAIF

  Namespace: /CMSOMAIF

  Interface Name: SBRC_R_IN

The SOAP Interface has the following basic components:

- SubscriptionContract, SubscriptionContractItem: The subscription contract document ID and the subscription document Item.
- SubscriptionContractChangeProcessReleaseChange: A control parameter that allows you to control if the "release order" step of the BTMF change process framework is triggered.
- SubscriptionContractChangeProcessType: The name of the change process as defined in customizing for the BTMF framework.
- SubscriptionContractActivationDate: A timestamp for activating the changes (UTC).
- SubscriptionContractChangeProcessAttributes: A table of name-value pairs for passing to process the implementation.

You must set up the Business Transactions Maintenance Framework (BTMF) as required. You have to define and implement the relevant BTMF process. A process that can be used by the API must ensure that all required input data is provided by the API interface. A generic collection of name-value pairs is available for this. This data is accessible from within the process using a global data container (BTMF container).

Requirements and Limitations

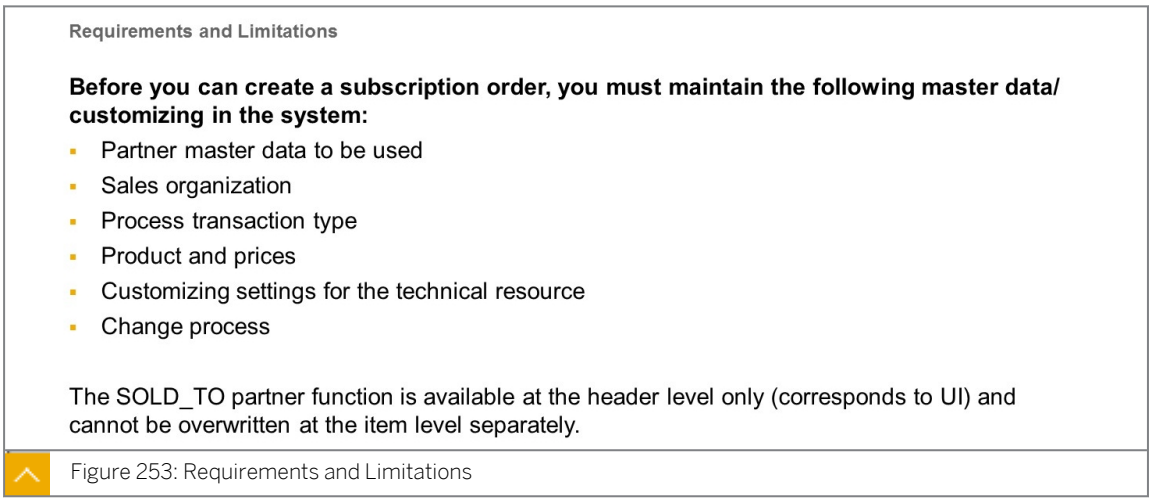

Figure 253: Requirements and Limitations

Enhancement:

Field extension can be done using SAP S4/HANA Extensibility Tools, to extend the SOAP service. The service must be enabled in the tool. For more information, see the product assistance for the *UI Technology Guide for SAP S/4HANA → Extensibility*.

In addition, the enhancement spot CRM_ISX_BTX_API is created to give the customer the option to define the behavior of fields that are not part of the API interface. For more information, see the description of the BAdI.

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Subscription Contract API

Key Facts About the Subscription Contract API

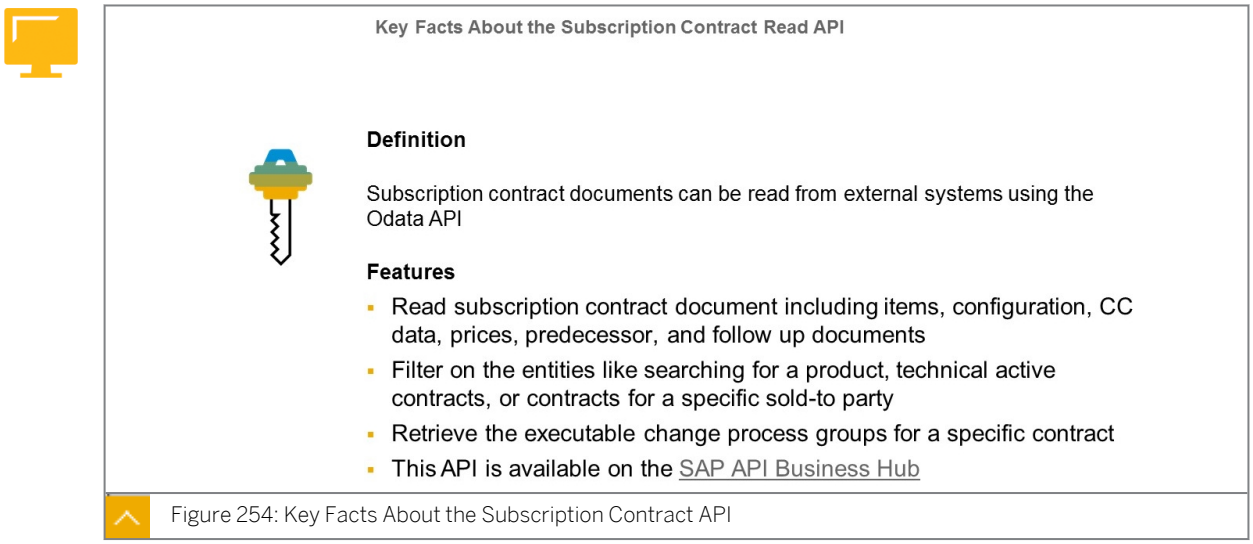

Figure 254: Key Facts About the Subscription Contract API

This API is available on the SAP API Business Hub.

OData Service API_SUBSCRIPTIONCONTRACT

Figure 255: OData Service API_SUBSCRIPTIONCONTRACT

Segments in the Subscription Contract API:

- Contract Header
  - SubscriptionContractHeader
  - SubscriptionContractText
- Contract Item

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- SubscriptionContractItem
- SubscrpnContrItmText

- Configuration
  - SubscrpnContrItmConfign
- Price
  - SubscrpnContrItmPrElmnt
- Technical Resource
  - SubscrpnContrItmTechRsce
- CC Parameters
  - SubscrpnContrItmDteParam
  - SubscrpnContrItmStrParam
  - SubscrpnContrItmNmrcParam
  - SubscrpnContrItmTblParam
- Predecessor Solution Quotation / Subscription Order
  - SbscrContrPrdcssrBusSolnQtn
  - SbscrContrItmPrdcssrQtnItm
  - SbscrContrPrdcssrSubscrpnOrd
  - SbscrContrItmPrdcssrOrdItm
- Follow Up Subscription Order (Change Order)
  - SbscrContrFUPSubscrpnOrd
  - SbscrContrItmFUPOrdItm

Searching for Contracts and Retrieving Allowed Change Process Groups

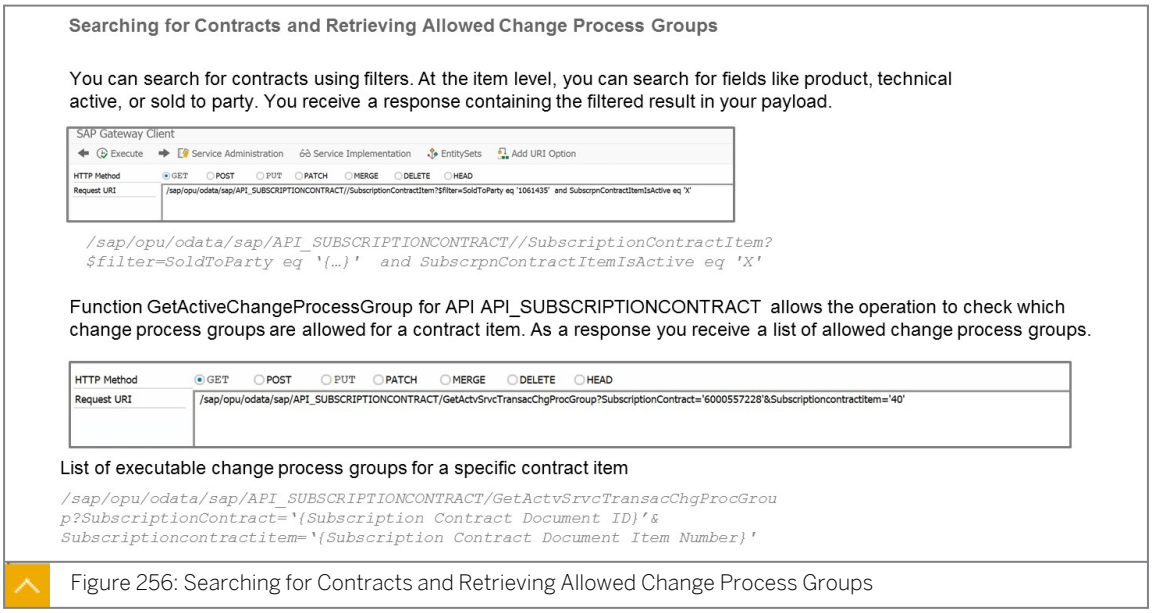

Figure 256: Searching for Contracts and Retrieving Allowed Change Process Groups

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Product Master API

Key Facts About the Product Master Replicate API

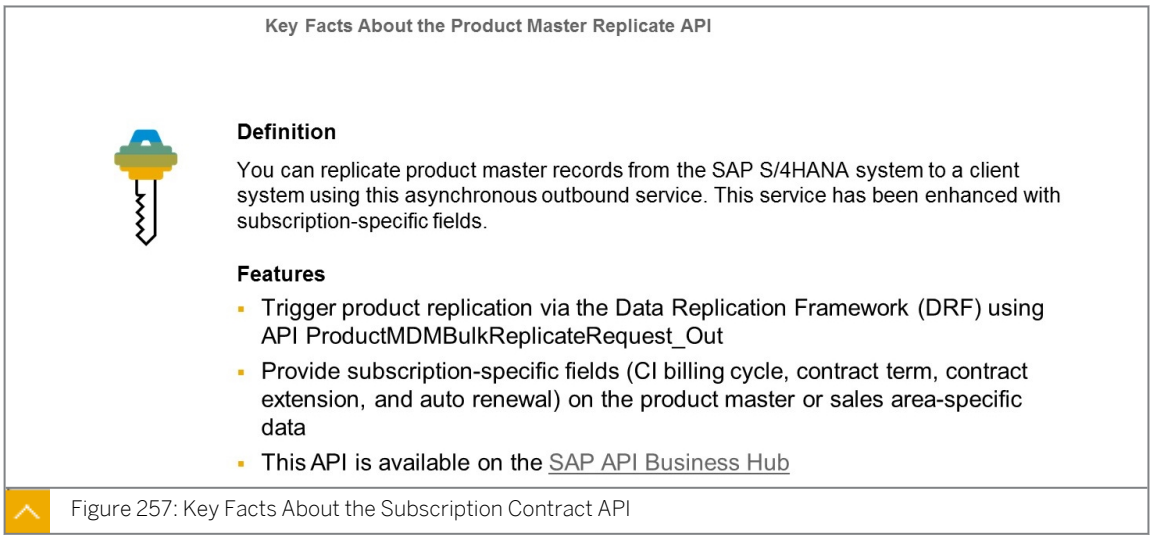

Figure 257: Key Facts About the Subscription Contract API

Use Cases:

- Migration
- Creating product data in another application and updating the subscription specific data for such products

General Notes:

- Classes that can be used in custom specific implementations
- A prerequisite is that the product master already exists in the system including the required distribution chain data

This API is available on the SAP API Business Hub.

Product API Classes

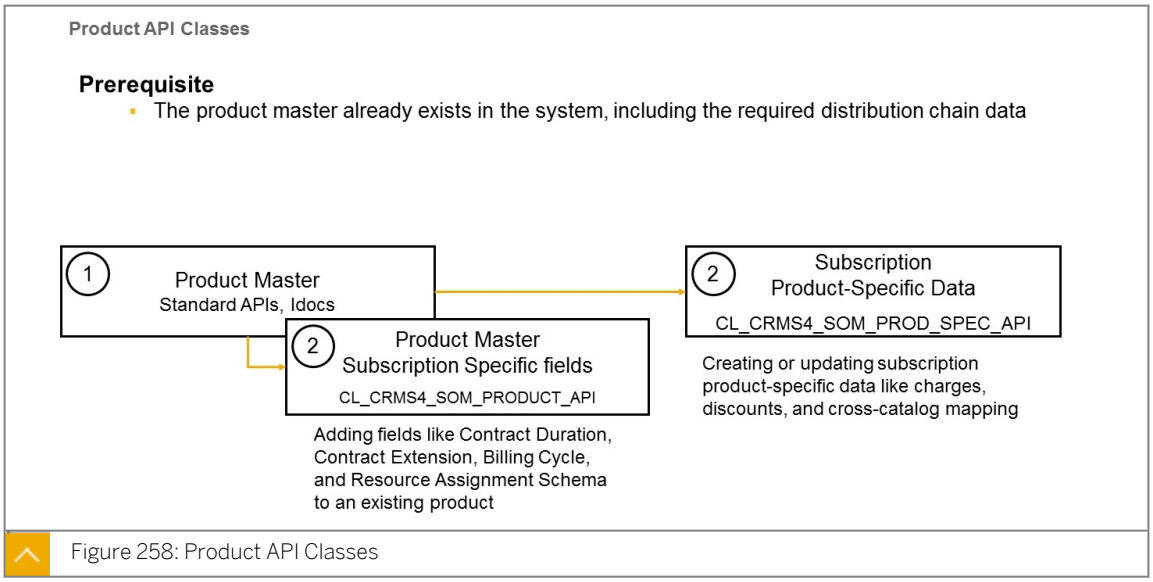

Figure 258: Product API Classes

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Subscription Fields of the Product Replicate API

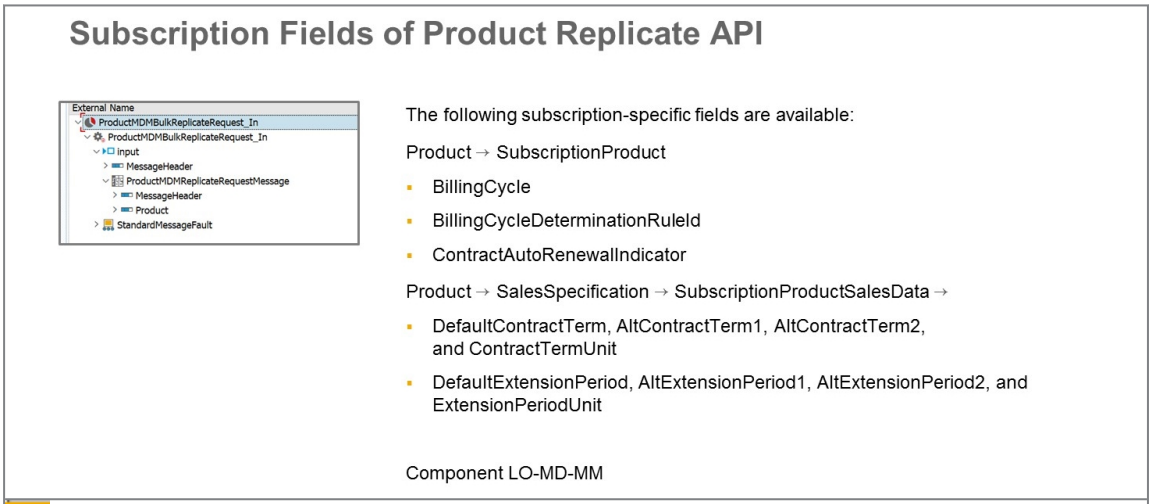

Figure 259: Subscription Fields of the Product Replicate API

Solution Quotation API

Key Facts About the Solution Quotation API

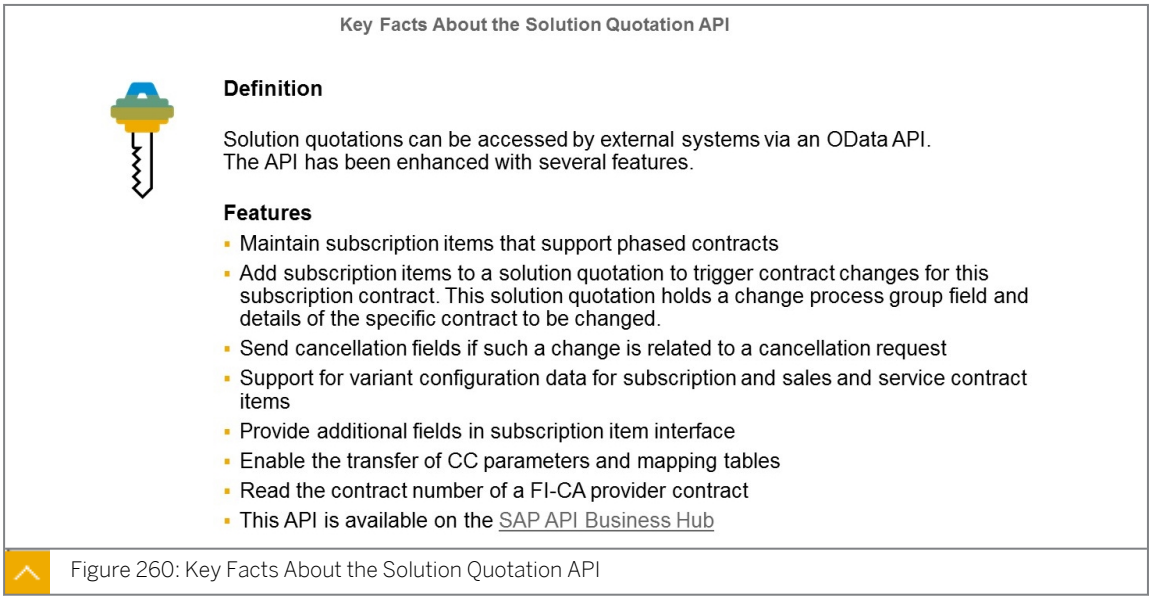

Figure 260: Key Facts About the Solution Quotation API

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OData URL and Sample Payloads

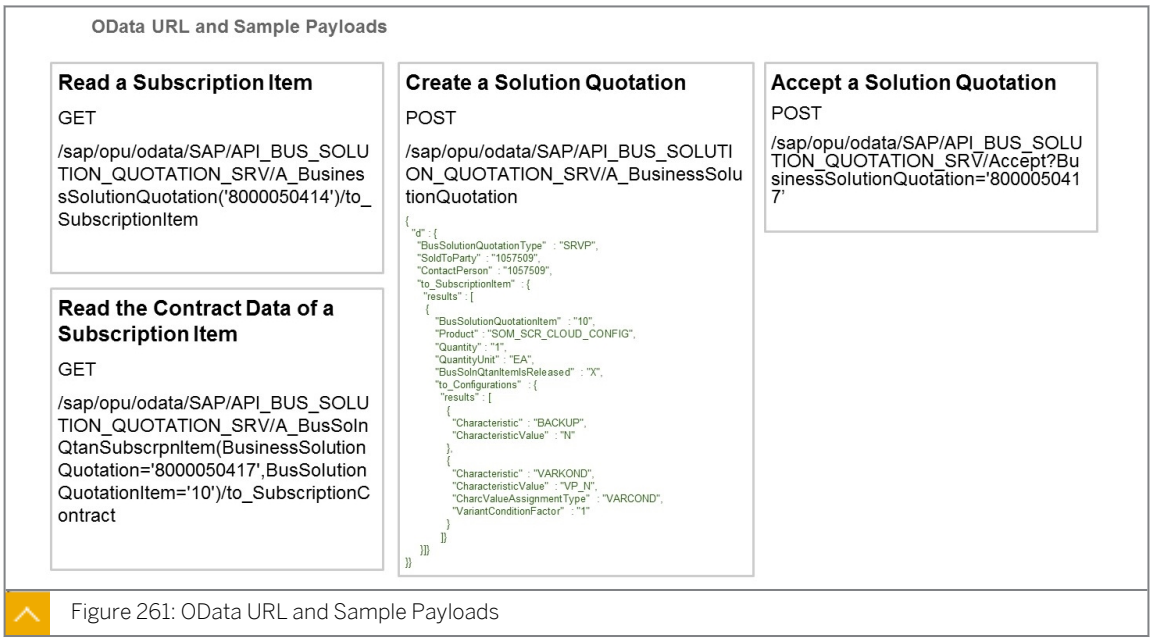

Figure 261: OData URL and Sample Payloads

OData Service Important Segments for Subscriptions

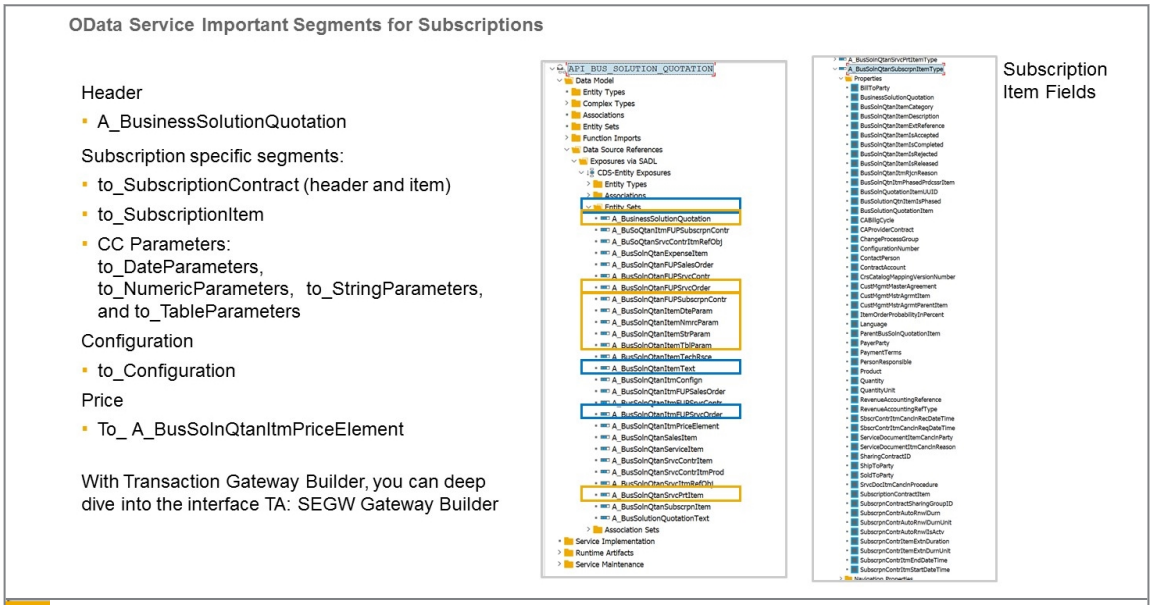

Figure 262: OData Service Important Segments for Subscriptions

New Additional Fields:

- Read Contract Number of FI-CA Provider Contract: SubscriptionContractExternalID
- CC Parameters (Number, Date, String) and Mapping Tables (CRUD), CCM Version Number required for CC Parameter
- Master Agreement (CRUD)
- Sharing Group (CRUD)

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- Auto Renewal (CRUD)
- Configuration Data for Subscription, Sales and Service Contract Items (CRUD)
- Phased Contract: Predecessor item number for previous phase (item number within Solution Quotation), Mark item as a phased item, Start and end of time slice for an item representing one phase
- Change Process Group: FICA provider contract number and item number from subscription contract document, Change process group
- Cancellation Change Process: Cancellation party, reason and procedure, Cancellation receive date, Cancellation request date

Update / Delete Limitations for Configuration

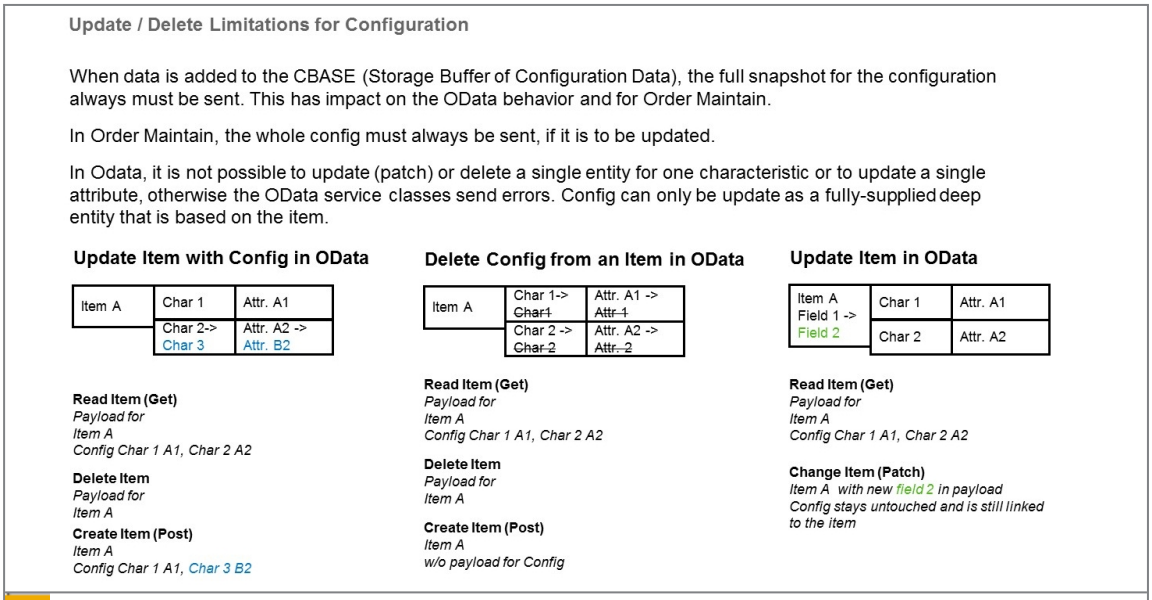

Figure 263: Update / Delete Limitations for Configuration

Variant Configuration and Pricing

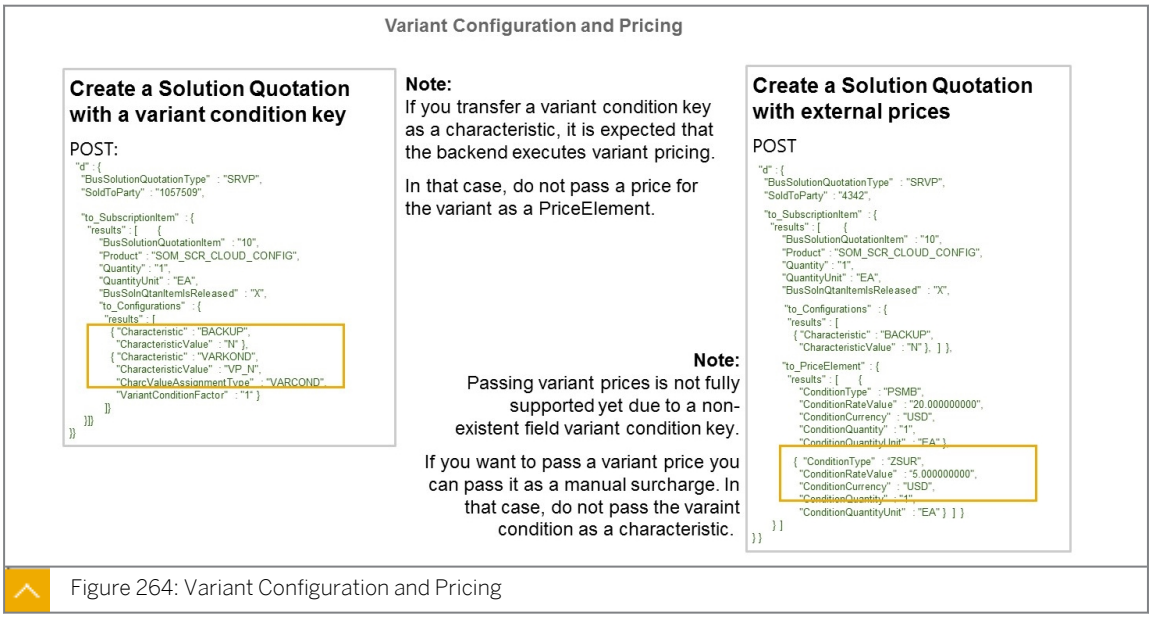

Figure 264: Variant Configuration and Pricing

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Contract Life Cycle Using the API

Key Facts About Change Process Group and the Change Process Group API

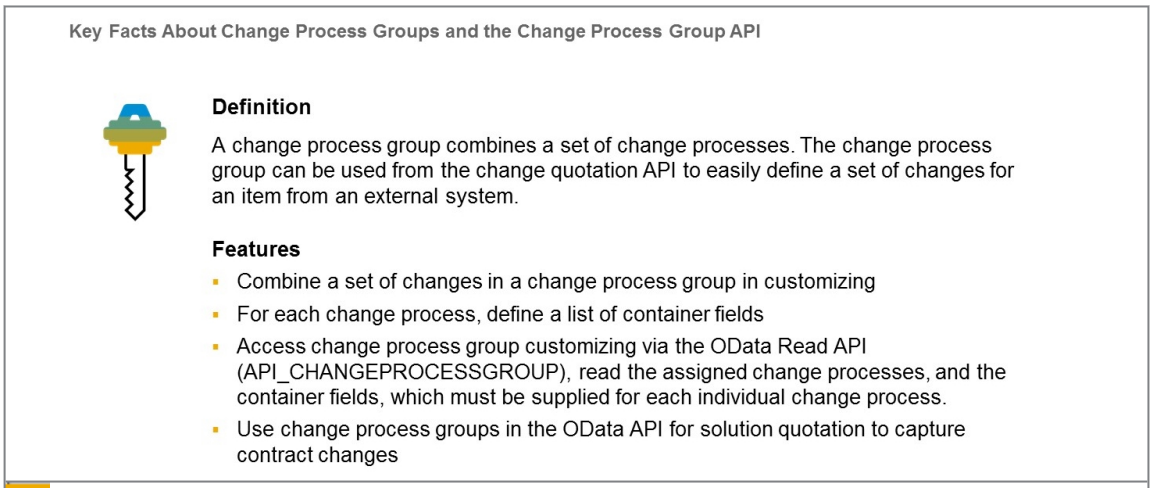

Figure 265: Key Facts About Change Process Group and the Change Process Group API

This API is available on the SAP API Business Hub.

Contract Lifecycle Using the API from an External Application

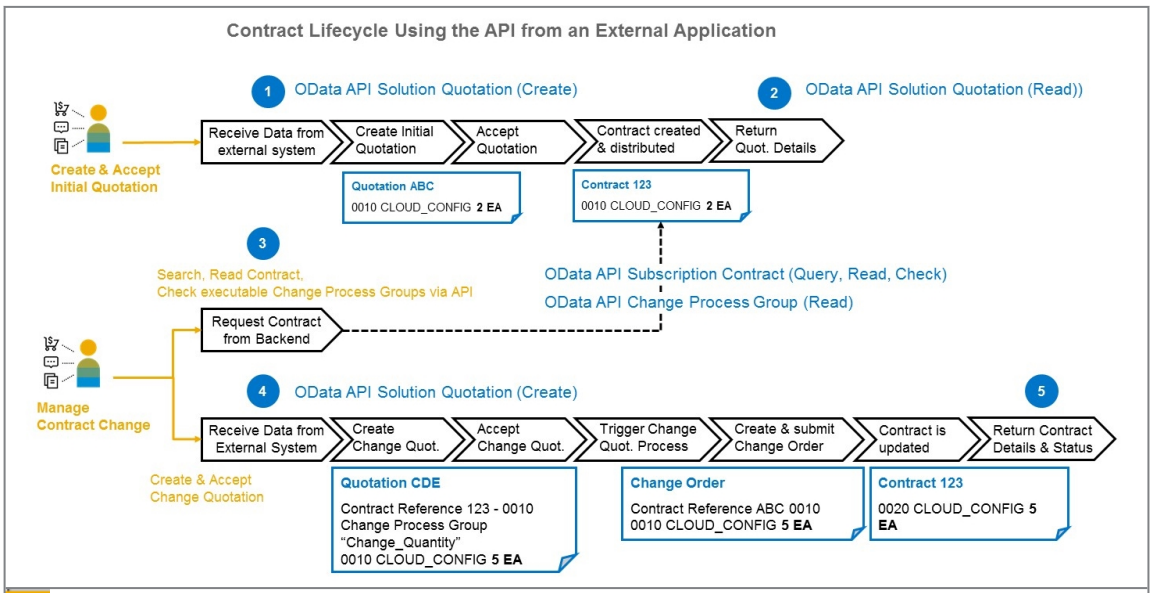

Figure 266: Contract Lifecycle Using the API from an External Application

The sequence of steps is as follows:

- Steps 1 and 2 create the initial quotation and return the quotation status and details.
- Step 3 searches and reads the contract.
- Steps 4 and 5 process and changes the contract and returns the contract status and details.

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BTMF Customizing Change Process Group

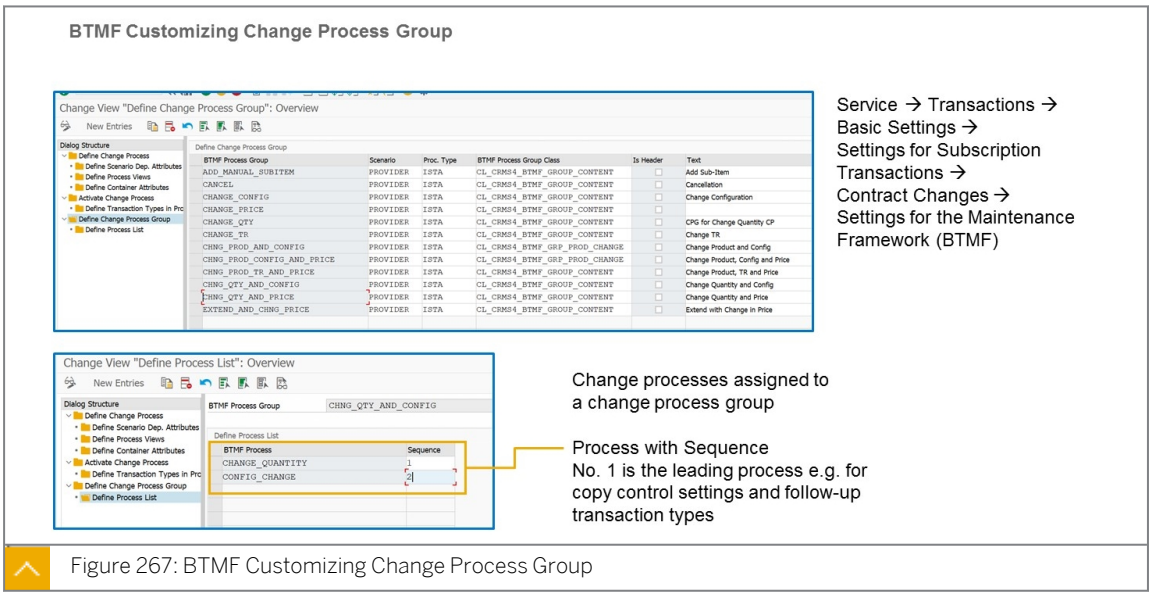

Figure 267: BTMF Customizing Change Process Group

Definition

Change process group customizing defines which changes processes are bundled together in group and are therefore executed together. The change process group API exposes these settings to external applications.

Features

- Get all change process groups defined in customizing using the API.
- Retrieve change processes assigned to the group and the fields that must be supplied to trigger a change process.
- This API is available on the SAP API Business Hub.

Service Name: API_CHANGEPROCESSGROUP

- Change Process Group
  - ChangeProcessGroup
- Change Process group Texts
  - _ChgProcGrpText
- Grouped Change Processes
  - _GrpChangeProcess
- Change Process Attributes
  - _ProcAttributes

Sample URI 1: Get the change processes inside a change process group

```
/sap/opu/odata4/sap/api_changeprocessgroup/srvd_a2x/sap/
ChangeProcessGroup/0001/
ChangeProcessGroup(SrvcTransChangeProcessGroup='CHANGE_QTY')/
_GrpChangeProcess
```


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Sample URI 2: Get the Change Process Attributes for a specific change process

```
/sap/opu/odata4/sap/api_changeprocessgroup/srvd_a2x/sap/
ChangeProcessGroup/0001/
ChangeProcessGroup(SrvcTransChangeProcessGroup='CHANGE_QTY')/
_GrpChangeProcess(SrvcTransChangeProcessGroup='CHANGE_QTY',SrvcTransCha
ngeProcess='CHANGE_QUANTITY')/_ProcAttributes
```

BTMF Customizing Change Process Container Field Mapping

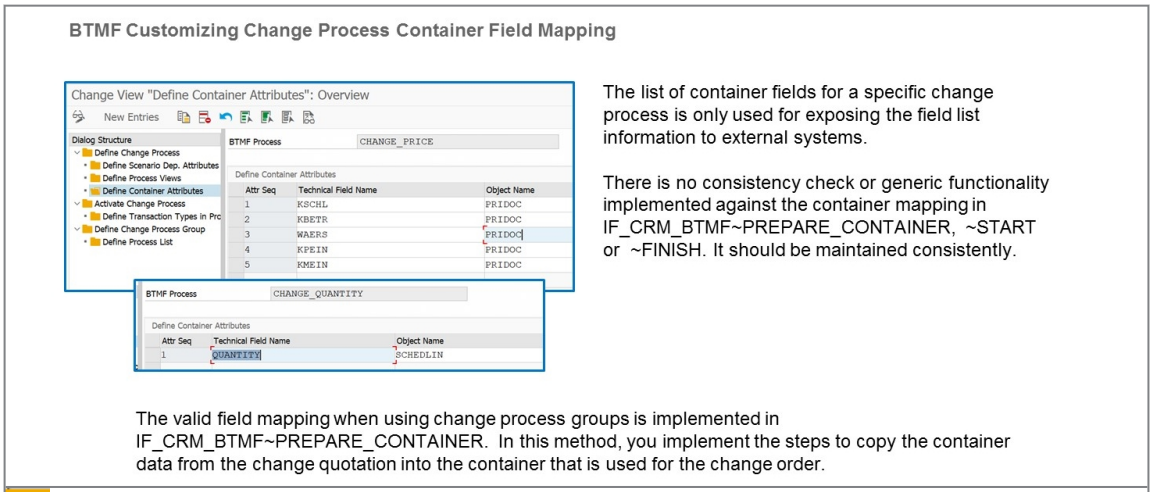

Figure 268: BTMF Customizing Change Process Container Field Mapping

Change Process Group Read API

Key Facts About Process Contract Changes Using the API

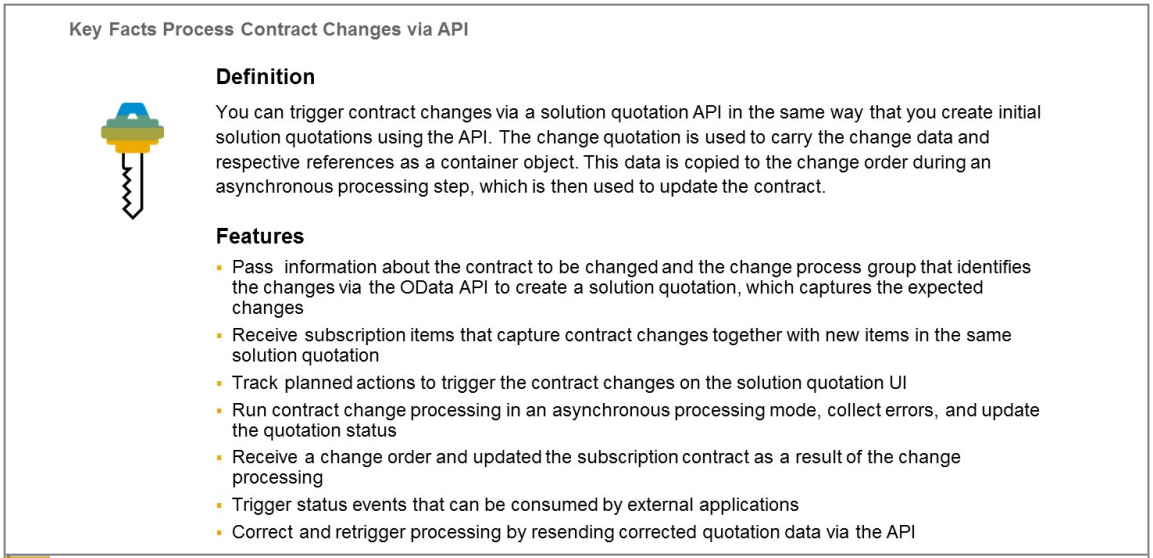

Figure 269: Key Facts About Process Contract Changes Using the API

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Detail Steps During Change Process Group Background Processing

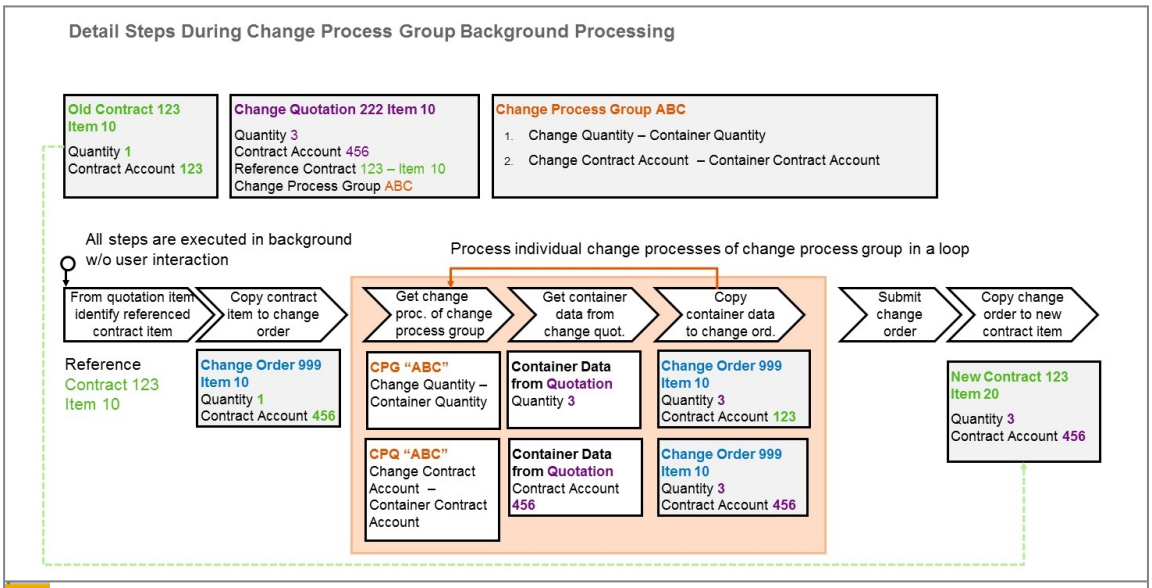

Figure 270: Detail Steps During Change Process Group Background Processing

Trigger Change Quotation Processing

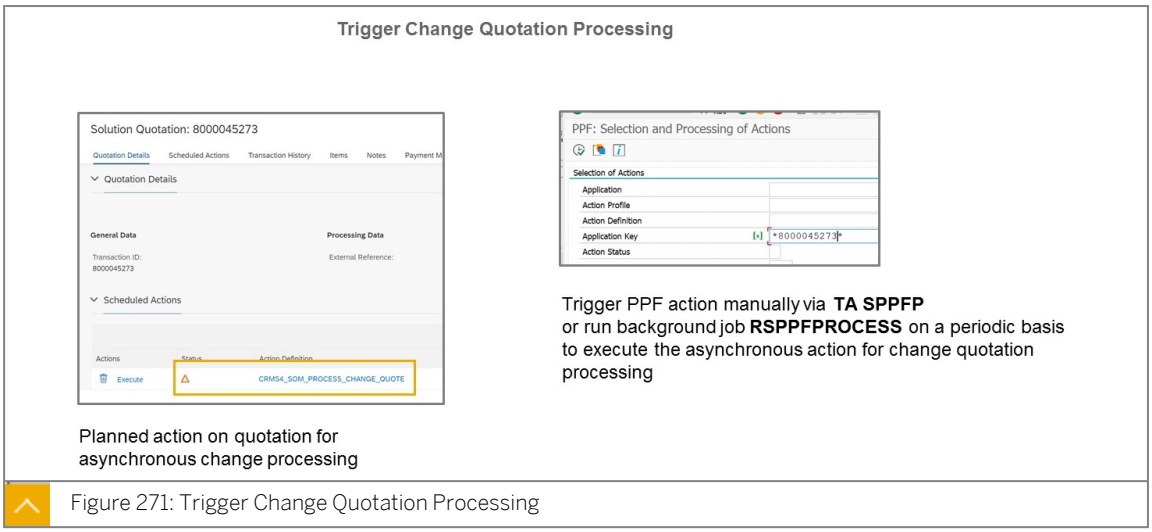

Figure 271: Trigger Change Quotation Processing

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Receive Updated Contract

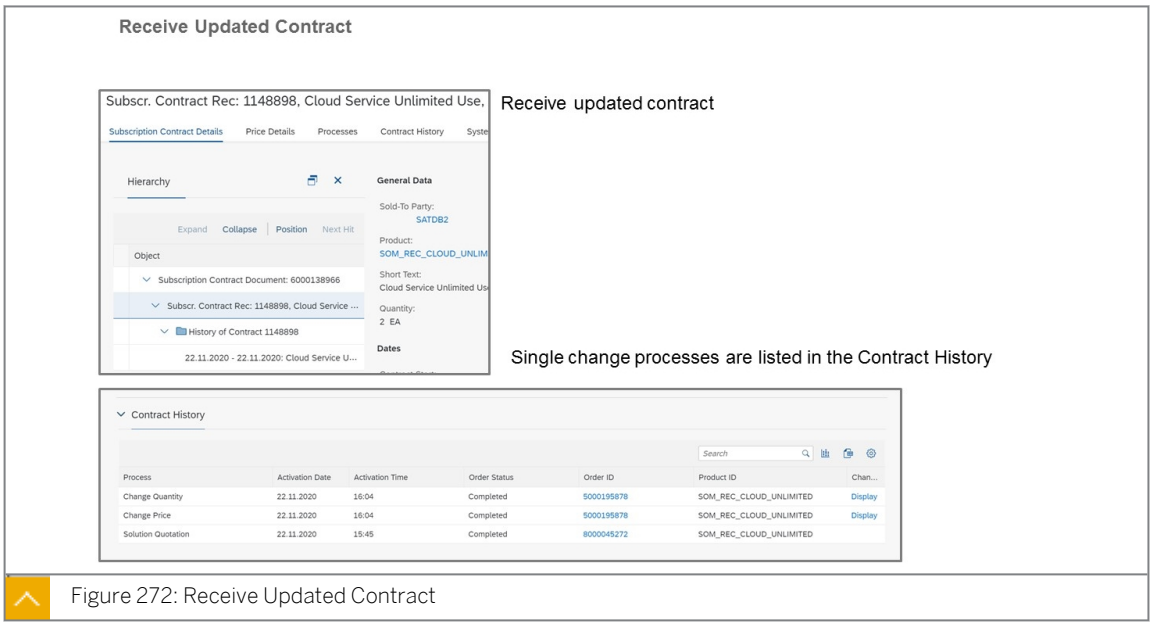

Figure 272: Receive Updated Contract

Status Handling Change Quotation and Reprocessing in the Case of Error

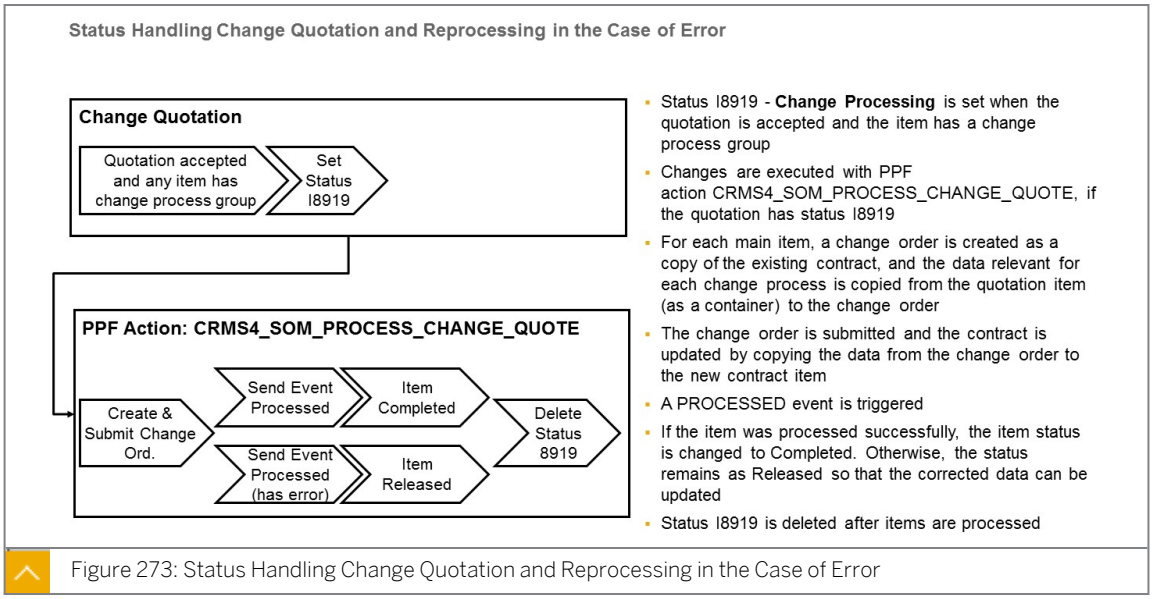

Figure 273: Status Handling Change Quotation and Reprocessing in the Case of Error

Event

- After processing a main item, an event is raised to signify whether the item and its sub-items were processed successfully or if an error has occurred
- You can register a receiver for this event in transaction `SWE2`.
- Object Category: CL – ABAP Class

  Object Type: CL_CRMS4_SOM_CHNGQUOTE_BEH

  Event name: PROCESSED (with parameter has error)

Log

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- A log of the complete processing can be found in the quotation for the executed PPF action.
- Additionally, it is stored separately for each main item in the Business Application Log (BAL), under object SOM, sub-object QCP, and quotation item GUID as the external ID.

LESSON SUMMARY

You should now be able to:

- Understand the Order and Contracts API
- Understand the Subscription Contract API
- Understand the Product Master API
- Understand the Solution Quotation API
- Understand how the contract lifecycle can be managed using the API
- Understand how contact changes are done using the API

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Learning Assessment

1. Which of the following objects can be created using the SOM API?

   Choose the correct answers.

   - [ ] A subscription order
   - [ ] B subscription solution
   - [ ] C master agreement
   - [ ] D subscription contract
   - [ ] E product bundles

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Learning Assessment - Answers

1. Which of the following objects can be created using the SOM API?

   Choose the correct answers.

   - [X] A subscription order
   - [X] B subscription solution
   - [ ] C master agreement
   - [X] D subscription contract
   - [ ] E product bundles

   That is correct! You can use the SOM API to create a subscription order, a subscription solution, and a subscription contract. It is not possible to use the SOM API to create a master agreement or product bundles.

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